



**GREEN
CLIMATE
FUND**

Meeting of the Board

15 – 18 July 2024

Songdo, Incheon, Republic of Korea

Provisional agenda item 10

GCF/B.39/02/Add.11

24 June 2024

Consideration of funding proposals – Addendum XI

Funding proposal package for FP236

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Basin Approach for Livelihood Sustainability through Adaptation Strategies (BALSAS)";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Funding Proposal

Project/Programme title:	Basin Approach for Livelihood Sustainability through Adaptation Strategies (BALSAS)
Country(ies):	Mexico
Accredited Entity:	International Fund for Agricultural Development (IFAD)
Date of first submission:	[YYYY/MM/DD]
Date of current submission	[YYYY/MM/DD]
Version number	[V.001]



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

A. PROJECT/PROGRAMME SUMMARY				
A.1. Project or programme	Project	A.2. Public or private sector	Public	
A.3. Request for Proposals (RFP)	Not applicable			
A.4. Result area(s)	Check the applicable GCF result area(s) that the overall proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of GCF and Co-financiers' contribution devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financiers' contribution respectively.			
		GCF contribution	Co-financiers' contribution	
	Mitigation total	22 %	30 %	
	☐ Energy generation and access	Enter number %	Enter number %	
	☐ Low-emission transport	Enter number %	Enter number %	
	☐ Buildings, cities, industries and appliances	Enter number %	Enter number %	
	☒ Forestry and land use	22 %	30 %	
	Adaptation total	78 %	70 %	
	☒ Most vulnerable people and communities	6 %	35 %	
	☒ Health and well-being, and food and water security	18 %	12 %	
	☐ Infrastructure and built environment	Enter number %	Enter number %	
☒ Ecosystems and ecosystem services	54 %	23 %		
A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)	3 926 876 CO _{2eq} (in 20 years) ¹	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)	3'309 211 total beneficiaries	
			201 867 Direct beneficiaries	3' 107 344 Indirect beneficiaries
			6.5% of the population of prioritized watersheds and indigenous territories	9.25% of the population of the Balsas Project Area
A.7. Total financing (GCF + co-finance ²)	90.953 million USD	A.9. Project size	Medium (Up to USD 250 million)	
A.8. Total GCF funding requested	39.5 million USD			
A.10. Financial instrument(s) requested for the GCF funding	Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8. ☒ Grant 19,500,000 ☐ Equity <u>Enter number</u> ☒ Loans 20.000.000 ☐ Results-based payment <u>Enter number</u> ☐ Guarantee <u>Enter number</u>			

A.11. Implementation period	8 years¹	A.12. Total lifespan	20 years
A.13. Expected date of AE internal approval	<i>This is the date that the Accredited Entity obtained/will obtain its own approval to implement the project/programme, if available.</i> <u>Click or tap to enter a date.</u>	A.14. ESS category	<i>Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.</i> B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	<i>Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.</i> Yes ☒ No ☐		
A.20. Executing Entity information	National Forestry Commission of México (CONAFOR) Nacional Financiera, S.N.C., Banca de Desarrollo (NAFIN) Ministry of Finance (HACIENDA)		

¹ The total project duration is 8 years, with GCF proceeds entering the project at year 3. Hence the total duration of the GCF Loans and grant is 6 years.

A.21. Executive summary (max. 750 words, approximately 1.5 pages)

The Balsas Basin is one of the two regions in Mexico with the highest level of climate vulnerability. This is exacerbated by high levels of poverty and marginalization across the rural population. These conditions are worsened by negative dynamics which are eroding vulnerable rural livelihoods and productive systems, which have yet to adapt to climate variability and change. Severely increasing temperatures, anomalous and more intense rainfall seasons, and longer dry seasons as a result from Climate Change and the El Niño Southern Oscillation (ENSO) continue to aggravate social vulnerability in the Balsas Basin by reducing the productivity of agricultural and natural resource-based livelihoods, and adversely affecting already dwindling ecosystem services. With the length of the dry season anticipated to increase, areas of the Basin with high annual precipitation will experience shorter but more intense rainfall events. This implies that flooding, crop damage and soil erosion will be compounded in the rainy season while extended droughts may be more frequent during the rest of the year. Furthermore, in already arid and semi-arid zones, the lengthening of the dry season will imply a decrease in the productivity of the vegetation.³ Finally, increasingly dry soils will take longer to recover their humidity, so growing seasons will shorten, reducing crop yields.⁴ If landscapes and ecosystems are not restored and their capabilities to buffer these extremes improved, the positive changes in the wet season do not carry over to the dry season because excess water is not stored.⁵ Addressing these problems is, however, restricted because currently local capacities are insufficient and institutional modalities do not consider climate change adaptation criteria in prioritizing forest landscape restoration incentive payments in the most vulnerable areas and communities.

This cross-cutting project (80% adaptation, 20% mitigation) aims to increase the climate resilience of the most vulnerable communities and ecosystems, and reduce emissions from production systems in the Balsas Project Area, via: i.) operationalizing the use of climate change criteria in current institutional modalities for the prioritization of rural development incentives through a community-centric bottom-up approach; ii.) investing in the climate-sensitive management of ecosystems, and climate-resilient productive systems and value chains for adaptation and mitigation benefits; and iii.) supporting a climate information and risk management system, and the systematic management of knowledge and lessons learnt.

The objective of the project is to enable community-level investment in adaptation and mitigation initiatives, through the climate-focused transformation of rural development incentives for the conservation of forest resources and the strengthening of local institutions and capacities to bring under sustainable management, conservation or restoration 324,578 ha of land, contributing to Target 5 on the Updated Strategic Plan of the GCF for 2024-2027. This objective will also be achieved by the project's strong focus on bottom-up decision-making and value chain development, ensuring social inclusion, benefiting and empowering women, youth and indigenous peoples. The proposed project will shift current sectoral rural development support allocation systems used by rural development institutions to a new integrated territorial development approach which:

- Uses climate change-related criteria for selecting and prioritizing the most vulnerable beneficiaries, interventions, and sites.
- Strengthens local watershed governance and participation of women, youth, and indigenous peoples at the sub-basin level to create incentives, reinforce inter institutional coordination, and foster a bottom-up approach for the sustainable use, management, and conservation of crucial ecosystems and ecosystem services to be piloted in the Balsas Project Area and replicated at national level. As part of the project's bottom-up approach, the participation of Agrarian Communities and Producer Organizations in value chains of timber and non-timber forest products (NTFPs) will also be strengthened, with the aim of increasing the incomes of the most vulnerable households, thereby building their adaptive capacity.
- Disburses incentive payments for forest ecosystem and landscape conservation multi-annually under a progress-based payment system, specifically rewarding performance of adaptation and mitigation initiatives. The multi-annual approach and annual evaluation of progress was first piloted by CONAFOR for Payment for Environmental Services (PES) schemes only but will be replicated to other forestry development incentive schemes.

The National Forestry Commission of Mexico (*Comisión Nacional Forestal* CONAFOR), Nacional Financiera S.N.C., Banca de Desarrollo NAFIN and the Ministry of Finance (HACIENDA) are the Executing Entities, whereby CONAFOR will undertake all operational responsibility for the implementation of the Project and NAFIN and HACIENDA will only channel funds. NAFIN is accredited as Direct Access Entity (DAE) to the GCF and through its participation in this Project, it will strengthen its capacities and experience to prepare and execute GCF co-financed climate change adaptation and mitigation projects in the future. Thus, the Accredited Entity IFAD, due to its long-term presence in the region and its multi-year international experience with several similar projects with the GCF, wants to capitalize on its previous successful experiences with GCF projects and provide innovative solutions to this project and strengthen the capacities of the partner organizations involved.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

General context

1. The Balsas Basin (Hydrological-Administrative Region RHA IV) and part of the South Pacific Hydrological-Administrative Region (RHA V) adjacent to it (henceforth "Project Area" PA) encompasses the states of Guerrero, Jalisco, México, Michoacán, Morelos, Oaxaca, Puebla, and Tlaxcala, which together cover 695 municipalities in south-west Mexico. The predominant climate is warm subhumid, with an average annual temperature of 22 °C and a regional average annual rainfall of 1,027 mm, varying considerably depending on the orography. The highest rainfall occurs in the Sierra Madre del Sur and the Transverse Neo-volcanic System, while the Inter-montane depressions suffer from dry to arid climate. The rainy season occurs from June to September, which concentrates more than 80% of the annual precipitation⁶. The PA is one of the two regions in Mexico with the highest level of climatic vulnerability⁷. The Balsas basin has a hydraulic infrastructure of great relevance at the national level, especially for hydropower generation, which uses 78% of the water volume produced in the basin. The Balsas basin hosts the second largest dam in Mexico: "El Infiernillo", with a storage capacity of 12,500 hm³.

2. The population of the PA is vulnerable to the impacts of climate change because of its high levels of poverty and low social development relative to the rest of Mexico, which is reflected in significant deficiencies in health, food, education, social security, housing, drinking water and basic sanitation, as well as limited income, access to decent employment and lack of productive options. This situation has generated adverse land use practices which threaten the sustainability of natural resources and the most vulnerable livelihoods, putting high pressure on soils, ecosystems, and ecosystem services, particularly water regulation. Lack of employment opportunities in rural areas disproportionately affects rural women, youth and indigenous peoples. Extreme poverty is primarily, though not exclusively, a rural phenomenon. The concentration of poverty shows a persistent pattern: it is mainly located in the southern and southeastern regions of Mexico and in states with a large presence of indigenous peoples. The states that make up the PA widely exceed the national average for poverty and extreme poverty: 4'935,00 people live in moderate and 2'131,800 people in extreme poverty conditions which represent 20% of all Mexicans living in these conditions⁸. The highest levels of poverty by municipalities occurs in those with the largest indigenous population: in 8 out of 10 municipalities that have the highest poverty rate 97% or more of their population speaks an indigenous language.

3. Mexico has been suffering from violence and insecurity, mainly due to the fight against organized crime contributing to a context of fragility in rural areas. Some micro-regions within the states of Morelos, Guerrero, Michoacán and Jalisco in the Balsas Basin are experiencing severe situations of social fragility and violence, and their homicide rates were above the national average in 2019 and 2020. For this reason, during the last years, the Government has transferred a more important role to the armed forces for the maintenance of public security. Violence is one of the common reasons why (young) people decide to migrate and is also a hindrance to achieve better women's participation and gender transformation. Strengthening of the social fabric and the dynamization of sustainable and inclusive local economies are important contributions of this project to governmental efforts to confront violence and migration.

Figure 1: Percentage of population living in poverty (dark blue > 51.1%): Guerrero 66.4%, Oaxaca 61.7%

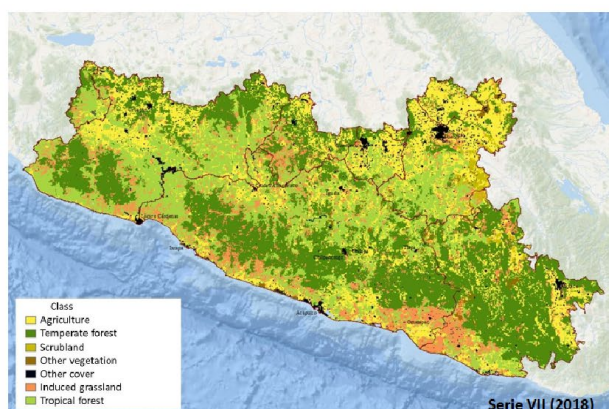


Source: CONEVAL 2020

4. The usual inequity faced by women to access different services and assets have been aggravated by the effects of climate change. On the one hand, the activities traditionally assigned for backyard orchard subsistence production (traspato), housework and family care, mostly managed by women, become more burdensome due to the increase in temperatures, water scarcity or water excess, in addition to the absence of public drinking water service. And on the other hand, the use of non-forest timber products (palm, aromatic and medicinal plants, fruits, etc.), which are also often conducted by women, are affected by the loss of vegetation due to droughts, fires, or other extreme weather events, such as hurricanes and torrential rains. Disasters, such as landslides and floods, affect houses and cause losses to fruit production, farm animals and domestic goods, situations that must be faced, mainly by the women in charge of their families. In addition to the economic losses, the climatic events mentioned before usually contribute to illnesses and accidents, which also lead to additional burden for the women who usually take care of their relatives⁹. In addition to the increasing household burdens women are more prone to gender-based violence in emergency and disaster contexts.

5. While rainfed agriculture¹⁰ (mainly maize, beans, vegetable and fruit trees) and pastoralism (mainly cattle and goats) are the main sources of subsistence, forestry plays also a fundamental role in the livelihoods of the rural population of the PA. Of the total area of approximately 18 million hectares, 11.4 million hectares (62%) are covered with forest ecosystems that belong mainly to *ejidos*¹¹, indigenous and agrarian communities, all henceforth called Agrarian Communities (ACs)¹². These territories offer a huge potential for forest landscape restoration and sustainable forest management including timber and non-timber forest products (NTFPs) for domestic and international markets. In 2021, 450 ACs obtained a permit from the Ministry of Environment and Natural Resources (SEMARNAT) to harvest timber on an area of 375,000 ha and 54,500 ha for NTFPs. The latter include pine resin, *olinalá* essence (*Bursera linaloe*), oak charcoal, palm fiber (*Brahea dulcis*) for production of handicrafts, aromatic and medicinal plants (including *Damiana*, *Guachalalate*, *Oregano*) and wild agaves to produce syrup, fibers and artisanal mezcal.

Figure 2: Land use in the Balsas and Pacific Coastal Plains Area

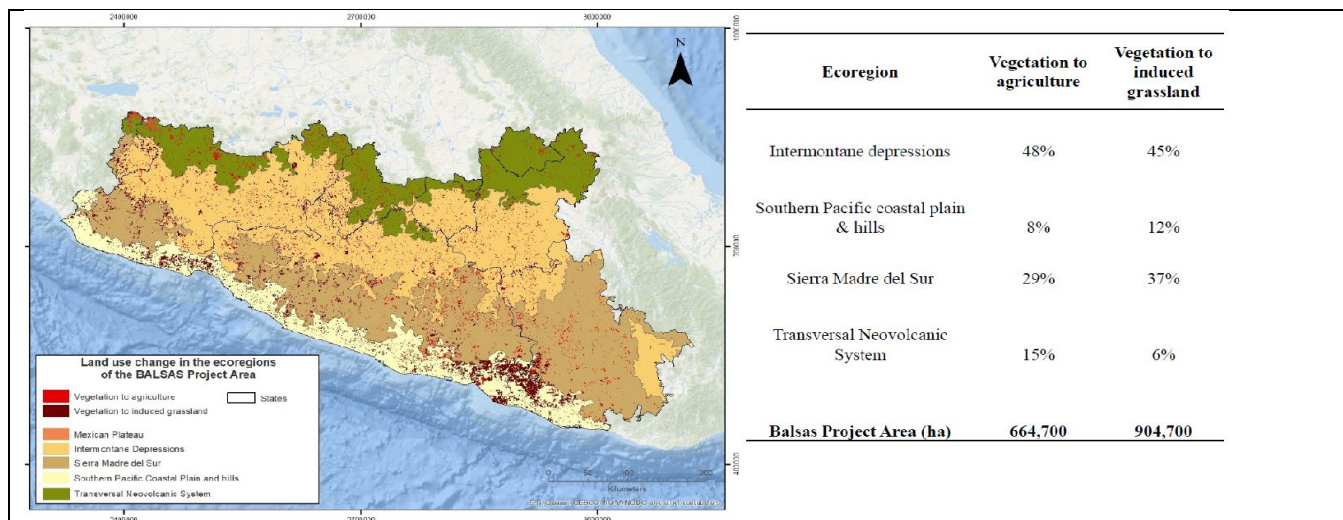


Balsas Project Area		
Land use, land cover	Hectare (2018)	%
Agriculture	4,133,000	23%
Induced grassland	2,044,100	11%
Temperate forest	6,325,200	35%
Tropical forest	4,902,100	27%
Scrublands	179,100	1%
Other vegetation	185,900	1%
Other cover	424,900	2%
Balsas Project Area	18,194,300	

Source: Authors with data from INEGI, Serie VII 2018

6. Forest and natural vegetation, which are crucial for the conservation of water flow regulation and biodiversity conservation have been under severe pressure in the PA: the main drivers for deforestation and forest degradation have been unsustainable agriculture and extensive livestock activities accounting to 664,700 and 904,700 ha between years 2002 and 2018, respectively.

Figure 3: Land use changes in the Balsas Project Region, years 2002-2018



Source: Authors with data from INEGI, Serie VII 2018

7. Approximately 15.6 million young people (4 out of every 10) live in rural areas in Mexico. Of these youths, 25.2% of young women and 7.6% of young men neither study or work and the proportion involved in agriculture has been falling. Not so in the PA were young people make up from 44% to 55% of the population. Rural youth represent more than 60% of the total population in Oaxaca and Guerrero. About 67% of these young people in Oaxaca and 36% in Guerrero are indigenous. 13 In small and isolated rural communities, youths aged 15-19 may drop out of or leave school. These school dropouts may then work as day-laborers (1.5%) or in underpaid agricultural work and other similar activities (0.9%). Multidimensional poverty among rural youth (60%) is nearly twice as high as among urban youth (36%). A lack of opportunities for rural youth is linked to their difficulty in gaining access to the factors of production (including land rights and land ownership), labor markets, education, and financial resources. In the absence of economic opportunities, or the possibility of participating in community decision-making and governance systems, young people in rural areas are more likely to migrate than adults and are at a greater risk of being co-opted or victims of crime (organized, common crime, human trafficking).

8. In the Balsas Basin 1'891,985 people speak and indigenous language which represents more than a quarter (25,7%) of the total indigenous peoples in México ¹⁴. Afro-descendent people make up more than 10% of the population of the states Guerrero and Oaxaca¹⁵. Thus, Oaxaca is the state with the highest percentage of indigenous people (43%) at the national level and in the project area: 16.3% of all indigenous people in the country live in Oaxaca. In Guerrero, there are 600,000 indigenous peoples who comprise 19.3% of the total state population. Historically, indigenous and afro-descendent peoples have the highest rates of poverty in Mexico. In 2016, 72% (8.2 million) of indigenous people lived in poverty. The Human Development Index (HDI) for the indigenous population (0.6761) is lower than that for the non-indigenous population (0.7628) and shows a gap for multiple socioeconomic indicators when compared to national averages. The situation is even more adverse for indigenous women living in rural areas (8 out of 10 live in poverty and 4 out of 10 in extreme poverty).¹⁶ The Indigenous Peoples Plan for this Project is included in Annex 6b.

9. In Mexico only 30.9% of rural households have an adequate level of food security, 40.7% suffer mild food insecurity, 17.7% moderate insecurity and the remaining 11.2% high insecurity. States with higher levels of poverty also record the highest level of food shortages and are all located in the southern and southeastern regions of the country. Data from the pandemic period in 2020 show that the proportion of people suffering from severe food insecurity increased from 9.2% in 2018 to 13%.¹⁷ The states with the highest levels of poverty also had the highest level of food shortages and are all located in the southern and southeastern regions of the country. Currently, 44% of households benefit from public food aid programs, which provide greater assistance to both indigenous and low-income households (70% of both receive assistance), as well as those suffering from moderate or severe food insecurity levels.¹⁸ The number of beneficiary households of some food aid programs is even higher in areas of the PA, for example reaching 65.8% of the population in the state of Guerrero.

10. The ongoing health crisis resulting from COVID-19 has affected Mexico, exacerbating climate change vulnerabilities in the country. The United Nations Economic Commission for Latin America (ECLAC, 2020) estimates that the 9% observed drop in output linked to the pandemic will result in 9.5 and 7.9 million persons falling into poverty and extreme poverty, respectively.¹⁹ The Mexican National Council for the Evaluation of Social

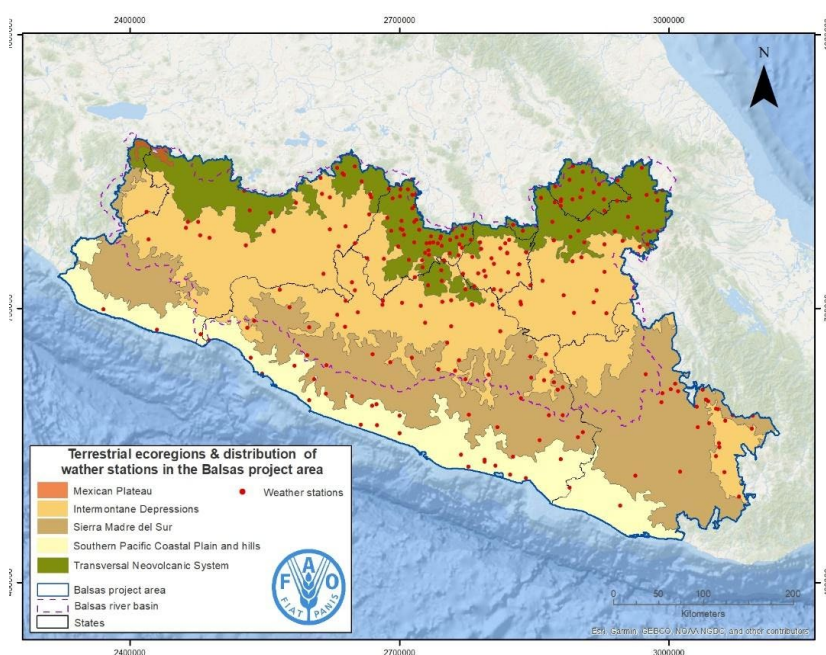
Development Policy (CONEVAL, 2020) also estimates 8.9 to 9.8 million new persons falling into poverty and 6.1 to 10.7 million into extreme poverty due to the pandemic.²⁰

Overview of climate trends affecting the Project Area

Ecoregional climate analysis of the Project Area

11. The Project Area is divided by two mountain ranges and includes two different lowlands. It can be divided in at least four ecoregions (Fig. 4) with different climate: a.) The Inter-montane Depressions (DI), inner lowlands characterized by low deciduous forest and xerophilous scrub vegetation, that cover an area of 7 million hectares; b.) The Southern Pacific Coastal Plains and hill landscape (PCP) with an area of 2.5 million hectares; c.) The temperate mountain ranges of the Sierra Madre del Sur (SMS) with an area of 5.7 million hectares and; d.) The Transverse Neovolcanic System (TNS) temperate mountain ranges of with nearly 3 million hectares.

Figure 4: Four main ecoregions in the Project Area, and location of selected weather stations used for the analysis



Source: Authors (see Annex 2b)

12. To gain a more differentiated insight on the impact of climate change and climate variability on rural livelihoods and vulnerable ecosystems, daily climatological data from weather stations located in the Project Area were obtained from the Hydrological Information System (SIH) of the National Water Commission CONAGUA²¹. As year 2018 was the last year with sufficient data for the stations that had information since 1970, 2000-2018 was defined as the study period and 1970-1999 as the reference period. Data from 1,251 stations were reviewed, of which 262 were selected within the four ecoregions as these had complete data for both time periods. The distribution of these weather stations by eco-region within the PA is shown through red dots in the map in Figure 4 above, and its number by eco-region the Table 1 below:

Table 1: Number of weather stations from which data could be analyzed for the project climate rationale

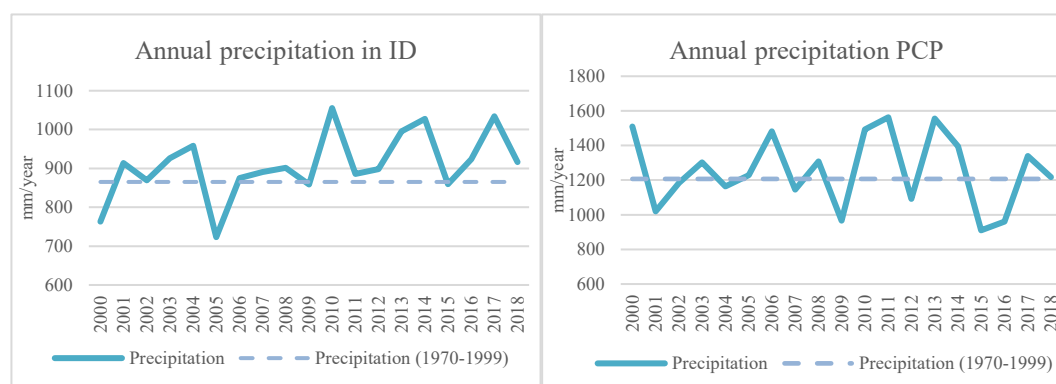
Terrestrial ecoregion	Number of weather stations
Intermontane depressions DI	106
Southern Pacific coastal plains / hills PCP	27
Sierra Madre del Sur SMS	37
Transversal Neovolcanic System TNS	92

Table 2: Observed climatic trends between the reference period (1970-1999) and study period (2000-2018)

Variable	Trends in Intermountain depressions (ID)	Trends in South Pacific Coastal Plain and Lomerío Hills (PCP)	Trends in Sierra Madre del Sur (SMS)	Trends in Transverse Neovolcanic System (TNS)
Precipitation – annually accumulated	5.00%	No trend	17.40%	9.90%
Dry days – accumulated during the dry season	-3 days	-3 days	-6 days	-7 days
Dry days – accumulated during the wet season	-2 days	-6 days	-4 days	-8 days
% of abnormally wet months ²	-0.5% points	+1.3% points	+7.5% points	+5% points
% of abnormally dry months ³	+3.3% points	-2.7% points	-7.1% points	-3.4% points
Temperature, maximum – annual mean	+0.17°C	+0.43°C	+0.29°C	+0.46°C
Temperature, minimum – annual mean	-0.15°C	+0.53°C	+0.57°C	+0.50°C
SPEI - % of wet months	+8.8% points	+6.9% points	12.7% points	+3.4% points
SPEI - % of severe wet months	+24% points	-26% points	+10% points	+8% points
SPEI - % of extremely wet months	-13 % points	+0% points	+16% points	+21% points
SPEI - % of dry months	-11.2% points	-6.3% points	-22.0% points	-12.6% points

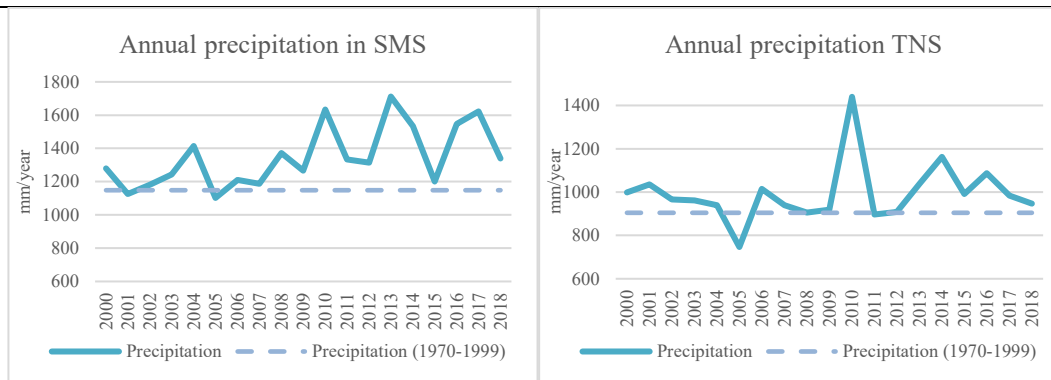
13. Observed increase in precipitations - Precipitation dynamics within the PA are predominantly influenced by the El Niño-Southern Oscillation (ENSO) System. During the study period (2010-2018), the region experienced the "La Niña" phase of ENSO, leading to heightened rainfall levels across the area. Consequently, a marked increase in precipitation was observed across most ecoregions, ranging from +5.0% of precipitation in ID to +17.4% of precipitation in SMS. Over the same timeframe, there was a reduction in the frequency of dry days, with decreases ranging from -3 to -7 days during the dry season and from -2 to -8 days during the wet season. Regarding extreme rainfall occurrences, the percentage of abnormally wet months (defined as months with a precipitation anomaly exceeding 50 mm) increased across most ecoregions of the PA, with increments ranging from +1.3% points in PCP to +7.5% points in SMS. Conversely, the percentage of abnormally dry months (months with a precipitation anomaly below -50 mm) decreased, ranging from -2.7% points in PCP to -7.1% points in SMS. The only exception was observed in the ID, where the percentage of abnormally wet months remained stable (-0.5% points), while the percentage of abnormally dry months increased by +3.3% points.

Figure 5: Trends in annual precipitation in the 4 ecoregions



² % of months when monthly anomaly > 50 mm

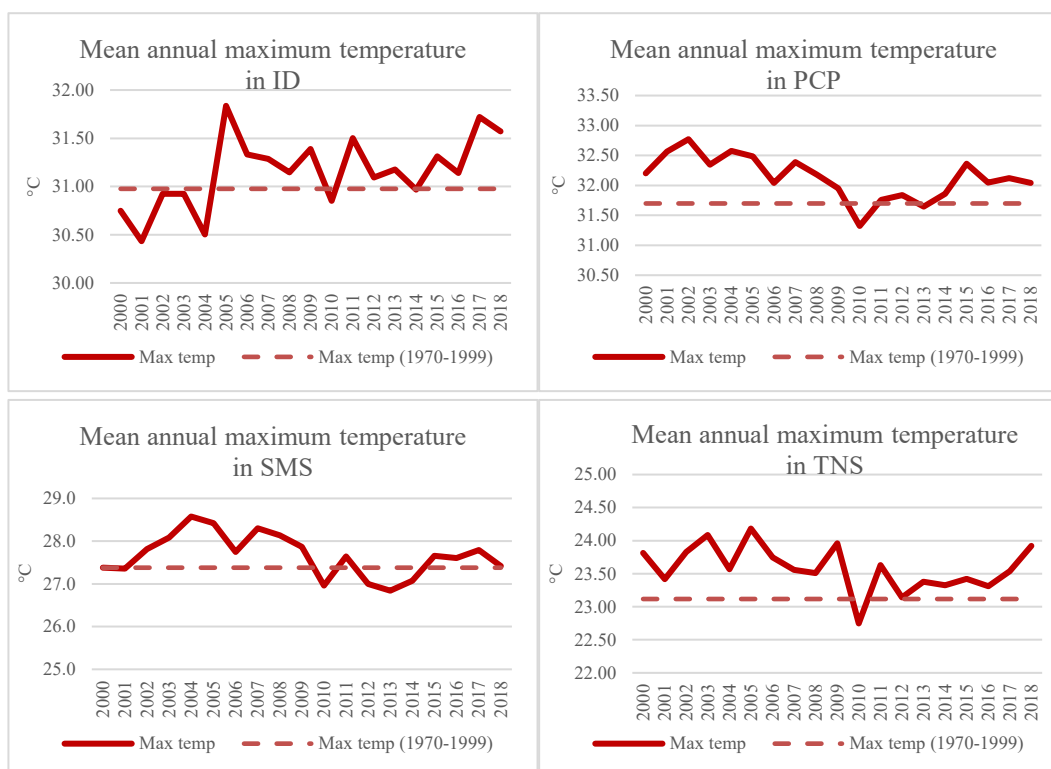
³ % of months when monthly anomaly < -50 mm



Source: Authors (see Annex 2b)

14. Observed Increase in temperatures - A notable rise in temperature was observed across the PA between the study and reference period. Maximum temperatures saw an increase ranging from $+0.17^{\circ}\text{C}$ in ID to $+0.46^{\circ}\text{C}$ in TNS, while minimum temperatures showed an increase from 0.50°C in TNS to $+0.57^{\circ}\text{C}$ in SMS. Here also, a notable exception must be mentioned in ID, where the minimum temperature decreased between the two period.

Figure 6: Trends in annual maximum temperature in the 4 ecoregions



Source: Authors (see Annex 2b)

15. Observed Increase in wet months occurrences - The analysis of dry and wet month occurrences in the PA was investigated using the Standardized Precipitation Evapotranspiration Index (SPEI)⁴. Across all ecoregions, there was a noticeable increase in the percentage of wet months, varying from 3.4% points in TNS to 12.7% points in SMS. Moreover, except for PCP, all ecoregions witnessed an increase in the proportion of wet

⁴ SPEI is a comprehensive metric that integrates both precipitation and evapotranspiration data to assess the moisture conditions within a particular climate, relative to the average conditions for the corresponding month. Furthermore, SPEI incorporates information from preceding months to provide a holistic understanding of the current conditions within a broader temporal context. The SPEI can be used to assess the moisture or dryness levels of a given month, using thresholds as follow:

Wet month: $\text{SPEI} > 1$; **Dry month:** $\text{SPEI} < -1$.

Moderately wet month: $1.5 > \text{SPEI} > 1$; **Moderately dry month:** $-1.5 < \text{SPEI} < -1$

Severely wet month: $2 > \text{SPEI} > 1.5$; **Severely dry month:** $-2 < \text{SPEI} < -1.5$

Extremely wet month: $\text{SPEI} > 2$; **Extremely dry month:** $\text{SPEI} < -2$

months classified as severe, ranging from +8% points in TNS to +24% points in ID. Furthermore, both SMS and TNS experienced a rise in the proportion of months classified as extremely wet, with increments of +16% points and +21% points respectively. During the same timeframe, there was a decline in the proportion of dry months, ranging from -6.3% points in PCP to -22.0% in SMS.

16. Comparison with the regional context - To provide regional context, we compared our results to the results based on the CRU TS observational dataset obtained from the IPCC WGI interactive atlas, in Southern Central America, for the period 1980-2015. Regional data indicates an increase in precipitation of +0.06 mm/day per decade, together with an increase of +0.15°C per decade. These results confirmed the direction of the trends we found in the PA.

Overview of the projected climatic trends

Table 3: projected climatic trends between the study period (2000-2018) and projection period (2041-2060)

Variable	Trends in Intermountain depressions (ID)	Trends in South Pacific Coastal Plain and Lomerío Hills (PCP)	Trends in Sierra Madre del Sur (SMS)	Trends in Transverse Neovolcanic System (TNS)
Precipitation, projection (2041-2060) – annually accumulated	SSP2-4.5: +0.1% SSP5-8.5: -4.7%	SSP2-4.5: +0.3% SSP5-8.5: -1.9%	SSP2-4.5: -14.2% SSP5-8.5: -18.2%	SSP2-4.5: +0.5% SSP5-8.5: -7.1%
Temperature, maximum, projection (2041-2060) – annual mean	SSP2-4.5: +3.05°C SSP5-8.5: +3.60°C	SSP2-4.5: +2.03°C SSP5-8.5: +2.70°C	SSP2-4.5: +2.43°C SSP5-8.5: +3.20°C	SSP2-4.5: +2.40°C SSP5-8.5: +3.30°C
Temperature, minimum, projection (2041-2060) – annual mean	SSP2-4.5: +3.18°C SSP5-8.5: +4.00°C	SSP2-4.5: +1.76°C SSP5-8.5: +2.20°C	SSP2-4.5: +2.90°C SSP5-8.5: +3.50°C	SSP2-4.5: +2.56°C SSP5-8.5: +3.08°C

17. Projected Increase in temperatures - A significant increase in temperatures is expected between the study period (1970-1999) and the projection period (2041-2060), across all ecoregions. Under the mitigation scenario (SSP2-4.5), an increase in maximum temperature ranging from +2.03°C in PCP to +2.43°C in SMS is expected, while for minimum temperature an increase ranging from +1.76°C in PCP to +3.18°C in ID is expected. Under the business-as-usual scenario, this increase becomes more pronounced. Maximum temperatures are projected to rise between +2.70°C in PCP to +3.60°C in ID, while minimum temperatures are expected to rise from +2.20°C in PCP to +4.00°C in ID.

At the monthly scale, under the mitigation scenario, maximum temperature increases are expected to range between +2.5°C and +3.7°C, with the largest increases during the wet season, while minimum temperature increases are expected to range between +2.4°C and +3.9°C, with the largest increases during the dry season. Under the business-as-usual scenario maximum temperature increases are expected to be in the range of +3°C and +4.2°C, with the largest increases at the very end of the wet season, while minimum temperatures increases are expected to range between +3.07°C and +4.9°C, with the largest increases during the dry season.

18. Divergent projected precipitation patterns – Compared to the studied period, precipitation projections indicate two divergent paths depending on the scenario under consideration. Within the mitigation scenario (SSP2-4.5), precipitation is expected to remain relatively stable, with minimal increases ranging from +0.1% in the ID to +0.5% in the TNS. An exception to this trend is however notable within the PA: SMS is projected to experience a significant -14.2% decrease in precipitation under the mitigation scenario. Under the business-as-usual scenario (SSP5-8.5), a widespread decrease in precipitation is expected, ranging from -1.9% in PCP to -18.2% in SMS.

19. Comparison with regional context - To provide regional context, we compared our results on climate projections to CORDEX downscaled data obtained from the IPCC WGI interactive atlas, for land in Southern Central America and the projection period of 2041-2060, compared to the reference period of 1995-2014. Regional data indicates that the increase in maximum temperature is expected to be approximately +1.3°C under SSP2-4.5 and +1.9°C under SSP5-8.5. For minimum temperatures, regional findings show an increase of +1.1°C under SSP2-4.5 and +1.4°C under SSP5-8.5. Regarding precipitation, regional data anticipates a decrease of -4.8% under SSP2-4.5 and -3.9% under SSP5-8.5. These results underscore a notable contrast between our findings in the PA and the projected climate of the surrounding areas, with higher rates of increase in both maximum and minimum temperatures, and a more nuanced pattern in projected precipitation. It's however important to note that CORDEX data covers a larger geographical area and doesn't consider the specific

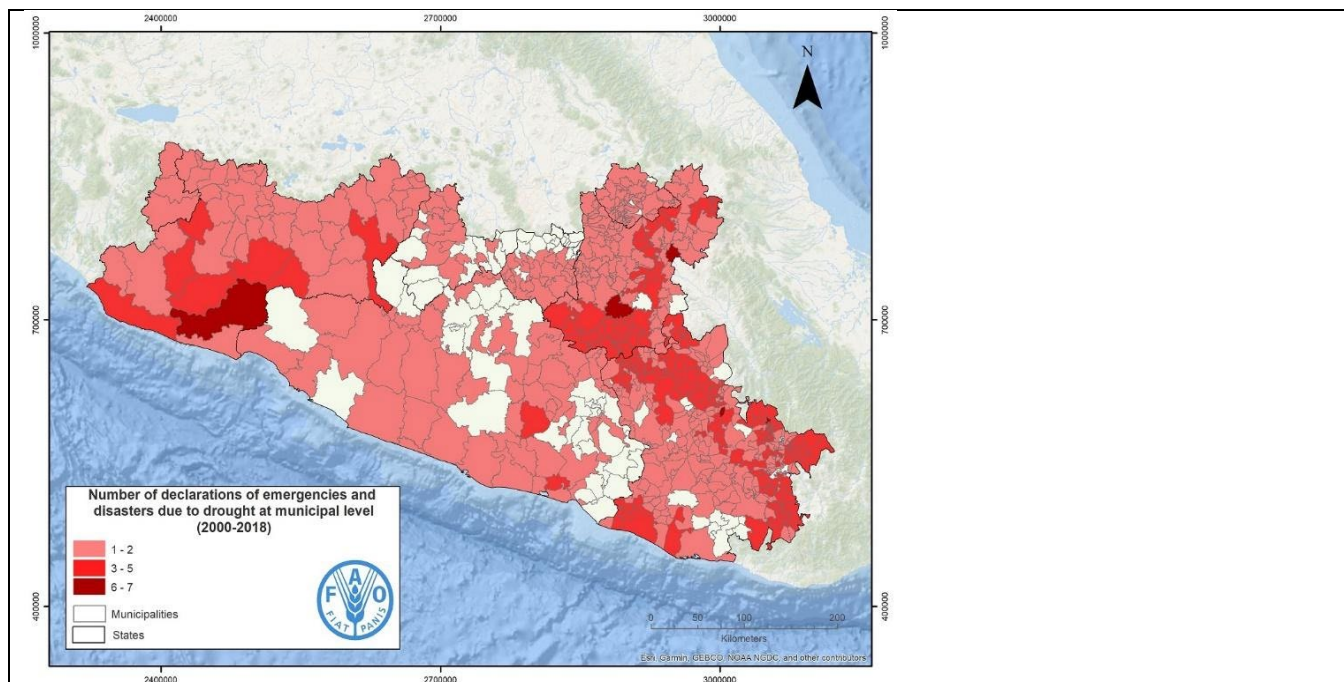
characteristics of each ecoregion. Moreover, it uses a different reference period (1995-2014 for CORDEX, 2000-2018 in our analysis). Thus, this comparison should be approached with caution.

20. Local sensitivity to climate change exposure - Given that any long-term variation in temperature has a significant impact on crop production²², these results are of great relevance to the project area, where most of the rural population's livelihoods depend on agriculture. High temperatures alter enzyme function within a leaf and cause changes in the growth and development stages of the crop, leading to reduced yields. In the case of minimum temperature, several studies have addressed the effects of increased minimum temperature (night temperature) on crop physiology, which alters leaf enzyme function, decreases photosynthetic function, increases respiration rate, suppresses flower bud development, and accelerates crop maturity, which causes a decrease in yields²³. Furthermore, the rates of temperature increase in tropical and sub-tropical areas are significantly higher in deforested areas, more than 50% higher than average temperature changes²⁴. These changes will affect the suitability of the main crops that support rural livelihoods in the PA, especially livestock and coffee, increasing especially the vulnerability of poor families that depend on these for daily subsistence. The project will support vulnerable households to adapt to these higher temperatures supporting the transition from traditional cropping systems to silvopastoral and agroforestry systems with much more shade trees, crop species diversity and varieties better adapted to higher temperatures and drought. Moreover, the project will make efficient use of the ability of ecosystems and landscapes to buffer more frequent and more intense rainfall events and to store excessive rainwater masses through nature-based landscape restoration and conservation solutions.

Impact of drought

21. According to the National Centre for the Prevention of Disasters CENAPRED²⁵, during the period 2000-2018, 1,337 emergency and disaster declarations at municipal level due to drought were registered in the PA. The municipalities that registered the highest number of declarations for these phenomena were in the DI and the PCP. Ninety-five per cent of these events were recorded between 2000 and 2009. The years with the highest number of declarations for this phenomenon were 2002, 2003, 2005 and 2009. In 2002, 311 declarations were recorded, most of which were issued at the expected beginning or during the rainy (planting) season and ended at the end of the rainy season. With respect to ENSO dynamics, during this period the transition from the neutral phase to El Niño was observed, which is associated with dry events during the rainy season in the PA. In 2003, 135 emergency and disaster declarations were issued due to drought, which were effective in September and October (the second planting season). During this period, ENSO dynamics were in a neutral phase. In 2005, 269 drought emergency and disaster declarations were registered, 89% of which started in April and May, the months corresponding to the dry season. However, in most cases, their duration extended into September and even November. In 2005, ENSO dynamics were in transition from neutral to La Niña. In that sense, the combined effects of both phases can be associated with dry events during the rainy season (planting) and extend into the dry season at the end of the year. In 2009, 216 warnings were recorded, most of which were issued at the beginning of the rainy season and remained in effect until August and September. The ENSO dynamics for this period were in transition from La Niña to El Niño, so the dry events are attributed to this phenomenon.

Figure 7: Declarations of emergencies and disasters due to droughts at municipal level in the Project Area



Source: CENAPRED, elaboration authors - Annex 2.b

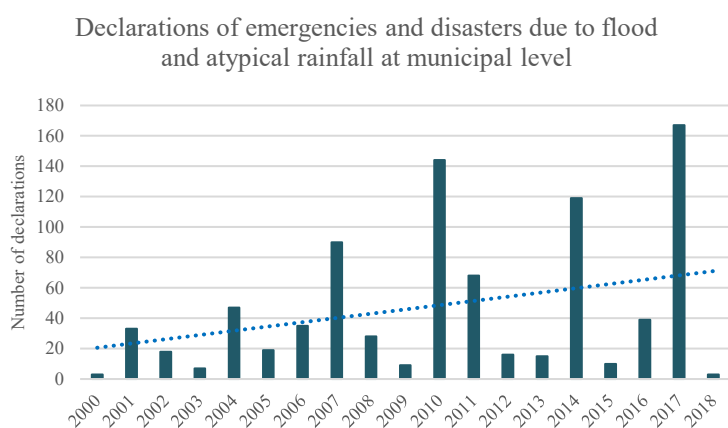
22. The climatic changes presented above will continue to have a direct impact on agriculture and forestry. Moreover, if no action is taken an expected 6.8% to 15.8% reduction in primary soil productivity could happen²⁶ due to prolonged dry season, more intense and erratic precipitation during the rainy season, and rising temperatures. Combined with all the economic and technological challenges to agriculture production in Mexico and, especially in PA, climate hazards have already imposed severe restrictions on farmers. For maize, Gómez et al. (2019) confirm that a lack of precipitation during July, combined with temperatures over 38°C, kills emerged seedlings. Along with this, drought for more than 21 days during August constituted a primary indicator of high climatic hazard to produce maize. In addition, this study evaluated 18 years (1999 to 2016) and concluded that the years with more than ten days with a temperature higher than 38°C matched with the years of low production and high climatic hazard. These findings are concerning when analyzing the observed changes in the PA, as also other similar findings reported²⁷.

23. Ecosystems will also see other changes such as increase in plant diseases, and significant decline in mangrove areas and temperate forests as well as an increase in arid and semi-arid areas or dry forests. These changes will subsequently affect the provision of ecosystem services, especially water regulation and provision.²⁸

Impacts of floods, landslides and atypical heavy rains

24. During the period 2000-2018, 870 municipal declarations of emergencies and disasters due to floods, flashfloods, landslides, and atypical rains were registered in the PA. The municipalities with the highest number of declarations for these phenomena were in the PCP and SMS. Two thirds of these events were recorded between 2010 and 2018. The years with the highest number of declarations for this phenomenon were 2007, 2010, 2014 and 2017. In 2007, 90 warnings were issued in three monthly periods: May (7), July (13) and August (70). The ENSO dynamics during these periods, corresponding to the rainy season, correspond to La Niña, whose effects are observed in increased rainfall during the rainy season. In 2010, 144 emergency and disaster declarations were issued for floods and atypical rains in five monthly periods, of which two correspond to the dry season: January (29), February (75); and three to the rainy season: July (11), August (22) and September (7). During this period, ENSO dynamics were in a transition phase from El Niño to La Niña, characterized by intense precipitation events in the rainy season. In 2014, 119 warnings were issued in three monthly periods, most of them at the beginning of the rainy season: June (112), July (1) and September (6). The ENSO dynamics during these periods correspond to the neutral phase. In 2017, 167 warnings were issued in three periods: one during the dry season in March (4) and two during the rainy season, in June (42) and September (121). The ENSO dynamics for this period were in the La Niña phase, so rainfall in the rainy season was more intense.

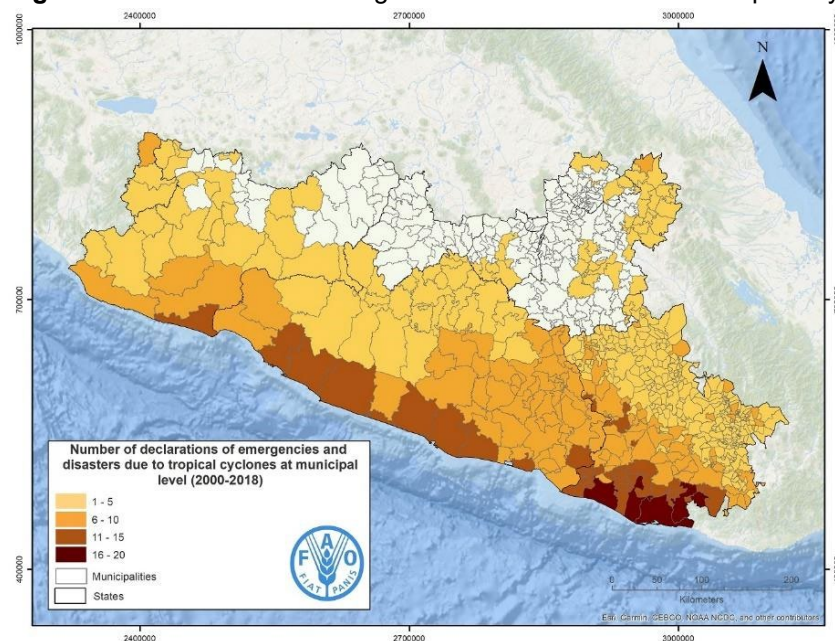
Figure 8: Flood and atypical rainfall emergency declarations in the Project Area years 2000-2018



Impact of tropical cyclones

25. During the period 2000-2018, 2,090 municipal declarations of emergencies and disasters due to tropical cyclones were registered in the PA. The municipalities that registered the highest number of declarations for these phenomena were in the PCP and SMS. Eighty-two per cent of the declarations were registered in five years: 2000, 2005, 2012, 2013 and 2014. In 2000, 760 emergency declarations were issued, the highest number recorded during the study period. This atypical number of declarations can be associated with the occurrence of two tropical storms (Norman and Rosa) that made landfall in the areas of the PA where the municipalities with the highest risk of tropical cyclones are located²⁹. During these months, ENSO dynamics were in the La Niña phase.

Figure 9: Declarations of emergencies and disasters due to tropical cyclones in the Project Area



Source: CENAPRED, elaboration authors - Annex 2.b

26. More than half of the municipalities in the PA have some level of vulnerability to climate change. In the case of hazards and risks due to hydrometeorological events, these do show patterns associated with the ecoregions. The municipalities located in the PCP and DI have a higher level of danger due to drought and floods, while those located in the PCP and SMS have a higher level of risk from tropical cyclones than the rest of the ecoregions. The most notorious patterns of the effects of climate change and variability can be seen in disaster declarations due to droughts and floods. In the case of drought declarations, these were most frequent between 2000 and 2009 during the ENSO El Niño phase. However, between 2010 and 2018, La Niña and the neutral phase predominated, resulting in fewer drought declarations, but more declarations for floods and atypical rains. In the case of tropical cyclone declarations, there is an association between the months in which the declarations were issued and the presence of La Niña and ENSO neutral phase.

27. In terms of water resources, climate change will exacerbate shortages that are already being experienced by rural ACs in the PA, both at the household level (drinking water) and the on-farm level (water for irrigation). The impacts of climate change on the water cycle will not be uniform, revealing significant vulnerability and risk exposure to certain ecosystems and many human activities (domestic and agricultural) given climate variability. For freshwater, renewable surface and groundwater is expected to decline during the 21st century, especially in dry subtropical regions, increasing competition among users and reducing supplies to ACs. In addition, the effects of climate change will be accentuated in areas with rapid urbanization processes. The impacts will also be felt in rural areas due to the reduced availability of water and changes in temperature, causing displacement of agricultural areas and, consequently, affecting both the rural population and food security in general.

Combined economic impacts

28. The hydrometeorological events described above have resulted in both the loss of human life and enormous financial impacts. Recent adverse climatic events have affected 2.4 million people in the region and caused 561 deaths. These events also damaged 147,000 homes, 1.4 million ha of cropland and 14,000 km of roads. The concurrent economic losses have been estimated at USD 4,279 million. All these factors, thus, increase the vulnerability of the Basin's populations and ecosystems. In fact, the Balsas Basin is one of the two main areas in Mexico with the highest rates of climatic vulnerability. 68.3% of the Basin (12.2 million ha) experiences average vulnerability, 27.4% (4.9 million ha) high vulnerability and only 4.3% (0.78 million ha) low vulnerability.³⁰ The interaction between social, economic, and environmental vulnerability in the Basin converges in Guerrero and Oaxaca, being the most affected states in southern Mexico. Most (97.7%) of the Basin's natural and productive centers with medium to high climate vulnerability are affected by water issues, especially problems linked to water distribution. 89.6% of agricultural areas are also affected by soil-related vulnerabilities (particularly inadequate soils for growing rainy season crops). 88% of forest areas are vulnerable in terms of their supporting capacity or for having ecological niches with representative species. 78% of livestock areas are vulnerable in terms of their naturally available animal food supplies. Finally, 62% of biodiversity areas are vulnerable due to the impacts of climate change on their plant communities.³¹

Mitigation rationale: GHG emissions profile

29. According to Mexico's Third Biannual Report to the United Nations Framework Convention on Climate Change³² and to Mexico's Greenhouse Gas Inventory 1990-2019³³ in 2019 the total estimated net emissions in all sectors at the national level were 534,688.6 MtCO₂e. The "Agriculture, Forestry and Other land Uses" sector contributed 19.12%, second after the Energy sector (63.52%). In contrast reforestation and preserving forests, farmland, or grasslands helped absorb CO₂ from the atmosphere to release oxygen and generate biomass to store 201.94 million tons of CO₂ in the year 2019.

30. For the specific case of the Project Area and focusing on the forestry sector, gross emissions during the period 2001-2018 were estimated at 76.1 MtCO₂eq³⁴. Of these emissions, conversion from forest to grazing land accounted for 72.2% (54.9 MtCO₂eq); 26.3% corresponded from forest conversion to agricultural land (20.1 MtCO₂eq) and the remaining 1.1% of emissions came from conversion from forests to human settlements and other types of land use. Emissions linked to the expanding agricultural frontier in the Basin are occurring in a context in which most or 65% (11.7 million ha) of the total land area is held by ACs.³⁵

Situation of deforestation and FREL in the Balsas Basin

31. In line with the definitions used in the Second FREL for the period 2007-2016 submitted by Mexico in 2020,³⁶ an assessment of the annual forest loss in the Balsas Basin was undertaken through 9,200 plots within the grid of the National Forest Inventory (INFyS, as to its acronym in Spanish). While the gross annual deforestation for the whole period between the years 2001-2012 was of around 32,675 ha / year, annual deforestation increased by 16% for the period 2013-2018 to reach 37,856 ha / year. The total forest area lost between years 2001 and 2018 was 616,595 ha, of which 63.8% corresponded to dry forests and 36.2% to temperate montane forests. More than 70% of these forests were lost to pastureland and the remainder mainly for other agricultural uses during that period.³⁷ Additional details on deforestation in the Balsas Basin are presented in Annex 2. It is to be noted that the Second FREL for the period 2007-2016 submitted by Mexico in 2020 includes two of the five REDD+ activities: reduction of emissions from deforestation, and partial reduction of emissions from forest degradation.

Alignment with the NDCs, the ENAREDD+, and other national priorities

32. The measures set out in Mexico's Nationally Determined Contributions³⁸ (NDCs) respond to mitigation and adaptation problematics in the Project Area. According to the NDCs, mitigation activity in the land use, land

use change and forestry sector (*uso del suelo, cambio de uso del suelo y silvicultura* USCUS) directly aligns with the overarching aim of ecosystem-based adaptation. The NDCs recognize the need to incorporate climate criteria into forestry and rural development programs. It also recognizes the need to implement environmental protection and land use standards. For this reason, the project framework aims to contribute to the NDCs mitigation and adaptation goals indicated in axes A, B, C and D.³⁹ Contributions will occur firstly in terms of **(i) Climate Change Adaptation**. This will involve activities focusing on both (a) social sector adaptation to climate change, (b) ecosystem-based adaptation and (c) water resource and supply management. Contributions will occur secondly in terms of **(ii) Climate Change Mitigation**. This will involve reducing emissions through activities focusing on (a) the land use sector via land use change and forestry activities (e.g., afforestation, reforestation, forest management, farmland management and pasture management) and (b) the agricultural sector via improved agricultural soil management, scheduled soil burns, improved silvopastoral practices, rotational grazing and manure management. A detailed matrix that shows the alignment of project outputs and activities with each of the axis and lines of action of Mexico's NDCs is presented in Annex 2a, Chapter 5.

33. The project will support the implementation of Mexico's National Strategy for Reducing Emissions from Deforestation and Forest Degradation 2017-2030 (*Estrategia Nacional de Reducción de Emisiones por Deforestación y Degradación Forestal* ENAREDD+)⁴⁰. This strategy aims to achieve the following milestones by 2030:

- Achieve 0% net deforestation rate by 2030.
- Significant reduction in the national rate of forest degradation compared to the baseline.
- Achieve an increase in sustainably managed forest area, natural and induced regeneration and forest conservation, with a consequent increase in carbon stocks.
- Maintain or improve the provision of environmental services of the territory by conserving biodiversity
- The continued development of the social and economic capital of rural communities.
- Increase the use of best practices of sustainable production at the rural landscape scale.

34. The Balsas Project will support the achievement of all the above mentioned six milestones of the ENAREDD+. Furthermore, the Project will implement most of the activities of the seven components and corresponding lines of action of the ENAREDD+ in the Project Area, as shown in the alignment matrix included in Annex 2a, Chapter 5.

35. The National Forest Programme (PNF) 2020–2024⁴¹ is a Special Programme derived from the National Development Plan 2019–2024, led by CONAFOR, with the participation of forest sector stakeholders and the approval of the National Forest Council. PNF's purpose is to contribute to the fulfillment of international commitments relating to forests that Mexico has undertaken by entering into a series of multilateral agreements. The PNF covers a total of 26 strategies and 188 specific actions, including the National Strategy for Community Forest Management, the ENAREDD+, the Fire Management Programme, the Programme to curb Illegal Logging and Illegal Trade in raw materials and forest products, the Strategy to promote active conservation through Payment for Environmental Services, the Strategy for Productive Restoration with an integrated land management approach, the Strategy for a Regulatory and Administrative Framework to facilitate Sustainable forest development, and the Strategy for Neighborhood Territorial Management, among others.

36. The PFN includes five priority objectives: 1.) Promote community forest management for the sustainable and diversified use of forest resources, as well as the integration and development of local networks of competitive value that trigger local economies to improve the quality of life of the population living in forest areas, in particular women, youth and indigenous peoples. 2.) Protect Forest ecosystems from factors that lead to deterioration in the vegetation cover to maintain the natural heritage and contribute to climate change mitigation, for the well-being of the population living in forest areas and of society in general, through territorial management. 3.) Conserve and restore the capacity to provide ecosystem services in strategic forest areas, through an inclusive and participatory approach that contributes to guaranteeing a healthy environment for the development and well-being of the population. 4.) Promote a new model of governance, plurality and effective and inclusive social and public participation in the forestry sector. 5.– Promote effective institutional coordination and improve institutional capacity-building for the forestry sector.

37. The BALSAS project will implement the mandate of the General Law for the Sustainable Forest Development and will be integrated within CONAFOR's "*Programa de Apoyos para el Desarrollo Forestal Sustentable para el Bienestar*" ([PADFSB](#)). This program supports Agrarian Communities to implement actions that contribute to the protection, conservation, restoration and incorporation into sustainable forest management of natural forests and forest land. The program also strengthens value chains, which in turn contribute to climate change adaptation and mitigation. It is made up of five PADFSB Incentive Schemes: I. Community Forest Management and Value Chains (MFCCV); II. Commercial Forest Plantations and Agroforestry Systems (PFCA); III. Forest Restoration of Micro-basins and Strategic Regions (RFM); IV. Environmental Services; (PSA) and V. Forest Protection (PF). All outcomes, in a cross-cutting way, will target women, youth and indigenous peoples, and propose specific activities for their empowerment (see Annex 8). Its approved budget for year 2021 was

825.5 million MXP. Additionally, compensation payments for authorized forest land use changes (CAxCUSF) that accrue into the Mexican Forestry Fund (*Fondo Forestal Mexicano*) will also co-finance project activities.

Link to other ongoing initiatives and projects in the project area

38. Early actions of REDD+, that started in 2012 with the support of the Forest and Climate Change project (World Bank/CONAFOR), were focused on the Yucatan Peninsula and coastal area of Jalisco. Other relevant initiatives are the Emission Reduction Initiative supported by the FCPF (2016)⁴². The document *Mapeo del Financiamiento para-REDD+ en México*⁴³ provides an overview of all REDD+ financing. Although these initiatives were not located in the Balsas basin CONAFOR's institutional capacities were strengthened, particularly environmental and social safeguard management, Payment for Environmental Services (PES) modalities and financial tools to strengthen AC inclusion in forestry value chains, on which this proposal is building upon.

39. The GEF co-financed GreenMex project (USD 10.1 million GEF Grant / 2022-2027) will mainstream biodiversity conservation, integrated landscape management and ecosystem connectivity into CONAFOR's PADFSB. The project will be implemented in forest and agroforestry landscapes within three areas of high biodiversity in Mexico, one of them the Balsas / South Pacific Region. GreenMex will have a synergistic effect with the BALSAS project that seeks to integrate climate adaptation criteria into the PADFSB and its institutional setting. CONAFOR as Executing Agency is being supported by FAO as GEF Implementing Agency and FAO will be invited to the Balsas Project Steering Committee CDAL, since one of the three Mexican regions covered by GreenMex is this same project area of the BALSAS Project. GreenMex will be financing nature-based solutions for biodiversity and landscape management to strengthen biological connectivity eight in bio-forestry corridors on an area of 2'712,633 ha in 116 municipalities in the Balsas Basin in the States of Michoacán, Guerrero and Oaxaca, in strong alignment and complementarity with the Balsas Project. At the institutional and project coordination level synergies will be sought between both projects.

40. Another initiative on which this project will build upon is the "Strengthening Entrepreneurship in Productive Forest Landscapes Project" (PROFOEM⁴⁴ - World Bank/CONAFOR, USD 66M loan, closing 09/2024) which has been piloting new schemes of PES and sustainable forest management. The new PES approach combines restoration and forest ecosystem management with PES through PADFSB program RFM.1: "Restauración forestal de microcuencas estratégicas". It supports ACs to restore ecosystems over several years ensuring a long-term commitment to improving soil and vegetation management if progress is verified in the field. At least 60% of the transferred funds to ACs must be used for forest ecosystem management and 40% can be between members of ACs. These advancements promote income generating activities for women, as they are not dependent on land ownership, as was the case with previous PES approaches.

41. Finally, the project "Connecting watershed health with sustainable livestock and agroforestry production" (CONECTA) aims to improve integrated landscape management and promote climate-smart production practices in selected watersheds, ensuring the social inclusion by targeting women, youth and indigenous peoples. Its implementation period is five years from July 2021 to July 2026. It is managed by the *Fondo Mexicano para la Conservación de la Naturaleza*, A.C. (FMCN), which is responsible for the fiduciary aspects, and the *Instituto Nacional de Ecología y Cambio Climático* (INECC), which is responsible for the technical leadership. In addition, nine federal agencies participate in the project through the Coordinating Committee, which seeks to scale CONECTA's impact through the alignment of both public and private investments. CONECTA will concentrate its interventions in 15 eligible watersheds in the states of Chiapas, Chihuahua, Jalisco and Veracruz, all outside of the Balsas PA. The watershed approach responds to integrated landscape management, which will be driven by the development of Integrated Watershed Management Action Plans (PAMIC) led by INECC. Three Regional Funds (RF) will support local coordination and implementation of investments. The Balsas Project team will seek to learn from experiences and good practices and to cross-fertilize with this project.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Project Theory of Change Statement

42. **IF** bottom-up participatory mechanisms for intersectoral coordination, and capacities to include climate change criteria in institutional structures and procedures, and participation in value chains are strengthened; and inclusive, participatory, multi-annual land use and investment plans within prioritized areas are implemented to strengthen the productive systems of ACs; and climate hazard early warning systems and knowledge are in place **THEN** climate resilient production systems and sustainable land use forms will improve the livelihood resilience of the most vulnerable rural households and communities, and ecosystems and ecosystem services will be conserved in the project area and the population of the Balsas Basin will be better informed about climate hazards **BECAUSE** government programs and community development plans integrate climate change criteria; forest landscapes are sustainably managed and better adapted to climate change and the population of the Balsas Basin has timely and relevant climate information for climate risk management, and disaster prevention.

43. The core medium- to long-term result of the project is related to the application of paradigm shifting innovations, first piloted in the municipalities prioritized through the Balsas project using GCF concessional finance to be scaled-up to the regional and national level. The average of the last five years of forest development finance provided by CONAFOR's PADFSB at the national level is equivalent to around USD 160 million per year. During the project lifespan of 20 years these would add up to a total of more than USD 1.54 billion (present value) of traditional forest development finance that will be transformed into sustainable forest development finance for climate change adaptation and mitigation in benefit of Mexico's most vulnerable communities and within these communities, target women, youth and indigenous peoples. There is immense potential for this leveraged finance to be increased further through the application of the same paradigm shifting innovations to other rural development finance sources in Mexico, such as CONAGUA's PROAGUA Program, through the strengthening of multisectoral coordination structures via the project.

44. This cross-cutting project (80% adaptation, 20% mitigation) aims to increase the climate resilience of the most vulnerable communities and ecosystems, and reduce emissions from production systems in the Balsas PA, via: i) operationalizing the use of climate change criteria in current institutional modalities for the prioritization of rural development finance through a community-centric bottom-up approach; ii) investing in the climate-sensitive management of ecosystems, and climate-resilient productive systems and value chains for adaptation and mitigation benefits; and iii) developing a climate information system, and implementation of knowledge management.

45. **The objective of the project** is to enable community-level investment in adaptation and mitigation initiatives, through climate-focused transformation of rural development incentives for the conservation of forest resources, and the strengthening of local institutions and capacities. This objective will be achieved by the project's strong focus on bottom-up decision-making and value chain development ensuring social inclusion, benefiting and empowering women, youth and indigenous peoples. The proposed project will shift the approach of the current rural development support allocation system used by rural development institutions to a new, transformative approach which:

- Uses climate change-related criteria in a participatory transparent way for selecting interventions and prioritizing the most vulnerable beneficiaries, and sites.
- Strengthens territorial governance and participation of women, youth, and indigenous peoples at the sub-basin level to provide incentives and strengthen inter institutional coordination, fostering a bottom-up approach for the sustainable use, management, and conservation of crucial ecosystems and ecosystem services to be piloted in the prioritized municipalities of the Balsas PA and replicated at regional and national level. As part of the project's bottom-up approach, the participation of Agrarian Communities and Producer Organizations in value chains of agroforestry, timber and non-timber forest products (NTFPs) will also be strengthened, with the aim of increasing the incomes of the most vulnerable households, thereby building their adaptive capacity.
- Disburses rural incentives multi-annually under a progress-based payment system, specifically rewarding performance of adaptation and mitigation initiatives. The multi-annual approach and annual evaluation of progress was first piloted by CONAFOR for PES schemes only and will be replicated by the project across other forest development support schemes under CONAFOR's PADFSB.

Outcomes

The three outcomes presented below contribute collectively to the achievement of the project's objective. These will be achieved via a results chain between activities and outputs as described below.

46. **Outcome 1: Government programs and community development plans integrate climate change criteria for allocation of supports.** This outcome will be achieved through:

- Strengthening mechanisms for intersectoral coordination, which will build the technical and organizational capacities of small producers and facilitate synergies between government institutions and the private sector (Output 1.1). This will include creating Forest Management Units (FMU) and the corresponding Regional Committees for the Integrated Management of the Territory (RCIMT) at basin, sub-basin and ethnic territory levels (Activity 1.1.1), and strengthening multi-sectoral coordination structures, operation rules and mechanisms for co-investment in basins (Activity 1.1.3). Additionally, priority intervention areas for climate change adaptation will be identified (Activity 1.1.2), while criteria for the selection of appropriate adaptation interventions will be integrated into relevant institutional territorial development procedures (Activity 1.1.4).
- Developing participatory climate-responsive (adaptation and mitigation) multiannual investment plans for prioritized areas (Output 1.2). This will include raising awareness amongst agrarian and indigenous

communities on how to benefit from incentive programs that promote climate resilience (Activity 1.2.1) followed by the development of investment plans at the sub-basin level (Activity 1.2.2).

- Building the capacities of local actors (including extension workers, local facilitators, and forest protection brigades (Activity 1.3.1) and CONAFOR's local extension offices, "*Promotorías Forestales*" (Activity 1.3.3) to plan and implement inclusive adaptation approaches that enhance resilience (Output 1.3). This will include supporting the establishment and operation of forest protection brigades and climate change resilience facilitators in the project's target municipalities (Activity 1.3.2).

47. **Outcome 2: Forest landscapes are sustainably managed and better adapted to climate change.**

This outcome will be reached through:

- Conserving, restoring and sustainably using forest landscapes by agrarian communities through the implementation of land-use plans and PES mechanisms (Activity 2.1.1; Output 2.1). Enhanced capacity and governance achieved via Outcome 1 will provide a platform to support the achievement of Output 2.1.
- Strengthening the climate resilience of agrarian communities' productive systems (Output 2.2), through the implementation of climate-smart forest landscape restoration in micro-watershed and strategic areas (Activity 2.2.1) and climate adapted commercial forest plantations and agroforestry systems (Activity 2.2.2).
- Enhancing the participation of agrarian communities and producer organizations, especially women and the youth, in forestry and agroforestry value chains and building the climate resilience of their livelihoods (Output 2.3). This will include building local capacity for and implementing climate-resilient sustainable forest management and ensuring inclusive access to timber and non-timber forest product value chains (Activity 2.3.1), supporting households to improve food and nutrition security (Activity 2.3.2), and establishing community forest protection brigades to monitor and ensure the sustainable climate-responsive use of forest resources (Activity 2.3.3).

48. **Outcome 3: The producers, inhabitants and authorities of the Balsas PA have timely and relevant climate information for climate risk management, disaster prevention.** This outcome will be achieved through:

- Supporting accessible climate hazard-related early warning and monitoring systems in the Balsas PA (Output 3.1) by developing a platform and applications to facilitate access to climate information and early warning systems (Activity 3.1.1), strengthening following of CONAFOR's crucial systems: i.) The Forest Fire Protection and Prevention System (SPPIF), ii.) The Forest Phytosanitary Surveillance (SIVICOFF) System (Activity 3.1.2), iii.) The Project-specific Monitoring, Reporting and Verification (MRV) System (Activity 3.1.3), and iv.) The Social and Environmental Safeguards Management System (Activity 3.1.4).
- Developing policy-relevant knowledge products through systematic documentation of good practices and intervention approaches of the project (Output 3.2), including the production of training and systemization materials to support ongoing capacity development (Activity 3.2.1), exchange of experiences, and South-South and Triangular Cooperation (SSTC) at the local, regional, national and international levels (Activity 3.2.2), and rolling out and operating a rigorous monitoring and evaluation (M&E) system for the project (Activity 3.2.3).

49. **Barriers:** The proposed project has been designed to respond to specific barriers to the achievement of its proposed outcomes and project objective. These barriers and the activities designed to address them are presented below.

50. **Barrier set 1.** A key factor shaping climate-related challenges is the weak level of existing climate governance for the design and implementation of climate change adaptation and mitigation initiatives, as well as the allocation of climate finance, since efforts to integrate a climate approach into public policies have only recently begun. Furthermore, institutional coordination among the involved actors, specialized technical assistance on climate change and involvement of community perspectives and participation have been limited. Specifically, the lack of technical assistance and coordination among actors, has delayed the roll out of climate change adaptation interventions across the Basin, because many of the actors are still used to business-as-usual development initiatives. Climate change adaptation interventions have therefore only been considered as an afterthought or co-benefit. As a result, there have been no clearly focused incentives encouraging concerted action for climate change mitigation and adaptation. These findings are supported by a recent independent

evaluation of CONAFOR's ongoing work⁴⁵ in the Balsas Basin which indicates a sectorized vision of the territory, generating a contradiction between initiatives, as well as scattered and disorganized efforts, deficient coordination between central offices and the local *Promotorías* in the design, definition, and implementation of the PADFSB. Regarding specialized technical assistance, the findings indicate little relationship in the development of support activities between the technical adviser and the beneficiaries, and shortcomings of technical advisers. Lastly, regarding community participation and the participation of women and youth, the reports indicate a negative and low perception of ownership of the planning, monitoring, and evaluation processes, and weak governance in ACs. A coordinated approach will ensure that climate change adaptation is front and center of development in the Basin, allowing for more effective budgeting and planning, and ultimately a greater impact on the climate resilience of local communities. These barriers will be addressed and overcome through the activities described below:

- Facilitate the creation of Forest Management Units (FMUs) and the corresponding Regional Committees for the Integrated Management of the Territory (RCIMT), the statutory natural resource and watershed management unit and bodies in each Basin, Sub-basin and ethnic territories prioritized by the Project (Activity 1.1.1) and agree on priority intervention areas (Activity 1.1.2).
- Strengthening multi-sectoral coordination structures, operation rules and mechanisms for co-investment in the prioritized basins (Activity 1.1.3).
- Raise the awareness amongst agrarian and indigenous communities on how to benefit from incentive programs that promote climate resilience (Activity 1.2.1).
- Developing participatory climate-responsive (adaptation and mitigation) multiannual investment plans at the sub-basin level with active involvement of women and youth (Activity 1.2.2).
- Developing the capacities of local actors (including extension workers, local facilitators, and forest protection brigades (Activity 1.3.1) and of CONAFOR's *Promotorías Forestales* (Activity 1.3.3) to plan and implement inclusive adaptation approaches that enhance resilience, including supporting the establishment and operation of forest protection brigades and climate change resilience facilitators in the project's target municipalities (Activity 1.3.2).

51. Barrier set 2. The main institutions working in the Balsas PA (CONAFOR, CONAGUA, and AGRICULTURA) have been developing advanced information systems related to weather forecast, forest fires, medium/long term evolution of climate patterns and climate risk prevention. However, climate information is hosted in different systems and platforms that are not interconnected, and therefore is not easily accessible by communities, municipalities and even extension services. The local population cannot access this information as they rely on cellphones for information sharing, while climate information is normally only available through computers, which many community members do not have access to. Such information remains mainly used by research and national institutes (e.g., *Instituto Nacional de Investigaciones Forestales, Agrícolas y Pecuarias* - INIFAP), but its accessibility by ACs, municipalities and local technicians remains difficult. Consequently, climate risks and impacts cannot be effectively integrated into local planning processes, with appropriate consideration of local priorities, context and knowledge. Evidence of this barrier is presented in several recent CONEVAL assessments (2019, 2021 and 2022)⁴⁶. This includes: i) information systems that do not inform integral territorial development support budgets and disbursements to vulnerable communities; ii) scarce guidance on the operation of the Information Systems which prevents information and data from being captured and used; iii) scarce information on climate change and the sustainable management of forest resources; iv) limited dissemination and use of the information generated by the National Monitoring, Reporting and Verification (MRV) System; v) a platform for registering PADFSB support requests online that is still in its infancy; and vi) little relation between the areas of forestry information, climate change, information technology, information, and operational areas to improve the efficiency of the systems. These barriers will be addressed through the activities described below:

- Developing a platform and applications to facilitate access to climate information and early warning systems (Activity 3.1.1).
- Strengthening CONAFOR's Forest Fire Protection and Prevention System (SPPIF), Forest Phytosanitary Surveillance System (SIVICOFF) (Activity 3.1.2), the Project-specific Monitoring, Reporting and Verification System (MRV) (Activity 3.1.3), and the Social and Environmental Safeguards Management System (Activity 3.1.4).

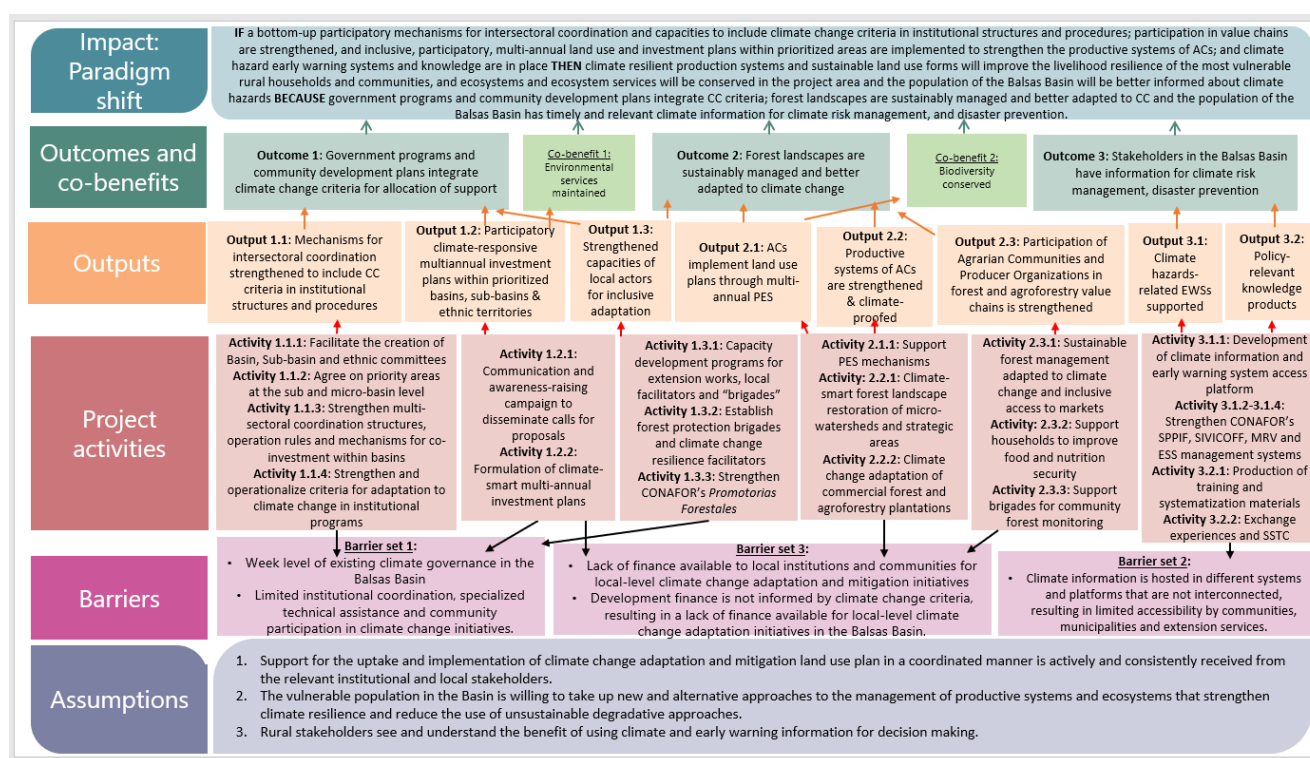
52. Barrier set 3. The restrictive impact of Barrier sets 1 and 2 on a coordinated and informed approach to climate change adaptation and mitigation across the Balsas Basin is further exacerbated by the lack of finance available to local institutions and communities for local-level initiatives. Currently, development finance is not informed by climate change criteria and therefore climate finance flows supporting the planning and implementation of climate change adaptation and mitigation initiatives in the Basin's most vulnerable communities are largely absent. This barrier will be addressed through following project activities:

- Communication and awareness-raising campaign to disseminate calls for proposals (Activity 1.2.1).

- Strengthening participatory climate-responsive multiannual financial investment plans at the sub-basin level (Activity 1.2.2).
- Forest landscape conservation will be supported through PES financed through the PADS, as well as through the Local PES Mechanism (Activity 2.1.1)

53. **Assumptions and Co-Benefits:** The assumptions underlying the achievement of the project's outcomes are described briefly hereafter: Assuming active and consistent support from all levels of institutional and local stakeholders, the uptake and implementation of climate-resilient forest landscape management in the Balsas PA will be achieved in a coordinated and sustainable manner. This will be anchored on the willingness of vulnerable agrarian communities (including local indigenous peoples) and groups (including women and youth) to take up new and alternative approaches to the management of productive systems and forest ecosystems that increase productivity, improve value chains and marked access, and adaptive capacity under conditions of climate change and reduce the use of unsustainable environmentally degradative practices. The uptake of such approaches will be fostered through an improved understanding of the use of climate and early warning information for decision making. The main co-benefits, which are described further below in more detail in Section D.3 are the following: i.) Environmental services maintained, ii.) Biodiversity conserved.

Figure 10: Diagram of the Theory of Change



B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☒	☒	☒	☐	☒
Outcome 2	☐	☐	☐	☒	☒	☒	☐	☒
Outcome 3	☐	☐	☐	☒	☒	☒	☐	☒

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1: Environmental services maintained	☒	☒	☐	☐	☐	☐
Co-benefit 2: Biodiversity conserved	☒	☐	☒	☐	☐	☐

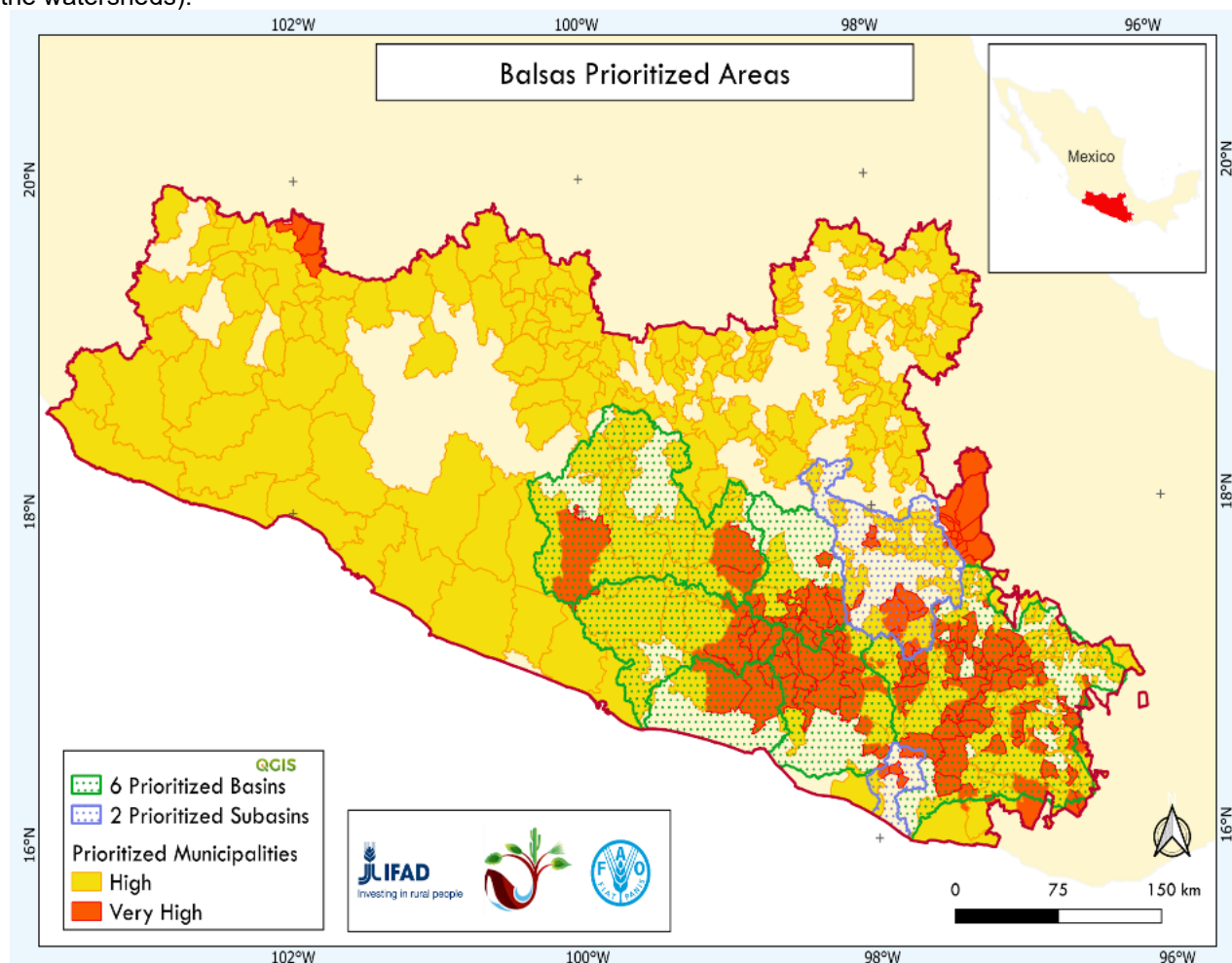
B.3. Project/programme description (max. 2500 words, approximately 5 pages)

54. Project Area Focalization: To specifically target the highest levels of social and environmental vulnerability in the Balsas Basin, the following targeting criteria were applied:

- Vulnerability to climate change: Municipalities with high and very high vulnerability to climate change according to seven vulnerability dimensions established by the SEMARNAT⁴⁷ and the National Institute of Ecology and Climate Change INECC⁴⁸.
- Poverty and social marginalization and prioritized indigenous territories: The level of poverty and social marginalization is being rated by the [National Council for the Evaluation of Social Policies CONEVAL](#).

55. As to these focalization criteria CONAFOR and CONAGUA identified 104 municipalities⁴⁹ with medium to high vulnerability and with high and very high poverty rates that are located in 8 watersheds. These watersheds have a high potential for forest landscape restoration and are important for the provision of environmental services, particularly soil protection for water flow regulation. The prioritized watersheds are: *Río Mezcala-Balsas*, *Río Tlapaneco*, *Río Atoyac*, *Río Ometepe o Grande*, *Río Nexpa*, *Río Papagayo*; and the Sub-watersheds *Río la Arena* and *Río Mixteco*. Additionally, 17 municipalities located in two indigenous territories, the *Xoxo Mixteca* (Puebla) and the *Meseta Purépecha* (Michoacán) were included, resulting in a total of 121 municipalities with the highest level of prioritization (orange color in Figure 11 below). Once piloted and successfully implemented in these municipalities, the project offers the opportunity for up scaling to 280 additional municipalities of high priority (yellow color, below) for a total of 401 municipalities in the Balsas Basin. The detailed analysis and the focalization strategy, as also the detailed list of the municipalities can be found in Annex 2a.

Figure 11: Priority pilot intervention zone of the BALSAS Project: 121 municipalities included in 8 prioritized watersheds, and the Xoxo Mixteca and Purépecha indigenous territories (the latter also marked in orange outside the watersheds).



Source: CONAFOR (2022)

56. The project aims to increase capacity for resilience, adaptation and climate risk management for approximately 211,633 people, women, children and men from vulnerable rural households, assuring social inclusion of women, youth and including indigenous communities. They represent 9.25% of the population of the 10 prioritized areas (8 basins and sub-basins and 2 indigenous territories). Among this target population, the project will directly benefit 40% women⁵, 58% indigenous peoples (*Nahua, Mixteco, Tlapaneco, Amuzgo*, among others) and at least 30% youths. The project will also include other vulnerable groups, mainly the Afro-Mexican population. Detailed project beneficiary calculations are presented in Annex 23. The project is structured into three interrelated, complementary, and mutually reinforcing outcomes as described below.

Outcome 1: Government programs and community development plans integrate climate change criteria for allocation of support (USD 10'091 726)

⁵ Recalling the national context in which women with ejido or communal ownership rights represent about 23%, and that CONAFOR's programs are essentially aimed at those who have rights over the land and associated natural resources. Through a new special sub-program for settled women (see rules of operation), CONAFOR seeks to increase benefits to this sector, but its ability to submit proposals for such support is limited. The latest figures from CONAFOR in the communities supported by the PROFOEM project (World Bank), showed that the percentage of women beneficiaries was around 26% (this would be the baseline). In other words, traditionally the participation of women is much lower than 40%. This has been also confirmed during consultation activities during preparation. Hence, the project proposed an ambitious, yet realistic target of 40% in the project, which it thrives to achieve through its intense gender participation approach.

Output 1.1: Mechanisms for intersectoral coordination are strengthened to include climate change criteria in institutional structures and procedures.

57. Activity 1.1.1. Facilitate the creation of Forest Management Units (FMUs) and Regional Committees for the Integrated Management of the Territory (RCIMTs).

The General Law of Sustainable Forest Development⁵⁰ mandates that CONAFOR delimits Forest Management Units (FMUs), based on physical, environmental, social, and economic aspects, with the purpose of managing forest landscapes sustainably, securing orderly planning of activities and an efficient management of the natural resources. As FMUs in the Balsas have yet to be created the Project will support the establishment of FMUs in the project area in the prioritized basins, and sub-basins, and the two identified indigenous territories. The project will also facilitate the development of Regional Committees for the Integrated Management of the Territory (RCIMTs). These innovative representation schemes will allow for meaningful participation of agrarian communities, indigenous communities, producer organizations, women, and youth, in addition to CONAFOR, local representatives, institutions of the states, federal entities and the private sector.

58. At the level of each state, CONAFOR has created consultation bodies “*Consejos Forestales Estatales*” (CFE), to provide guidance and advice on forest management programs and the development of value chains for timber and non-timber forest products. Nevertheless, at present there are no similar coordination and governance structures at the “lower”, basin and sub-basin levels, where the sustainable use and conservation of forest landscapes, the provision of environmental services, river basin management and hydrometeorological risks are dealt with in a comprehensive way. A new FMU approach that integrates forestry, sustainable territorial development within a watershed and climate change adaptation approach is needed. To advance towards this objective solid territorial governance and decision-making structures need to be in place within the FMUs, where all relevant institutions and actors contribute and take decisions in a coordinated and synergistic way. Inter-institutional and multisectoral interventions to address climate change within an integrated territorial development approach will be one of the key innovations introduced by the Balsas Project, a paradigm changing innovation to be replicated in other regions of the country. The project will support the creation of 10 basin, sub-basin and ethnic territories FMUs and the corresponding committees (RCIMTs) through intensive technical and facilitation assistance that will be provided to CONAFOR’s local extension offices, the “*Promotorias Forestales*” by two specialists, a senior watershed planning and a junior social participation specialist, that will provide technical, communication and facilitation to support the formalization and official registration of these statutory entities.

59. Activity 1.1.2. Agree on priority intervention areas at the sub and micro-basin level. Within each Regional Committee for the Integrated Management of the Territory (RCIMTs), and with project technical assistance and the cartographic support from CONAGUA’s, AGRICULTURA’s and CONAFOR’s experts, priority intervention areas for landscape restoration and risk management within each basin, sub-basin and indigenous territory will be identified through a bottom-up, participatory process. The prioritization criteria for both these dimensions, “landscape restoration need and potential”, as well as “risk management” shall be further developed within each RCIMT with the participation of stakeholders and included in an Operational Manual, in alignment with the improved prioritization criteria to be implemented during project implementation. Nevertheless, the basic elements and dimensions for this prioritization are included in paragraph 62, namely: incorporating climate change dimensions for sustainable forest landscape management and restoration; the potential for sustainable use, conservation and restoration of forests, its natural resources and environmental services, the identification of water supply and demand zones, among others. Detailed description of prioritization criteria for each of the incentive schemes of the PADFSB can be found in Sections 5.5.5 and 5.5.6 of the Feasibility Study, Annex 2.a to the FP; as well as in the Operation Rules of the PADFSB included in Section 6.

60. Building on CONAFOR’s “Regional Forest Management Programs” (RFMP) methodology and taking on board relevant elements and lessons learnt of the “Integral Basin Management Action Plan” (PAMIC) methodology⁵¹ developed by the SEMARNAT and the National Commission of Natural Protected Areas (CONANP), a participatory territorial planning process will be facilitated. This participatory prioritization and planning process will incorporate climate change dimensions for sustainable forest landscape management and restoration at the basin and sub-basin levels. This participatory territorial planning instrument encompasses sustainable use, conservation and restoration of forests, its natural resources and environmental services at the basin, sub-basin and ethnic territory levels. Project technical assistance will enable different stakeholders to develop land use zoning plans based on socio-economic and climate change vulnerabilities including higher rainfall intensity, flash floods, soil erosion and landslides, increasing temperatures and droughts.

61. Through this participatory territorial planning approach each basin, sub-basin and ethnic territory will be delimited into several prioritized micro-basins, where rural communities and municipal capitals share social and economic relations and together can take decisions related to the management of water, soil, forest, and other natural resources. This concept helps to identify water supply and demand zones and connects them through joint work with the municipalities, agrarian communities, and other state and federal government entities. Moreover, it harmonizes the cultural and the physical-natural areas that facilitate the social management of micro-

basins. In the Balsas PA, these prioritized areas may include three to five municipalities, and several agrarian communities.

62. The main elements of a stepwise participatory territorial management plan are the following:
- Identification of prioritized micro basins, in terms of hydrological functionalities and vulnerability to climate change
 - Assessment of ecosystems, water supply and demand zones (ground and surface water supply and sediment retention potential); and identification of communities that have no supply from irrigation systems or local drinking water networks
 - Participatory validation of technical models for sustainable management and conservation of ecosystems, developed according to the specific conditions of each basin and sub-basin and the most promising value chains in terms of the existing market, the inclusion of smallholders, women, young and indigenous peoples, as well as the participation of the private sector (leading to Outcome 2)
 - Culturally appropriate, gender inclusive communication and awareness raising activities about the Project, its objective, and its approach
 - Assignment of responsibility to a *Promotoría Forestal* to each RCIMT for facilitation support acting as technical secretariat
 - Discussion of proposals to strengthen technical capacities and organization for forest firefighting and prevention, dealing with extreme hydrometeorological events, as well as adequate access to information systems (leading to Outcome 3)
 - Disseminate information about the incentive programs offered by CONAFOR, CONAGUA, AGRICULTURA and BIENESTAR
 - Sharing of information about institutional programs and budgets to strengthen complementarities and synergies at all 3 levels of government, with the participation and oversight of local actors and the Agrarian Communities (leading to Activities 1.1.3 and 1.1.4)
 - Explore opportunities for involvement and additional support from other actors and the private sector.

63. Activity 1.1.3. Strengthen multi sectoral coordination structures, Rules of Operation and mechanism for co-investment within the basins. CONAFOR's *Promotorías forestales* will support and facilitate the planning and execution of participatory territorial management plans. For the development of participatory territorial management plans at each basin, sub-basin and ethnic territories, CONAFOR, CONAGUA, local governments, research centers, and other relevant institutional actors will collaborate through the corresponding RCIMTs to identify the prioritized actions and work plans and promote the participation of grassroots organizations, women, youth, indigenous peoples, and the private sector. Traditional knowledge of agrarian communities and indigenous peoples for the sustainable management of the forest landscapes and the adoption of agroecological best practices will be identified and incorporated. The private sector and public companies will also be approached, and their active involvement sought (please see also Annex 7c Stakeholder Engagement Plan). Once a plan has been drafted, municipalities, agrarian communities, and grass-root organizations will be invited to prepare and submit their investment proposals to CONAFOR and CONAGUA to prepare more detailed investment plans to be financed by the institutional incentive programs PADFSB and PROAGUA (refer to the activity 1.2.2). With project support and facilitation, it is expected that this collaborative and participatory planning process will improve inter-institutional and multisectoral coordination, foster the active participation of grassroot organizations and Agrarian Communities, and lead to formal agreements (LoA's, MoU's and others) and an active involvement of the private sector. The preparation of proposals by groups with broad participation of women and young people will be encouraged and formulation supported. Private companies, municipalities and other state and non-state entities will be invited to participate in innovative financing mechanisms for forest landscape conservation, like the Local Mechanisms for PES.

64. Activity 1.1.4 Strengthen and operationalize criteria for adaptation to climate change in institutional program procedures. The Rules of Operation ("*Reglas de Operación*") that norm the PADFSB⁵² regulate the allocation of financial incentives for forest landscape restoration, forest management and value chain improvement. Its prioritization and eligibility criteria are based on socio-economic and ecological aspects. Nevertheless, these criteria do not always target the most vulnerable rural communities and groups such as women and youth and fail to include clear criteria for climate change adaptation that are oriented to improve current forest landscape restoration and forest management practices and at the same time, increase the livelihood resilience of the most vulnerable rural communities. Through this activity and with project technical assistance, CONAFOR and CONAGUA will adaptively improve, simplify, and adjust the operational procedures

and criteria contained in the *Reglas de Operación*, based on the experience and the feedback from field operations, to make sure its operational procedures prioritize projects that clearly improve livelihood and ecosystem resilience within an integrated Ecosystem and Community based Climate Change Adaptation approach (please refer also to Chapter 5, section 5.5.5 CONAFOR's *Operational Rules*, section 5.5.6 *Eligibility, prioritization and exclusion criteria according to ROP*, and the related Annex 2a in the Feasibility Study). This involves also the up take into the *Reglas de Operación* of *Principles and Criteria for Adaptive Silviculture* for awarding incentives for forest plantations and sustainable forest management (please refer also to Activity 2.2.2 and 2.3.1 below).

Output 1.2: Participatory climate responsive multiannual investment plans within prioritized basins, sub-basins and ethnic territories

65. Activity 1.2.1 Communication and awareness raising campaign to disseminate calls for proposals. While CONAFOR's *Promotorías forestales* disseminate the offer of PADFSB incentives, they do not have enough capacities and resources to reach all of the vast territory and the communities that they should be benefiting. Therefore, a strategy to target the prioritized watersheds and territories will be developed through an intensive communication and awareness raising campaign that has the objective to inform Agrarian and Indigenous Communities in these areas of the incentive programs and of their new modalities to improve resilience and adaptation to climate change. The campaign will be done through public websites and social media, but also in traditional communication forms in indigenous languages where appropriate, and specifically target women and youth. Climate change adaptation principles and criteria for landscape restoration practices will be communicated, CONAFOR and CONAGUA's incentive schemes explained, and examples of investment models will be disseminated. Economic, financial, and social benefits of climate adapted practices will be explained, discussed, and improved with feedback from the local actors. It is expected that this activity will trigger an increased interest of the most vulnerable and marginalized ACs and its members to submit Project Profiles to be discussed and validated by the RCIMTs. The most vulnerable groups (women, youth, and the indigenous population) will receive technical support and will have simplified formats for presenting their proposals.

66. Activity 1.2.2 Formulation of climate smart multi-annual investment plans based on micro-basin zoning plans and validated through FPIC. The Project Profiles submitted by AC's will be analyzed by the RCIMTs and CONAFOR's *Promotorías Forestales*. Those that are aligned with the Regional Forest Management Programs (RFMPs), are located within prioritized intervention zones, include eligible activities, and comply with climate change adaptation and livelihood resilience criteria will be prioritized. Special attention and support will be provided to proposals presented by groups of women, youth, and indigenous peoples. Once prioritized, CONAFOR, in accordance with the Rules for Operation of the PADFSB, will transfer financial resources to the benefiting organizations and AC's, to recruit accredited technical service providers. As these beneficiaries recruit technical services by themselves, double-dipping of technical support services is not possible. Project Profiles will then be developed into full proposals to be financed by different incentive schemes of the PADFSB⁵³ (please see detailed description of PADFSB incentive schemes also in Outcome 2 and Annex 2c), including local zoning plans and multi annual investment plans. Project experts will assist CONAFOR's local technicians to evaluate proposals making sure that they are technically viable and economically sound, comply with climate change criteria and are subject to a Free, Prior and Informed Consultation (FPIC) process involving all relevant stakeholders.

Output 1.3: Strengthened capacities of local actors for climate change adaptation, resilience, and inclusiveness

67. Activity 1.3.1: Capacity development programs for extension workers, local facilitators, and forest protection brigades. While the impacts of rising temperatures and much stronger and erratic rainfall are clearly felt and known, local extension workers, technical service providers, facilitators and the members of AC's do not have access to the knowledge and to concrete instruments and practical measures to better plan and adapt to the impacts of climate change and variability. Through this activity specific curricula will be developed in which clear and concrete guidance and best practices to better adapt to climate change, improve livelihood resilience, and restore forest ecosystem services will be offered. These curricula will be developed with IPs, women and youths and include training on gender equality and social inclusion and will be specific to the socio-economic and ecological conditions of the Balsas PA, and will be culturally appropriate, translated into the indigenous languages, including *Mixteco*, *Nahuatl* and *Purepecha*. Positive action will be taken to make sure that at least around 50% of trainees are women and youth; and 50% indigenous or afro descendant. Subject of the training curricula will include, but not be limited to:

- Basic principles of adaptation to climate change, the process, and its phases
- Ecosystem services and community-based adaptation practices for forest landscape restoration (see also examples of best practices described in Outcome 2)

- Sustainable, Adaptive Forest Management
- Water harvesting, management and irrigation practices for households and orchards
- Food security and quality, nutrition, and health
- Access to weather and early warning information systems and platforms
- Market access and value chain improvement
- Microcredit, community-based small businesses, and startups
- Forest and watershed protection (forest fire prevention and management, forest health)
- Community monitoring of environmental and social safeguards, forest and ecosystem monitoring
- Instruments and techniques of participatory planning and community organization
- Gender and social inclusion
- Other relevant subjects depending on demand

68. Activity 1.3.2: Support and creation of forest protection brigades and climate change resilience facilitators.

The project will support: i) the creation and strengthening of existing local forest protection brigades (forest fires, pests and diseases) and; ii) the incorporation of women, youth, and indigenous people as local facilitators who participate in actions oriented to improve climate change resilience. These groups will be created at local level to serve the 121 municipalities of the highest priority but should be able to be deployed also to the other 280 municipalities of high priority within the Balsas PA. The Project will support their participation in the formulation of local land use plans and multi-annual investment plans, climate, and hazard risk management, as well as discussions to develop or improve local community' by-laws. Project support will include uniforms, boots, tools, operational expenses for mobilization, control and monitoring rounds, and material for training and extension, among others. In both cases support by the project to these groups will have to be validated by the RCIMTs, under the facilitation and final decision of CONAFOR's local *Promotorías forestales*. Positive action should be applied in both cases to secure the participation of women and young people. It is expected that every year 38 local facilitator groups and brigades will be supported.

69. Activity 1.3.3: Strengthening of CONAFORs' *Promotorías Forestales*. CONAFOR operates 26 regional and local offices in the Balsas Basin called "*Promotorías forestales*" of which 7 are responsible to support sustainable forest development and forest landscape restoration in the prioritized basins, sub-basins and indigenous territories (see Figure 12). These *promotorías* have very limited resources and capacities, which are insufficient to promote the paradigm shift that this project strives to bring about, and to intervene in the most socially and environmentally critical areas. Therefore, CONAFOR will commit substantial public resources to strengthen these *promotorías* through the recruitment of 26 additional local technicians, which will be trained and supported by the thematic specialists recruited by the Project. Merit, but also positive action should be practiced hiring more women and local indigenous people that speak an indigenous language and are active in their communities, guaranteeing the presence of women in all *promotorías* as a mechanism to facilitate the approach of Project initiatives to the female target population. Current and new personnel will be subject to intensive training and capacity development to put them in the capacity to use the training material, promote the effective participation of women, youth and Indigenous peoples, and to profit from experience exchanges, access to information platforms and early warning systems (Outcome 3), as well as the use of safeguard instruments. The *promotorías* will be provided with equipment and material to undertake their duties adequately financed by CONAFORs regular budget, which grants its sustainability and its maintenance in the long term.

Outcome 2: Forest landscapes are sustainably managed and better adapted to climate change (USD 73'496 484)

70. Forest ecosystems and forest landscape that are crucial for the provision of environmental services and to sustain the livelihoods of the most vulnerable households in the Balsas PA have been subject to deforestation and forest degradation through unregulated over-exploitation, the forest fires and extensive overgrazing. As Outcome 1 will support bottom-up participatory processes and technical assistance to improve territorial governance and investment planning on the prioritized watersheds, this second outcome will provide the capacities and investments needed to implement climate adapted landscape restoration and conservation practices and systems, and will to bring under sustainable management, conservation or restoration 324,578 ha of land, contributing to Target 5 on the Updated Strategic Plan of the GCF for 2024-2027.

71. A fundamental paradigm shift that this project will bring about is that incentive payments that CONAFOR grants to Agrarian Communities through the PADFSB comply with principles, criteria and indicators for effective climate change adaptation and improved resilience of the most vulnerable households and communities. The sub-activities described below are examples of how PADFSB's incentive payments can be transformed from BAU to finance for climate smart practices that support climate change adaptation of vulnerable communities and ecosystems. Nevertheless, as the payment of incentives follows a demand driven, bottom-up approach that

responds to the needs and capacities of each Agrarian Community as project proponent, the models and practices described below will surely be adjusted and improved and will differ from case to case, depending on the specific situation and the solution proposed by each Agrarian Community.

Output 2.1: Agrarian communities implement land-use plans through multi-annual payment of environmental services (PES)

72. Agrarian communities are owners and custodians of very large areas of forests and forest landscapes that provide essential ecosystem services for the Basin. Rainwater regulation, storage and erosion control is probably one of the most important ecosystem services not only for the livelihood of Agrarian Communities, but also because the *El Infiernillo* dam, which is the second most important for hydroelectric production in Mexico gets its water from the Balsas Basin. The forest landscapes and ecosystems of the Balsas Basin, but particularly their organic soil components need to be protected as they have the capacity to store rainfall water and release it slowly, which is crucial vis a vis the increasing and more intense and erratic precipitation patterns that have been observed in the last decades, especially in the Sierra Madre del Sur and in the Sistema Neovolcánico Transversal mountain ranges. Ensuring forest landscape conservation is an effective adaptation strategy, which will be supported through PES to be financed through PADFSB Component IV “Environmental Services for Wellbeing” (SA.1.1 *Servicios Ambientales*), as well as through Local PES Mechanisms LM-PES (SA2.MLPSC-FC *Mecanismos locales de pago por Servicios Ambientales a través de Fondos Concurrentes*). Please see also Annex 2.a) The latter can involve state and municipal governments, the private sector, non-governmental organizations (NGO) and communities and will be supported and expanded as a result of Outcome 1 of this Project.

73. Activity 2.1.1 Payment of Environmental Services (PES). For the strategic areas for the provision of environmental services, particularly micro-watersheds identified within the participatory planning and decision-making process promoted through the activities in Outcome 1, Agrarian Communities will be put in the capacities to present proposals and investment plans to CONAFOR to receive multi-annual PES financed by the PADFSB as to its Rules for Operation. In contrast to past practices where only forest ecosystem protection was supported by CONAFOR and bringing on board the innovative approach developed through its PROFOEM Project, CONAFOR will now finance PES not only for forest protection, but also conservation, sustainable use and restoration activities. This is a fundamental change. Additionally, AC are now allowed to spend 50%⁵⁴ of PES transfers in community development, women projects, *traspas* improvements of other activities of their choice. Through this mechanism vital ecosystem services of forests and natural ecosystems on forest landscapes that are in threat of degradation through unsustainable use of fire and land use change / deforestation will be conserved. Additionally, these investments will leverage conservation finance from the private, state and NGO sector to be channeled through LM-PES that will secure the long-term financial sustainability and smooth transition from catalytic public (PADFSB) to sustained long term regional and private finance (LM-PES). To be eligible, Agrarian Communities will have to comply with the eligibility, prioritization and exclusion⁵⁵ criteria included in CONAFOR’s Rules for Operation⁵⁶ and include in their proposals for areas of reasonably well conserved forests and forest landscapes that will have to be prioritized by the RCIMT to secure adaptation benefits and alignment with the RFMPs.

Output 2.2: The productive systems of Agrarian Communities are strengthened and better adapted to climate change and variability

74. The main productive systems that sustain the livelihoods of rural households of Agrarian Communities in the project region are threatened by changes and variability of climate regimes as described in detail in Section B.1 above and in Annex 2b. Improvements to forestry and agricultural practices and management systems will be introduced and implemented by these communities to help them adapt better to significantly higher temperatures and the rise and variability of rainfall patterns to secure their livelihood resilience and to conserve vital ecosystem services. The expected main adaptation impact of the activities described below are the following: (i) reduce soil erosion to decrease sediments, improve water quality and diminish silting of watercourses and flooding; (ii) increase of the time that water remains within the landscape and the watershed, decreasing the force and speed of runoff, as well as increasing infiltration; (iii) conserve soil for productive activities; (iv) moderate extreme temperatures thanks to vegetation coverage and tree shade.

75. To this end, CONAFOR will grant incentive payments to finance the investment plans developed by Agrarian Communities to cover costs for technical assistance, tools, and inputs, as also for labour and time invested by its members. These incentive payments (“*apoyos*”) will be done through the following components of the PADFSB (please see also Annex 2.a, Section 5.5.6): a.) Forest Landscape Restoration of Micro-watersheds and Strategic Areas “*Restauración Forestal de Microcuencas y Regiones Estratégicas*” (RFM); b.) Commercial Forest Plantations and Agroforestry Systems “*Plantaciones Forestales Comerciales y Sistemas Agroforestales*”

(PFCFA); and c.) Community Forest Management and Value Chains “*Manejo Forestal Comunitario y Cadenas de Valor*” (MFCCV). Through project activity 1.1.4 these incentive schemes of the PADFSB, which pursue the objective of forest landscape conservation and the wellbeing (“*para el Bienestar*”) will be improved to include climate adaptation criteria for enhanced ecosystem and livelihood resilience of vulnerable rural population.

76. Activity 2.2.1 Climate Smart Forest Landscape Restoration of Micro-watersheds and Strategic Areas (RFM). This activity has the objective to support members of Agrarian Communities to implement forest landscape restoration practices on degraded areas to recover soil and ecosystem productivity, generate employment and wellbeing and increase ecosystem services and livelihood resilience. The following activities exemplify three possible soil and water conservation options out of a multiplicity of variants and their combinations, that members of Agrarian Communities may want to include in their Proposals and Investment Plans to be submitted to CONAFOR for financial support.

77. Sub-activity 2.2.1.1 Establish climate resilient agroforestry systems. Cocoa and coffee are important cash crops that sustain vulnerable rural livelihoods in the project area, both highly dependent on ambient temperatures and rainfall patterns. Moreover, high value Arabica coffee, especially the type that meets the qualifications of more lucrative specialty markets, requires lower temperatures, as higher ambient temperatures speed up the ripening of coffee berries, leading to poorer cup quality. Rainfall in the Balsas PA will increase and be erratic, exhibiting greater extremes, which will significantly impact coffee, whose production cycle is highly dependent on rainfall patterns. Coffee flowering is triggered by the first rainfalls at the onset of the rainy season, but if precipitation drops off or becomes too heavy, both coffee flowers and fruits may drop from the tree. This stunted fruit growth would result in fewer, smaller beans of lower quality, which would in turn fetch lower prices⁵⁷. Additionally, higher temperature and humidity are expected to increase the spreading of phytopathogens and disease development, such as coffee rust, especially at higher altitudes⁵⁸.

78. Through this activity the agrarian communities will be put in the capacity to restore degraded forest landscapes and pastures into biodiverse coffee and/or cocoa agroforestry systems (AFS). They will be provided with the financial resources to procure technical assistance from CONAFOR’s accredited service providers, and to buy inputs and materials to establish AFS and restore degraded pasture, rangeland, or renew old degraded orchards with varieties better adapted to the changing climate. One of several possible highly biodiverse tree / crop combination within an AFS as presented in Annex 2a, Chapter 3, would include corn intercropped with 3 500 arabica coffee and 467 cocoa plants, 350 cardamom plants, 35 banana and 66 melliferous timber trees or other fruit trees in an area of one hectare. Of especial relevance is the introduction of improved crop varieties more adapted to rising temperatures and heavier rainfall, and more resistant to pest and diseases, with higher productivity and high organoleptic quality. The combination of banana, fruit and native timber tree species and annual crops create a highly biodiverse agroecosystem that will sequester more carbon, regulate the micro-climate, and lower ambient temperature as a result of increased shade. I will protect the soil and water sources, enhance family asset due to timber tree species and ensure food security through crop diversification. Overall, integrative management of AFS will improve social, economic, and environmental sustainability. In addition to these adaptation benefits the GHG mitigation potential of this AFS has been estimated to be - 5.8 t CO₂ eq/ha/year.

79. Sub-activity 2.2.1.2 Establish agave – forest water / soil retention terraces. Increasing and more intense rainfall on heavily degraded areas located on moderate to steep slopes are subject to increased soil erosion and the formation of gullies and ravines. Unregulated grazing, exploitation of remaining trees for firewood and predatory removal of vegetation, including wild agaves, exacerbate the degradation process caused by extreme rainfall and rising temperatures, causing barrages, catastrophic flashfloods, and landslides. Landscape restoration practices to slow down runoff, enhance rainwater infiltration, retention and storage in the soil, as well as the formation of natural terraces along the contour lines are needed to restore landscapes and ecosystem services.

80. Approximately 50 m wide fringes of densely planted agave (incl. *A. cupreata*, *A. augustifolia* and others) will be established along the contour lines, followed up-hill with fringes of biodiverse, naturally regenerated forests (50 m) and enrichment plantings of native tree species (*Cedro*, *Tepeguaje*, *Guachichimil*, *Encino*, *Chiquil*, others). These fringes will be protected through fences during the first 5 years to prevent goats and cattle to destroy the young growing tree seedlings and the agave plants. After this time the densely planted agave fringes, once big enough, will form thorny fences that will hinder goats and cattle to enter the naturally regenerating forest fringes. As agaves mature irregularly, proper management of regeneration and selective harvesting will assure the permanence of these natural agave barriers. Over time the agave fringes will retain and accumulate sediments and organic matter which will form natural terraces contour lines against the slope. These natural terraces will

slow down the runoff during extreme rainfall events, diminishing soil erosion and improving water infiltration, storage and allow for slow water release.

81. As documented in Annex 2 the ample offer of products from the wild agave species (syrup, fibres, artisanal mezcal, others) have a rising demand and are important as income source for households of Agrarian Communities. On the other hand, timber and especially firewood that will be produced in the naturally regenerated and sustainably managed forest fringes are important for the provision of timber and non-timber forest products (fibres, species, medicinal herbs, aromatic essences, among others), and especially firewood for household use and to supply the demand for artisanal mezcal production. The adaptation benefits of landscape restoration through natural fringes and terraces for water regulation has an important GHG mitigation benefit that has been estimated at -26.2 t CO_{2eq}/ha/year.

82. Sub-activity 2.2.1.3 Establish forest water/soil retention terraces. This sub-activity parts from a similar baseline situation as the one described above of degraded areas located on moderate to steep slopes to be restored to forest landscapes for sub-activity 2.2.1.2: The difference consists in the interest of AC and the local markets conditions to plant, process and sell agave; or to rather continue with livestock activities and to transform them into a sustainable, carbon neutral income source. To this end a transformation from extensive unregulated grazing, to regulated rotational grazing on areas of communal use will be incentivized. To achieve this, approximately 50 m wide fringes of forests will be established through natural regeneration and enrichment plantings along the contour lines against the slope, leaving 100 m wide fringes of fenced pastureland free between the next forest fringe, again 50 m wide. These forest fringes will accumulate the sediments and organic matter transported downhill by the water flows foreseen due to increasing and more extreme precipitation, therefore improve rainfall infiltration in the soil, and successively form natural terraces. Native tree species to be used include *Tepeguaje*, *Guachichimil*, *Encino*, *Chiquil*, *Cahuesquitl*, *Guaje* *Leucaena* sp and others. Enrichment plantings and management of *Bursera linaloe* to produce *olinalá* essence, *Cuachalalate* and other medicinal plants and non-timber forest products (*Orégano*, *Palma soyate*, *palma dulce*) will generate additional income opportunities, especially to benefit women.

83. The separated and individualized fringes of grassland will allow for better livestock management by through rotating grazing with mobile electric fences operated with mobile solar panels. Rotational grazing produces less soil compaction and allows the pasture ecosystem to regenerate during the time it is not used. Dispersed naturally regenerated native trees within the pastures fringes (including *samán*, *pochote*, *colorín*, *native guaje* and other *Leucaena* species, *mezquite*, *huizatche*) further improve the soil through nitrogen fixation and gradually incorporate nutrients into the soil-grass system⁵⁹. Furthermore, these trees enhance the nutrition of the herd through protein rich pods, providing shade that improve animal wellbeing and productivity. In dry tropical climate regions in Colombia which are similar to the Intermontane Depressions (ID) of the Balsas PA, Murgueitio et al⁶⁰ found that the presence of shade trees in appropriate density in silvo-pastoral systems reduce high temperatures and maintain soil moisture which resulted in reduction of the average annual temperature between 2-3 degrees Celsius, and had a cooling effect of up to 13 degrees in the hottest days. In addition, relative humidity increased by 10-20% and evapotranspiration was reduced by approximately 1.8 mm/day. This is particularly relevant in the ID Ecoregion of the Basin PA where extensive livestock herding is practiced, and already very high mean annual temperatures are expected to increase even more from the year 2041 onwards.

84. GHG emission from this semi extensive silvopastoral production system is mitigated through tree growth of the forest fringes and biomass accumulation: the net GHG sequestration has been estimated at -26.2 t CO_{2eq} / ha / year. The additional adaptation benefits as well as the environmental, economic, and social benefits from sustainable silvopastoral systems (SPS) are documented in Annex 2a.

85. Activity 2.2.2 Commercial Forest and Agroforestry Plantations Adapted to Climate Change (PFCA). While Activity 2.2.1 (RFM) has the objective to restore degraded forest landscapes, this activity aims to improve climate change adaptation and the productive potential of commercial forest plantations and agroforestry systems and to increase the sustainable production of forest timber and non-timber forest products.

86. Sub-activity 2.2.2.1 Establish climate adapted planted forests (PFCA.1). Due to climate change and variability, forest managers seek to generate tools to prepare the forests they manage for a future climate that may be very different from the one in which these forests were established. The principles and strategies⁶¹ for Adaptive Silviculture explained further below will help maintain production and ecological balance, especially in productive ecosystems that will not be able to be maintained under future climatic conditions. Adaptive silviculture takes advantage of the different coping strategies with which ecosystems deal with changes⁶², among the main ones are:

- Resistance: the capacity of forests to maintain conditions relatively unchanged, this capacity is useful when ecosystems face small changes. Forest managers can help forest's resistance by improving their defenses maintaining the original ecosystem conditions⁶³.

- Resilience: the ability of forests to return to the previous condition after a medium size disturbance, such as wildfire, deforestation, natural pests, or not too extreme climate changes. To improve the resilience more diversity and changes of composition of the forest species should be considered.
- Response options: if climate changes or disturbances are strong and constant, ecosystems can make a transition and respond to the new conditions. In this case foresters should manage ecosystems to adapt to the new conditions.

87. Adaptive forestry is the combination of applied ecology with applied genetics that seeks to foster a system to address the functionality of forest ecosystems under conditions of environmental change to meet future needs. The most important components of adaptive silviculture are:

- Tree density management: having an optimal density reduces risks such as wildfires, water stress by reducing competition for water and nutrients between trees. It is also important to have control over the density of shrubs, as this can interfere with natural regeneration and can be combustible material which may increase the risk to forest fires⁶⁴.
- Stand composition management: maintaining tree diversity, in terms of species, age and size⁶⁵, adding species that are better adapted to future conditions⁶⁶, or more resilient. Planting a wider variety of species and genotypes allows expanding the genetic pool and therefore, improving the adaptive capacity of the forest.
- Assisted migration of species that do not inhabit the area, but are very likely to do so, due to future conditions⁶⁷.
- Threat control: through activities such as cutting and controlled “cold” burning it is possible to limit the effect caused by biotic threats such as pests, diseases, and invasive species; or removing physical threats such as ignition sources⁶⁸.

88. Symptoms of forest decline have been detected in the forests of the Trans-Mexican Volcanic Belt of the Balsas Basin and northwestern temperate forest of Mexico due to an increase in temperature and a decrease in precipitation causing a significant loss of habitat for Mexico’s conifer and pine-oak biomes. Furthermore, as a collateral effect of climate change, the populations of *Pinus hartwegii* have been affected by different species of mistletoe, and the populations of *Abies religiosa*, of the Monarch Butterfly Natural Reserve and threatened species like *Picea martinezii* T.F. Patt., and *Picea mexicana* have reduced their size drastically⁶⁹.

89. Together with national academic research partners, including the Forest Faculty of the University of Durango, the project will support Agrarian Communities to establish and to manage forests applying the principles and the management practices of Adaptive Silviculture mentioned above, support the adaptive adjustment of the Rules for Operation of CONAFOR’s PADFSB and to monitor its impacts. This will be also supported with intensive training and capacity building of CONAFOR’s local technicians of the *Promotorías Forestales*, technical service providers, forest managers and members of Agrarian Communities, the forest protection brigades, local leaders and facilitators. The mitigation impact of climate adapted forest plantations in the Balsas PA has been estimated to be -20.1 t CO_{2eq} / ha / year.

90. *Sub-activity 2.2.2.2 Establish climate-smart silvopastoral systems (PFCA.2)*. In contrast with the pasture-forest fringes described for Sub-activity 2.2.1.3, which is designed for extensive degraded areas of communal use land of Agrarian Communities, this sub-activity consists in supporting individual families or groups of ACs to improve their livestock management on areas of individual or family use of an area of at least 20 ha / household located on moderate to low slopes, in relative proximity to the house, settlement, or town. These systems can therefore be managed in a much more intense way.

91. The baseline situation is extensively managed pasture land with very few or no shade trees that is at risk of further degradation through soil compaction, erosion and consequently, low rainwater infiltration leading to the formation of ravines and gullies. Foreseen higher precipitation volumes increase the risk for landslides and flash floods. On the other hand, low feed quality contributes to higher methane emissions from enteric fermentation while carbon storage into the aerial and soil biomass decreases as soils are being degraded by mismanagement, higher temperatures, and more erratic precipitation patterns.

92. The specific adaptation benefits of silvopastoral systems for the project region are similar to those described above for Activity 2.2.1.3, which are the following: i.) Separated and individualized fringes of grassland will allow for better livestock management by rotating the herd between different fringes /or their compartments formed by mobile electric fences; ii.) Rotational grazing produces less soil compaction and allows the pasture ecosystem to regenerate during the time it is not used and; iii.) Dispersed naturally regenerated native trees within the pastures fringes (including *saman*, *pochote*, *colorín*, *native guaje* and other *Leucaena* species, *mezquite*, *huizatche*) further improve the soil through nitrogen fixation and gradually incorporate nutrients into the soil-grass system⁷⁰. Furthermore, these trees enhance the nutrition of the herd through protein rich pods, providing shade that improve animal wellbeing and productivity.

93. Silvo-pastoral systems in combination with improved forage grasses increase high quality feed availability, thus lower methane emissions, allow for soil restoration, increase resilience to extreme weather events (drought, excess rainfall), provide firewood and contribute to household food security. Apart from providing shade and animal feed, the trees provide additional income (which can be substantial) through the sale of timber of valuable trees like *guanacaste* (*Enterolobium cyclocarpum*), *ramón* or *tillo blanco* (*Brosimum alicastrum*), *cedro* (*Cedrela odorata*), among various others. Because of the importance of livestock production in the Balsas PA, massive adoption can have a profound impact, if some rather basic improvements to actual BAU practices are introduced.

94. As there are multiple alternatives and combinations of the before mentioned elements for a SPS which depend on the local condition and the preferences of each farmer, one SPS model with its technical specifications and an economic and financial analysis specifically developed and proposed by Simmonit et al. (2020) for the project region is included in Annex 2⁷¹. It is estimated that family income will increase up to 2.5 times within the lifespan of these productive projects and that the net GHG mitigation potential of the introduction of silvo-pastoral systems will be approximately -13.25 t CO_{2eq} / ha / year.

Output 2.3: Participation of Agrarian Communities and Producer Organizations in forest and agroforestry value chains is strengthened.

95. Activity 2.3.1 Sustainable Forest Management adapted to climate change, inclusion, and access to markets (MFCCV). This Component of the PADFSB has the objective to support Agrarian Communities that own natural forests to implement actions that enable them to strengthen their forest governance, including their social, technical and cultural capacities to manage, utilize and market their forest resources in a sustainable way. Activities will include elements to make traditional sustainable forest management better adapted to climate change and promote greater inclusion and empowerment of the most vulnerable, women and youth in production processes as well as inclusive, fair trade and certified value chains.

96. *Sub-activity 2.3.1.1 Manage forests sustainably adapted to climate change (MFCCV.1).* During the project appraisal mission several Agrarian Communities that were visited and local CONAFOR technicians reported that natural forests are under stress due to climate change and variability as resin production has declined or had climate related yield variability. Resin yield of *Pinus oocarpa* and *pseudostrobus* forests being traditionally harvested by Agrarian Communities are highly sensitive to temperature extremes and show increased susceptibility to pests and diseases, particularly to bark beetles (*Dendroctonus* sp.), which have already led to catastrophic forest loss in broad areas of Central America, particularly in neighboring Guatemala and Honduras. On the other hand, natural forest of *Bursera linaloe* traditionally used by Agrarian Communities for their aromatic timber and aromatic oil had much lower seed production in years of longer dry periods and years with irregular rainfall.

97. Through this sub-activity the project will support Agrarian Communities to plan forest management, strengthen governance structures and to implement cultural and harvesting practices applying principles, methods and best practices of adaptive silviculture described for Sub-activity 2.2.2.1 The objective is to promote sustainable forest management that improves the resilience of the forest ecosystems which are an important source of income for the most vulnerable rural households and on the other side provide vital ecosystem services, especially water regulation and soil protection. Furthermore, the project will support community forest monitoring and impact evaluation for this activity, in collaboration with academic partners as the Forest Faculty of the University of Durango.

98. Agrarian Communities will be supported with incentive payments included in PADFSB MFCCV.1 to obtain technical support from CONAFOR's accredited technical assistance service providers, to formulate or improve forest management plans, acquire tools and inputs, pay for the labour invested and strengthen their organization and technical capacities to undertake climate adapted, sustainable management and harvesting of timber and non-timber products.

99. Apart from evident adaptation benefits, resin (rosin) production will be maintained, and climate adapted diversified sustainable forest management will produce additional timber and firewood that will increase income for Agrarian Communities. As the baseline situation in this case is traditionally managed natural forests, the additional mitigation impact of this activity has been estimated to be of only -0.88 t CO_{2eq} / ha / year.

100. *Sub-activities 2.3.1.2 and 2.3.1.3 Sustainable use and added value to timber and to non-timber forest products (MFCCV.2).* Agrarian Communities in the Project Area frequently depend on the use and commercialization of forest products to sustain their livelihoods. For produce sold in raw form or poorly transformed low prices are obtained and the producers stay at the lowest link of the supply chain. This often causes overexploitation and forest degradation, putting vulnerable rural households and forest ecosystems in even greater peril and perpetuating the poverty-resources degradation vicious cycle. To overcome marked

access burdens, financial incentives and technical assistance will be provided to Agrarian Communities and producer organizations to undertake strategic planning, organization, value chain development and marketing for sustainably harvested timber and non-timber forest products. Supply, processing, marketing and certification of forest products will also be supported with the objective to access more sophisticated, better paying markets for forest products with a low carbon footprint and socially responsible, fair traded production process. Since value addition activities are an important space for the participation of women, the provision of financial incentives and technical assistance to groups with broad female participation will be encouraged and prioritized.

101. *Sub-activity 2.3.1.4 Support women and youth in benefitting from forest-based value chains (MFCCV.3).* Financial and technical support will be provided specifically to trigger entrepreneurship and projects designed, implemented, and led by groups of women and youth. This specific funding window of the PADFSB will support women-managed productive projects and enterprises with the objective to support the empowerment of women within forest based productive processes and strengthen their voice and decision making within their communities. These projects may involve any step within the forest production chain, as well as any product and service development based on the sustainable use of forests and forest landscapes. These initiatives may include the use of pine woods, pine resin, *olinalá* essence (*Bursera linaloe*), oak charcoal, palm to produce handicrafts (*Brahea dulcis*), a diversity of fruits, aromatic plants and medicinal herbs (including *guachalalate*, *damiana*, *orégano*) among others. Some groups of women in the Project Area have developed nurseries to produce seedlings for forest and fruit trees, as well as non-timber forest plants like native agaves. The project will search to improve and scale-up these initiatives as the demand for seeds and seedlings with a wider genetic diversity and resilience for climate adapted silviculture will rise; as also to support landscape restoration terraces with agave and native tree saplings. Projects will be implemented under PMFSB incentive line MFCCV.3.1 for productive projects for women and youth which additionally supports women group initiatives to lighten the burden of care and household tasks, so that women can dedicate more time to these productive activities.

102. *Sub-activity 2.3.1.5 Capacity building for local producers in sustainable forest management adapted to climate change (MFCCV.4).* In complementarity with the activities to improve sustainable production of forest goods and services, transformation and added value of forest raw materials members of Agrarian Communities and Producer Organizations will receive intensive training and capacity building. Training in gender and inclusion will be part of the themes to create socio-cultural spaces that are more receptive to the participation of women and youth. In case of women, the strengthening of capacities in organizational, managerial and marketing aspects is also contemplated. This training will involve courses within the Agrarian Communities, exchanges of experiences between Agrarian Communities and forest practitioners and groups of women to share best practices and experiences, as well as scholarships for community members.

103. *Activity 2.3.2 Support households to improve food security and nutrition.* Higher temperatures and more erratic rainfall patterns that have been identified in the project region pose an additional hazard to these already highly vulnerable communities that can be mitigated through targeted strategies to improve the adaptation of the *traspatio* (backyard) and the *milpa*⁷² food production systems, which are the main sources of food for these vulnerable households.

104. *Sub-activities 2.3.2.1 – 2.3.2.3 Traditional milpa (maize/beans) and traspatio orchards are transformed to diversified milpa with trees, improved irrigation and water solutions for improved food quality and nutrition.* Vulnerable households will be supported in improving the *milpa* and *traspatio* house-near orchards through targeted technical assistance for women, youth, and indigenous peoples, to build capacities in food security and nutrition, conserve and multiply native varieties, introduce new crops more resistant to higher temperatures, drought and flooding, (e.g., *pithaya*, *amaranthus*, *aloe vera*, others), increase the crop diversity to reduce dependence and spread risks, and introduce more fruit and firewood producing trees. Other measures to increase the livelihood resilience and reduce resources use include water solutions like drop irrigation, rainwater harvesting solutions and improved cooking stoves that consume less firewood.

105. The project will specifically support the improvement of the traditional *milpa* (maize and beans) and *traspatio* production system into diversified agroforestry adapted to climate change. A basic but fundamental technological innovation to be introduced is that furrows will be established along the contour lines to reduce soil erosion and water run-off from rain that is increasing in volume and intensity. Additionally, a wide diversity of fruit trees (e.g., lemon in warmer zones, peaches in the mountains) and other native trees (firewood) will be planted every 10 meters along the contour lines to form natural terraces which will improve soil formation and protection, nutrient cycling, and rainwater infiltration. Additional specific technical assistance will be provided to support the participatory design and implementation of water harvesting and storage solutions, including collector roofs, collector tanks (*"sombrecitos"*) and reservoirs, among others. Due to the relevance of women in *traspatio*

production, they will be the main stakeholders and recipients of the services and infrastructures that the project will support.

106. Activity 2.3.3. Brigades to strengthen adaptation capacity in Agrarian Communities (Community Forest Monitoring). Traditionally the measurement for the National Forest Inventory and for the National Measurement, Reporting and Verification System (MRV) for REDD+ has been outsourced by CONAFOR to private service providers. This activity has the objective to pilot community forest resources monitoring to empower the Agrarian Communities not only to manage their forest resources sustainably, but also to monitor the forest cover and the forest condition. To this end, a total of 38 community forest brigades (MRV and forest protection/adaptation) will be formed in the project region that will be put together of 154 community members with a strong participation of women and young people.

Outcome 3: The producers, inhabitants and authorities of the Balsas PA have timely and relevant climate information for climate risk management, disaster prevention and knowledge management (USD 3'648 721)

107. **Output 3.1: Climate hazards related early warning and monitoring systems supported.** The main institutions working in the Balsas PA (CONAFOR, CONAGUA, and AGRICULTURA) have been developing advanced information systems related to weather forecast, forest fires, medium/long term evolution of climate patterns and climate risk prevention. However, climate information is hosted in different systems and platforms that are not yet interconnected. On the other hand, this information is accessed and used mainly by research and national institutes (e.g., *Instituto Nacional de Investigaciones Forestales, Agrícolas y Pecuarias* - INIFAP) and is not easily accessible to communities, municipalities, extension services and to the leaders and households of Agrarian Communities.

108. Activity 3.1.1 Development of a platform and applications to facilitate access to climate information and early warning systems. The Project will support the development of an online platform to facilitate access of vulnerable communities, extension workers, the *Promotorías forestales*, municipalities and other regional and local actors to existing information on weather forecasts and early warning about hazards from extreme / anomalous climate events to improve their readiness and preparedness. To this end, an initial participatory diagnostic of the regional and local reach and the barriers of the existing climate information systems and risk management of the National Disaster Prevention System CENAPRED will be carried out to understand to what extent it is used by municipalities, Agrarian Communities, indigenous communities, and households in the project region, considering the importance of women having access to the information provided through the platform. As a result, an online digital platform and its applications will be designed in coordination between CENAPRED, CONAGUA and CONAFOR. The platform and its applications will be disseminated at regional level, and capacities of the members of local institutions, Agrarian Communities and households will be strengthened to manage agroclimatic risks, including the use of participatory mechanisms like "talking" agroclimatic risk maps, planting and harvesting calendars, and participatory vulnerability analysis tools. Considering the availability of mobile phone by the population, access to the platform will be promoted through these devices to facilitate access for women and young people. The experiences of participatory Agroclimatic Roundtables and [Agroclimatic Bulletins](#) piloted in the State of Chiapas will be replicated and scaled up in the project area. This process will be done in close coordination with the capacity building activities 1.3.1, 1.3.2 and 1.3.3 involving the brigades and the local facilitators, including groups of women and youth, in coordination with the regional and local risk management entities (CENAPRED). Local committees / brigades and extension workers will be supported to draft contingency plans to deal with climate related emergencies and disasters, response preparedness and management at community level, as well as to use the community tools for the evaluation of damages and losses.

109. Activity 3.1.2 Strengthen CONAFORs Forest Fire Prediction and Prevention (SPPIF) and Forest Phytosanitary Surveillance (SIVICOFF) Systems. As in other regions of the North American West, forest fires pose an increasing risk to natural forest ecosystems and communities in Mexico and in the Balsas Basin. This is clearly due to higher temperatures that, especially during the dry season, increase the susceptibility and exposure to forests fires. The Project will strengthen the existing Forest Fire Prediction and Prevention System (SPPIF) to make it more reliable and effective to provide timely information and generate data on areas at higher exposure, risks and on burned areas. A diagnostic of existent resources and capacities will be carried out and supplementary IT equipment procured to make sure that the information generated by the SPPIF and the SIVICOFF can be accessed and used by the local brigades (see also Activity 2.3.3).

110. Activity 3.1.3 Strengthen CONAFOR's Measurement, Reporting and Verification System (MRV). The Project will strengthen the existing MRV System through additional GIS specialists and acquisition of equipment.

The MRV field activities that traditionally have been outsourced by CONAFOR to consultants and technical service providers will take a new approach and will be contracted to Agrarian Communities to undertake community forest monitoring. Community forest monitoring will help to empower community members in the sustainable use and conservation of their forest resources and include especially women and the youth. This pilot to be tested in the Balsas PA will involve local facilitators that will support Agrarian Communities to form the forest monitoring brigades described in Activity 2.3.3. This approach of community forest monitoring with support of local facilitators and brigades will be piloted and tested in the Balsas PA, and lessons learnt capitalized to up-scale community forest monitoring to other States of the Mexican Federation. CONAFOR's MRV System will be strengthened with IT equipment. Community Forest Monitoring Brigades will be supported by the purchase of vehicles and field equipment for forest measurement (including notebooks, GPS, measuring tapes, other). This equipment will be kept and used at CONAFOR Promotorías Forestales and maintained as per CONAFOR's standard equipment maintenance protocols after project completion.

111. Activity 3.1.4 Strengthen CONAFOR's social and environmental safeguards management system. While environmental and social safeguards are fully integrated within CONAFOR's rules and procedures, its understanding and operational implementation still needs to be reinforced, especially at the level of its field offices, the *Promotorías forestales*. The project will provide intensive training and capacity building to CONAFOR's *Promotorías forestales*, but also to the technical service providers and local facilitators to make sure environmental and social safeguards, and especially the project Gender Action Plan and the Indigenous Peoples Planning Framework IPPF are understood and duly implemented.

Output 3.2: Policy-relevant knowledge products from the systematic documentation of good practices and intervention approaches of the project

112. Activity 3.2.1 Production of training and systematization materials. With project support material for capacity development, knowledge products and communication products will be developed. Relevant themes include, but will not be limited to basin intervention approach, forestry adapted to climate change and adaptation models, forest ecosystem monitoring system in the field, gender empowerment and youth inclusion, among others. Successful experiences will be systematized and documented in at least 4 relevant knowledge products to be disseminated at the regional and national level and will also be available on CONAFOR's webpage, including successful experiences of women in climate resilient activities, relevant for the exchange of knowledge, the adoption of new policies and institutional strategies.

113. Activity 3.2.2. Exchange of experiences, and South-South and Triangular Cooperation (SSTC). The Project will facilitate the organization of local, national and international exchanges of experiences on themes including, but not limited to adaptive forestry and silviculture, indigenous knowledge on forestry and food production, local vulnerability assessment, basin management, information systems and community MRV, inclusive value chains for women and youth to foster the exchange of experiences between different Agrarian Communities and actors in the Balsas PA. Face to face visits and virtual experience exchange will be facilitated, as also, if required, South-South exchange events with other countries facilitated and supported.

114. Activity 3.2.3 Monitoring and evaluation. Monitoring, reporting and evaluation will be conducted in accordance with IFAD and GCF procedures by the project team. A project-specific monitoring, evaluation and reporting system will be set-up by the PMU to carry out regular project monitoring activities, as also midterm and final project evaluations. A M&E specialist will be recruited to provide overall coordination of monitoring, evaluation and reporting related activities, and an external mid-term and final evaluation team hired at the end of the Project. The possibility of incorporating the IFAD women empowerment index for measuring results on women empowerment will be considered.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

115. The institutional arrangements for decision-making and project operational guidance have been developed based on discussions with the CONAFOR, the National Water Commission (“CONAGUA”), the Ministry of Finance (“HACIENDA”), and the Ministry of Agriculture and Rural Development (“AGRICULTURA”) under the guidance of the Office for the Coordination of Advisers to the Presidency of the Republic (*Coordinación de Asesores de la Presidencia de la República* - CAP). The CAP will support CONAFOR in facilitating dialogue and coordination with other institutions at the federal and state levels, such as the Ministry of Environment and Natural Resources (“SEMARNAT”), which will support CONAFOR in formalizing agreements with other institutions or ministries.

116. For the financing of the Project, GCF will enter into a Funded Activity Agreement (“FAA”, trust account) with IFAD, whereby the nature of the funding for the Project will be a combination of GCF Loans, notably a 50/50 mix of a loan with high concessionality (“GCF Loan A”) and a loan with low concessionality (“GCF Loan B”) ⁶, and a GCF Grant under which IFAD, in its capacity as Accredited Entity (the “AE”), shall administer the relevant GCF Proceeds.

117. IFAD will, therefore, channel such GCF Proceeds through the Executing Entities (EEs – see below), in accordance with the Accreditation Master Agreement (AMA), the FAA and in line with the terms and conditions broadly described in the Term Sheet.

118. IFAD will enter into the following subsidiary agreements:

i) one Subsidiary Agreement with the Ministry of Finance (“HACIENDA”) for the GCF Loans (the “Subsidiary Loan Agreement”). HACIENDA will subsequently enter into a Mandate and Implementation Agreement to assign: a) Nacional Financiera, S.N.C., Banca de Desarrollo (“NAFIN”) to act as Intermediary on behalf of the Federal Government, which will serve as the Executing Entity (“EE2”) and the fiduciary of the Project’s loans b) The National Forestry Commission (“CONAFOR”) to act as the executing entity for the project, serving as the Executing Entity (“EE3”) with overall responsibility for project execution and coordination. By designated authority from HACIENDA, NAFIN will channel funds to CONAFOR upon request. NAFIN will administer the GCF Loans but will not execute any specific activities.

ii) one Subsidiary Agreement with CONAFOR for the GCF Grant (the “Subsidiary Grant Agreement”). CONAFOR will act as the EE and will bear overall responsibility for the execution and coordination of the Project and will create a sub-account in the Fondo Forestal Mexicano (“FFM”) to manage the GCF Grant.

119. Under both Subsidiary Agreements, CONAFOR will undertake all operational responsibility for the implementation of the Project in the country.

120. The Subsidiary Agreements shall detail specific roles and responsibilities of the relevant ministries involved in the implementation of the Project (CONAFOR) and in the channeling of funds (CONAFOR for the grant and NAFIN and HACIENDA for the loans). The Subsidiary Agreements shall establish the terms for the financing and implementation of Outcomes, Outputs, and Activities, and include the relevant obligations of the Executing Entities in respect of the implementation of the overall Funded Activity, including compliance with the eligibility criteria (as set out in Annex II, the “Eligibility Criteria”) and the Project Implementation Manual (the “PIM”). Further details on the Subsidiary Agreements and other implementation arrangements are set out in the Term Sheet.

121. The Subsidiary Agreements will be legally binding and will pass down the GCF policy requirements as set out in the FAA. CONAFOR will select the beneficiaries of all Activities of the Project (the “Final Beneficiaries”) and procure service providers for specific Project Activities in accordance with the Eligibility Criteria and the PIM and subject to IFAD’s no objection⁷ in accordance with the provisions of the Subsidiary Agreements. CONAFOR will enter into contracts for procured services with the selected services providers, subject to IFAD’s no objection in accordance with the provisions of the Subsidiary Agreements.

The Project parties for Balsas are:

IFAD (GCF Accredited Entity AE)

122. As GCF AE, IFAD will be responsible for overall technical, financial, and administrative supervision, and compliance with GCF reporting and safeguards requirements as defined in the Accredited Master Agreement (AMA) and the Funded Activity Agreement (FAA) to be signed. Project operations will be in compliance with the following instruments: Subsidiary Loan Agreement and Subsidiary Grant Agreement; IFAD General Conditions for Agricultural Development Financing and its disbursements, financial reporting and audit procedures; Government of Mexico regulations for financing state entities with external resources, specifically on issues

⁶ The detailed conditions of the GCF Loans are presented in the Term Sheet.

⁷ IFAD does not have a say in the selection of the beneficiaries and service providers. The no-objection is provided to ensure that the procurement and selection process are followed as per the Subsidiary Agreements.

related to budget allocations and disbursements through the reimbursement mechanism, financial statements and external audits and external audits; and CONAFOR's regulations on financial management systems. The GCF grant and loans will be made available to the Government of Mexico under two financing agreements as described above. IFAD Client Portal will be the online platform used by the Government of Mexico to submit requests for disbursements to IFAD. Project supervision will be carried out directly by IFAD, independent from other national control and oversight bodies. From the financial perspective, it will consist of onsite supervisory missions, which will assess financial management arrangements, identify areas requiring improvement, fiduciary risks, and corresponding mitigation measures. Financial supervision will also be conducted through office reviews of periodic financial and annual consolidated audit reports. The Accredited Entity (AE) shall require the Executing Entity (EE) to report on co-financing by the EE to the satisfaction of the AE. The AE will not be in a position to validate the data at the time of the submission of the semi-annual or annual reports. The EE's auditor will certify the accuracy and reliability of financial information contained. Such audit reports will be submitted by the EE to the Accredited Entity within 6 months from the end of each the fiscal year. After such 6 months period, any subsequent disbursement from AE to EE shall be subject to AE having received to its satisfaction the evidence of the EE co-financing during the preceding fiscal year(s). An interim and final evaluation of the project will be undertaken aiming at assessing: (i) progress in relation to achievement of results and disbursement performance; (ii) effectiveness of the institutional and financial arrangements; (iii) the financial procedures manual, suggesting revisions and updates; (iv) the results of execution and financial management; and (v) achievement of Annual Work Plans and Budgets (AWPBs). All IFAD reviews, as well as the audits, will be guided by IFAD's procurement and eligibility policy and policies to fight fraud and corruption according to Section D (iii) of the IFAD Policy on Preventing Fraud and Corruption in its Activities and Operation (2018) which regulates anti-corruption arrangements and responsibilities of IFAD and recipient.

HACIENDA (Executing Entity 1 – "EE1")

123. HACIENDA is the political and operational government focal point for international climate finance agencies. It also administers and defines the federal budget and expenditures for implementing public programs, including those related to climate. The mandate and implementation contract is a legal contract required by the Mexican legislation by which HACIENDA grants the mandate to a Development Banking Institution (in the case of Balsas: NAFIN), to act on behalf of the Federal Government as Intermediary between the International Financial Institutions (in the case of Balsas: IFAD) and the Agencies and Entities executing programs and projects (in the case of Balsas: CONAFOR) financed with external credit in terms of Preparation, Negotiation; Monitoring and compliance with contractual commitments; verification of procurement processes; training and technical assistance, Disbursements of eligible expenses; financial monitoring and debt servicing.

124. Within HACIENDA, the International Financial Affairs Unit (UAIH, by its acronym in Spanish) is in charge of: i) formulating and undertaking the Public Finance policy in its relations with foreign countries, as well as negotiations with international financial agencies; ii) exercising the rights, and managing the fulfillment of the obligations with banks and international organizations; iii) advising on new project proposals linked to external credit and; iv) negotiating and signing grants and non-reimbursable technical cooperation agreements with foreign agencies and institutions, as well as coordinating and following up on them.

125. The Treasury of the Federation (TESOFE) is an administrative Unit of the HACIENDA in charge of the financial management of the Federal Government's resources and securities. In the Balsas project, TESOFE transfers budgetary resources to CONAFOR, which advances the expenses of the Project activities funded by the IFAD-GCF Loans. As such, HACIENDA will be an Executing Entity, but it will not implement any Funded Activity.

126. As HACIENDA is the authorized Representative of the Borrower, it will sign the Subsidiary Loan Agreement with IFAD. The UAIH designated NAFIN as the financial agent and CONAFOR as the executing agent for the GCF Loans. As the proceeds of the GCF Loan A and the GCF Loan B will be disbursed by IFAD to the Financial Agent (NAFIN), which channels the funds to the Treasury of the Federation (TESOFE) within HACIENDA, and TESOFE subsequently channels it to CONAFOR, HACIENDA will be an Executing Entity, but it will not implement any Funded Activity.

NAFIN (Executing Entity 2 – "EE2")

127. Nacional Financiera, S.N.C., Banca de Desarrollo (NAFIN) is an accredited national Direct Access Entity (DAE) based in Mexico. As mentioned above, NAFIN will act as the fiduciary of the loans of the Project and will channel the funds from IFAD to CONAFOR, as CONAFOR cannot directly receive external loan financing. As such, NAFIN will be an Executing Entity, but it will not implement any Funded Activity. NAFIN confirms the disbursement and ensures that the funds have been credited to its account. Subsequently, NAFIN auctions the funds in the money market, and once the auction is completed, it transfers the equivalent amount in pesos to the

Treasury of the Federation. This includes informing HACIENDA of the value date and the amount in pesos and dollars of the officially discounted disbursement request. The funds are disbursed in their equivalent in U.S. dollars, with the TESOFE being notified of the expected receipt date two days prior to the deposit. The proceeds of GCF Loan A and GCF Loan B, disbursed by IFAD, pass to CONAFOR through the Financial Agent (NAFIN) and the Treasury of the Federation (TESOFE). NAFIN will be responsible for managing disbursements to IFAD using the IFAD Client Portal (ICP) already installed at IFAD.

CONAFOR (Executing Entity 3 – “EE3”)

128. The Secretariat of Environment and Natural Resources (SEMARNAT) has been designated to be formally responsible for implementing the project, and has delegated responsibility for project execution to the National Forest Commission (CONAFOR), a Decentralized Public Entity of SEMARNAT responsible for implementing the country's public policy on forestry and with extensive experience in operating projects with international financing.

129. CONAFOR is mandated by the General Law of Sustainable Forest Development to administer the National Forest Fund (FFM). The National Forest Fund manages all the resources destined for the beneficiaries of the PADFS (including PES) and other minor CONAFOR programs. CONAFOR will create a sub-account in the Fondo Forestal Mexicano (FFM) to transparently manage the GCF Grant and GCF Loan resources. The GCF Loans and GCF grant will have different codes in the SAI system to allow tracking expenditure. Hence, the GCF Loan and GCF grant will have their respective sub-accounts. A Technical Project Team (ETP) will be created within CONAFOR, with CONAFOR technical staff who will carry out the comprehensive management of the technical, administrative, and fiduciary and procurement aspects of project implementation, especially with regard to the loans administration. CONAFOR has an organic structure and governing board that supervises its institutional work. Based on its central structure in Zapopan (Jalisco), CONAFOR has an extensive representation network in the states of Guerrero, Oaxaca and Michoacán. This will ensure the participation of the federal institutions active in the Balsas Basin as well as its facilitating role of municipal-level participation.

130. CONAFOR has gained extensive experience in executing internationally funded projects by working on the design and implementation of diverse programs and activities with IFAD, the World Bank, the Forest Investment Program⁷³, the Bio Carbon Fund, and diverse bilateral cooperation projects managing, and operating loans of over USD 500 million and grants of approximately USD 25 million. CONAFOR has a 15-year history of ongoing successful cooperation with IFAD, focused on forest areas and vulnerable populations subject to exclusion due to their marginalization, poverty, and climate vulnerability. Throughout the formulation and implementation of these initiatives, CONAFOR has adapted and improved the PADFSB to support sustainable forest management and ecosystem conservation and restoration by AC, including through Payment of Environmental Services (PES). CONAFOR has also built its capacity for measurement, reporting and verification (MRV) and the implementation of Mexico's REDD+ Strategy. Building on these CONAFOR has consolidated the FFM, which is a financial instrument to transfer funds to the beneficiaries of the PADFSB allowing for multi-annual incentive payments for PES and RFM, and the capitalization of public and private financial resources (among other for the development of Local Mechanisms for PES).

Management, coordination and supervision

The Steering Committee of the Balsas Project (CDAL)

131. At the federal level, a High-Level Steering Committee (*Comité Directivo de Alto Nivel* or CDAL) will be created to facilitate inter-institutional coordination of the federal institutions involved in the project implementation. The CDAL will provide strategic guidance and oversight for project implementation. It will be chaired, hosted, and convened by CONAFOR, with equal participation of SEMARNAT, CONAFOR, CONAGUA, AGRICULTURA and CAP. IFAD and HACIENDA will be permanent guests, with voice but without voting rights, while FAO will cover a technical advisory role when invited to participate. Depending on the needs and issues to be addressed, additional institutions, representatives from Federal State administrations or other guests may be invited. Within the CDAL, the co-chairing institutions will be represented at the highest level (National Commissioner, National Representative, Director General or Executive Director) or by a delegate, if the high-level decision maker is unable to attend. Decisions of the CDAL will be taken by consensus. It will meet at least twice a year to: (i) approve the Annual Work Plans and Budget (AWPB); (ii) monitor results, approve quarterly technical and financial project reports; (iii) follow-up on the co-financing for the project from institution and federal state governments contributions; (iv) provide for up scaling and replication of project results; (v) provide strategic guidance to the CONAFOR and co-executing and implementing partners; viii) approve the communication strategy of the Project. The meetings of the CDAL will be held face to face, virtually, or in a hybrid manner. The CDAL will also be responsible for the coordination and technical and operational orientation of the Project among the priority

programs of SEMARNAT's Directorate General for Climate Policy and Action, CONAFOR, CONAGUA and AGRICULTURA.

The Project Coordination Unit (UCP)

132. The Project Coordination Unit (UCP for its Spanish acronym) will be based in CONAFOR's headquarters and will be under the supervision of CONAFOR's Project Technical Team (ETP). The UCP will prepare the reports that will be presented by CONAFOR to the CDAL for regular feedback on project implementation. The UCP will consist of 10 people, of which only 4 will be hired by the Project: the Lead Technical Adviser, one financial and one procurement specialists, and a Monitoring and Evaluation Specialist (M&E). The main functions of the UCP will be to ensure efficient project management, coordination, implementation, and monitoring through the effective execution of the AWPB, support and supervision of field personnel and experts, quality assurance and safeguard compliance. The UCP will be also contributing to the financial management of GCF and IFAD financial resources in accordance with CONAFOR's and IFAD's regulations and procedures. The UCP will be in charge of (i) integrating the AWPB to be presented to the CONAFOR and CDAL, based on the investment plans developed in prioritized sub-basin and territories; (ii) follow-up on support to RCIMTs; (iii) contributing to the capacity development of CONAFOR's Promotorías and extension workers; (iv.) facilitating the access to technical assistance to the communities and ensuring the technical supervision of all project activities; (v) facilitating the contribution of CONAFOR's different Technical Divisions, including GreenMex's project team with project activities in the prioritized project areas, and the alignment of activities with the PADFSB; (vi) preparing the quarterly technical and financial reports; (vii) monitoring project indicators in accordance with the logical framework; (viii) ensuring the development of the project baseline, midterm review, and final evaluation; ix) ensure close collaboration between the project activities and other relevant partners at the national, state and municipality levels; and (x) preparation of Terms of Reference (ToR) for the recruitment of consultants and firms, which will be funded by GCF grant resources.

The Regional Committees for the Integrated Management of the Territory (RCIMT)

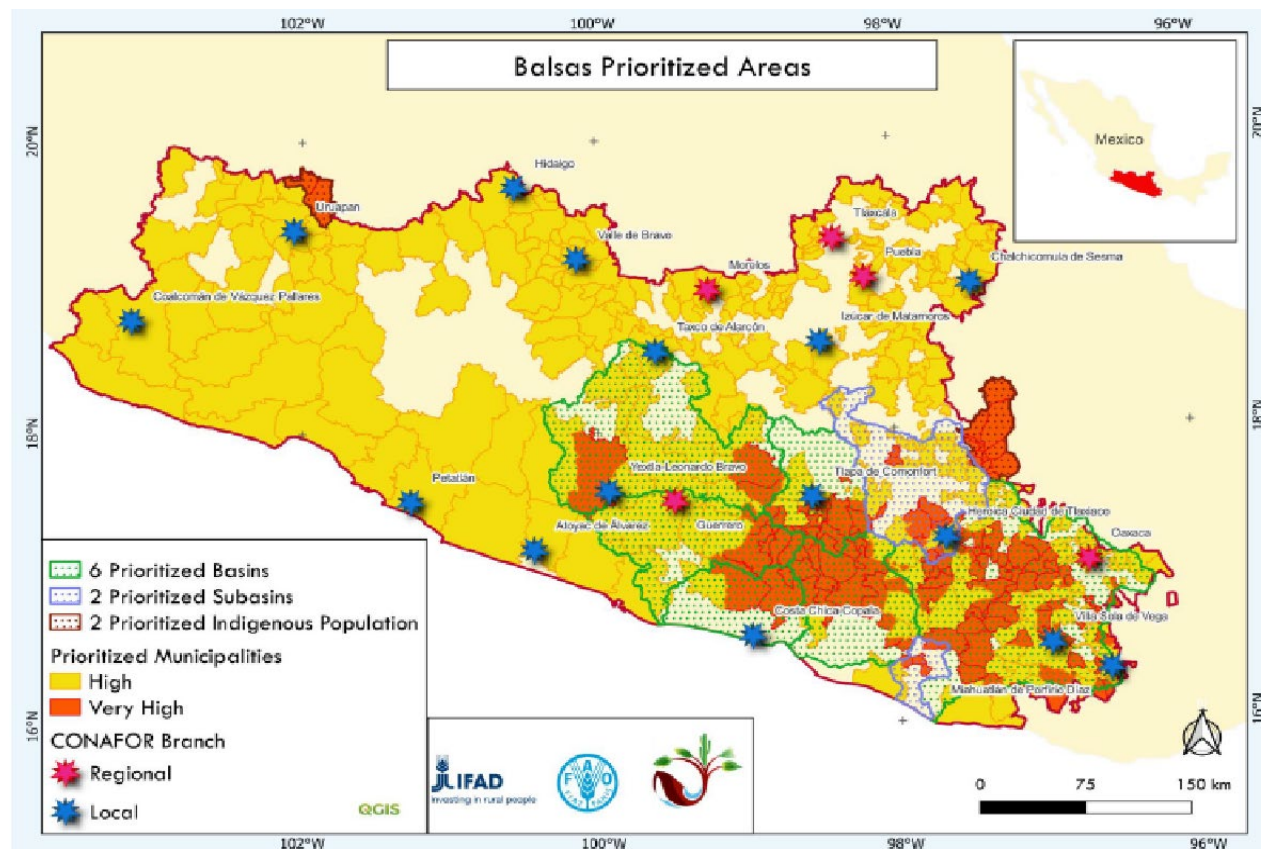
133. RCIMTs are multi-sectoral participation and consultation entities at Forest Management Unit (FMU) level. They will provide advice to CONAFOR and allow for the participation of the local stakeholders in the formulation of local strategies, prioritization of the most vulnerable areas, and monitoring of progress. Moreover, the RCIMTs will support a better coordination of interventions between institutions at the federal, state, and municipality levels. The committee's representatives will be selected by state governments, municipalities, agrarian communities, indigenous communities, organizations of farmers, private companies and other relevant sectoral representatives and actors. They will meet at least once each quarter to: (i) discuss on the Project intervention strategy in the FMU, (ii) approve the prioritization of micro basins; (iii) identify needed financial and technical support and the federal or state programs to cover their demand; and (iv) orient action plans to ensure that the different stakeholders coordinate their intervention and mobilize resources; among other duties. The Local Promotoría of CONAFOR will be responsible for the announcement, preparation of the agenda and elaboration of the minutes, as well as ensuring that the members of the RCIMT will have access to the needed information regarding the project design, implementation procedures and progress reports.

CONAFOR's Forestry Development Offices (*Promotorías forestales*)

134. CONAFOR operates Forestry Development Offices called "Promotorías Forestales" at the state (regional) and local levels to support the implementation of the PADFSB. At the state level, the main functions of the promotorías are the promotion of the PADFSB, identification of priorities in the state; development of coordination mechanisms with the State administration and especially with its environment-related entities; and participation in the selection of the applications for PADFSB incentives submitted by the agrarian communities and other forestry organizations. Within the Balsas Basin, there are seven State level promotorías but only those of Oaxaca, Guerrero, Puebla and Michoacán cover the 10 prioritized project intervention areas. At the local level, the promotorías forestales are located close to the agrarian communities where the PADFSB can be potentially implemented. In each of these small entities, there is a team of forestry facilitators recruited by CONAFOR, whose functions consist of disseminating information on the PADFSB and other initiatives of the environmental sector. They are also responsible for providing technical support to the agrarian communities in their efforts of planning and implementing sustainable forest management, restoration of the ecosystems, the provision of environmental services, and the development of agrarian and productive structures. The location of the existing local promotorías in the Balsas basin will allow allocating one for each sub-basin (see map below). The team of the promotorías forestales will assist the Project UCP lead adviser in facilitating RCIMT meetings to improve the coordination with other institutions and programs operating in the Sub-Basin. The Project will strengthen the promotorías through the recruitment of additional 26 forestry facilitators, and the acquisition of equipment. Technical assistance and training will be also provided by CONAFOR's team and experts recruited by the Project,

especially about adaptive forestry, utilization of different information systems, implementation of safeguard, gender, and social inclusion.

Figure 12 Map with the location of CONAFOR's regional and local *Promotorías Forestales* in the Balsas Project Area



Source: CONAFOR (2022)

Project coordination with other relevant institutions, programs, projects, and initiatives

135. One of the most important public policy challenges in rural Mexico is achieving greater efficiency in terms of quality and outcomes through the various priority programs concentrating the largest shares of government spending in rural areas. This raises the need for maximum synergy, inter-agency coordination and fostering converging actions across federal entities and territories.

136. SEMARNAT aims to incorporate in the different spheres of the Mexican society and the public sector criteria and instruments that ensure protection, conservation, and sustainable use of the country's natural resources, thus forming an integral and inclusive environmental policy that will allow to achieve sustainable development. In order to comply with the above, SEMARNAT, its three sub-secretariats and the various deconcentrated and decentralized entities, such as CONAFOR and CONAGUA, work on four priority aspects: i) The conservation and sustainable use of ecosystems and their biodiversity, ii) Pollution prevention and control, iii) Integral management of water resources, and iv) Combating climate change. As leader of the Environment Sector, SEMARNAT provides advice and supervises CONAFOR and CONAGUA participation in the Balsas PA, and revises the project budget to be submitted to HACIENDA. Especially, the Directorate General for Climate Policy and Action of SEMARNAT, which is the entity in charge of the National Institute of Ecology and Climate Change (INECC)⁷⁴, will support CONAFOR in assessing climate change and perception, diagnosis and vulnerability in the prioritized areas of the Balsas PA, contribute to adaptation measures and monitoring in the intervened areas, train different actors of the project on climate change issues, provide climate change scenarios and support in the calculation of GHG emissions and participate in different technical working groups, including the RCIMT's.

137. CONAGUA undertakes functions related to the national water policy and regulation for its use (concessions, permits, and sanctions for non-compliance), as well as technical and financial support to the states

and municipalities in the development of urban and rural hydraulic infrastructures. The Entity is also in charge of the National Meteorological Service (SMN) and collaborates in civil protection programs, addressing emergencies due to extreme weather events. Programs such as PROAGUA related to better management of water in the prioritized municipalities offers a significant opportunity for collaboration and co-financing. Within the project intervention area, CONAGUA operates two main Administrative Offices (Regions RHA IV; RHA V), each with its own units that contribute to the implementation of PROAGUA and other programs. The communities that do not have access to tap water or suffer from irregular provision will be identified in each municipality. Through the support of the RCIMTs, the Project team will seek synergy with the State administrations, municipalities, and the PROAGUA program to facilitate the necessary co-funding. The investment will consist of rainwater catchment for households, management of micro basins to build small hydraulic infrastructure and development of irrigation systems.

138. **AGRICULTURA** The Ministry of Agriculture and Rural Development leads the formulation, implementation, and evaluation of the general rural development policy. Among other activities, AGRICULTURA operates programs benefiting small farmers, through the transfer of subsidies and provision of technical assistance. At sub regional level, AGRICULTURA operates through the Rural Center for Rural Development (CADER). As co-implementing partners of the project, AGRICULTURA will support the following activities: i) identification of the interventions in the prioritized areas that are aligned with the project, ii) sharing of information on agricultural production, soils, the effects of climate change on agriculture, and productive re-conversion ongoing efforts; iii) technical assistance to the prioritized communities to strengthen capacities especially in agroforestry, silvo-pastoral systems and traspatio production; iv) coordination with CONAFOR to improve fire management and protection, and vi) facilitating collaboration with the sub-regional structures (CADER) and Research Centers that can provide technical assistance and financial support to the producers.

139. **State and municipality Administrations in CONSEJOS FORESTALES ESTATALES (CFE).** It is strategic for the Balsas Project to involve the State and municipal governments, especially in the prioritized sub-basin, mainly located in the states of Guerrero, Oaxaca, Michoacán, and Puebla. In that sense, the CFE promoted by CONAFOR, and the State administration are relevant entities to share information and seek synergies. The CFE, in which state authorities, forestry organizations and municipalities are represented, will serve to establish specific collaboration agreements and co-investments, facilitating, in the prioritized municipalities, the possibility for states and municipal governments to participate financially in the project initiatives, such as the Local Mechanisms for Payment for Environment Services, micro basin restoration, or the implementation of PROAGUA. In the recently elected state administrations of Oaxaca, Guerrero and Michoacán, the Project will promote the inclusion of the Balsas project objectives within the State Development Plans.

140. **Engagement of the private sector** The project will seek to create partnerships with business organizations and private companies, with the aim of both developing innovative schemes for the payment of environmental services and the integration of inclusive value chains. Private sector engagement could be oriented to:

- Create Local Mechanisms for Payment for Environment Services, based on the experience and instruments developed by CONAFOR, with the purpose of promoting the participation of private companies that use environmental services (water, landscape, carbon sinks, etc.).
- Foster the financial inclusion of forestry and other social enterprises with Multiple Purpose Financial Companies (SOFOMs in Spanish) that have already opened credit lines to finance green and inclusive business models.
- Support access to the Guarantees Funds that CONAFOR created to facilitate the access of credit to community forestry companies for the development of partnerships with private companies for timber and non-timber forest value chains⁷⁵. Emphasis will be given on facilitating access to credits for small-scale producers, especially women, young people, and indigenous producers to develop commercial agreements with the private sector in conditions of fairness and equity.
- Specific needs of industrial processes that allow the inclusion of women and youth in the forestry and agroforestry value chains, tapping into potential green and fair-trade markets. Products with a high potential for such agri-business partnerships are pine and other tree resin, timber, coffee, agave, and palm fibres.

Within this diverse and challenging institutional setting, CONAFOR and its partners have negotiated and agreed on roles and functions for its project partners and project entities that are systematized in the following matrix:

Table 4: Main functions of Project Entities and Partners

PROJECT ENTITY	INSTITUTION (S)	MAIN FUNCTIONS
STEERING COMMITTEE (CDAL)	SEMARNAT, CONAFOR, CONAGUA, AGRICULTURA, CAP	• Provide political decision-makers information on project progress, results, and impacts. • Provide political and strategic orientation • Secure good inter-institutional coordination • Provide transparency, accountability, and participation • Review and approve Annual Working Plan and Budget (AWPB) prepared by the UCP and CONAFOR • Review and approve the selection of Implementation Partners presented by the Executing Entity
EXECUTING ENTITY	CONAFOR	• Sign the Subsidiary Grant Agreement with IFAD, the Accredited Entity (AE) • Sign the Mandate and Implementation agreement with HACIENDA as responsible for the execution of the project (GCF Loans). • Request loan and grant resources once activities are under implementation • Request reimbursement to NAFIN once financial information is aggregated and consolidated • Procure the provision of technical services, goods and inputs needed to achieve Outputs according to the AWPBs approved by the Project Steering Committee (CDAL) according to its internal procedures • Prepare calls for proposals, select and contract services providers • Set up and operate CONAFOR's Technical Project Team ETP and the UCP of the Balsas Project • Select and contract implementing partners after approval of the CDAL • Sign collaboration agreements with CONAFOR and co-implementing partners (public, national and international) for synergistic and complementary actions in the PA • Transfer financial project resources to the National Forest Fund and then to the beneficiaries, according to the contracts celebrated with them • Monitor and evaluate the implementation performance and safeguard compliance of implementation partners • Prepare and monitor the execution of AWPB • Prepare and submit quarterly, bi-annually and yearly follow-up and monitoring reports to the AE • Prepare and submit specific reports on demand of AE • FFM within CONAFOR: Prepare and submit quarterly, bi-annual and yearly follow-up and monitoring reports to the AE for the GCF Grant
EXECUTING ENTITY and National Designated Authority to the GCF	HACIENDA	• Sign the Subsidiary Loan Agreement with the AE (IFAD), on behalf of the Government of Mexico • Sign the Mandate and Implementation Contract with CONAFOR and NAFIN • Transfer resources received from NAFIN to CONAFOR (EE3) through its Treasury of the Federation (TESOFE), according to the annual budget approved by the National Congress • Monitor execution of AWPB presented by CONAFOR • Participate as Observer in the Project Steering Committee (CDAL)

EXECUTING ENTITY, Financial Agent and National Direct Access Entity	NAFIN	• Ensure that the contractual conditions of the EE operations are aligned both with the GoM legal framework and with the Project Financial Agreement • Sign the Mandate and Implementation Contract for loan resources between HACIENDA, and NAFIN • Manage the disbursement of GCF Loans to TESOFE, according to the financial agreement and the financial and administrative reports submitted by CONAFOR • Review CONAFOR's statement of expenditures and submit them to IFAD, together with the next disbursement request for the GCF Loans • Transfer financial resources received from IFAD to the TESOFE in case of the expenditures related to the loans • Monitor execution of AWPB presented by CONAFOR • Prepare and submit quarterly, bi-annual and yearly follow-up and monitoring reports to the AE • Prepare / submit specific reports on demand of the AE • Supervise the financial and procurement management of project resources by the EE and revise audits to the EE • Participate as observer in the Project Steering Committee (CDAL) • Participate in supervision and implementation support missions
Orientation coordination and advise	Coordination of Advisors of the President (CAP)	• Participate in the Steering Committee (CDAL) • Provide technical and operational orientation to implement the priority programs of the Government of México • Support high-level discussions and decision-making process related to the long-term engagement of the government in the Balsas PA • Facilitate the development of inter-institutional coordination mechanisms in the Project Area • Oversee the preparation of the required financial and implementation reports for IFAD
Accredited Entity to the GCF	IFAD	• Sign the Funding Activity Agreement (FAA) with the GCF (GCF Loan A, GCF Loan B and grant tranches) • Sign the Subsidiary Loan Agreement with HACIENDA • Sign the Subsidiary Grant Agreement with CONAFOR • Appraise and finalize project implementation arrangements and advise Executing Entities (EE's) • Advise on / participate in project inception workshops • Assist the EE's to draft terms of reference (TOR) and provide advice on the selection of experts or implementation partners • Participate, as observer in the Project Steering Committee (CDAL) • Conduct at least one supervision mission per year, including briefing operational focal points on project progress. As necessary, include technical consultants during supervision missions to provide advice and assistance to EE's and implementing partners, as needed • Provide technical guidance and support for project implementation, as necessary • Disburse funds to the Financial Agent, monitor and review project expenditures and review financial reports • Oversee procurement and financial management to ensure implementation is in line with IFAD's policies,

		procedures, and timelines, ensuring that any entity relevant to the project complies with IFRS accounting standards. • Oversee the preparation of the required reports for submission to GCF Secretariat • Prepare periodic revisions and implementation support missions to oversee if changes have been made in annual expense category budgets • Assist/oversee the audit process throughout the project life cycle • Undertake the Project Interim evaluation • Oversee the preparation of the project completion report/independent final evaluation, and submit the report to the Secretariat of the GCF • Prepare project closing documents, including financial closure, for submission to the Secretariat of the GCF
Advisory entity	FAO Office Mexico	As implementing entity of the GREENMEX project, which includes the Balsas Basin, the FAO office in Mexico will: • Participate in the sessions of the CDAL as covering a technical advisory role and as invitees when deemed necessary • Assure synergies with the GREENMEX Project to ensure that support to the beneficiaries is complementarity • Contribute to capacity development and technical support to the beneficiaries • Provide information related to international experiences on best practices
Implementation partner	CONAGUA	• Participate in the CDAL and its CTO • Sign collaboration agreements with CONAFOR and co-implementing partners (public, national and international) for synergistic and complementary actions in the PA • Contribute to the climate information systems and Early Warning Systems, based on the information generated in the meteorological stations and the information generated by the National Meteorological Service • Support the creation and strengthening of the RCIMTs in the prioritized basins, sub-basin and territories • Support with information for the Basin and Sub-basin participatory planning and planning processes • Develop water collection and management systems for domestic and productive use, and in the development of hydraulic infrastructure in the project territories through PROAGUA and other special projects • Contribute to strengthening inter-institutional coordination between the entities of the three levels of government • Generate statistics on water availability in the municipalities considered to be of very high priority • Support Agrarian Communities in the management of requests for rehabilitation or opening of water wells for agricultural activities and commercial forest plantations • Through local governments, coordinate actions regarding water supply for productive agricultural activities promoted by AGRICULTURA and for domestic use
Implementation partner	Directorate General for Climate Policy and Action	• Participate in the CDAL Board and its CTO • Support CONAFOR to assess climate vulnerability in the prioritized areas of the Balsas PA

	of SEMARNAT (in charge of INECC)	• Contribute to the monitoring of change in the intervened areas. Train the different actors of the project on climate change issues • Support in the calculation of greenhouse gas emissions • Participate in work session with different technical working groups and the RCIMTs
Implementation partner	AGRICULTURA	• Participate in the CDAL Board and its CTO • Sign collaboration agreements with CONAFOR and co-implementing partners (public, national and international) for synergistic and complementary actions in the PA • Generate and share information on agricultural production, the soil situation, the effects of climate change, and productive re-conversion • Align agriculture public policies, strategies, and other instruments of attention to rural producers that have an impact on the areas defined in the project • Provide technical assistance to the Agrarian Communities to strengthen their capacities regarding climate adapted agriculture and agro-ecology, agroforestry and silvo-pastoral activities • Contribute to strengthening women and youth initiatives related to the development of agroforestry, <i>traspatio</i> production and inclusive value chains • Participate in the efforts to avoid forest fires, subsidies to the extensive cattle ranching in restored or conserved areas or the advancing of the agricultural frontier where there are forest ecosystems • Link CONAFOR and other co-implementing agencies with AGRICULTURA decentralized research and development entities (SENASICA, SNICS, INIFAP, among others)
Participatory multi-sectorial platforms for regional coordination, transparency and accountability	Regional Committees for the Integrated Management of the Territory (RCIMT)	• Participate in the Project, to improve coordination, participation, consultation, transparency and accountability to the CONAFOR • Encourage the coordination of federal, state and municipal government, with the participation of rural communities, indigenous peoples, grassroots' organizations, and private sector • Jointly with the <i>Promotorías Forestales</i>, promote the project objectives and modality of support within the Balsas PA • Contribute to the identification and communication effort of the risks and alternatives related to climate change and social vulnerability in the basins (including local and traditional knowledge) • Participate actively in the development of the participatory territorial plans and identification of the prioritized micro-basins • Participate in the preparation of regional studies for the preparation of Regional Forest Investment Programs • Provide advice on the project profiles and concept notes submitted by the Agrarian Communities and grassroots organizations to receive support from CONAFOR, CONAGUA, AGRICULTURA; as well as state, municipality of private programs • Contribute to monitor the progress of the project in the field

Figure 13: Diagram of Institutional Arrangements

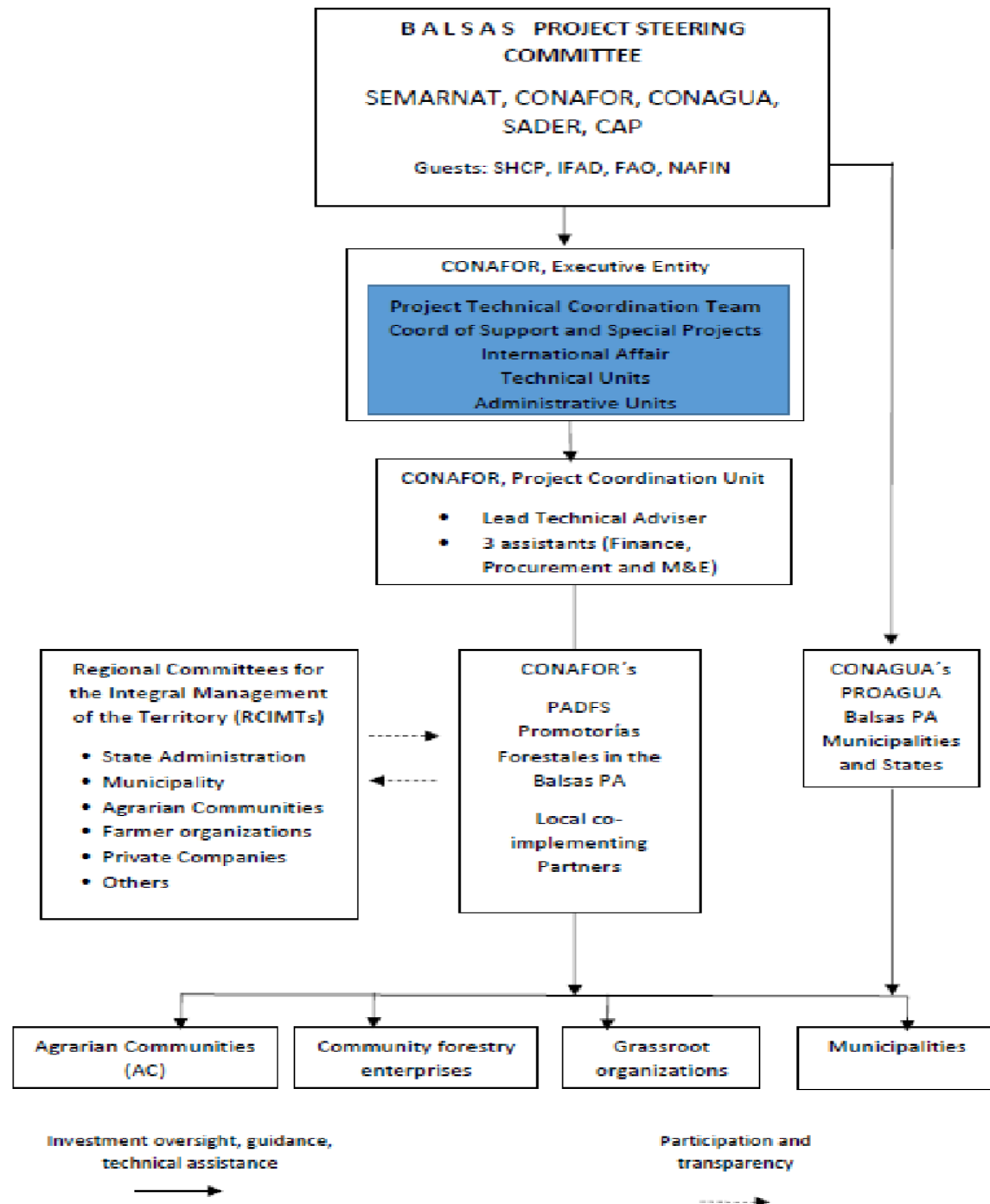
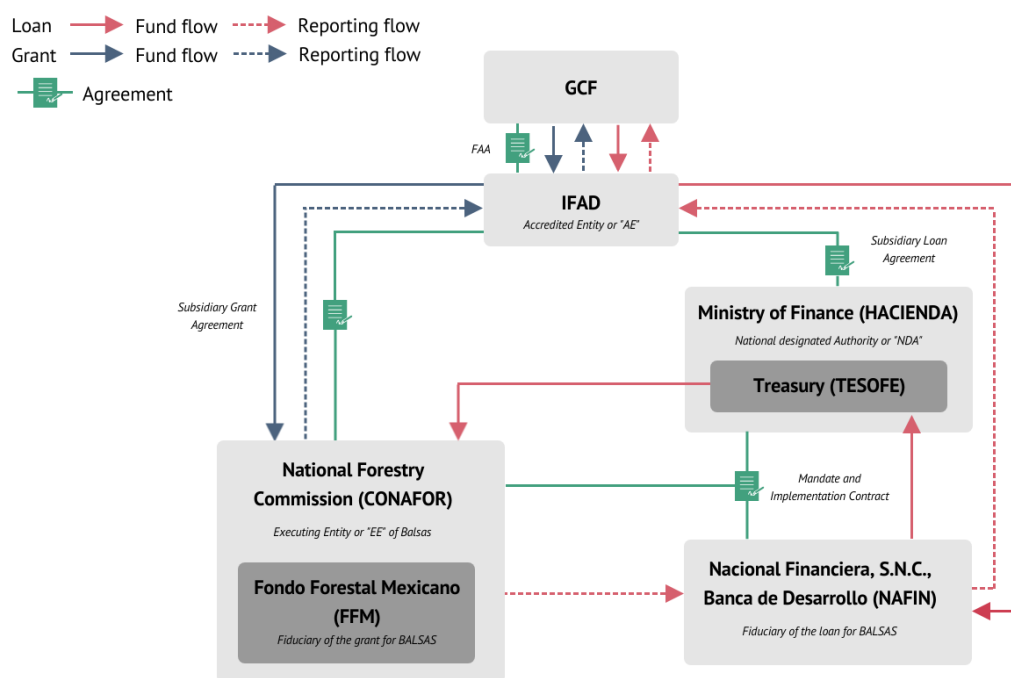


Figure 14. Schematic diagram of financial flows



141. Fund flow for the GCF Loans – red arrows (Figure 14):

- GCF transfers GCF Loan A and GCF Loan B resources to IFAD in accordance with signed financial agreement between both entities.
- IFAD channels (or "reimburses") eligible funds to a bank account managed by NAFIN.
- NAFIN channels the resources received to the TESOFE.
- The Treasury of the Federation (TESOFE) managed by HACIENDA channels budgetary resources to CONAFOR (and in particular the Fondo Forestal Mexicano (FFM)), which pays for the expenses of the Project activities funded by the IFAD-GCF Loans.
- (a) CONAFOR spends the budgetary funds of the Project and processes the payments destined for goods and works, consultancy services, non-consultancy services, training, and incremental costs of the project; (b) CONAFOR selects the beneficiaries, signs the agreements to deliver the financial incentives to the eligible beneficiaries, and instructs BANORTE, financial agent of the FFM, to process the payments.
- The beneficiaries send reports and information to CONAFOR, in accordance with the provisions of the PADFSB Operating Rules and the agreement signed between the parties.
- Based on the reports submitted by the beneficiaries and the payments processed to consultants and suppliers, CONAFOR aggregates and consolidates financial information in the Statements of Expenditures (SOE), and formally sends them to NAFIN.
- NAFIN reviews the Statements of Expenditures and sends them to IFAD, together with the next disbursement requests for the GCF Loans and grant.
- IFAD reviews and verifies those Statements of Expenditures and prepares the Annual Performance Reports and semi-annual financial reports in line with the AMA and FAA.
- .
- Fund flow for the Grant (see Figure 14):
- GCF transfers grant resources to IFAD in accordance with signed financial agreement between both entities.

- IFAD channels resources to CONAFOR (and in particular the Fondo Forestal Mexicano (FFM), in accordance with the total amount of the activities planned in the Annual Operating Plan presented by this entity.
- (a) CONAFOR spends the grant and processes the payments destined for direct supports, goods and works, consultancy services, non-consultancy services, training, and incremental costs of the project; (b) In the case of direct supports, CONAFOR selects the beneficiaries and the supported ACs' initiatives, according to the approved operational guidelines, signs the agreements to deliver the financial incentives to the eligible beneficiaries, and instructs BANORTE, financial agent of the FFM, to process the payments
- The beneficiaries send reports and information to CONAFOR, in accordance with the provisions of the special guidelines that the Project prepare and the agreement signed between the parties.
- Based on the reports submitted by the beneficiaries and the payments processed to consultants and suppliers, CONAFOR aggregates and consolidates financial information in the Statements of Expenditures (SOE), sending the proofs of the eligible expenses to accredit the expenses made. The SOE is reported to IFAD.
- IFAD reviews and verifies those Statements of Expenditures and prepares the Annual Performance Reports and semi-annual financial reports in line with the AMA and FAA.

No transfers of funds will be made to any other participating institutions identified in the Funding proposal.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

142. Justification The project GCF financing is made up of GCF Loans worth USD 20 million, which consist of a USD 10 million loan with high concessionality (GCF Loan A) and a USD 10 million loan with low concessionality (GCF Loan B), and a USD 19.5 million grant that, together with an IFAD loan of USD 38.453 million aim at consolidating and scaling up innovative forest development financing lines for climate change adaptation and mitigation under CONAFOR's PADFSB and the programs of the other agencies involved. It also promotes a scheme of inter-institutional convergence of priority budgetary actions and programs of the Government of Mexico in the Balsas PA with lines of environmental financing and climate change adaptation and mitigation, coordinating actions and maximizing impacts. The investment loans from external sources will make it possible for CONAFOR to maintain the minimum investment needs of the PADFSB in the PA in a context of fiscal restraint.

143. The project contemplates a national counterpart of USD 13 million that mainly includes CONAFOR's operating expenses, funding for land use change compensation (CA_CUS) and CONAGUA's investments from its PROAGUA program. The GCF grant will be essential to provide the necessary technical assistance to promote the paradigm shift, inter-institutional convergence and to generate innovative productive models for adaptation to climate change, as also to finance productive models that are not financed within the PADFSB (such as MIAF systems or support for backyard agriculture), and the financing of forest monitoring brigades of young people and women. In contrast to the loan resources, under Mexican regulations, the grant will secure financial resources additional to those allocated to CONAFOR by the Ministry of Finance from the fiscal budget.

144. The Balsas Project targets the poorest and most vulnerable rural communities in the Balsas basin, including indigenous peoples, women, and youth. Of the highly prioritized municipalities, there are 121 municipalities of the Project which have very high and high vulnerability and very high and high poverty levels and include 17 municipalities located in two indigenous territories.

145. The poorest and most vulnerable rural communities of those prioritized municipalities have very limited capital available and lack access to finance, making them dependent on short-term returns to meet their immediate needs. Investing in forestry and/or agroforestry will not yield any positive financial returns in the first 3-5 years due to the significant upfront costs and the time it takes for trees to reach maturity and generate revenue (negative cashflows in first years). Hence, the investments and activities related to forestry and agroforestry within the BALSAS project would not occur without the required level of concessional support from the GCF (in particular the grant, but also GCF Loan A). This is also why HACIENDA on behalf of the Government of Mexico will be taking on GCF Loan A and GCF Loan B and will be reimbursing the GCF Loans to the GCF via IFAD. The beneficiaries, in turn, will receive the project financing from the Government of Mexico in the form of non-reimbursable grants.

146. Importantly, it is worth noting that in the absence of concessional support from the GCF, the paradigm shift is unlikely to occur. The current government programs would continue to be implemented in their current form and would neither reach the highly vulnerable communities identified by the GCF Balsas project, nor

would they allow for climate change criteria to be included for innovative production models for adaptation to climate change to be financed within the PADFSB.

147. IFAD and the Government of Mexico carefully costed the project in a way that concessional resources (GCF Grant and GCF Loan A) will be non-revenue generating, while the least concessional resources (Loan B) will be used for revenue-generating activities.

- USD 19.5 million GCF grant resources will be used to finance exclusively non-revenue generating activities and are either:
 - o tailored to cover specific activities where the benefits and repayment of the investment are less evident and would therefore only be viable with high levels of concessionality, such as institutional strengthening activities, technical assistance, monitoring and evaluation and project management (Activities 1.1.1., 1.1.2, 1.1.3, 1.2.1, 1.2.2, 1.3.1., 1.3.2, 1.3.3., 2.3.3., 3.1.1, 3.1.2, 3.1.3, 3.1.4, 3.2.2, 3.2.3), through e.g. Activity 1.3.1: Capacity development programs for extension workers, local facilitators, and forest protection “brigades”, which essentially is TA for agroforestry and afforestation.
 - o tailored to incremental costs in the case of Activity 2.3.2, where the grant serves the crucial purpose of providing necessary injections to kickstart the market for an activity which would otherwise not occur as it is financially not viable (financial internal rate of return of 0.19% is less than the financial discount rate of 6%). In the absence of the GCF grant, the diversified milpa systems with agroforestry wouldn't be established as they are currently not part of PADFS support and there are no alternative sources of finances. In the hypothetical case of access to credit, high interest rates would hinder any positive scenario on return on investment. Besides, the vulnerable communities don't have the means to repay loans.
- USD 20 million of GCF Loans consist of a USD 10 million GCF Loan with high concessionality (Loan A) and a GCF Loan with low concessionality (GCF Loan B).
 - o USD 10 million GCF Loan A resources will be used to finance a PES scheme to reach the prioritized poor and vulnerable communities in the Balsas basin (Activity 2.1.1). A highly concessional loan for the Balsas PES scheme (Activity 2.1.1) is justified as it is non-revenue generating and avoids potentially higher costs of addressing deforestation-related problems in the future. Compared to other sectors (e.g. energy), Forest Lands and ecosystems usually receive a much higher proportion of grant finance (e.g. RIOS project in Mexico with 100% grant). Using a GCF Loan for a PES scheme is a key innovation that fosters improved incentives for scaling up forest establishment and maintenance activities as it brings co-responsibility for natural resources conservation.
 - o USD 10 million GCF Loan B resources will be used to finance all revenue-generating activities of the project (activities for which the financial internal rate of return exceed the discount rate of 6%), i.e. Activities 1.3.2, 2.2.1 and 2.2.2. Activity 1.3.2 (Forest Protection Programs) in Outcome 1 has a mix of grant and loan financing, with grants allocated to the non-revenue generating part of this activity (Local consultants and travel to support positive action measures to incorporate women, youth and indigenous peoples). The loan resources will be allocated to public expenditures cost-saving activities (as it will reduce the probability of outbreaks involving forest fires & pests or diseases for agricultural practices, that would require the government to provide emergency response).

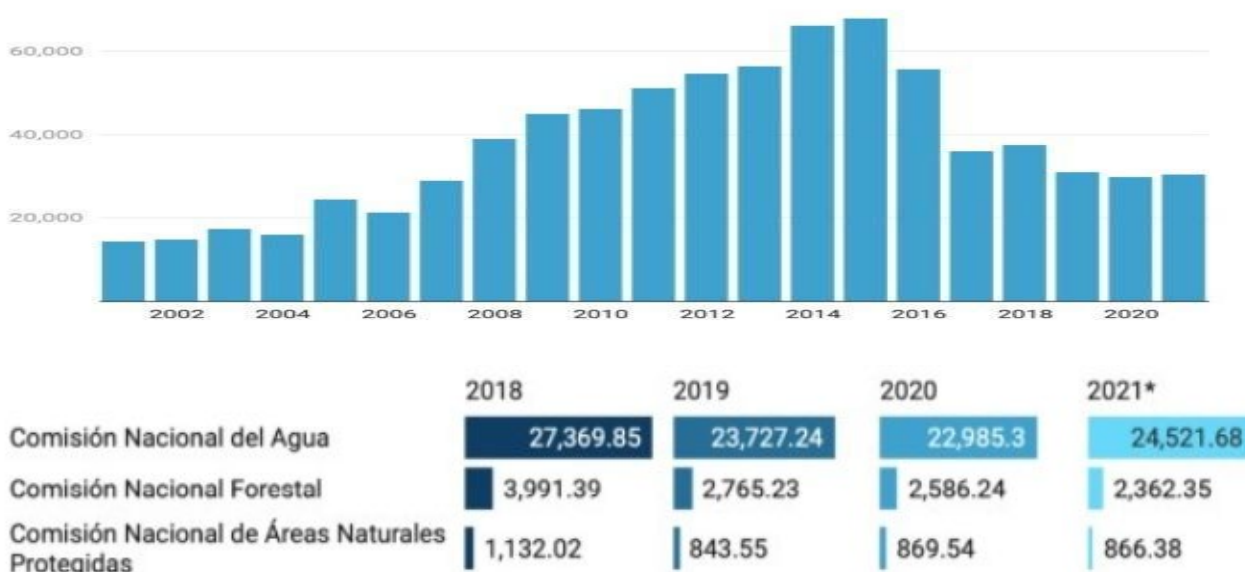
148. Concessionality Transfers of project resources to beneficiaries are carried out under strict targeting criteria to benefit the poorest and most vulnerable populations. These beneficiaries face steep barriers to credit access. Although CONAFOR manages two funds for forestry project loans (FONAFOR and FOSEFOR), none of them has been able to place their funds. This is due to the long project-cycle of the forest sector characterized by a long time (in years) from the investment until the first benefits occur, and even longer until the investment has a financial return. Such long periods are impossible to bear for the targeted population, if financial incentives are not provided. Second, most of these beneficiaries lack a collateral, which excludes them from most if not all existing credit sources, including CONAFOR's credit funds. In this sense, the grant resources are required to make this investment proposal viable.

149. In accordance with national regulations, external credit financial resources (loans) must be included within the yearly budget allocated to an institution. This implies that CONAFOR must include within its annual budget ceiling the necessary resources to execute the external credit loans it is responsible for. In terms of their execution, in accordance with national policy, the loans are implemented through the incentives granted by CONAFOR through its incentive program PADFSB. Loan resources cannot be used to finance operating expenses or incremental costs associated with the project. In contrast, as CONAFOR is a decentralized public body of the Federal Public Administration, with its own legal personality and assets, grants from international funding are additional to its regular budget. Non-reimbursable resources represent funding for the operation and implementation of the project. Therefore, grant resources play a key role in the successful implementation of the

project, as they allow to pay for specialized technical assistance, innovative studies and external support that are not covered in the regular operations. It also allows for the financing of capacity building for both beneficiaries and institutions that facilitate the implementation and achievement of the project objective. In view of the above, as well as in view of COVID-19 related fiscal limitations, grant resources as an addition to the current budget will be essential for the project to be implemented appropriately and effectively.

150. **Additionality** GCF funds will fill the existing climate finance gap to contribute to meeting the Nationally Determined Contributions (NDCs) in the Balsas PA. To meet fiscal targets the government budget linked to the environmental sector has been lowered by 32% from years 2016 to 2019, which puts at risk the fulfilment of the Paris Agreement targets on climate change⁷⁶. Figure 15 below shows that between years 2015 and 2020 the budget for the environmental sector has been halved, with sensitive reductions to the budgets of the CONAFOR (70%) and CONAGUA (60%). At constant prices, in the year 2019 Mexico's investments for environment and climate change was as high as during the year 2001. This situation is being exacerbated by the COVID-19 pandemic, as public resources are reallocated to emergency response and care, altering medium-term priorities and targets for climate finance. GCF funds will leverage public funding and funding from other agencies, including IFAD, the GEF and others, to meet the targets set.

Figure 15: Federal budget assigned to the institutions of the environmental sector in Mexico (in millions MXN)



Source: El Economista, México (09/2019)⁷⁷

151. According to the database on the budgetary situation of HACIENDA⁷⁸, between 2019 and 2020, budgetary revenues decreased by 1.3% and remained stagnated between 2020 and 2021 (0.1% increase) while primary expenditure shows a constant increase from 2019 to 2022 (0.1% in 2020, 6.3% in 2021 and 5.1% in 2022). Primary balance resulted in expenditures exceeding revenues for 0.3% of GDP in 2021. This allows us to anticipate limited availability of public funding for this type of investment due to the fiscal situation and the priorities of emergency attention in response to the consequences and recovery of COVID19.

152. The project combines sources of finance to transform production systems and land use into production systems that are more adapted to climate change in one of the poorest regions of the country, which is among the most vulnerable to climate change and variability nationwide. This raises the perceived risk for the private sector to invest. Through GCF, IFAD and GEF funding, joint and coordinated actions for better governance, mitigation and adaptation to climate change will be implemented, which will reduce the identified risk, leverage more public and private investment, and scale up existing best practices and business models. The concessionality with which resources are transferred to beneficiaries will be determined by their level of vulnerability. Grant resources are required to enable specific investments perceived as higher risk by beneficiaries and the government, as well as to cover a portion of incremental project operating costs.

The project has a high potential for replicability in other regions of the country or can even be a model that can be applied to other Latin American countries that may present similar conditions. CONAFOR will be able to replicate the models, practices and criteria generated by this project for its application to the PADFSB at the national level and for cross-fertilization with similar public forest landscape restoration programs within South-South cooperation projects.

B.6. Exit strategy (max. 500 words, approximately 1 page)

153. The project's sustainability and replicability depend on transforming institutional modalities (the Operational Rules of the PADFSB) used to allocate traditional forest development incentives, re-aligning them as climate finance for adaptation and mitigation initiatives. While the PADFSB is a national program, the approach will be piloted in the most vulnerable areas of the Balsas PA. As described in Section B.2 above and D.2 further below, the project's financial exit strategy is built into this paradigm-shift. Through this approach an average of USD 160 million per year⁷⁹ of traditional forestry incentives at the national level is expected to be re-aligned towards integral, climate adapted territorial development finance which adds-up to more than USD 1.5 billion nationwide during the project lifespan of 20 years (please see Figure 16 below). The piloting of this approach by CONAFOR through intensive capacity building and technical assistance in the Balsas Basin provides a model that shall be replicated and scaled-up across Mexico's climate-vulnerable areas without further external financial contributions.

154. The proposed model builds on the existing experiences and relationships which exist between IFAD, CONAFOR and the Government of Mexico. It also integrates lessons learned from the previous projects in terms of project design and implementation. The project will seek to strengthen institutional and multi-sectoral coordination at the state and basin level. It also aims to incorporate a climate approach into the design and implementation of government priority programs which are currently contributing to the NDCs commitments. The medium and long-term sustainability of the project is also based on a Memorandum of Understanding to be signed between IFAD and CONAFOR for the medium and long term sustainability of the project after its natural End of 8 years with the commitment to further:

- Establishing participatory mechanisms for local and micro-regional planning and setting up social rules that support territorial land-use management while taking into account the threats and opportunities linked to climate change.
- Developing forestry, agroforestry and adaptive forestry system technologies which promote native tree species better adapted to climate change.
- Encouraging existing rural extension systems to incorporate a climate dimension into their technical assistance services and strengthening organizational and technology transfers, especially for the most vulnerable populations.

155. The project has strong potential for replicability in other regions of the country. It may even serve as an applicable model in other Latin American countries or areas with similar conditions to the project area. CONAFOR may further develop the models and practices emerging from this project to apply them at the national level.

C. FINANCING INFORMATION						
C.1. Total financing						
(1) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	39.5			million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans (GCF Loan A + GCF Loan B)	20					
(i.a) GCF Loan A	10	40 years	10 years	0.00 %		
(i.b) GCF Loan B	10	20 years	5 years	0.75 %		
(ii) Subordinated loans	Enter amount	Enter years	Enter years	Enter %		
(iii) Equity	Enter amount			Enter % equity return		
(iv) Guarantees	Enter amount	Enter years				
(v) Reimbursable grants	Enter amount					
(vi) Grants	19.5					
(vii) Results-based payments	Enter amount					
(b) Co-financing information	Total amount			Currency		
	51.453			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
GoM / CONAFOR	Grant	7	million USD (\$)	Enter years Enter years	Enter%	Options
GoM / CONAGUA	Grant	6	million USD (\$)	Enter years Enter years	Enter%	Options
IFAD	Senior Loans	38.453	million USD (\$)	24.5 years 5 years	Variable	Options
(c) Total financing I = (a)+(b)	Amount			Currency		
	90.953.-			Million USD		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	Synergies and complementarities will be established with the GREENMEX Project: Green and Inclusive Recovery in Mexico (GEF financing) implemented by FAO (USD 3 million) in the Balsas PA. The Government will also be in charge of taxes for all purchased goods and services, even those directly imported such as vehicles and equipment.					

Table 5: Financing by Outcome / Output / Activity and Funding Source

C.2 Financing by Outcome / Output / Activity	Total	GCF Loan A (High Concessionality)	GCF Loan B (Low Concessionality)	GCF Grant	Gov MEX Cofinance	IFAD Loan
Outcome 1. Government programs and community development plans integrate climate change criteria for allocation of supports	10,091,726		2,492,891	6,128,047	1,470,788	
Output 1.1 Mechanisms for intersectoral coordination are strengthened	474,656			252,656	222,000	
Activity 1.1.1 Facilitate the creation Forest Management Units (FMUs) and Regional Committees Integrated Management of the Territory (RCIMTs).	146,656			146,656		
Activity 1.1.2 Agree on priority intervention areas at the micro-basin level	50,000			50,000		
Activity 1.1.3. Strengthen multisectoral coordination structures, operation rules and mechanism for co-investment within the basins	167,000			56,000	111,000	
Activity 1.1.4. Strengthen and operationalize criteria for adaptation to climate change in institutional program procedures	111,000				111,000	
Output 1.2. Participatory climate responsive multiannual investment plans within prioritized basin, sub-basins and ethnic areas	1,736,576			708,576	1,028,000	
Activity 1.2.1 Communication and awareness campaign to disseminate calls for proposals	169,120			111,120	28,000	
Activity 1.2.2 Formulation of climate smart multiannual investment plan based on micro-basin zoning plans and validated through FPIC	1,567,456			567,456	1,000,000	
Output 1.3 Strengthened capacities of local actors for climate change adaptation, resilience, and inclusiveness	7,880,494		2,492,891	5,166,815	220,788	
Activity 1.3.1 Capacity development programs for extension workers, local facilitators and "brigades"	251,188			30,400	220,788	
Activity 1.3.2: Support and creation of forest protection brigades and local climate change resilience facilitators	4,254,971		2,492,891	1,762,080		
Activity 1.3.3: Strengthening the CONAFOR Promotorias forestales	3,374,335			3,374,335		
Outcome 2. Forest landscapes are sustainably managed and better adapted to climate change	73,496,484	10,000,000	7,507,109	9,462,362	8,074,012	38,453,000
Output 2.1 Agrarian communities implement land use plans through multiannual payment of environmental services	19,383,799	10,000,000				9,383,799
Activity 2.1.1 Payment of environmental services	19,383,799	10,000,000				9,383,799
Output 2.2 The productive systems of Agrarian Communities are strengthened	21,954,109		7,507,109		1,500,000	12,947,000
Activity 2.2.1 Climate Smart Forest Landscape Restoration of Micro-watersheds and Strategic Areas (RFM)	15,320,000		1848,000		1,500,000	11,972,000
Activity 2.2.2 Commercial Forest and Agroforestry Plantations Adapted to Climate Change (PFCA)	6,634,109		5,659,109			975,000
Output 2.3: Participation of Agrarian Communities and Producer Organizations in forest and agroforestry value chains is strengthened	32,158,575			9,462,362	6,574,012	16,122,201
Activity 2.3.1 Sustainable Forest Management adapted to climate change, inclusion, and access to markets (MFCCV)	16,122,201					16,122,201
Activity 2.3.2 Support households to improve food security/nutrition	12,864,390			6,864,390	6,000,000	
Activity 2.3.3 Brigades to strengthen adaptation capacity in Agrarian Communities (Community Forest Monitoring)	3,171,984			2,597,972	574,012	
Outcome 3. The producers, inhabitants and authorities of the Balsas PA have timely and relevant climate information for climate risk management, disaster prevention	3,648,721			3,320,721	328,000	
Output 3.1: Climate-hazards-related early warning and monitoring systems supported	1,848,756			1,848,756		
Activity 3.1.1 Development of a platform and applications to facilitate access to climate information and early warning systems	150,000			150,000		
Activity 3.1.2 Strengthening of prediction and prevention systems for fire (SPPIF) and forest phytosanitary risks	780,000			780,000		
Activity 3.1.3 Strengthening CONAFOR's Monitoring Report and Verification System (MRV)	762,420			762,420		
Activity 3.1.4 Strengthening CONAFOR's social and environmental safeguard management system	156,336			156,336		
Output 3.2: Policy-relevant knowledge products elaborated	1,799,965			1,471,965	328,000	
Activity 3.2.1 Production of training and systematization materials	120,000				120,000	
Activity 3.2.2 Exchange of experiences, and South-South and Triangular Cooperation (SSCT)	320,000			320,000		
Activity 3.2.3 Monitoring and evaluation	1,359,965			1,151,965	208,000	
Project Management (4, %)	3,716,070			588,870	3,127,200	
Indicative Total Cost (USD)	90,953,000	10,000,000	10,000,000	19,500,000	13,000,000	38,453,000
	(100%)	(11%)	(11%)	(21.4%)	(14.3%)	(42.3%)

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)																																																																																																											
C.3.1 Does GCF funding finance capacity building activities?						Yes ☒ No ☐																																																																																																					
C.3.2. Does GCF funding finance technology development/transfer?						Yes ☒ No ☐																																																																																																					
156. The total GCF financing for capacity building and technology transfer is estimated at US\$ 38,911,130 (98.5% of the total GCF funding). The table below identifies the volume of financing in each outcome that is allocated for capacity building and technology transfer from the total budget and GCF. Table 6: Financing for Capacity Building and Technology Transfer <table border="1"> <thead> <tr> <th rowspan="3"></th> <th colspan="5">Total Cost</th> <th colspan="5">GCF Financing</th> </tr> <tr> <th>Total Amount (US\$)</th> <th>Capacity Building (US\$)</th> <th>%</th> <th>Technology Transfer (US\$)</th> <th>%</th> <th>Total Amount (US\$)</th> <th>Capacity Building (US\$)</th> <th>%</th> <th>Technology Transfer (US\$)</th> <th>%</th> </tr> <tr> <th></th> <th></th> <th></th> <th></th> <th></th> <th></th> <th></th> <th></th> <th></th> <th></th> </tr> </thead> <tbody> <tr> <td>Outcome 1</td> <td>10.091.726</td> <td>10.091.726</td> <td>100</td> <td></td> <td>0</td> <td>8.620.938</td> <td>8.620.938</td> <td>85</td> <td></td> <td>0</td> </tr> <tr> <td>Outcome 2</td> <td>73.496.484</td> <td>6.346.530</td> <td>9</td> <td>65.392.754</td> <td>89</td> <td>26.969.472</td> <td>2.597.972</td> <td>4</td> <td>24.371.499</td> <td>33</td> </tr> <tr> <td>Outcome 3</td> <td>3.648.721</td> <td>930.000</td> <td>25</td> <td>440.000</td> <td>12</td> <td>3.320.721</td> <td>1.471.965</td> <td>40</td> <td>1.848.756</td> <td>51</td> </tr> <tr> <td>PMU</td> <td>3.716.070</td> <td></td> <td>-</td> <td></td> <td>-</td> <td>588.870</td> <td></td> <td>0</td> <td></td> <td>0</td> </tr> <tr> <td>Total</td> <td>90.953.000</td> <td>17.368.255</td> <td>19</td> <td>65.832.754</td> <td>72</td> <td>39.500.000</td> <td>12.690.875</td> <td>32</td> <td>26.220.255</td> <td>29</td> </tr> <tr> <td colspan="7">Total GCF Capacity Building and TT</td> <td>38.911.130</td> <td></td> <td></td> <td></td> </tr> </tbody> </table>												Total Cost					GCF Financing					Total Amount (US\$)	Capacity Building (US\$)	%	Technology Transfer (US\$)	%	Total Amount (US\$)	Capacity Building (US\$)	%	Technology Transfer (US\$)	%											Outcome 1	10.091.726	10.091.726	100		0	8.620.938	8.620.938	85		0	Outcome 2	73.496.484	6.346.530	9	65.392.754	89	26.969.472	2.597.972	4	24.371.499	33	Outcome 3	3.648.721	930.000	25	440.000	12	3.320.721	1.471.965	40	1.848.756	51	PMU	3.716.070		-		-	588.870		0		0	Total	90.953.000	17.368.255	19	65.832.754	72	39.500.000	12.690.875	32	26.220.255	29	Total GCF Capacity Building and TT							38.911.130			
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D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

157. This cross-cutting project will be contributing to the goals of Mex'co's NDCs in terms of mitigation by reducing emissions linked to the agriculture, forestry, and other land use sector (AFOLU). But the main contribution will be in terms of adaptation under the NDC's: axis-A - preventing and responding to the negative impacts on the territory and human population; axis-B - resilient production systems and food security; axis-C - conservation, rehabilitation and the sustainable use of biodiversity and ecosystem services; and axis-D - integrated water resource management with a climate change approach. The mitigation and adaptation lines of action in the NDCs for the AFOLU sector are fully integrated and aligned with each other. The project's specific contributions and alignment with the NDCs can be found in Annex 2 (Feasibility Study).

158. The proposed project, although cross-cutting, is primarily focused on adaptation. The project will foster climate change adaptation impacts in the 10 prioritized areas of the Balsas PA in four ways:

- Basin ecosystems will generate environmental services, strengthening the local population's capacity for adaptation
- Groups most vulnerable to climate change will increase their income, food security and access to water
- Climate risks affecting the population and their productive systems will be reduced; and
- Dry season water supplies will be supplemented through investments in small-scale rainwater harvesting and water supply infrastructure.

Adaptation

159. The project beneficiaries will be primarily vulnerable rural small producers and households living in poverty and indigenous peoples present in the intervention area including: (i) agrarian communities (ACs) mainly owned by indigenous peoples; (ii) farmer organizations within the ACs (cooperatives and other formal and informal figures); (iii)

Mitigation

161. The Balsas Project will contribute to the achievement of all of the seven milestones set by the ENAREDD+ for the year 2030 (see also paragraph 35 above). Furthermore, the Project supports the implementation of all the seven components and most of its action lines and activities of the ENAREDD+, as shown in the table included in Annex 2a. It is important to note that the ENAREDD+ sees new models addressing the causes of deforestation and forest degradation as pivotal to reaching zero deforestation by 2030, a goal also specified in the NDCs. The project responds to this requirement. It also outlines a path that will help the Government of Mexico develop these necessary new models. These models will help address existing challenges by applying a cross-cutting watershed management approach to mitigation and adaptation. They may also generate a mechanism for distributing resulting benefits to the most vulnerable population.

163. The average of USD 160 million per year that the project aims to realign has been calculated from two CONAFOR programmes: Support for Sustainable Forest Development (S219) and Forest Protection (E014). These initiatives were allocated on average USD 94,805,800 (S219) and USD 67,330,400 (E014) per annum over the years 2017-2021, which represents a combined national average of USD 164,740,565 per year between the two initiatives (see Table 7 below). Given the consistency of the average annual budget allocations across these two initiatives over these 5 years, it is assumed that at least USD 160 million has the potential to be reassigned as climate finance per year via the project's paradigm-shifting (a financial analysis depicting this is presented in Figure 16 below). The annual average of USD 160 million sums up to the total amount of USD 1.5 billion in 20 years of re-aligned traditional forestry incentives, towards climate finance. This value was derived as net present value using a conservative social discount rate of 10% (as defined by the Government of Mexico in 2022 — see Annex 10). The sensitivity analysis of the re-aligned climate finance during the project's lifespan to different discount rates is also presented in Table 8 below.

Concept	2014	2015	2016	2017	2018	2019	2020	2021	Average annual budget (last 5 years)
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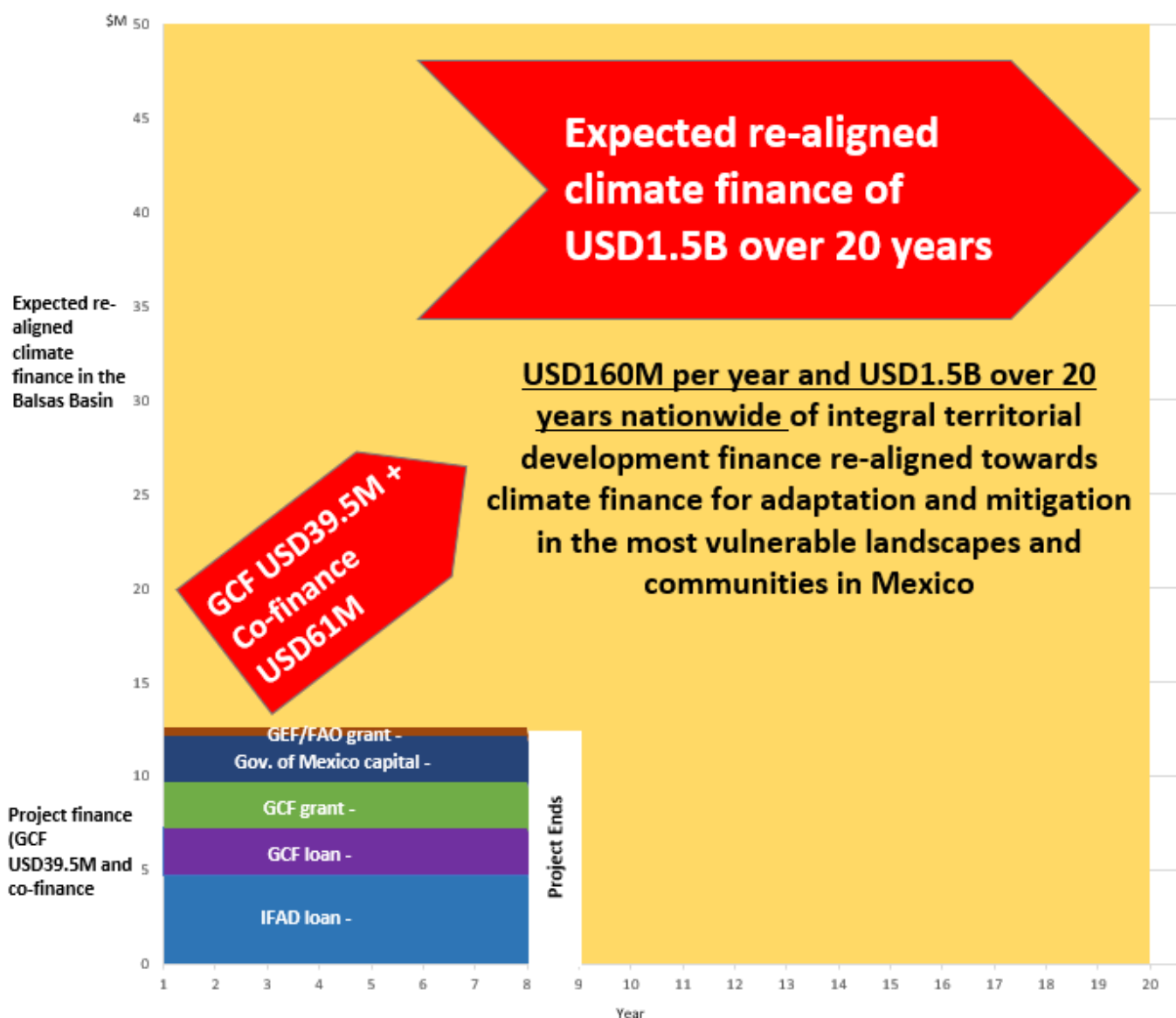
S219 Support for Sustainable Forest Development	135,500,000	128,551,000	240,584,000	103,137,000	112,007,000	73,171,000	61,312,000	124,402,000	94,805,800
E014 Forest Protection	105,451,000	110,160,000	96,169,000	83,083,000	82,569,000	60,724,000	52,166,000	58,110,000	67,330,400
Total	240,951,000	238,711,000	336,753,000	186,220,000	194,576,000	133,895,000	113,478,000	182,512,000	164,740,565

Source: CONAFOR (Sistema de Administración Integral)

Table 8. Sensitivity of potential re-aligned climate finance over project lifespan to varying social discount rates

Discount rate	Year representative of future budget — billion USD								Average annual budget (last 5 years)
	2014	2015	2016	2017	2018	2019	2020	2021	
0%	7.25	6.02	7.22	3.94	4.04	2.78	2.11	3.60	3.29
2%	6.05	5.02	6.02	3.28	3.37	2.32	1.76	3.00	2.75
6%	4.41	3.66	4.39	2.39	2.46	1.69	1.28	2.19	2.00
8%	3.84	3.19	3.83	2.09	2.14	1.47	1.12	1.91	1.75
10%	3.40	2.82	3.38	1.84	1.89	1.30	0.99	1.69	1.54

Figure 16: Expected re-aligned traditional forest incentives towards leveraged climate finance (nationwide)



164. A continued flow of resources towards USD 1.5 billion in re-aligned, leveraged climate finance is expected to be guaranteed by the resources budgeted by the Government of Mexico for integral territorial development over the next 20 years. The origin of this territorial integral development finance, disbursed by CONAFOR, is the *Fondo Forestal Mexicano* (FFM), which manages all the resources destined for the beneficiaries of the PADFSB (including PES) and other minor CONAFOR programs. Approximately 48% of this territorial integral forest development finance includes resources from various national sectors that pay the Government of Mexico a compensation fee related to land use change caused by development in the respective sectors. The remaining 51% is from government fiscal resources, including a contribution from CONAGUA transfers (~USD 30 million/year) deriving from the management of dams. Given the origin and the continuity during the last decade of the integral forest development funds described above, there is a high potential for the estimated total of USD 1.5 billion of re-aligned climate finance at national level to be achieved, or even surpassed, because of the very conservative discount rate chose for calculating its net present value.

Project design aspects that contribute to the paradigm shift

165. The project will transform production systems and land use in the Balsas PA region, targeting the most climate vulnerable and poor rural communities in the country, specifically the most vulnerable municipalities with the highest levels of poverty and extreme poverty. This cross-cutting project (80% adaptation, 20% mitigation) aims to increase the climate resilience of the most vulnerable communities and ecosystems, and reduce emissions from production systems in the Basin, via: i) operationalizing the use of climate change criteria in current institutional

modalities for the prioritization of rural development support in the Balsas PA through a community-centric bottom-up approach; ii) investing in the climate-sensitive management of ecosystems, and climate-resilient productive systems and value chains for adaptation and mitigation benefits; and iii) developing a climate information system, and implementation of knowledge management. The proposed project will shift the approach of the current forestry development support allocation system used by rural development institutions to one which:

- Uses climate change-related criteria for selecting and prioritizing the most vulnerable beneficiaries, interventions and sites.
- Strengthen local watershed governance and participation of women, youth, and indigenous peoples at the sub-basin level to grant incentives and for interinstitutional coordination and a bottom-up approach for the sustainable use management and conservation of crucial ecosystems and ecosystem services to be piloted in the Balsas PA and replicated at national level. As part of the project's bottom-up approach, the participation of Agrarian Communities and Producer Organizations, especially women and youth, in value chains of timber and non-timber forest products (NTFPs) will also be strengthened, with the aim of increasing the incomes of the most vulnerable households, thereby building their adaptive capacity.
- Disburses territorial development and forest landscape conservation incentives multi-annually under a progress-based payment system, specifically rewarding performance of adaptation and mitigation initiatives. The multi-annual approach and annual evaluation of progress was first piloted by CONAFOR for PES schemes only. This will be replicated by the project across other forestry development support programs.

These three paradigms shifting levers will be actioned through the following project approach:

166. Piloting of climate-sensitive approach to financial incentives for forestry and rural development (Outcome 1)

The three proposed paradigm-shifting innovations described above will be first piloted within CONAFOR's incentive payments for forestry development through the prioritization criteria of the Operational Rules of the PADFSB. Although these already include climate change criteria, these have been poorly operationalized to prioritize interventions, beneficiaries, and sites for rural development support because of lack of capacities at the local level and poor integration into planning instruments and coordination between the responsible institutions. The prioritization and selection of interventions and sites using climate change adaptation criteria will inform climate-focused support in rural areas with the highest vulnerability, while the multi-annual progress-based payments system will allow for the renewal of support in the form of incentives in these areas, increasing the scale and sustainability impacts of the climate-focused initiatives. In addition, specialized technical assistance will be provided by the project to facilitate access of those most poor and vulnerable communities that traditionally have had less or difficult access to CONAFOR's support programs through a demand driven, participatory approach. This will be complemented by the implementation of a pilot program to increase ease of access to PADFSB support by the most vulnerable communities in the Balsas PA.

167. Bottom-up participatory territorial planning and allocation of incentives. To ensure that the paradigm shift is achieved at a decentralized level and not top down, Regional Committees for the Integrated Management of the Territory (RCIMTs) will be established at Forest Management Unit level to foster the participation of local communities and all levels of government (municipality, state and national) to create awareness of climate change impacts and co-develop adaptation solutions. Furthermore, the project will support existing and create new groups of local facilitators, including women and youth, to strengthen community-level adaptation and mitigation initiatives and monitor ecosystem health. This local-level bottom-up approach will promote more flexible procedures for the channeling of development finance towards climate change adaptation and mitigation initiatives. Agrarian communities and producer organizations will prepare climate change project profiles which will be reviewed for approval for financing by RCIMTs and CONAFOR. Once approved, the Project will provide technical assistance to beneficiaries to design their own climate-responsive land use and investment plan. Promoting capacity building and converging action among the involved institutions and programs is important for achieving the desired transformational change.

168. Project's community-centric bottom-up approach and value chain support (Outcomes 1 and 2) The project will support bottom-up participatory processes and provide technical assistance at the community-level to improve territorial governance and investment planning instruments for climate change adaptation in the Balsas PA, in particular but not limited to facilitating the creation of the RCIMTs, implementing capacity development programs for extension workers, and local facilitators and the creation of "climate change adaptation brigades", with a focus on women and youth. In addition, the project will build the capacities and make the investments needed to implement climate-resilient production practices at the community level, including the enhancement of gender-inclusive value chains and market access to support their livelihoods and raise their incomes.

169. Scaling up of the climate-sensitive rural development support prioritization process across the Balsas PA (Outcomes 2 and 3). The project will promote the mainstreaming of the new climate-based approach into the main rural development institutions (and their initiatives) operating in the Balsas PA (CONAFOR, CONAGUA and AGRICULTURA/AGRICULTURA), through the multi-sectoral coordination platforms. This will ensure that rural

development support in the Balsas adopts a decentralized, coordinated and climate-responsive approach across all sectors, and presents a model that can be scaled up across Mexico's rural areas.

170. **Lessons learned and innovation.** The project has significant potential for generating lessons learned and developing new models that integrate climate change mitigation, adaptation, livelihood resilience and climate risk management into key forestry and agricultural policy instruments. This integration, however, will be done gradually. It will be parallel to ongoing monitoring and evaluation process aimed at continuous improvement. These efforts will lead to field-proved, replicable, and scalable models for climate smart forest conservation and forestland restoration practices, which can be scaled up at the national level and in other countries in the Latin American Region.

D.3. Sustainable development (max. 500 words, approximately 1 page)

171. The proposed project builds upon the Balsas-South Pacific Basin Regional Development and Welfare Program included in the 2019-2024 National Development Plan (NDP). Project outcomes, outputs and activities will contribute to reducing emissions linked to agriculture, forestry, and land use change, while simultaneously strengthening the provision of ecosystem services and increasing the target population's resilience to climate change. These actions are directly linked to Sustainable Development Goal (SDG) 13 – Climate action, which aims to "take urgent action to combat climate change and its impacts". Project activities will contribute towards achieving a further 6 of the 17 SDGs, namely: SDG 1 – No poverty; SDG 2 – Zero hunger; SDG 3 – Good health and wellbeing; SDG 5 – Gender equality; SDG 6 – Clean water and sanitation; and SDG 15 – Life on land. In addition, interventions under the project will produce multiple environmental, social, and economic co-benefits, being the following most relevant:

- **Maintenance of environmental services.** A crucial environmental co-benefit of this project is the retention, regulation and release of rainwater flows within the ecosystems and the landscapes. Prolonged draught periods on one hand, and more severe and heavy rainfalls on the other hand pose important hazards for local communities and especially for the most vulnerable rural livelihoods that depend on forest products and agriculture for which a regulated water flow and soil humidity is vital.
- **Conservation of biodiversity.** Within the Basin, 62% of biodiversity areas are vulnerable due to the impacts of climate change on their plant communities⁸⁰, while the expansion of agricultural activities into forested areas is further negatively impacting biodiversity in the region. The promotion of climate-adapted forestry and natural resource management under the project will promote the conservation of biodiversity and forest resources, resulting in increased available habitat for wildlife, the preservation of culturally and aesthetically important areas, and improved ecosystem service supplies, such as disease regulation.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

Socioeconomic status

172. Despite being among the 15 largest economies in the world and second largest economy in Latin America and the Caribbean, and having strong macroeconomic institutions, Mexico has underperformed in terms of growth, inclusion and poverty reduction compared to similar countries over the last three decades. Average economic growth per year between 1980 and 2018 was low at 2%, limiting progress relative to high income economies. After contracting by 8.2% in 2020, Mexico's economy grew by 4.8% in 2021⁸¹.

173. In 2018, poverty and extreme poverty affected 41.9% (52.4 million) and 7.4% (9.3 million) of the population, respectively⁸², according to the multidimensional poverty measure used by the Government of Mexico⁸³. It is estimated that the COVID-19 crisis resulted in a further 9.5 and 7.9 million people falling into poverty and extreme poverty, respectively⁸⁴. In addition to economic inequality and poverty, Mexican populations suffer from significant food insecurity and nutrition challenges – among Mexico's 7.8 million rural households, only 30.9% have an adequate level of food security. Data from the pandemic period in 2020 show that the proportion of people suffering from severe food insecurity increased from 9.2% in 2018 to 13%⁸⁵. According to the National Council for the Evaluation of Social Development Policy (CONEVAL), the states that make up the Balsas PA widely exceed the national average for poverty and extreme poverty, in particular those prioritized by the proposed project (Oaxaca, Guerrero, Michoacan and Puebla).

174. The population of 15.8 million people living in the Balsas PA is extremely vulnerable to the impacts of climate change because of its high poverty rates and limited social development levels relative to the rest of Mexico. Existing inequities faced by women in the Balsas PA in accessing different services and assets are exacerbated by the effects of climate change. For example, activities such as subsistence production (*traspato*), housework and family care are made more difficult due to the increase in temperatures, the frequency and intensity of extreme weather events (such as floods), water scarcity and the absence of public drinking water services. Economic activities undertaken by women, such as the management and harvesting of non-timber forest products, are affected by the loss of vegetation due to droughts, fires, hurricanes, and torrential rains. In addition, many of the women *ejidatarios*, not being owners

of the land, cannot access programs for equipment, infrastructure, credits, or economic support for payments of ecosystem services. Women are thus highly vulnerable to climate change impacts expected in this region.⁸⁶ 175. Young people in the Balsas PA make up 44-55% of the population, with rural youth representing more than 60% of the total population in the States Oaxaca and Guerrero. These rural young people lack economic opportunities or the possibility of participating in community decision-making, leaving them with limited mechanisms to increase their resilience to the impacts of climate change.⁸⁷ Indigenous people historically have the highest rates of poverty in Mexico: In 2016, 72% (8.2 million) of Mexico's indigenous people lived in poverty. In 2020, over 9.6 million indigenous people were living in poverty in the municipalities where data was available⁸⁸. The situation is even more adverse for indigenous women living in rural areas (8 out of 10 live in poverty and 4 out of 10 in extreme poverty)⁸⁹. Despite their great cultural wealth and their natural resources, the indigenous and traditional peoples have the lowest monetary income and access to services, and have high rates of food and nutritional insecurity¹⁵, leaving them extremely vulnerable to climate change impacts. Based on an evaluation of climate vulnerability and levels of marginalization in the Balsas PA, the project will prioritize eight watersheds and two indigenous territories for a total of 121 municipalities and respond to the needs of ACs, small producers, women, youth, people with disabilities and indigenous peoples, helping them to better confront the harm, risks and effects they are exposed to because of extreme weather events and climate change.

Absence of alternative sources of finance

176. As already described in Section B.5., while the Government of Mexico (CONAFOR in particular) continue to invest in the country's forestry sector there are several obstacles preventing the flow of tailored funding for climate change adaptation and mitigation in the Balsas PA. These include: i) current financial support is not being specifically focused on the most vulnerable communities and ecosystems, and is not informed by climate scenarios to support climate adaptive sustainable forest ecosystem management; ii) limited access to credit for promoting the inclusion of small producers (especially women and youth who are usually not the owners of land in agrarian communities) in value chains related to timber, non-timber and agroforestry products; iii) lack of inclusion of women and youth in ecosystem management and value chain activities, which negatively affects the sustainability of medium- to long-term adaptation needs; and iv) despite the current PES support provided by CONAFOR and CONAGUA, the end users of ecosystem services do not sufficiently contribute to the maintenance of the relevant ecosystems. External investment is therefore required to provide the concessionally needed and to finance the existing funding gap for adaptation and mitigation activities in the Balsas PA.

Institutional capacity needs

177. There is currently a weak level of existing climate governance in the Balsas PA. This is partly because efforts to integrate a climate approach into public policies have only recently begun. Furthermore, institutional coordination among the involved actors and specialized technical assistance on climate change have been limited. As a result, there have been no clearly focused incentives encouraging concerted action for climate change mitigation and adaptation. The main institutions working in the Balsas PA (CONAFOR, CONAGUA, and AGRICULTURAAGRICULTURA) have been developing advanced information systems related to weather forecast, forest fires, medium/long term evolution of climate patterns and climate risk prevention at central level. However, capacity building at regional and local level is needed to improve climate information sharing and uptake by communities, municipalities, and extension services.

D.5. Country ownership (max. 500 words, approximately 1 page)

178. This project has been developed in collaboration with multiple institutions and stakeholders under the lead of the CONAFOR and the guidance of the NDA, the Mexican Ministry of Finance and Public Credit (*Secretaría de Hacienda y Crédito Público* SCHP). The Office for the Coordination of Advisers to the Presidency of the Republic (CAP) has facilitated smooth inter-institutional collaboration with CONAGUA and AGRICULTURAAGRICULTURA, while IFAD and the FAO Investment Centre have supported the drafting of this project that has involved a participatory, consultative, and multi-sectoral planning process. Through this process, a paradigm shift to current institutional modalities and development models have been identified that foster low emissions and increase adaptation capacity and resilience of the most vulnerable rural livelihoods. The Project is completely aligned with and has been designed to implement Mexico's climate change, environmental and social policies, and priorities, specifically:

- Mexico's Nationally Determined Contributions: 2020 update⁹⁰ (NDCs) respond to both mitigation and adaptation problems experienced in the country across multiple sectors. Using a cross-cutting approach, this project has the potential to contribute concurrently to both climate change mitigation and adaptation targets in Mexico's NDCs. According to the NDCs, mitigation activity in the land use, land use change and forestry sector (*uso del suelo, cambio de uso del suelo y silvicultura* USCUS) directly aligns with the overarching aim of ecosystem-based adaptation (EbA). The NDCs recognize the need to incorporate climate criteria into

rural development programs, as well as to implement environmental protection and land use standards. The project framework aims to contribute to the NDCs' goals firstly in terms of Climate Change Adaptation, specifically in axes A, B, C and D⁹¹. This will involve activities focusing on: i) social sector adaptation to climate change — all project outputs; ii) EbA — Output 2.1 (climate-sensitive management of ecosystems); and iii) water resource and supply management (Output 2.2). Contributions will occur secondly in terms of Climate Change Mitigation. This will involve reducing emissions through activities focusing on the: (a) land use sector via land use change and forestry activities (e.g., afforestation, reforestation, forest management, farmland management and pasture management) — Outputs 2.1 and 2.2 (increased commercialization and added value of climate-resilient forestry, agroforestry, silvo-pastoral and backyard production systems and their products); and (b) agricultural sector via soil management, scheduled soil burns and manure management — Output 2.2. A detailed matrix that shows how project activities are aligned within the NDCs is presented in Annex 2.

- Mexico's National Climate Change Strategy (2013)⁹² (*Estrategia Nacional de Cambio Climático / ENACC*) sets the long-term planning for climate change in the country. The proposed project will contribute to the following strategic adaptation (A) and mitigation (M) axes in the ENACC: i) reducing vulnerability and increasing social sector resilience to climate change (lines of action A1.7 and A1.11) — all project outputs; ii) reducing vulnerability and increasing strategic infrastructure and production-system resilience to climate change (line of action A2.6) — Outputs 2.1 and 2.2; iii) sustainably using and protecting ecosystems while preserving the environmental services they provide (lines of action A3.1, A3.8 and A3.9) — Outputs 2.1 and 2.2; and iv) promoting agricultural and forestry best practices in order to increase and sustain natural carbon sinks (lines of action M4.1, M4.2, M4.7, M4.8, M4.10, M4.11, M4.12 and M4.13) — Outputs 2.1 and 2.2.
- The Special Program on Climate Change 2021–2024⁹³ (*Programa Especial de Cambio Climático/PECC*) relates to short-term planning for climate change in the country. This document includes targets for 2030 to be achieved by the development and implementation of 169 specific actions under four priority objectives: adaptation; mitigation; co-benefits derived from synergies between adaptation and mitigation; and climate governance. The proposed project aligns with the PECC through the following lines of action: 1.1.9, 1.3.6, 1.3.7, 2.7.1, 2.7.2, 2.7.3, 2.7.6, 3.1.8, 4.2., 4.3.8 and 4.6.8.
- Mexico's National REDD+ Strategy for Reducing Emissions from Deforestation and Forest Degradation 2017–2030⁹⁴ (*Estrategia Nacional de Reducción de Emisiones por Deforestación y Degradación Forestal/ENAREDD+*) has the following objectives: (i) to increase forest area via sustainable management, natural and induced forest regeneration and forest conservation; (ii) to continue developing the social and economic capital of rural communities; and (iii) to increase sustainable best-practices in production at the rural landscape level. The proposed project will contribute to these objectives through improved climate-resilient and low-emission management of ecosystems to enhance service supplies, as well as through climate-sensitive production systems and value chains to diversify. This will include improved protection of forests and prioritization of rural agrarian communities. The Project will support the implementation of all seven components of this Strategy, as shown in the matrix included in Annex 2a..
- Project activities will directly support the National Forestry Program 2020–2024⁹⁵ (*Programa Nacional Forestal 2020–2024*), which has the main objective to promote community forest management for sustainable and diversified use of forest resources, as well as the integration and development of local networks and competitive values that trigger local economies to improve the quality of life of the population living in forest areas. Activities under Outputs 2.1 and 2.2 will contribute directly to the program's objectives.
- The General Law on Climate Change (2012)⁹⁶ (*Ley General de Cambio Climático / LGCC*) aims to align activities on mitigation and adaptation between different government entities. The law established the following institutions: the National Institute of Ecology and Climate Change; the Climate Change Council; and the Inter-Secretariat Commission for Climate Change. It also created a fund to channel financial resources (public, private, national, and international) to support actions to combat climate change. The law was amended in November 2020 to comply with the Paris Agreement and the NDCs. The proposed project will comply with all directives included in the LGCC, to ensure that adaptation and mitigation activities implemented in the Balsas PA are effectively coordinated between government actors at all levels. This will include the strengthening of institutional coordination mechanisms for climate-driven rural development in the Balsas PA (Output 1.2).
- This cross-cutting project aligns with the goals and commitments in Mexico's most recent National Development Plan 2019–2024⁹⁷ (NDP) (*Plan Nacional de Desarrollo 2019–2024*), both in terms of mitigation and adaptation⁹⁸. One of the main principles of this plan is the promotion of the efficient and responsible use of resources, which activities under Outputs 1.2 and 2.1 will contribute to. In addition, activities under Output 2.2 will directly contribute to specific objectives in the NDP, including: i) the stimulation of innovation, competition and integration of all value chains and the generation of higher value-added activities in all productive sectors of the economy in an environmentally conscious and sustainable manner; and ii) the

development of sustainable and inclusive agriculture and aquaculture industries that serve the broader Mexican nation, as well as the country's indigenous communities. Further detail on the project's alignment with the NDP is presented in Annex 2.

Institutions involved and stakeholder engagement

179. The capacity, experience, and the roles of IFAD as the GCF Accredited Entity (AE), CONAFOR as the Project Executing Entity undertaking all operational responsibility for the implementation of the Project (EE), and NAFIN and HACIENDA as Executing Entities channeling the funds to CONAFOR as well as the other project partners have been extensively described in section B.4 Implementation Arrangements above and will therefore not be repeated in this section.

180. Under the guidance of CONAFOR and the CAP a mission to México City, Guadalajara and the Balsas PA was undertaken by an IFAD/FAO project appraisal and preparation team in June 2022. The objective of the mission was to gather relevant information required for the preparation of the Funding Proposal with contributions from CONAFOR, CAP, HACIENDA, CONAGUA, and other government and civil society institutions operating in the project area. In addition, the mission provided the opportunity to identify and visit a diversity of actors, communities and experiences in the Balsas PA that could inform best practices for stakeholder engagement and participation in the project. Mission participants visited several *ejidos* and indigenous communities who have already received support from CONAFOR's PADFSB. These visits provided a better understanding of how the PADFSB operates, as well as an opportunity to gather diverse views from different rural development actors about community enterprises, livelihoods, development possibilities and the effects of climate change at local level. Specific focus was given to initiatives prioritizing women and youth, as the active and visible participation of these groups is recognized as an ongoing challenge. Through meetings held at CONAFOR headquarters in Guadalajara, the organizational structure, strategic priorities, and implementation arrangements for the project were discussed and agreed upon and a Stakeholder Participation Plan designed (see Annex 7.b).

181. During the months of September and October 2022 several meaningful consultation workshops were organized by CONAFOR with indigenous communities and local actors in the project region. These consultation meetings have been documented as part of in the Indigenous Peoples Consultation and Participation Plan included in Annex 6 and are summarized in the following table.

Table 9: Consultations with Indigenous and Agrarian Communities during Funding Proposal Preparation

Date	Participating communities, ejidos (" <i>bienes comunales</i> ") / municipalities	Location / territory	Number of representatives
September 2, 2022	Cochoapa el Grande, Acatepec, Atlamajalzingo del Monte, Malinaltepec, Alcozauca de Guerrero, Zapotitlán Tablas, Atlixac People: Tlapaneco, Mixteco y Nahuá	Tlapa, Guerrero	19
September 29, 2022	Cherán, Nahuatzen People: Purépecha	Uruapan, Michoacán.	29
October 1, 2022	San Juan Numí, Santiago Nundiche, San Cristóbal Amoltepec, Santo Tomás Ocotepec, Magdalena Peñasco, Santa Cruz Nundaco, San Antonio Sinicahua, San Mateo Peñasco, San Juan Mixtepec, San Juan Teita, Santa María Yucuhiti, Santa María Tataltepec, Santiago Nuyoó, San Esteban Atlatlahuca, Santa Catarina Yosonotú, San Pablo Tijaltepec, Santa Lucía Monteverde, Santa Cruz Tacahua People: Mixteca	Tlaxiaco Oaxaca.	21

D.6. Efficiency and effectiveness (max`. 500 words, approximately 1 page)

182. The Economic and Financial Analysis (please see also Annex 3) shows that this project is economically viable as a whole. Main sources of quantifiable benefits result from the investments in Outcome 2 and include the increase in productivity, resilience of the productive systems to climate change, and mitigation benefits from GHG emission reductions. Main sources of benefits in Outcomes 1 and 3 include enabling and supporting activities for the correct implementation and monitoring of the project investments, which also increase the adaptation benefits of the project, at a wider scale, namely the watershed scale, and are not directly quantified in this analysis, rendering its results rather conservative.

183. The benefit analysis focuses on the investments, where forest landscapes are sustainably managed and better adapted to climate change for the provision of environmental services and increased livelihood resilience and its activities include: 2.1.1 Payment of Environmental Services (PES), 2.2.1 Climate Smart Forest Landscape Restoration of Micro-watersheds and Strategic Areas (RFM), 2.2.2 Commercial Forest and Agroforestry Plantations Adapted to Climate Change (PFCA), 2.3.1 Sustainable Forest Management adapted to climate change, inclusion, and access to markets (MFCVV), and 2.3.2 Support households to improve food security and nutrition. Models for

each one of these activities were constructed based on business plans provided by the CONAFOR, secondary information, and validated through fieldwork. The following table shows the financial returns for the individual models with project as compared to the without project scenarios as well as the scale and number of beneficiaries estimated to each within the project. While most of the activities render positive IRRf the supporting households to improve food security and nutrition has an internal return rate of 0.19% rendering it unattractive if any potential investor would be available.

Table 10 Financial profitability (measured as Financial Internal Rate of Return *IRRf) and scale

Models	Units	Scale	Beneficiaries	Lifetime	Year>@6%	IRRf
B. Activity 2.2.1 Climate Smart Forest Landscape Restoration of Micro-watersheds and Strategic Areas (RFM)						
Sub-activity 2.2.1.1 Establish climate resilient agroforestry systems	Ha	2,000	3,061	10	NA	4.9%
Sub-activity 2.2.1.2 Establish agave – forest water / soil retention terraces	Ha	1,000	1,530	20	18	7.4%
Sub-activity 2.2.1.3 Establish forest – pasture water / soil retention terraces	Ha	1,400	2,143	5	12	10.7%
C. Activity 2.2.2 Commercial Forest and Agroforestry Plantations Adapted to Climate Change (PFC)						
Sub-activity 2.2.2.1 Establish climate adapted planted forests (PFC 1)	Ha	4,353	978	10	10	19.0%
Sub-activity 2.2.2.2 Establish silvopastures (PFC 2)	Ha	1,500	337	20	12	11.6%
D. Activity 2.3.1 Sustainable Forest Management adapted to climate change, inclusion and access to markets (MFCCCV)						
Sub-activity 2.3.1.1 Manage forests sustainably adapted to climate change (MFCCV.1)	Ha	120,487	8,112	20	19	15.3%
Sub-activity 2.3.1.2 Sustainable use and add value to timber forest products (MFCCV.2)	Proj.	30	4,376	5	7	15.9%
Sub-activity 2.3.1.4 Support women in benefitting from forest based value chains (MFCCV.4)	Proj.	20	1,347	5	5	12.1%
E. Activity 2.3.2 Support households to improve food security and nutrition						
Sub-activities 2.3.2.1 – 2.3.2.3 Traditional milpa (maize/beans) and traspatio orchards are transformed to diversified milpa with trees, improved irrigation and water solutions for improved food quality and nutrition	Ha	941	3,763	5	0	0.19%

The economic results at the aggregate level of the project are estimated based on the area (in hectares) or projects estimated to be completed by the project. Importantly, this step also includes the benefits related to climate change mitigation through the impact on GHG emissions reduction estimated with the Ex-Act tool for the different models (see Annex 22). Three price scenarios were used to include the mitigation benefits from the World Bank guide for the social valuation of carbon, starting at approximately USD 42/tCO_{2eq} for the low-price scenario up to USD 84/tCO_{2eq} for the high scenario. The discount rates used to estimate the economic net present value (NPVe) were 6% and 10%. Table 11 summarizes the results for the carbon price scenarios, as well as for the 2 discount rates. Although the important adaptation benefits are not included, the economic rate of return (IRRe) as well as the net present value for the discount rates used are positive.

Table 11: Economic analysis indicators by scenario

Million USD	NET ec BENEFITS without GHG	NET ec Benefits + GHG Market	NET ec Benefits + GHG Low social price	NET ec Benefits + GHG Median social price	NET ec Benefits + GHG High social price
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NPVe@6%	52	60	118	151	216
NPVe@10%	25	30	68	89	132
B/Ce@6%	1.40	1.46	1.90	2.15	3.39
B/Ce@10%	1.24	1.28	1.65	1.85	2.88
IRRe	16.6%	17.7%	24.8%	28.3%	34.6%

184. **Sensitivity analysis.** A sensitivity analysis was carried out for following different scenarios: (i) a reduction in net benefits due to the combination of risks in sale prices; (ii) productive yields; (iii) levels of adoption; (iv) survival of the ventures; (v) an increase in costs due, for example, to the increase in prices of inputs and increases in expenses; (vi) delays in starting the project or in incorporating beneficiaries; and (vii) mixed scenarios of these phenomena. The results of the sensitivity analysis indicate that the economic results of the project remain attractive even under the simulated unfavorable conditions, showing a positive NPV in all the scenarios considered. Detailed assumptions and calculations are attached in Annex 3. Table 12 below presents the main results of the sensitivity test.

Table 12: Summary of the sensitivity analysis

Sensitivity analysis					IRRe	NPVe@6%		
GHG baseline scenario (low social price)	Δ%	Risk			24.8%	\$117,950,202		
Project benefits	-10%	Combination of risks in sale prices, returns, adoption levels, survival of the ventures			21.9%	\$98,303,494		
Project benefits	-20%				18.9%	\$78,656,785		
Costs	10%				22.2%	\$110,098,514		
Costs	20%	Increase in prices of inputs and expenses			19.9%	\$102,246,826		
Delay of Benefits of 1 year	Start-up delay / Delay in the incorporation of beneficiaries				19.9%	\$106,829,424		
Delay of Benefits of 2 years					16.9%	\$96,338,123		
Mixed scenarios	Costs	Δ%	Benefits	Δ%	IRRe	NPVe@6%		
					-10%	19.5%	\$90,451,805	
					-20%	16.7%	\$70,805,097	
					-20%	14.9%	\$62,953,408	
					-30%	12.2%	\$43,306,700	
					-10%	17.4%	\$82,600,117	

185. Regarding other efficiency and effectiveness indicators, the co-financing ratio of the of the GCF funds versus the government and IFAD funds in the project is 1:1.3, which is pertinent for the country and is evidence that the GCF is not the major source of funding for the project. The cost for the project per direct beneficiary amounts to 364 USD, while the costs per indirect beneficiary is about 24 USD. Although the project is not directly focused on mitigation it still achieves a cost of 18.7 USD/ton CO₂eq a value that is almost half of the lowest social price of carbon.

E. LOGICAL FRAMEWORK

This section refers to the project/programme's logical framework in accordance with the GCF's Integrated Results Management Framework to which the project/programme contributes as a whole, including in respect of any co-financing.

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

- ☒ Reduced emissions (mitigation)
- ☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

This section of the logical framework is meant to help a project/programme monitor and assess how it contributes to the paradigm shift described in section D.2 above by applying three assessment dimensions - scale, replicability, and sustainability.

Accordingly, for each assessment dimension (see the definition per assessment in the accompanying guidance note), describe the current state (baseline) and the potential scenario (target) and rate the current state (baseline) by using the three-point-scale rating (low, medium, and high) provided in the guidance note. Also describe how the project/programme will contribute to that shift/ transformation under respective assessment dimensions (scale, replicability and sustainability). In doing so, please refer to section B.2(a) (theory of change).

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Currently, a transition to climate-resilient and low-emission production systems in the Balsas Project Area (PA) is not prioritized as forest development incentives invested via institutional modalities (e.g. PADFSB,) is not based on climate change adaptation and	<u>Low</u>	The project will transform production systems and land use forms in the Balsas PA into systems which are climate resilient and low in GHG emissions, targeting the most climate vulnerable and poor rural communities in the country, specifically the most vulnerable municipalities and the areas of the PA with the highest levels of poverty and extreme poverty.	The project will promote the mainstreaming of the new climate-based approach into the main rural development institutions (and their incentive programs and initiatives) operating in the Balsas PA (CONAFOR, CONAGUA and AGRICULTURAAGRICULTURA) fostering multi-sectoral coordination at Basin level. This will be achieved through: i.) operationalizing the use of climate change criteria in current institutional support modalities for the prioritization of integral territorial development through a community-

	mitigation needs. Consequently, climate-resilient and low-emission development in the Basin PA is not scalable.			centric, bottom-up approach and; ii.) investing in the climate-sensitive management of ecosystems, and climate-resilient productive systems and value chains for adaptation and mitigation benefits. As a result, integral territorial development support in the PA will adopt a decentralized, coordinated and climate-responsive approach across all sectors, and presents a model that can be scaled up across Mexico's rural areas.
Replicability	An explicit model for climate-resilient and low-emission development for production systems in the forestry and agriculture sectors in Mexico's has not yet been proven. As such, there is no proof of concept for replication by multisectoral coordination structures. W	<u>Low</u>	The project's paradigm shift has the potential to be replicated across Mexico: i.) Through the operationalization of climate change criteria in the prioritization of rural development incentives and; ii.) through the strengthening of multisectoral coordination structures to re-align other sources of national development finance towards climate finance, such as those of CONAGUA. These approaches are the basis of the project's financial exit strategy and its potential for replicability in Mexico.	Piloting of an incentivized climate-sensitive approach to integral territorial development will first take place within CONAFOR's <i>Reglas de Operación</i> . In addition, specialized technical assistance will be provided by the project to facilitate access of those most poor and vulnerable communities that traditionally have had less or difficult access to CONAFOR's support programs through a demand driven, participatory approach. This will be complemented by the implementation of a pilot program to increase ease of access to PADFSB support by the most vulnerable communities in the Basin. To ensure that the paradigm shift is achieved at a decentralized level and not top down, multisectoral platforms will be established at the sub-basin and ethnic territory level to foster the participation of local communities and all levels of government. Furthermore, the project will support existing and create new local facilitators. This local-level bottom-up approach will promote more flexible procedures for the channeling of development finance to climate change initiatives via Basin and Sub-Basin

				Committees (SBCs). The piloting of such a cross-cutting approach by CONAFOR in the Balsas Basin provides a model that can be replicated across sectors in Mexico's climate-vulnerable areas without further external financial contributions.
Sustainability	There is currently no consistent long-term budget allocation to climate finance within Mexico's development framework. Financial flows are mainly directly to traditional development projects in the country's rural areas.	<u>Low</u>	The core medium- to long-term result of the project is related to the application of the paradigm shifting innovations introduced above, first piloted in the Balsas PA using GCF concessional finance, that will have the potential to realign an average of USD 160 million per year of forest development finance provided by CONAFOR towards climate finance (for adaptation and mitigation) in Mexico's most vulnerable communities. This amount of USD 160 million per year will result in the re-alignment of USD 1.5 billion climate finance during the lifespan of the project (20 years). The continued flow of resources required for the projected USD 1.5 billion in leveraged climate finance, is guaranteed by the resources budgeted by the Government of Mexico for integral territorial development over the next 20 years.	The project's sustainability depends on transforming institutional modalities used to allocate incentives for forest development initiatives to include climate change criteria, re-aligning forestry and forest landscape finance as climate finance for adaptation and mitigation initiatives in the most vulnerable areas of the Balsas PA. The financial exit strategy is built into this paradigm-shifting approach. This will ensure that re-aligned climate finance flows into adaptation and mitigation initiatives in Mexico are sustained in the medium to long term. The prioritization and selection of interventions and sites using climate change criteria will inform climate-focused support in rural areas with the highest vulnerability, while the multi-annual progress-based payments system will allow for the renewal of support in the form of incentives in these areas, increasing the scale and sustainability impacts of the climate-focused initiatives. Sustainability will also be supported by: i) Establishing participatory mechanisms for local and micro-regional planning and setting up social rules that afford territorial land-use management while taking into account the threats and opportunities linked to climate change; ii) Developing forestry, agroforestry and

				adaptive forestry system technologies which promote native tree species and its spatial arrangements and combinations better adapted to climate change; and iii) Encouraging existing rural extension systems to incorporate a climate dimension into their technical assistance services and strengthening organizational and technology transfers, especially for the most vulnerable populations.
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E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).						
GCF Result Area	IRMF	Means of Verification (MoV)	Baseline			Assumptions / Note
	Indicator			Mid-term	Final[1]	
MRA4 Forestry and land use	Core 1: GHG emissions reduced, avoided or removed/sequestered	PDR 3.2.1/Meta: Local monitoring of the project supported investments in ecosystem management (brigades) as input for the Ex Act calculator (adapted by CONAFOR to Mexican conditions). CONAFOR's MRV System will process information that will be independently checked at Midterm and final independent evaluation Based on ML PDR (Meta Mitig.) (Ton CO2eq)	0	(Ton CO2eq)	(Ton CO2eq)	The ecosystems of the Balsas are not drastically affected by catastrophic events as massive earthquakes and landslides, forest fires, pests, or hurricanes.
				726,327.3	1,644,348.5	
Total beneficiaries	Core 2: Direct and indirect beneficiaries reached	CONAFORs' gender disaggregated register of direct beneficiaries (D,W) of the PADFSB "Sistema Integral de Información de Apoyos" (SIAC); and total population of the watersheds and indigenous territories for indirect beneficiaries (I)	0	Total: 1,323,685 (W: 683,143) D: 80,747 (W:32,299) I: 1,242,938 (W: 650,844)	Total: 3,309,211 (W: 1,707,856) D: 201,867 (W: 80,747) I: 3,107,344 (W: 1,627,109)	Note: Beneficiaries are counted in number of individuals. T: Total D: Direct I: Indirect W: Women
ARA1 Most vulnerable people and communities	Supplementary 2.5: Beneficiaries (female/male) adopting innovations that strengthen climate	CONAFOR's registers of participants and beneficiaries of capacity development activities (Sub-Activity 1.3.1.1), forest brigades and facilitators (Sub-Activity 1.3.2.1). Independent evaluations at Midterm and final phase of the Project (SIAC)	0	D=T: 19,231 (W: 7,692)	D=T: 48,078 (W: 19,231)	Priorities of the Government are aligned with the participatory, bottom-up approach of the project and the Balsas Project Area is considered a national priority.

ARA2 Health, well-being, food and water security	<u>Supplementary 2.1:</u> Beneficiaries (female/male) adopting improved and/or new climate resilient livelihood options (number of individuals)	CONAFOR's list of direct beneficiaries of Sub-activities 2.1.1.1, 2.2.1.2-3, 2.2.2.1, 2.3.1.1, 2.3.1.3, 2.3.1.5, 2.3.3.1. (SIAC) Independent evaluations at midterm and final phase of the Project	0	T: 799,955 (W:415,109) D: 30,529 (W:12,212) I: 769,426 (W: 402,897)	T: 1,999,886(W: 1,037,771) D: 76,322 (W:30,529) I: 1,923,564 (W: 1,007,242)	Project interventions for sustainable landscape conservation and enhanced climate adapted production systems will render its positive financial results for beneficiaries and are maintained in the medium to long term.
ARA2 Health, well-being, food and water security	<u>Supplementary 2.2:</u> Beneficiaries (female/male) with improved food security	CONAFOR's list of direct beneficiaries of Sub-activities 2.2.1.1, 2.2.2.2, 2.3.1.4, 2.3.2.1. which finance climate adapted food production systems (SIAC) Independent household survey at midterm and final phase of the Project	0	T: 497,611 (W: 257,587) D: 24,099 (W: 9,640) I: 473,512 (W: 247,947)	T: 1,244,027 (W: 643,966) D: 60,247 (W: 24,099) I: 1,183,780 (W: 619,867)	Agroforestry production remains financially attractive, especially to the youth, and CONAGUA and state government ensure that women have water access to produce food in their <i>traspatio</i> parcels.
ARA1 Most vulnerable people and communities	<u>Supplementary 2.4:</u> Beneficiaries (female/male) covered by new or improved early warning systems	CONAFORs registries of beneficiaries of Output 3.1, including Sub-activity 3.1.1.1, 3.1.2.1, 3.2.2.1, 3.2.2.2. (SIAC) Independent household survey at Midterm and final phase of the Project	0	T: 729,888 (W: 381,342) D: 6,888 (W:2,755) I: 723,000 (W:378,587)	T: 1,824,719 (W: 953,355) D: 17,220 (W: 6,888) I: 1,807,499 (W: 946,467)	The available information from the warning system can be transmitted to the most vulnerable areas, and women have access to cell phone in the same condition as men.
ARA4 Ecosystems and ecosystem services	Core 4: Hectares of natural resource areas brought under improved low-emission and/or climate-resilient management practices	CONAFOR registries of all activities which involve climate resilient project interventions in the field and data from SIAC	0	Hectares 162,289	Hectares 324,478	It is assumed that the interventions financed through the PADFSB will be financially viable for the beneficiaries and will be maintained for a period of time long enough to provide improved
ARA4 Ecosystems and ecosystem services	<u>Supplementary 4.1:</u> Hectares of terrestrial forest, terrestrial non-forest, freshwater and	CONAFORs registries of area benefitted through Payment of Environmental Services (PES, Sub-activity 2.1.1.1.1) (SIAC)	0	Hectares 96,919	Hectares 193,838	

	coastal-marine areas brought under restoration and/or improved ecosystems					ecosystem services, particularly water regulation, storage and release.
ARA4 Ecosystems and ecosystem services	<u>Supplementary 4.1:</u> Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal-marine areas brought under restoration and/or improved ecosystems	CONAFORs registries of area benefitted through all productive landscape restoration activities other than PES (Sub-activity 2.1.1.1.1) (SIAC)	0	Hectares 65,370	Hectares 130,740	

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E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8, as applicable)

Core Indicator	Baseline context	Rating for current state (baseline)	Target scenario		Coverage
	(description)		(description)		
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	There are policy frameworks and strategies for ecosystems management taking into consideration the Climate Change in the CONAFOR programs, however the current instruments to prioritize vulnerable areas and engage the different stakeholders are weak or nonexistent. In the field, CONAFOR and CONAGUA don't integrate their support to strengthen the resilience of rural communities, taking into consideration the water management within the watershed	medium	Participatory Climate resilient planning for forest landscapes and water is developed at the prioritized basin and sub-basin, with the participation of rural communities, municipalities, State Institutions, Private Sector CONAGUA, and CONAFOR.		Multiple sub-national areas within a country
			CONAFOR will pilot a new approach to making the participation of the most vulnerable communities in PADFSB easier		

Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation	The current forestry management and restoration models do not include the climate change dimension, especially for better adaptation. The national systems to identify and alert about forest fire and pest need to be improved and linked with the rural communities better. Rural communities are not involved in forest and ecosystem monitoring	medium	CONAFOR and CONAGUA support programs promote and incorporate adaptive forest and ecosystem restoration models, which are adopted by rural communities.		National level (one country)
			Systems to detect forest fires and pests and monitor them are strengthened in CONAFOR headquarters. An application to monitor management results and the situation of forests and ecosystems is developed. Local authorities, extension workers and Brigades of community members access and use these systems and report most efficiently strategic information for the decision-making process.		
Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level	Small farmers and communities that manage timber and non-timber forest are barely involved in value chains, and sell their product to local middlemen. The added value for products provided from managed forest remains low.	low	CONAFOR provides technical assistance to the groups of farmers and forest producers to improve their internal organization, collaboration within other communities in the basin prioritized, access to technology and resources (working capital and assets).		Single sub-national area within a country

Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards	Farmers and extension workers are lacking the knowledge to adapt forests management, agroforestry systems and soil management to ensure sustainable management of forest landscapes and stabilize water availability at the community and farm levels, in the context of climate change. ;. These issues, among other, contribute to the high adaptation deficit of agriculture's stakeholders in project areas	low	Farmers, women, youth and extension workers will have access to enhanced climate adaptive knowledge and use of information systems through different sources: Tailored training and capacity development processes, webinars, and community to community seminars		Multiple sub-national areas within a country
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E.5. Project/programme specific indicators (project outcomes and outputs)

This section should list out project/programme-specific performance indicators (outcomes and outputs) that are not covered in sections above (E.1-E.4). List down tailored indicators to monitor /track progress against relevant project/programme results (outcomes/outputs). AEs have the freedom to decide against which outcomes they would like to set project/programme specific indicators. If any co-benefits are identified in sections B.2(a)(b), and D.3, AEs are encouraged to add and monitor co-benefit indicators under the "Project/programme co-benefit indicators" section in table below. Add rows as needed.

Please number each outcome and output as shown below to indicate association of outputs to the contributing outcome. The numbering for outputs under this section should correspond to the output numbering in annex 4 (detailed budget plan).

Project/programme results (outcomes/ outputs)	Project/programme specific indicators	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	

Outcome 1: Government programs and community development plans integrate climate change criteria for allocation of supports	PDR Policy 3: Number of existing/new laws, regulations, policies or strategies proposed to policy makers for approval, ratification or amendment	Interinstitutional collaboration agreements of institutions in the Balsas PA , and other official documents approved by corresponding jurisdictions	0	2	6	Balsas region remains a priority for the National Government and the related Government State aligns their priorities and strategies
	New: Percentage of new laws, regulations, policies or strategies existing or proposals that incorporate affirmative action to improve women participation in production of the value chains		0	50%	50%	
	PDR SF.2.2: % of Households reporting they can influence decision-making of local authorities and project-supported service providers	Independent household survey at Midterm and final phase of the Project (using IFAD/COI Methodology)	0	15	30	Agrarian Community structures remain strong and internal conflicts are avoided, which enable women and youth to participate
Output 1.1 Mechanisms for intersectoral coordination are strengthened to include climate change criteria in institutional structures and procedures	New: Number of approved participatory Basin, Sub-basins and ethnic territorial plans that incorporate climate change criteria and involvement of more than one federal institutions	<i>Approved plans by Sub-basin and Ethnic RCIMTs</i>	0	8	10	State institutions and municipalities participate in a regular manner to the committee meetings
			0	50%	50%	Private companies are interested in ensuring the sustainable provision of Environmental Services and contribute to the government and communities' efforts
Output 1.2 Participatory climate responsive multiannual investment plans within prioritized basin, sub-basins and ethnic territories	Number of investment plans developed and presented and discussed at the RCIMTs <i>PDR Product 1.1 and GP: % of women population participating in updating the multi-annual investment plans</i>	Registries of CONAFOR at RCIMT level	0	90	185	Agrarian Community structures remain strong and internal conflicts are avoided, which enable women and youth to participate

	<i>PDR Product 1.1 and GP: % of women population participating in updating the multi-annual investment plans</i>	CONAFOR registry of participants in the elaboration of these plans. Independent household survey at Midterm and final phase of the Project	0	20	40	
Output 1.3 Strengthened capacities of local actors for climate change adaptation, resilience, and inclusiveness	<i>New, Number of people incorporated in Brigades and extension services that strengthen the local capacity to improve climate change adaptation, resilience, and inclusiveness</i>	CONAFOR's Integrated Forest Incentive Information System (Sistema de Información Integrado de los Apoyos de la CONAFOR SIIAC)	Person	Person	Person	Agrarian Communities structure remain strong and internal conflicts are avoided, which enable women and youth to participate
			0	203	508	
			Women	Women	Women	
			0	93	232	
			Youth	Youth	Youth	
			0	173	432	
Outcome 2. Forest landscapes are sustainably managed and better adapted to climate change.	<i>PDR 2.1. Environmental recovery areas with an increase in the Normalized Difference Vegetation Index (NDVI) [Number of hectares]</i>	MRV Brigades.	0	162,289	324,578	The ecosystems of the Balsas are not drastically affected by external event as massive forest fires, pests or hurricanes
		External verification at mid-term and final evaluation				
	<i>PDR 1.2.4. Percent of households reporting an increase in production</i>	Independent household survey at Midterm and final phase of the Project (IFAD/COI Methodology)	0	30	30	
Output 2.1 Agrarian communities implement land-use plans through multiannual payment of environmental services (PES)	<i>PDR 3.1.4. Hectares of land brought under climate-resilient management</i>	SIIAC, MRV Brigades.	0	96,919	193,838	Agrarian Communities structure remain strong and internal conflicts are avoided
	<i>PDR 3.1.1. Groups supported to sustainably manage natural resources and climate-related risks</i>	External verification at mid-term and final evaluation	0	80	210	

	<i>Percentage of indigenous communities whose land is managed under climate resilience criteria/aspect/approach</i>			30	30	
Output 2.2 The productive systems of Agrarian Communities are strengthened and better adapted to climate change and climate variability	<i>PDR 3.1.4. Hectares of land brought under climate-resilient management</i>	SIIAC, MRV Brigades.	0	65,370	130,740	
	<i>PDR 1.1.4: Persons trained in production practices and/or technologies</i>	Recorded projects support in SIIAC system	Person	Person	Person	The ecosystems of the Balsas are not drastically affected by external event as massive forest fires, pests, or hurricanes
			0	15,614	31,228	
		Independent household survey at Midterm and final phase of the Project	Women	Women	Women	
			0	8,089	16,178	
			Youths	Youths	Youths	
			0	3,904	7,807	
Output 2.3: Participation of Agrarian Communities and Producer Organizations in forest and agroforestry value chains is strengthened	<i>PDR 2.1.3 Number of Rural producers' organizations supported</i>	Recorded projects support in SIIAC system	Org./AC	Org./AC	Org./AC	The national and international market for the proposed value chains the project remains stable
			0	25	50	
			Indig.Org. 0	Indig.Org. 12	Indig.Org. 25	
	<i>% of women that receive additional incomes either from transformation or commercialization from supported value chains</i>	Independent household survey at Midterm and final phase of the Project (IFAD/COI Methodology)	Women	Women	Women	
			0	30	60	
	<i>GP Supported technologies and work practices that reduce the workload of women</i>	Recorded projects support in SIIAC system	0	5	10	

	<i>PDR 2.1.3 (modified) Number of both organizations and AC that access new financial services</i>	Recorded projects support in SIAC system	Org./AC	Org./AC	Org./AC	
			0	10	25	
Outcome 3: The producers, inhabitants and authorities of the Balsas PA have timely and relevant climate information for climate risk management, disaster prevention	<i>Based on PDR indicator of the Product 3.1 members of producers, inhabitants and authorities that use the system</i>	Independent household survey at Midterm and final phase of the Project	Total beneficiaries 0	Total beneficiaries	Total beneficiaries *	Inhabitants have access to cellphone network and households have at least one device connected. * Note: These are included in the sum of the indirect beneficiaries of ecosystem services in the prioritized basins and territories (3 '107 344 people)
				723,000	1'807,499	
Output 3.1: Climate-hazards-related early warning and monitoring systems supported	<i>PDR Direct Ef. 3.1: Num. of Local authorities, beneficiaries and institutions who use new or improved early warning systems for climate risks, forest fire and pest (Supplementary 2.4)</i>	Register of participant in SIAC	0	6,888	17,220	Major part of the local authorities and beneficiaries have access to a cellphone, including women
	<i>GP: % of women trained in the use of the integrated climate information system in relation to the total number of producers trained</i>	Independent household survey at Midterm and final phase of the Project	Women	Women	Women	
			0	25%	40%	
	<i>Number of national systems related to fire and forest disease and MRV are improved to provide more reliable and timely information</i>	Independent household survey at Midterm and final phase of the Project		2	3	
Output 3.2: Policy-relevant knowledge products elaborated from the systematic documentation of good practices and intervention approaches of the project	<i>Number of relevant knowledge products for approved new policies or procedures</i>	CONAFOR web-based platform where knowledge products are available	0	2	4	CONAFOR maintains internal capacity and communication channels to produce and disseminate information

Project/programme co-benefit indicators						
Co-benefits	Project specific indicator	Means of verification	Baseline	Target Mid-term	Target Final	Assumptions / Notes
<u>Co-benefit 1:</u> Environmental services maintained	Area (in hectares) within the prioritized Basins, Sub-Basins and Indigenous Territories of the Balsas Basin were EbA measures are implemented	CONAFOR registries of PADFSB implementation and Project reports	0	140,000	340,000	SEMARNAT and CONAFOR maintain its decision to implement the GreenMex Project in complementarity with the Balsas Project in the project target area.
<u>Co-benefit 2:</u> Biodiversity conserved	Area (in hectares) of natural ecosystems that are part of bio-forestry corridors, in complementary to GEF GreenMex project in the Balsas Basin	GreenMex and Balsas project reports	0	32,500	65,000	At least 50% of areas benefitting from the PES incentives of the Balsas Project are related to biodiversity corridors supported through the GreenMex Project
E.6. Project/programme activities and deliverables						
<i>All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in annex 5 implementation timetable. Add rows as needed.</i> <i>Please number the activities as shown below to indicate association of activities to the related outputs provided above in section E.5. Similarly, please number sub-activities as shown below to associate to the related activity.</i>						
Activities	Description	Sub-activities	Deliverables			
Activity 1.1.1. Facilitate the creation of Forest Management Units (FMUs) and Regional Committees for the Integrated	With the support of CONAGUA, CONAFOR and the project team facilitate the creation of new RCIMTs and facilitate the participation of local actors	Sub-activity 1.1.1.1 Recruitment of senior specialist in basin planning and Junior Specialists in social participation (social safeguard) and communication	1	Communication awareness campaign		

Management of the Territory (RCIMTs).		Sub-activity 1.1.1.2 meeting with key actors (3 levels of government, indigenous communities and organization, productive organizations; women organizations, private sector) and sensitization meetings and formalization of the RCIMTs	10	Formal creation of and renewal of Basin, Sub-basin and ethnic territory RCIMTs
Activity 1.1.2. Agree on priority intervention areas at the micro-basin level	RCIMTs, project team and experts identify the most vulnerable micro-basins, ecosystems and communities and agree on prioritization	Sub-activity 1.1.2.1 Participatory discussion with the RCIMTs to collect local knowledge and decision making	2	Priorities areas (several agrarian communities) are identified in each basin and sub-basin, at least
		Sub-activity 1.1.2.3: Cartographic and field investigations (CONAFOR's expert + Promotorías)		
Activity 1.1.3. Strengthen multisectoral coordination structures, operation rules and mechanism for co-investment within the basins	Institutions and municipalities identify climate change trends and risks, solutions, programs and budgets that are available to support the Project at the prioritized Basin levels	Sub-activity 1.1.3.1 Intervention plan and coordination mechanism among the 3 levels of government are established and sharing results of the basin diagnostic with key institutional actors	20	Intervention plans formulated in each basin and sub-basin
		Sub-activity 1.1.3.2 formulation of the intervention plans for the prioritized micro-basins (CONAFOR's expert +)	3	MoU signed among institution (Federal and state levels), at least
Activity 1.1.4 Strengthen and operationalize criteria for adaptation to climate change in institutional program procedures	Once innovations in intervention models are systematized, the institutions adjust their operation procedures (<i>Reglas y lineamientos de operación</i>) to enhance the climate change resilience of the most vulnerable communities and develop positive actions measures for indigenous people and women	Sub-activity 1.1.4.1 Monitor and systematize project intervention in the prioritized basins to identify strengths and weaknesses	2	Institutional programs adjust their operational procedures to include climate change adaptation criteria and equity measures for indigenous people and women, at least
		Sub-activity 1.1.4.2 Elaborate modifications in the Operating Rules & Guidelines		
Activity 1.2.1 Communication and awareness campaign to disseminate calls for proposals	The project team prepares a communication and awareness campaign at the basin and sub-basin levels to disseminate the calls for proposal	Sub-activity 1.2.1.1 Recruitment of a specialist in social participation and local organization	4	Biannual calls for proposals are disseminated in each prioritized basin
		Sub-activity 1.2.1.2 Develop a communication and awareness campaign (including indigenous languages) with field activities in the most vulnerable and prepare special calls for proposals to receive project profiles (simple concept notes in each prioritized basin) indigenous communities (Promotorías Locales CONAFOR)		
Activity 1.2.2 formulation of climate smart <u>multi annual investment plans</u> based on micro-	Communities interested in the call for proposals submit project profiles (Concept Notes). Those who are aligned with the basin	Sub-activity 1.2.2.1 Micro-basin zoning plans including FPIC: Communities formulate profiles and submit them	10	Multiannual investment plans (MIP) based on micro-basin zoning plans are formulated and validated by community assemblies (for 4 years,

basin zoning plans and validated through FPIC	intervention plan and prioritized zones will receive financial and technical support to develop local zoning plans, investment plans and technical proposals to be funded by institutional programs (PADFSB, ProAgua...etc)	through the CONAFOR Promotorías and RCIMTs Profiles are analyzed by CONAFOR Promotorías and selected by the RCIMTs FPICs are carried out to valid Zoning plans and Investments plans at the community level (CONAFOR safeguard specialist + Local Promotoría).		40 MIP) in basin, sub-basin and ethnic areas, at least
		Sub-activity 1.2.2.2: Basin restoration expert (2)		
		Sub- Activity 1.2.2.3. Formulate Investment Plans according to existing institutional instruments (Community Consultants)		
Activity 1.3.1: Capacity development programs for extension workers, local facilitators, and forest protection brigades.	The Project enhance the current capacity development program for extension workers and brigades in order to incorporate Climate Change adaptations aspects related to the management of forest landscapes, use of applications for sub-project, and forest monitoring, the inclusion of youth and women, social and environmental safeguards; food production and nutrition	Sub-activity 1.3.1.1: Design of the training curricula and support materials (Project team + recruited experts).	3,700	Extension workers and local facilitators are trained every year (30% are women and youths), and their direct families benefitted from their income
		Sub- Activity 1.3.1.2 Development and dissemination of the training programs for extension workers, local facilitators and brigades (CONAFOR trainers + recruited experts) (positive action measures to incorporate women, youth and indigenous peoples).		
Activity 1.3.2: Support and creation of forest protection brigades and local climate change resilience facilitators	The project supports i) the creation and strengthening of local brigades against forest fires and disease; ii) the incorporation of women, youth, and indigenous local facilitators who participate in actions oriented to the climate change resilience	Sub- Activity 1.3.2.1 Financial support of brigades and local facilitator group including: the promotion, selection process (validated by municipalities), operation and monitoring (positive action measures to incorporate women, youth and indigenous peoples).	9,420	Members of Brigades and local facilitator groups are supported during the project's lifespan, and their direct families benefitted from their income.
		Sub- Activity 1.3.2.2 Financial support for the operation of brigades against forest fires & parasites (Forest Protection Support in Operational Rules)		
Activity 1.3.3: Strengthening CONAFORs' promotorías forestales	The Project will strengthen the CONAFOR promotorías forestales to support the project implementation	Sub-activity 1.3.2.1: Recruitment of 26 additional local promotors (speaking an indigenous language where necessary)	26	CONAFOR local specialist are recruited, and capacities are strengthened in 26 Promotorías forestales
		Sub-activity 1.3.2.2: Acquisition of equipment and materials		
		Sub-activity 1.3.2.3: Training sessions and technical assistance related to the inclusion of women, youth and indigenous peoples, implementation of safeguards, development of climate resilient forestry and agroforestry systems, as well as Project M&E system		

Activity 2.1.1 Payment of environmental services	Forest landscape conservation will be supported through PES to be financed through PADFSB Component IV (SA), as well as through Local PES Mechanisms (LM-PES).	Sub-activity 2.1.1.1 PDFS PES projects are approved and financed (including local mechanisms promoted in prioritized basins and sub-basins)	193,838	ha of forests / forest landscapes conserved under PES
			11,264	people benefitting from PES
Activity 2.2.1 Climate Smart Forest Landscape Restoration of Micro-watersheds and Strategic Areas (RFM)	Incentive payments and support to Agrarian Communities to implement forest landscape restoration practices on degraded areas to recover soil and ecosystem productivity, generate employment and wellbeing and increase ecosystem services and livelihood resilience. These financial transfers to AC include the procurements of technical assistance, inputs and supplies and compensation of labor costs.	Sub-activity 2.2.1.1 Establish climate resilient agroforestry systems	2,000	ha of agroforestry systems established
			3,061	households applying climate adapted agroforestry practices
		Sub-activity 2.2.1.2 Establish agave – forest water / soil retention terraces	1,000	ha of agave / forest fringes
			1,530	households apply forest – pasture fringes
		Sub-activity 2.2.1.3 Establish Forest – pasture water / soil retention terraces or similar techniques	1,400	ha of forest – pasture fringes or similar
			2,413	households apply forest – pasture fringes or similar
Activity 2.2.2 Commercial Forest and Agroforestry Plantations Adapted to Climate Change (PFCA)	Incentive payments to improve climate change adaptation and the productive potential of commercial forest plantations and agroforestry systems in order to increase the sustainable production of forest timber and non-timber forest products. These financial transfers to AC include the procurements of technical assistance, inputs and supplies and compensation of labor costs.	Sub-activity 2.2.2.1 Establish climate adapted planted forests (PFCA 1)	4,353	ha of planted forests established
			978	households apply climate adapted methods for planted forests
		Sub-activity 2.2.2.2 Establish climate smart silvopastoral / agroforestry system (PFCA 2)	1,500	ha of climate adapted SPS established
			337	households apply climate smart methods for SPS establishment
Activity 2.3.1 Sustainable Forest Management adapted to climate change, inclusion and access to markets (MFCCV)	Provide financial incentives and support to Agrarian Communities that own natural forests to implement actions that enable them to strengthen their forest governance, including their social, technical and cultural capacities as to manage, utilize and market their forest resources in a sustainable way. Activities will include elements to make sustainable forest management better adapted to climate change and promote bigger inclusion of the most vulnerable, women and youth in production processes and inclusive, fair trade and certified value chains. financial transfers to AC include the procurements of technical assistance, inputs and supplies and compensation of labor costs.	Sub-activity 2.3.1.1 Manage forests sustainably adapted to climate change (MFCCV.1)	120,487	ha of forests managed under adaptive silviculture practices
			8,112	households apply climate adaptive silviculture
		Sub-activity 2.3.1.2 Sustainable use and add value to timber forest products (MFCCV.2)	7,474	people use better practices
		Sub-activity 2.3.1.3 AC improve their marked access for timber and non-timber forest products (MFCCV.3)	8,717	people benefitted
		Sub-activity 2.3.1.4 Support women in benefitting from forest-based value chains (MFCCV.4)	4,984	people benefitted
		Sub-activity 2.3.1.5 Capacity building for local producers in sustainable forest management adapted to climate change (MFCCV.6)	34,558	producers trained and their families benefitting

Activity 2.3.2 Support households to improve food security and nutrition	Financial transfers to AC for the procurements of technical assistance, inputs and supplies and compensation of labor costs	Sub-activity 2.3.2.1 Traditional milpa (maize/beans) are transformed to diversified milpa with trees, improved irrigation, crop renovation and family orchards / traspatio parcels are improved through crop diversification, improved food quality and nutrition	13,923	people benefitted
		Sub-activity 2.3.2.2 Community water harvesting projects, collector roofs, hats (sombros), reservoirs	1,882	hectares
		Sub-activity 2.2.2.3: Technical Specialist in Water Management and Irrigation, 2 Technical Specialists in Inclusive Value Chains, 1 Technical Specialist in Forestry and Agroforestry		
Activity 2.3.3. Brigades to strengthen adaptation capacity in Agrarian Communities (Community Forest Monitoring)		Sub-activity 2.3.3.1. Contribute to national MRV system, national forest inventory and project monitoring and evaluation system	400	beneficiaries
Activity 3.1.1 Development of a platform and applications to facilitate access to climate information and early warning systems	The Project develops a platform to facilitate access of vulnerable communities to existing information on weather forecasts and early warning information improve the preparedness and to prevent extreme event hazards	Sub- Activity 3.1.1.1 Recruitment of specialists to design platform and applications in coordination with CENAPRED, CONAGUA and CONAFOR: - Conduct participatory and technical diagnostic of the existent information system in CENAPRED and CONAGUA and the access for rural communities) - Disseminate the platform and applications through the RCIMTs, brigades and local facilitator groups and institutions (support of Local Promotorías)	1	Platform that facilitates access to strategic information for weather forecast and early warning systems
		Sub-activity 3.1.1.2: Purchase of equipment & software.	17,220	People directly trained and their families. 1'807 499 indirect beneficiaries (Same indicator as Supplementary 2.4 in Table E.3)
Activity 3.1.2 Strengthening of prediction and prevention systems for fire (SPPIF) and forest phytosanitary risk	The Project strengthens the existing systems to make them more reliable and effective to provide timely information and generate data on burned areas	Sub-Activity 3.1.1.1 Acquisition of IT equipment and enhancement of existing systems: - Diagnostic of the existent information CONAFOR systems - Piloting and finalizing the development of the systems	2	Systems of prediction and prevention for fire (SPPIF) and forest diseases are enhanced
Activity 3.1.3 Strengthening CONAFOR's Monitoring Report and Verification System (MRV)	The Project strengthens the existing MRV system to enable local communities to participate in the data collection of information (Ecosystems information and parcels supported by CONAFOR programs)	Sub-activity 3.1.3.1: Recruitment of SIG specialist: - Diagnostic of the existent MRV and plot monitoring systems - Enhancement of existing systems - Piloting the MRV systems in the field with the local facilitator groups and finalizing its development	1	System on MRV and plot monitoring are enhanced to allow local facilitator to collect field information
		Sub-activity 3.1.3.2 Acquisition of IT equipment		

Activity 3.1.4 Strengthening CONAFOR's social and environmental safeguard management system	The Project strengthens the existing system to monitor environmental and social safeguards through CONAFOR local Promotorías	Sub-Activity 3.1.4.1 Safeguards Specialist: - Diagnostic of the existent system - Enhancement of existing systems to incorporate the Project requirements in IPPF	1	Systems on environmental and social Safeguards is enhanced to allow the CONAFOR <i>Promotorías</i> to collect data and monitor project intervention
		Sub-Activity 3.1.4.2 Piloting and adjusting the Safeguards systems in the field with support of local Promotorías		
Activity 3.2.1 Production of training and systematization materials	The project develops material for capacity development, knowledge products and communication (Basin intervention approach, forestry adapted to climate change, Monitoring ecosystem in the field, Women and Youths inclusion...etc)	Sub-activity 3.2.1.1 Development and piloting of training systematization materials and systematization of experiences and technology adoption	4	Knowledge products and training materials produced an available in the web page of CONAFOR, at least
Activity 3.2.2. Exchange of experiences, and South-South and Triangular Cooperation (SSCT)	The Project facilitates the organization of local, national and international exchanges of experiences (adaptative forestry, Indigenous knowledge on forestry and food production; Basin management, information systems and MRV, inclusive values chains for women and youth etc)	Sub-Activity 3.2.2.1 Formulation of face to face or virtual exchanges of experiences and cooperation: - Systematization of experiences - Identification of local, national, and international experiences that are valuable for the project	6	Face to face and virtual events of exchange of experiences and SSCT are organized at the local, national, and international levels, at least
		Sub-Activity 3.2.2.2 Development of field visits, face to face events or webinar at the local, national and international levels		
Activity 3.2.3 Monitoring and evaluation	The Project will carry out a regular monitoring of the project and evaluations a midterm and final	Sub-activity 3.2.3.1 Recruitment of a M&E specialist	Reports provided through project M&E system:	
		Sub-activity 3.2.3.2 Preparation of quarterly M&E reports	32	Quarterly reports,
		Sub-activity 3.2.3.3 Preparation M&E Outcome studies and M&E quarterly reports	5	Outcome studies
		Sub-Activity 3.2.3.4 Recruitment of external evaluation for Baseline and Midterm review	2	Baseline and midterm review,
		Sub-Activity 3.2.3.5 Recruitment of external evaluation for project finalization	1	and final evaluation

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

186. During project implementation, CONAFOR will be responsible for overall Project monitoring. The project will recruit a full time Monitoring and Evaluation (M&E) expert who will collaborate with the UCP and ETP of CONAFOR. The Project's progress will be measured against the indicators included in the Project Logical Framework above. A project-specific Monitoring and Evaluation plan will put together according to the approved Logical Framework. The plan includes a household baseline and at least two external evaluations during Project implementation (at midterm and final review), a permanent strategy to collect and analyze information related to the implementation of the Project. The final review will assess the achievement of indicators, sustainability of results, and lessons learned. The parameters of such evaluations will be outlined in the Operation Manual.

187. Although impact indicators are essentially related to the improvement in the capacity in adopting new climate-resilient forest landscape management and livelihood strategies, the Logical Framework also includes substantial GHG emission reduction benefits. At the impact, outcome, and output levels, the indicators disaggregate targets by gender, age, and ethnic composition of the beneficiaries. The M&E system will also measure improvement in terms of the development of new institutional instruments and mechanisms oriented to mainstream climate change in policies, operational rules of national programs, and coordination efforts among the different levels of the Government. The Logical Framework also includes indicators that reflect the progress of the Project in terms of the forest landscape areas that are managed according to new climate change criteria and nature-based solutions. In addition, a set of deliverables for each activity is identified to enable the project team to follow up on the progress of the different outcomes of the Project. Each Multiannual Invest Plan supported in the prioritized areas will also include basic indicators, whose progress will trigger the corresponding annual financial and technical supports.

188. The collection of M&E data in the field and at the CONAFOR level will combine different systems and activities:

- The Integrated Information System of Supports of CONAFOR ("Sistema Integral de Información de Apoyos" SIAC for the Spanish acronym) collects and records information on the proposals formulated by potential beneficiaries, the evaluation of these proposals and the supported actions by the PADFSB in the rural communities, including the Multi- Annual Investment Plans. SIACC comprises a geospatial application that allows for the localization of the beneficiaries, the intervened polygons, and the delimitation of the rural communities. SIAC will be a major information system to collect, analyse, and report information related to the beneficiaries and actions supported by the project.
- Regarding overlapping / double counting of beneficiaries: To grant and incentive payment within the PADFSB, CONAFOR signs a legal agreement with the benefitting Agrarian Community or group. This legal agreement has to include a list of the heads of the household that are going to benefit from the supported activities, including their names and their personal identification number "*Clave Única del Registro de Población*" (CURP). This information is registered within CONAFORs' information system for PADFSB forest incentives, the SIICA, which allows CONAFOR for full traceability of the granted incentives. There are cases in which a household benefits from more than one activity, as some are synergistic and/or complementary. In those cases, CONAFOR eliminates any possible double counting or duplication by using the unique personal identification number CURP, as its holder is counted as only one beneficiary for the CONAFOR.
- Household survey. At the beginning and the final of the project, a household survey will be carried out to collect social, cultural, and economic information on a probabilistic sample of potential project beneficiaries, who will be later divided between control and treatment groups. This survey aims to obtain quantitative and qualitative information to measure improvement in climate resilience, livelihood, and adoption of new practices, as well as gender and ethnic disaggregated information.
- The project will follow the IFAD's Guidelines to measure several corporative Core Indicators (CI) that were incorporated in the Balsas logical framework. The CIs consist of 1 outreach, 20 output and 22 outcome indicators that are mapped to the strategic objectives (SOs) and areas of thematic focus of IFAD Strategic Framework 2016-2025; they are also aligned with the Sustainable Development Goals (SDGs) defined in the 2030 Agenda.
- The Local Brigades for Measuring, Reporting and Verification (MRV) is a new CONAFOR initiative that will be piloted through the Balsas project and then scaled up at the national level. Within the project area, 18 brigades formed by one local expert and two rural youth (men and women) will be recruited by CONAFOR to participate in the data collection for the National Forest Inventory, the estimation of the deforestation rate, as well as the

monitoring of a sample of plots where the Project supports the improvement of the management of forest landscapes. These MRV Brigades will report data to the CONAFOR MRV team through a new online application that will be developed by the Project (refer to Activity 3.1.3).

- The Project will also recruit independent consultants who will assess project progress in terms of main impacts, outcomes and output achievements and will verify whether the M&E system reports the targets properly. These consultants will also participate in field visits to organize focus group discussions and evaluate the level of satisfaction of the beneficiaries, as well as correct safeguards implementation.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Selected Risk Factor 1

Category	Probability	Impact
<u>Technical and operational</u>	<u>Low</u>	<u>Medium</u>

Description

Low annual project budget allocations and/or the failure to allocate resources in accordance with the project timeline can seriously disrupt disbursements and operations.

Mitigation Measure(s)

IFAD's experience in Mexico has shown national systems to be advanced and reliable, which will ensure that the probability of this risk occurring remains low. To further mitigate this risk and ensure that project implementation is not adversely affected, IFAD as AE will work closely with CONAFOR (EE) to promptly allocate sufficient budgetary resources over the project life cycle.

Selected Risk Factor 2

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>

Description

Lack of coordination between the various state agencies involved in project coordination (including CONAFOR, AGRICULTURA, SEMARNAT, BIENESTAR, FIRA and INPI amongst others). This is linked to a lack of platforms to foster dialogue.

Mitigation Measure(s)

The project will contribute to building the institutional capacities of these stakeholders by strengthening inter-institutional coordination at the federal, state and municipal level. In addition, the project will ensure regular engagements take place with the institutional stakeholders and ensure that platforms are in place to bring them together for dialogue and planning related to project implementation. The risk will also be mitigated through close coordination with and support from the Office for the Coordination of Advisers to the Presidency of the Republic, HACIENDA and the implementing partners.

Selected Risk Factor 3

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>

Description

Economic crises may arise as a result of the disasters such as the COVID-19 pandemic. This, in turn, could lead to reduced government revenue and thus, limited public investment in initiatives and incentives for rural production or forest landscape conservation activities, including PES.

Mitigation Measure(s)

In the case of unavoidable disasters, such as the COVID-19 pandemic, which may result in economic crises, investment in rural production and forest landscape conservation activities will be safeguarded through predetermined integral territorial budgets allocated in advance by CONAFOR. CONAFOR's integral territorial budgets are purposed to support and protect economic development in rural areas, including during economic crises (such as that caused by the COVID-19 pandemic), which will contribute to mitigating this risk.

Selected Risk Factor 4

Category	Probability	Impact
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<u>Forex</u>	<u>Medium</u>	<u>High</u>
Description		
Fluctuations in the exchange range, particularly devaluation of the Mexican peso relative to the US dollar, negatively impact project budget planning, disbursement and spending, which may disrupt the implementation of the project.		
Mitigation Measure(s)		
The GCF funding will be disbursed quarterly throughout the project's term, in the historically and presumptively more stable currency of the US Dollar, which will mitigate the impact of any devaluation of the Mexican peso relative to the dollar. In its Co-financing agreements, IFAD expects to require CONAFOR to hold this Co-financing in Mexican peso denominated accounts to allow for an onshore account. Although the project's financial model is articulated in US Dollars, most costs will eventually be incurred in Mexican pesos, with currency conversion to take place upon receipt by CONAFOR for project costs. The Operating Manual governing CONAFOR's management of these funds incorporates a strategy to mitigate exchange rate risk. The project financial model also includes a contingency fund that may be available to mitigate the impact of currency fluctuation.		
Selected Risk Factor 5		
Category	Probability	Impact
<u>Other</u>	<u>Low</u>	<u>Medium</u>
Description		
The project may face a change in the Federal Government, shifting priorities in the target region and resulting in varied institutional support during the implementation period.		
Mitigation Measure(s)		
This risk will be mitigated by aligning the project with state programmes aimed at climate change mitigation and poverty reduction — both during project development and implementation. This will ensure that the project is tightly aligned with government priorities at all times during the implementation period. Similarly, the project will align with other platforms and public contexts which may foster opportunities for relevant dialogue with stakeholder, to ensure continued support.		
Selected Risk Factor 6		
Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	<u>High</u>
Description		
Forest fires (caused by humans) and extreme weather events (such as floods) may result in damage to or loss of forest cover, negatively affecting the progress and success of reforestation activities.		
Mitigation Measure(s)		
Selected Risk Factor 7		
Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	<u>Medium</u>
Description		
Organized crime in the Project Region may hinder or delay project implementation and/or put project personnel at risk.		
Mitigation Measure(s)		
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Military intelligence of the Government of Mexico informs the CONAFOR, as well as all other government institutions, about the presence and the situation relative of organized crime in their respective region. CONAFOR's local field offices ("promotorías) obtain direct information about the security situation so that dangerous situations can be avoided and the security of CONAFOR's technical staff is granted. The project will reinforce the communication		

between CONAFOR headquarters and the promotorías through the establishment of the project UCP, the recruitment of additional facilitators in the field, (which will have to adhere to the security protocols), equipment (including communication) and intensive training of the technical team, particularly advanced security in the field training.

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

189. To identify environmental and social risks and propose appropriate mitigation measures, an Environmental and Social Management Framework (ESMF) has been prepared for the project (Annex 6). Based on the assessment conducted, the project is categorized as a Category B or 'moderate risk' according to the GCF project classification. This categorization was selected because any potential environmental and social impacts related to activities (especially under Outcome 2) will be localized and easily reversible through already known and commonly used mitigation measures. All sub-projects under Outcome 2 will be screened to ensure a Category B risk level or lower and abbreviated Environmental and Social Management Plans (ESMPs) to screen and monitor eventual impacts will be prepared.

190. CONAFOR has an adequate institutional framework to assess and manage potential social and environmental impacts. CONAFOR's operational rules (*Reglas de Operación* - ROs) and manuals (*guías*) are used to screen and monitor community project proposals and prevent the funding of any socially or environmentally harmful activities as per Mexican Law and safeguard requirements. The project will strengthen these ROs and CONAFOR's capacities. CONAFOR has an extensive experience in safeguards gained by working on the design and implementation of diverse programs and activities with IFAD, the World Bank, the Forest Investment Program, the Bio Carbon Fund, and bilateral cooperation with Norway and others. Its safeguard institutional framework is also aligned with the GCF's safeguards requirements. CONAFOR has successfully managed loans of over USD 500 million and grants of approximately USD 25 million without any significant safeguard issues.

191. The project seeks to increase the resilience of vulnerable forest-dependent communities and the ecosystems they depend on. In terms of Indigenous Peoples of the *Tlapaneco*, *Zapoteco*, *Amuzgo*, *Mixteco*, *Chatino*, *Tacuate*, *Triqui*, *Nahua* and *Purépecha* ethnic groups and other vulnerable groups, the greatest risk identified is the possibility that their positive and active role in climate action is not adequately recognized by focusing narrowly on them as mere beneficiaries rather than on their potential leadership. In this sense, the most significant risks revolve around:

- Inadequate indigenous and vulnerable groups' participation.
- Marginalization and erosion of Indigenous and vulnerable groups' aspirations, capacities, knowledge, practices, and perspectives; specifically:
 - Indigenous planning practices
 - Indigenous knowledge around land stewardship
 - Indigenous governance practices
 - Indigenous economies of reciprocity

192. The project's environmental risks and impacts on biodiversity and natural habitats are very small, introduction of invasive species, use of forbidden pesticides, undue use of natural resources are prevented at the inception phase of sub-project through the use of CONAFOR's screening tools and exclusion list. To ensure proper implementation of these restrictions project extension agents will receive additional capacity building for safeguards implementation and monitoring.

193. A Grievance Redress Mechanism (GRM) to handle eventual questions, complaints, suggestions, access to information requests, comments and grievances is already in place at CONAFOR (called the *Mecanismo de Acción Ciudadana*, MAC). The operation of the MAC is carried out in three different areas that have their own channels, regulations, and attention protocols, depending on the nature of the matters to resolve and/or attend, namely: (i) Internal Control Body (includes the Social Comptrollers); (ii) Liaison Unit - INAI (National Institute for Access to Information); and (iii) Citizen Information and Attention Service (SIAC). If any project-related grievances cannot be resolved by CONAFOR's MAC, they will be elevated to the level of IFAD's GRM/Complaints procedure. IFAD Complaints Procedure exists to ensure that appropriate mechanisms are in place to allow individuals and communities to contact IFAD directly and file a complaint if they believe they are or may be adversely affected by the project not complying with IFAD's social and environmental policies. Complainants are thus encouraged to first bring the matter to the attention of CONAFOR as the EE but may bring the issue directly to IFAD if they feel the EE has not adequately responded or if they feel they might be subject to retaliation if they approach the EE directly.

194. Coordination of safeguards monitoring will be the responsibility of the *Gerencia de Biodiversidad y Cambio Climático* (GBCC). In order to complement GBCC supervision capacity at the field level, safeguards monitoring will also be undertaken by the *Promotorías Forestales* (PF) and its staff, who will receive capacity building for this purpose. The GBCC will be in charge of overseeing training and capacity building at the PF to ensure proper safeguards implementation and monitoring. The PF will monitor safeguards implementation at the field level ensuring compliance with safeguard requirements.

195. Relevant documentation related to safeguards will be disclosed in a timely manner. For Category B projects, disclosure shall be at least 30 days before the expected date of approval of the GCF in accordance with the requirements of the GCF Disclosure Policy. Sub-projects at the state level will disclose all relevant documentation at least 30 days before the expected date of approval by CONAFOR. Documents disclosed will be presented in an

accessible and culturally appropriate manner, accounting for the specific needs of Indigenous and community groups that may be affected by the implementation of the project (such as literacy, gender, language differences, accessibility of technical information or connectivity).

196. During the consultation process with the representatives of the Agrarian Communities (ACs) of the prioritized indigenous municipalities of the *Purépecha*, *Amuzgo*, *Tlapaneco*, *Mixtec* and *Nahua* peoples, emphasis was placed on mechanisms and forms of resolution of possible conflicts generated from the implementation of the project to safeguard the communities. The participants highlighted the importance of the community assembly as the highest institution for resolving internal conflicts and decision-making, as well as the importance of ensuring the participation of institutions such as CONAFOR in the community assemblies to clearly inform communities about the objectives and activities of the project. To do so, a complete set of activities, including a free and prior informed consultation (FPIC) plan, has been developed (see Annex 7a) to ensure Indigenous communities' participation in project development and to prevent any potential conflict during implementation.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

197. In line with GCF's gender policies, the project stimulates climate investments that contribute to reducing gender gaps and promotes equality through climate change mitigation and adaptation measures. To this end, a Gender Assessment and Action Plan (GAP) (please see also Annex 8) was carried out that shows, that in the Mexican countryside and specifically in the project area women are more affected by poverty than men, as 56% of women are poor⁹⁹. Women represent only 26% of landowners and gender stereotypes and *machismo* are determining factors in their living conditions and their possibilities for economic and social empowerment. Among the most relevant gender gaps are the long working hours, the invisibility of their contributions to production, conservation of the environment, nutrition, and the family economy. Women have higher levels of illiteracy¹⁰⁰, are less connected than men to the mass media¹⁰¹ and hold much fewer¹⁰² decision-making positions in *ejidos* and communities. Only one third of women in agriculture are paid for their work and have less access to credit and technical assistance.

198. Women are more vulnerable to the effects of climate change. Household activities become more burdensome due to rising temperatures and scarcity (or excess) of water, increasing the difficulties of washing, cleaning, and cooking. The occurrence of pathogenic diseases and the limited availability of food for the family also add to women's psychological and work burdens. *Traspasito* subsistence production and economic activities such as harvesting of non-forest timber products, which are also often carried out by women, are affected by the loss of vegetation. More women are the heads of households due to the labour migration of men and are in charge of agricultural production, but they lack recognition of their labour and land ownership. Inequity in access to services and goods for production and resilience to climate change, affect women's performance and increase their vulnerability.

199. Based on these findings, recommendations were formulated in the GAP to ensure women's participation in project activities and benefits by defining quotas or participation targets, capacity building for gender equality for the different groups of project stakeholders. The GAP describes existing national laws and policies, strategies and plans of the institutions involved in the project. The GAP is elaborated based on the project's expected results and outputs framework and defines activities, indicators, and targets for women's participation. Through its implementation, women are expected to be active participants and beneficiaries of increased social and climate resilience, improved climate adaptation of productive ecosystems and rural community development. Furthermore the GAP:

- convenes and strengthens women's participation in decision-making in RCIMTs for the prioritization of intervention areas as well as in the elaboration of participatory multi-annual investment plans
- promotes greater gender equity in institutional work by adjusting sectoral and/or institutional policies, strategies, and regulations
- includes women as beneficiaries of investment plans in basins and micro-basins and for this purpose develops communication and awareness campaigns to disseminate calls for projects with inclusive language and based on profiles of the target population according to gender
- prepares simplified formats for the submission of project proposals
- proposes training in gender and inclusion for extensionists, local facilitators, *brigadistas* and *promotores forestales*, and it's up-take into CONAFOR's capacity building programme
- proposes to prioritize the participation of women in *ejido* and communal authorities and to use selection criteria for Payment of Environmental Services (PES) projects that favours women's participation
- promotes that women participate in projects to improve the productive systems of the farming communities, be trained in production practices and technologies adapted to climate change and variability, and benefit from the financial transfers made by the project to AC's for the hiring of technical assistance, provision of inputs and compensation of labour costs
- proposes that climate-resilient sustainable forest management activities open-up opportunities for women's participation in value chains, especially in economic activities linked to sustainable use and value addition to timber and non-timber forest products, strengthening their capacities on gender, sustainable and climate-resilient production practices, organizational, managerial, and marketing strengthening

- proposes to implement comprehensive productive projects for women that include affirmative actions with a gender perspective and support backyard food production through productive diversification and improved irrigation
- proposes to improve women's access to climate information and grant its equitable use, including training in the use of smartphones
- proposes to carry out workshops for the exchange of experiences and good practices implemented by women in the framework of the project

200. The GAP includes indicators and targets for monitoring and evaluation. It also provides for the availability of specialized gender support for the implementation of the Plan and gender mainstreaming throughout the project.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

201. The project's financial management (FM) will be carried out in accordance with the legal framework governing public administration in Mexico. FM will be the responsibility of the International Affairs and Financial Promotion Unit (UAIFF) of CONAFOR; the UCP will contribute to FM and will have a Finance specialist who will work closely with the UAIFF.

202. NAFIN, as the Financial Agent designated by the Ministry of Finance, will be responsible for the administration of the financing. It will follow-up on audits, managing disbursements, and will be in charge of loan disbursement processes that the GCF will channel through IFAD via NAFIN and TESOFÉ to CONAFOR.

203. CONAFOR with its Fondo Forestal Mexicano will follow-up on audits, managing disbursements, and will be in charge of grant disbursement processes that the GCF will channel through IFAD to CONAFOR.

204. For disbursements of GCF Loan A and GCF Loan B, the reimbursement method will be used given that project expenses are pre-financed by the Government. For grants, a replenishment mechanism could be used based on an advance payment. The bank accounts will be those defined by NAFIN, from which funds are transferred to the Treasury of the Federation (TESOFÉ) in the case of the loans, and to CONAFOR in the case of the grant. The flow of funds to beneficiaries is controlled by CONAFOR Integrated Support System (SIIAC) and by Integral administration System (SAI) for other expenses. IFAD's Client Portal will be the online platform used by NAFIN and CONAFOR to submit disbursement requests of the GCF Loan A, GCF Loan B and GCF grant proceeds, respectively, to IFAD.

205. The internal audit will be carried out by the Internal Control Organ (OIC) of CONAFOR, and the external audit will be carried out by private firms to be appointed by the SFP applying the terms of reference agreed with the WB and IDB, accepted by IFAD. IFAD's responsibility will consist of on-site supervision missions to assess FM arrangements, identify areas requiring improvement, fiduciary risks, and corresponding mitigation measures.

206. Procurement activities will be carried out in accordance with Mexico's national procurement system. The procurement methods will be those established in those regulations and the maximum thresholds per method will be those established for each fiscal year in the Federal Expenditure Budget (PEF). IFAD will maintain the review prior to the processes based on the thresholds resulting from the application of the Procurement Risk Matrix (PRM). A risk assessment was undertaken in August 2021 during the design and it concluded that Mexico's procurement rules are aligned with international standards. This was confirmed in the last supervision mission for the IFAD loan to Mexico in 2023.

207. Procurement planning will be done through IFAD's current model procurement plan (PP), which will indicate, among others, the selection methods, and the type of review by IFAD (prior or post-review) to which the processes will be subject. The first PP will be for the first year of operation of the project and subsequent PPs will be annual. The PIM includes the need of providing an 18-months PP in the first year of the project.

208. Requests for non-objection, both to the PP and its modifications, as well as to the procurement processes established in the PP as subject to prior review, will be processed through the No Objection Tracking Utility System (NOTUS). Contracts arising from procurements raised in the PP will be registered in IFAD's Client Portal (ICP) Contract Management Tool (CMT) platform. Procurement management will be reviewed and evaluated by IFAD during supervision missions which will be carried out periodically, but at least twice a year.

G.4. Disclosure of funding proposal

☒ **No confidential information:** The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ **With confidential information:** The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and

- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes:

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 2a Feasibility study; 2b Climate Rationale
- ☒ Annex 3 3a Economic and/or financial analyses; 3b in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 7a IPPF; 7b Summary of IP consultations; 7c Stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)¹⁰³
- ☒ Annex 23 Estimation of Beneficiaries

- ¹ The total lifespan of the project/programme is defined as the maximum number of years over which the outcomes of the investment are expected to be effective. This is different from the project/programme implementation period.
- ² Refer to the Policy of Co-financing of the GCF.
- ³ Trejo Vazquez, R. I. 2016. Updating and analyzing climate impact in bioclimatic areas of Mexico with new climate change scenarios/Actualización y análisis del impacto climático en zonas bioclimáticas de México con nuevos escenarios de cambio climático. United Nations Development Programme. Mexico, p. 95.
- ⁴ Gómez Días, J. D. 2016. Assessment of the productive potential of Mexico's soils considering climate change scenarios/Evaluación del potencial productivo de los suelos de México considerando escenarios de cambio climático. United Nations Development Programme. Mexico, p. 129.
- ⁵ Trejo Vazquez, R. I. 2016. Updating and analyzing climate impact in bioclimatic areas of Mexico with new climate change scenarios/Actualización y análisis del impacto climático en zonas bioclimáticas de México con nuevos escenarios de cambio climático. United Nations Development Programme. Mexico, p. 95.
- ⁶ Cuervo-Robayo AP, Téllez-Valdés O, Gómez-Albores MA, Venegas-Barrera CS, Manjarrez J, et al. 2014. An update of high-resolution monthly climate surfaces for Mexico. *International Journal of Climatology* 34: 24272437.
- ⁷ Information presented in INECC, 2020. Atlas of climate vulnerability/Atlas de vulnerabilidad climática. <https://atlasvulnerabilidad.inecc.gob.mx>
- ⁸ National Council for the Evaluation of Social Development Policy (CONEVAL 2020)
- ⁹ <https://www.pincc.unam.mx/publicaciones/cambio-climatico-miradas-de-genero/>
- ¹⁰ GFSAD1KCD Earth Map accessed 24 March 2024: GFSAD data were produced for nominal 2010 by overlaying the five dominant crops of the world produced by Ramankutty et al. (2008), Monfreda et al. (2008), and Portman et al. (2009) over the remote sensing derived global irrigated and rainfed cropland area map of the International Water Management Institute (IWMI; Thenkabail et al., 2009a, 2009b, 2011, Biradar et al., 2009) to ultimately create eight classes of crop dominance. The GFSAD1KCD nominal 2010 product is based on data ranging from years 2007 through 2012.
- ¹¹ An ejido is a form of land tenure that combines communal ownership with individual use. The ejido consists of cultivated land, pastureland, other uncultivated lands, and the town. (Being part of an ejido no longer necessarily involves communal ownership. After the reform of article 27 of the Constitution and the PROCEDURE program many ejidos have been privatized some or all of their land, but, unless they are formally dissolved as such, they continue to exist legally as ejidos.)
- ¹² Project interventions will be carried out mainly at the level of the "núcleos agrarios (NA)", which are Agrarian Communities (AC): these constitute the basic territorial unit, whose members can make decisions on the management of their resources. This territorial governance unit is also a criteria for targeting of project interventions. An in depth description and analysis of Agrarian Communities in Mexico and the Balsas Basin are provided in Annex 2: Feasibility Study.
- ¹³ Diagnosis of rural youth in Mexico, 2015. (Diagnóstico de la juventud rural en México, 2015). CEDRSSAR Report
- ¹⁴ INEGI. Censo de Población y Vivienda 2020. According to the National Institute of Statistics and Geography (INEGI), based on the results of the last population census of 2020, and the indigenous household criterion, the indigenous population consist on 7 million 364 thousand people , which represents 6.1% of the total population of the country.
- ¹⁵ CONAPRED, CNDH, INEGI. Perfil sociodemográfico de la población afrodescendiente en México. 2017
- ¹⁶ FIDA- FAO. 2018. Cerrar las Brechas: Nota de Política Pública para la inclusión de la perspectiva de género e interculturalidad en la agricultura y el Desarrollo rural. <https://www.ifad.org/en/web/knowledge/publication/asset/41195962>
- ¹⁷ Méndez-Pérez, M, et al. 2021. Food and nutritional insecurity before and during the COVID-19 pandemic in the households of women belonging to a social program. *Food and Nutrition Sciences*. Vol. 12 No. 7. <https://www.scirp.org/journal/paperinformation.aspx?paperid=110655>
- ¹⁸ <https://idea.usaid.gov/cd/mexico/hunger-and-food-security#tab-food-security>
- ¹⁹ <https://statistics.cepal.org/portal/cepalstat/national-profile.html?theme=1&country=mex&lang=en>
- ²⁰ <https://www.coneval.org.mx/Medicion/Paginas/Pobreza-estados-en.aspx>
- ²¹ CONAGUA (sf). Sistema de información hidrológica. Retrieved from: <https://sih.conagua.gob.mx/>
- ²² Mohammed, A. R. , & Tarpley, L. (2011). Effects of High Night Temperature on Crop Physiology and Productivity: Plant Growth Regulators Provide a Management Option. In (Ed.), *Global Warming Impacts - Case Studies on the Economy, Human Health, and on Urban and Natural Environments*. IntechOpen. <https://doi.org/10.5772/24537>
- ²³ Mohammed & Tarpley, 2011; Loka & Oosterhuis, 2010; Mohammed & Tarpley, 2009a; Mohammed & Tarpley, 2009b ; Turnbull et al., 2002
- ²⁴ Gourdj S; Läderach P; Martínez Valle A; Zelaya Martínez C; Lobell D. 2015. Historical climate trends, deforestation, and maize and bean yields in Nicaragua. *Agricultural and Forest Meteorology* 200:270-281.
- ²⁵ Centro Nacional de Prevención de Desastres CENAPRED (2012). Grado de peligro por sequía por municipio', scale: 1:20000. 1 ed. Retrived from: http://www.conabio.gob.mx/informacion/metadatos/gis/pelsequigw.xml?_httpcache=yes&_xsl=/db/metadatos/xsl/fgdc.html.xsl&_indent=no
- ²⁶ Gómez Días, J. D. 2016. Assessment of the productive potential of Mexico's soils considering climate change scenarios/Evaluación del potencial productivo de los suelos de México considerando escenarios de cambio climático. United Nations Development Programme. Mexico, p. 129.
- ²⁷ Pavia et al. (2009) and Cuervo-Robayo et al. (2020).
- ²⁸ Trejo Vazquez, R. I. 2016. Updating and analyzing climate impact in bioclimatic areas of Mexico with new climate change scenarios/Actualización y análisis del impacto climático en zonas bioclimáticas de México con nuevos escenarios de cambio climático. United Nations Development Programme. Mexico, p. 95.
- ²⁹ CONAGUA^a (2000) Tropical Storm Norman. Tropical Cyclones 2000. Retrived from: <https://smn.conagua.gob.mx/tools/DATA/Ciclones%20Tropicales/Ciclones/2000-Norman.pdf> CONAGUA^b (2000) Tropical Storm Rosa. Tropical Cyclones 2000. Retrived from: <https://smn.conagua.gob.mx/tools/DATA/Ciclones%20Tropicales/Ciclones/2018-Rosa.pdf>
- ³⁰ INECC, 2020. *Atlas Nacional de Vulnerabilidad al Cambio Climático*. <https://atlasvulnerabilidad.inecc.gob.mx>. Retrieved 20 September 2020.
- ³¹ UNAM, 2021. Monterroso R.A, Fernández E, R.I Trejo V. Conde A, J Escandón C., L. Villeres R y C. Gay 2014. *Vulnerabilidad y adaptación a los efectos del cambio climático*. <http://atlasclimatico.unam.mx/VyA/-34>. Retrieved 16 January 2021.
- ³² Gobierno de México. Secretaría de Medio Ambiente y Recursos Naturales. (2022). México: Tercer Informe Bienal de Actualización ante la Convención Marco de las Naciones Unidas sobre el Cambio Climático (2022).
- ³³ Gobierno de México. Secretaría de Medio Ambiente y Recursos Naturales. e Instituto Nacional de Ecología y Cambio Climático (2022). México: Inventario Nacional de Emisiones de Gases y Compuestos de Efecto Invernadero, 1990-2019
- ³⁴ FIDA 2021. Proyecto BALSAS. Informe de Diseño. Párrafo 39 y Anexo 5 Evaluación social y ambiental (párrafos 129-142).
- ³⁵ 7.2 million ha belong to the following indigenous peoples, the: Nahuatl, Tarasco/Purépecha, Amuzgo, Mazahua, Chatino, Matlazinca, Mazateco, Mixe, Mixteco, Otomi, Popoloca, Tlahuica/Ocuilteco, Tlapaneco, Totonaco, Triqui and Zapoteco groups.
- ³⁶ https://redd.unfccc.int/files/nref_2020_modificado_mexico_23072021_nt_20220103_clean.pdf
- ³⁷ CONAFOR, 2020. Preliminary estimation of gross deforestation rates in the Balsas Basin/Estimación preliminar de las tasas de deforestación bruta en la Cuenca del Balsas para el período 2000-2018. National Forestry Commission/Comisión Nacional Forestal. Calculations done for this project.
- ³⁸ Gobierno de México. Secretaría de Medio Ambiente y Recursos Naturales e INECC. Contribución Determinada a Nivel Nacional. Actualización 2022 <https://unfccc.int/documents/624282>
- ³⁹ Axis A: Prevention and attention to the negative impacts on the human population and the territory; Axis B: Resilient production systems and food security; Axis C: Conservation, rehabilitation and the sustainable use of biodiversity and ecosystem services; and Axis D: Integrated water resource management with a climate change approach (*Nationally Determined Contribution – NDCs. 2020 Update*).
- ⁴⁰ Comisión Intersectorial de Cambio Climático. Estrategia Nacional de Reducción de Emisiones por Deforestación y Degradación Forestal/ENAREDD+ 2017-2030. <http://www.enaredd.gob.mx/wp-content/uploads/2017/09/Estrategia-Nacional-REDD+-2017-2030.pdf>
- ⁴¹ Programa Nacional Forestal 2020-2024. Diario Oficial de la Federación 31/12/2020. https://www.dof.gob.mx/nota_detalle.php?codigo=5609275&fecha=31/12/2020#gsc.tab=0
- ⁴² https://www.forestcarbonpartnership.org/system/files/documents/_ENGLISH_6november_2017_Mx.pdf
- ⁴³ <https://www.unredd.net/documents/redd-papers-and-publications-90/un-redd-publications-1191/17249-mapeo-de-finanzamiento-de-redd-en-mexico.html>
- ⁴⁴ <https://documents1.worldbank.org/curated/en/670141517454051085/pdf/Mexico-PAD-01092018.pdf>

- 45 Ficha de Monitoreo 2020-2021 del Consejo nacional de Evaluación de la Política de Desarrollo Social, see: https://www.gob.mx/cms/uploads/attachment/file/673527/FMyE_P1_16_S219.pdf
- 46 Ficha de Monitoreo 2020-2021 del Consejo nacional de Evaluación de la Política de Desarrollo Social, see: https://www.gob.mx/cms/uploads/attachment/file/673527/FMyE_P1_16_S219.pdf
- 47 [National Atlas of Vulnerability to Climate Change](#) (ANVCC) SEMARNAT: The vulnerability dimensions are: i.) Vulnerability of human settlements to flooding, ii.) Vulnerability of human settlements to landslides, iii.) Vulnerability of the population to increase in dengue fever, iv.) Vulnerability of forage production to water stress, v.) Vulnerability of livestock production to water stress and vi.) Vulnerability of livestock production to flooding and vii.) Change in current distribution of priority species. For further detail, please refer also to Annex 2 Feasibility Study and to the Atlas referred to.
- 48 Instituto Nacional de Ecología y Cambio Climático (INECC). 2021 González Terrazas D., Vermonden Thibodeau A., Gress Carrasco F., Municipios Vulnerables al Cambio Climático con base en los resultados del Atlas Nacional de Vulnerabilidad al Cambio Climático. pp.60
- 49 Please find more details on this ratings and the list with the 104 rated municipalities in Annex 2, Feasibility Study
- 50 Ley General de Desarrollo Forestal Sustentable. Artículo 105. De las Unidades de Manejo Forestal: La Comisión, en coordinación con las Entidades Federativas, delimitarán las unidades de manejo forestal, tomando como base las semejanzas físicas, ambientales, sociales y económicas de un territorio, con el propósito de lograr una ordenación forestal sustentable, una planeación ordenada de las actividades forestales y el manejo eficiente de los recursos forestales. <https://www.diputados.gob.mx/LeyesBiblio/pdf/LGDFS.pdf>
- 51 See PAMIC information in <https://panorama.solutions/en/solution/un-instrumento-dinamico-y-operativo-para-el-uso-integrado-del-suelo-y-la-gestion-del-agua>
- 52 Diario Oficial de la Federación 30/12/2023. Reglas de Operación del Programa de Apoyos para el Desarrollo Forestal Sustentable para el Bienestar 2024. https://www.dof.gob.mx/nota_detalle.php?codigo=5713386&fecha=30/12/2023#gsc.tab=0
- 53 The incentive schemes of the PADFS to applied and improved by this project are: a.) Payment for Environmental Services, to be financed through PADFS Incentive Scheme IV “*Servicios Ambientales*” (SA, for its acronym in Spanish), as well as through Local PES Mechanisms (ML-PSA) b.) Forest Landscape Restoration of Micro-watersheds and Strategic Areas “*Restauración Forestal de Microcuencas y Regiones Estratégicas*” (RFM, for its acronym in Spanish); c.) Commercial Forest Plantations and Agroforestry Systems “*Plantaciones Forestales Comerciales y Sistemas Agroforestales*” (PFCA, for its acronym in Spanish); and d.) Community Forest Management and Value Chains “*Manejo Forestal Comunitario y Cadenas de Valor*” (MFCCV, for its acronym in Spanish). Please see also an unofficial English Translation of PADFS ROP 2024 in Annex 2.a
- 54 Through ROP 2024 this amount was raised from 40 to 50% (ROP 2024, page 52)
- 55 In addition to specific exclusion criteria for each incentive program or support of the PADFS included in its Rules of Operation, CONAFOR runs an information systems and database with beneficiaries to make sure there is no double dipping from other support programs
- 56 CONAFORs’ Operational Rules of the PADFS (2024) and specifically, its Technical Annex pp. 47ff. (please see also pages 51-66 Annex 2c of this FP) includes a detailed description of prioritization criteria, payment modalities, beneficiaries, amounts, eligibility and exclusion criteria for PES and LM-PES.
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- 59 MAHECHA, L., 2002.- El silvopastoreo: una alternativa de producción que disminuye el impacto ambiental de la ganadería bovina. Rev. colomb. cienc. pecu., 15 (2): 226-231.
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- 70 MAHECHA, L., 2002.- El silvopastoreo: una alternativa de producción que disminuye el impacto ambiental de la ganadería bovina. Rev. colomb. cienc. pecu., 15 (2): 226-231.
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- 78 Secretaría de Hacienda y Crédito Público 2022, Estadísticas Oportunas de Fianzas Públicas. Consulted on 04/01/2022: <http://presto.hacienda.gob.mx/EstoporLayout/estadisticas.jsp>
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- 80 UNAM, 2021. *Atlas of climate vulnerability*. <http://atlasclimatico.unam.mx/VyA/> - 34. Retrieved 16 January 2021.
- 81 Source: <https://www.worldbank.org/en/country/mexico/overview>
- 82 Source: https://databankfiles.worldbank.org/data/download/poverty/33EF03BB-9722-4AE2-ABC7-AA2972D68AFE/GlobaI_POVEQ_MEX.pdf

- ⁸³ In 2008, Mexico adopted an official multidimensional poverty measure that combines income-based monetary poverty with six non-monetary dimensions of wellbeing called social deprivations – education, health, food, social security, quality and space of the dwelling, and basic services in the dwelling.
- ⁸⁴ Source: <https://statistics.cepal.org/portal/cepalstat/national-profile.html?theme=1&country=mex&lang=en>
- ⁸⁵ Méndez-Pérez, M, et al. 2021. Food and nutritional insecurity before and during the COVID-19 pandemic in the households of women belonging to a social program. Food and Nutrition Sciences. Vol. 12 No. 7. <https://www.scirp.org/journal/paperinformation.aspx?paperid=110655>
- ⁸⁶ Source: <https://www.gob.mx/inmujeres/es/articulos/cambio-climatico-y-perspectiva-de-genero?idiom=es>
- ⁸⁷ Diagnosis of rural youth in Mexico, 2015. (Diagnóstico de la juventud rural en México, 2015). CEDRSSAR Report
- ⁸⁸ CONEVAL. 2020. Poverty by population groups at the municipal level 2010, 2015 and 2020. Source: https://www.coneval.org.mx/Medicion/Paginas/Pobreza_grupos_poblacionales_municipal_2010_2020.aspx
- ⁸⁹ FIDA- FAO. 2018. Cerrar las Brechas: Nota de Política Pública para la inclusión de la perspectiva de género e interculturalidad en la agricultura y el Desarrollo rural. <https://www.ifad.org/en/web/knowledge/publication/asset/41195962>
- ⁹⁰ Government of Mexico. 2020. Nationally Determined Contributions: 2020 Update. Available online at: <https://unfccc.int/sites/default/files/NDC/2022-06/NDC-Eng-Dec30.pdf>
- ⁹¹ Axis A: Prevention and attention to the negative impacts on the region and population; Axis B: Resilient production systems and food security; Axis C: Conservation, rehabilitation and the sustainable use of biodiversity and ecosystem services; and Axis D: Integrated water resource management with a climate change approach (Nationally Determined Contributions – NDCs. 2020 Update).
- ⁹² Government of Mexico. 2013. National Climate Change Strategy/Estrategia Nacional de Cambio Climático. Available online at: <https://www.gob.mx/cms/uploads/attachment/file/41978/Estrategia-Nacional-Cambio-Climatico-2013.pdf>
- ⁹³ Secretaría de Medio Ambiente y Recursos Naturales. 2021. Special Program on Climate Change 2021–2024/Programa Especial de Cambio Climático 2021–2024. Available online at: https://www.gob.mx/cms/uploads/attachment/file/685848/SEMARNAT_081121_EV.PDF
- ⁹⁴ Comisión Intersecretarial de Cambio Climático. 2017. National Strategy for Reducing Emissions from Deforestation and Forest Degradation 2017–2030/Estrategia Nacional de Reducción de Emisiones por Deforestación y Degradación Forestal 2017–2030. https://redd.unfccc.int/files/estrategia-nacional-redd_-2017-2030_mexico.pdf
- ⁹⁵ Secretaría de Medio Ambiente y Recursos Naturales. 2020. National Forestry Program 2020–2024/Programa Nacional Forestal 2020–2024. https://www.conafor.gob.mx/transparencia/docs/2021/Programa_Nacional_Forestal_PNF_2020-2024.pdf
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- ⁹⁷ Government of Mexico. 2019. National Development Plan 2019–2024/Plan Nacional de Desarrollo 2019–2024. Available online at: https://www.effectivecooperation.org/system/files/2022-02/10042_0.pdf
- ⁹⁸ The *Parliamentary Gazette/Gaceta Parlamentaria* cites the Balsas-South Pacific Basin Regional Development and Welfare Programme as a priority within the programmes devolving from the 2019–2026 NDP.
- ⁹⁹ RIMISP. 2021. *El papel de las mujeres en la ruralidad mexicana: Retos y futuro de las nuevas generaciones*. <https://www.rimisp.org/noticia/el-papel-de-las-mujeres-en-la-ruralidad-mexicana-retos-y-futuro-de-las-proximas-generaciones/>
- ¹⁰⁰ General illiteracy in Mexico is 3.8% for men and 6.1% for women, but in the state of Guerrero women illiteracy is 13.6% and in Oaxaca 13.3%. Encuesta Intercensal, 2015. Consultado en CNDH. México. <https://desca.cndh.org.mx/indicadores/Educacion>.
- ¹⁰¹ In Mexico women are connected to the internet 2 hours per week less than men, but in the State of Oaxaca, the difference raises to 4 hours a week. INEGI, INMUJERES. *Encuesta Nacional sobre Uso del Tiempo 2019*. Presentación de resultados. Segunda edición.
- ¹⁰² At the national level only 22% of public decision makers are women, but in the project region this figure goes down to 7.4% in Puebla and 2.8% in the State of Guerrero. Instituto Mexicano para la Competitividad. IMCO.2020. *Mujer rural en la economía*.
- ¹⁰³ Annex 22 is mandatory for mitigation and cross-cutting projects.



HACIENDA
SECRETARÍA DE HACIENDA Y CRÉDITO PÚBLICO

Subsecretaría de Hacienda y Crédito Público
Unidad de Asuntos Internacionales de Hacienda
Dirección General

To: The Green Climate Fund ("GCF")

Mexico City, 06 March 2023.

Re: Funding proposal for the GCF by the International Fund for Agricultural Development (IFAD) regarding the Basin Approach for Livelihood Sustainability through Adaptation Strategies (BALSAS) project.

Dear Madam, Sir,

We refer to the project titled Basin Approach for Livelihood Sustainability through Adaptation Strategies (BALSAS) in Mexico as included in the funding proposal submitted by the International Fund for Agricultural Development (IFAD) to us on 24 February 2023.

The undersigned is the duly authorized representative of the Ministry of Finance and Public Credit, the National Designated Authority for the United Mexican States.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- a) The government of the United Mexican States has no-objection to the project as included in the funding proposal;
- b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of the United Mexican States;
- c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

THE DIRECTOR GENERAL

LAURA ELISA AGUIRRE TÉLLEZ
NATIONAL DESIGNATED AUTHORITY FOR MEXICO



Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Basin Approach for Livelihood Sustainability through Adaptation Strategies (BALSAS)
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	International Fund for Agricultural Development (IFAD)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	United Mexican States Balsas Basin (Hydrological-Administrative Region RHA IV) and part of the South Pacific Hydrological-Administrative Region (RHA V) adjacent to it encompassing the states of Guerrero, Jalisco, México, Michoacán, Morelos, Oaxaca, Puebla, and Tlaxcala.
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Friday, May 31, 2024
Language(s) of disclosure	English and Spanish
Explanation on language	Spanish is the official language of Mexico.
Link to disclosure	English: https://www.ifad.org/documents/d/guest/mexico-balsas-emsf_e Spanish: https://www.ifad.org/documents/d/guest/mexico-balsas-emsf_s
Other link(s)	N/A
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the "Environmental and Social Management Framework (ESMF)". The project Indigenous Peoples Plan is included as an annex.
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Friday, May 31, 2024
Language(s) of disclosure	English and Spanish
Explanation on language	Spanish is the official language of Mexico.
Link to disclosure	English: https://www.ifad.org/documents/d/guest/mexico-balsas-emsf_e Spanish: https://www.ifad.org/documents/d/guest/mexico-balsas-emsf_s

Other link(s)	Government of Mexico websites: https://snif.cnf.gob.mx/2023/07/31/balsas/ https://www.conafor.gob.mx/apoyos/docs/externos/2024/Salvaguardas_ambientales.pdf
Remarks	An ESMP consistent with the requirements for a Category B project is contained in the “Environmental and Social Management Framework (ESMF)”.
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity’s website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report/disclosure on accredited entity’s website	Indigenous Peoples Plan / Friday, May 31, 2024
Language(s) of disclosure	English and Spanish
Explanation on language	Spanish is the official language of Mexico.
Link to disclosure	English: https://www.ifad.org/documents/d/guest/mexico-balsas-design_e Spanish: https://www.ifad.org/documents/d/guest/mexico-balsas-design_s
Other link(s)	Government of Mexico websites: https://snif.cnf.gob.mx/2023/07/31/balsas/ https://www.conafor.gob.mx/apoyos/docs/externos/2024/Consulta_pueblos_indigenas_Balsas.pdf
Remarks	Findings of the meaningful consultation at design stage are included as an annex to the Indigenous Peoples Plan.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, May 31, 2024
Place	<u>CONAFOR Central Office</u> Periférico Poniente 5360, Col. San Juan de Ocotán, Zapopan, Jalisco C.P. 45019. Teléfono: (333) 777-7000 Ext: 2617. <u>CONAFOR Puebla</u> C. 5 Sur 5118-int. 9, Residencial Boulevares, 72440 Heroica Puebla de Zaragoza, Pue. Teléfono: 222 240 0581

	<u>CONAFOR Guerrero</u> Insurgentes 25, Benito Juárez, 39010 Benito Juárez, Gro. Teléfono: 747 471 1744
	<u>CONAFOR Oaxaca</u> Priv. de Almendros 106, Reforma, 68050 Oaxaca de Juárez, Oax. Teléfono: 951 518 7210
	<u>CONAFOR Estado de México</u> Km. 53, Carretera México Toluca, Col. Los Patos, C.P. 52000, Lerma de Villada, Estado de México. Teléfono 728 282 1925
	<u>CONAFOR Michoacán</u> Periférico Independencia No. 4193, Fraccionamiento Ana María Gallaga, C.P. 58195, Morelia, Michoacán. Teléfono: (333) 777-7000
	<u>CONAFOR Morelos</u> Río Balsas 31, Colonia Lienzo Charro, 62130 Cuernavaca, Mor. Teléfono: 777 145 5017
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, July 15, 2024

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP236

Proposal name:	Basin Approach for Livelihood Sustainability through Adaptation Strategies (BALSAS)
Accredited entity:	International Fund for Agricultural Development (IFAD)
Country:	Mexico
Project/programme size:	Medium

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
This project introduces a new integrated territorial development approach by using climate change related criteria for selecting and prioritizing the most vulnerable beneficiaries, interventions and sites.	The project has the potential to unlock up to USD 1.5 billion in domestic public finance by aligning federal rural development subsidies with climate priorities. However, achieving the USD 1.5 billion figure depends on the federal government scaling up the new approach to allocation of rural development funds, which lies outside the scope of the project.
The BALSAS project's sustainability will be increased by using and strengthening existing public domestic rural extension and development channels.	
Its potential leverage ratio is 1:40 thanks to the possibility of scaling up the realignment of domestic public finance flows with climate objectives (USD 1.5 billion over 20 years).	
The project focuses on one of Mexico's poorest and most vulnerable regions, the population of which includes Indigenous Peoples.	

II. Summary of the Secretariat's assessment

2. The Board may wish to consider approving this funding proposal in accordance with the term sheet agreed between the Secretariat and the accredited entity, and, if considered appropriate, subject to the conditions set out in Annex II of document GCF/B.39/02.

2.1 Project background

3. The Balsas Basin in Mexico, a region highly vulnerable to climate change, faces exacerbated challenges as one of the country's poorest and most marginalized areas. Climate effects such as increased temperatures, erratic rainfall, and extended dry seasons are reducing agricultural productivity and degrading natural resources. These changes lead to more frequent flooding, crop damage and droughts, further reducing crop yields. Local efforts to combat these issues are hindered by inadequate capacities and institutional frameworks that fail to prioritize climate adaptation in vulnerable areas.
4. The cross-cutting project (78 per cent adaptation, 22 per cent mitigation) aims to increase the climate resilience of the most vulnerable communities and ecosystems, and to reduce emissions from production systems in the BALSAS project area by:
 - (a) Operationalizing the use of climate-change criteria in current institutional modalities for the prioritization of rural development incentives through a community-centric bottom-up approach;
 - (b) Investing in the climate-sensitive management of ecosystems, and climate-resilient productive systems and value chains for adaptation and mitigation benefits; and
 - (c) Supporting a climate information and risk management system, and the systematic management of knowledge and lessons learned.
5. The cost for the proposed project is estimated at USD 90,953,000, of which GCF investment is requested of USD 20,000,000 in loans and USD 19,500,000 in grants.
6. The project is expected to achieve a mitigation outcome of 3,926,876 tonnes of carbon dioxide equivalent (t CO₂ eq) over 20 years and to increase the climate resilience of 3,309,211 people, including 201,867 (6.5 per cent of the of the population of prioritized watersheds and Indigenous territories) as direct beneficiaries.
7. GCF rates the programme as category B (medium) for environmental and social safeguards.

2.2 Component-by-component analysis

8. The project is split into three components leading to the following outcomes:
 - (a) Outcome 1: Government programmes and community development plans integrate climate-change criteria for the allocation of support;
 - (b) Outcome 2: Forest landscapes are sustainably managed and better adapted to climate change for the provision of environmental services and increased livelihood resilience; and
 - (c) Outcome 3: The producers, inhabitants and authorities of the BALSAS project area receive timely and relevant climate information for climate risk management, disaster prevention, and knowledge management.

Component 1: Operationalization of a system that prioritizes sites, beneficiaries and interventions for rural development support in the Balsas Basin using climate-change criteria (Total cost: USD 10,091,726; GCF cost: USD 2,492,891 in loans and USD 6,128,047 in grants)

9. The outcome of component 1 is that government programmes and community development plans integrate climate-change criteria for the allocation of support.
10. The activities of component 1 are:
 - (a) Activity 1.1.1. Facilitate the creation of forest management units and regional committees for the integrated management of the territory;

- (b) Activity 1.1.2. Agree on priority intervention areas at the micro-basin level;
- (c) Activity 1.1.3. Strengthen multisectoral coordination structures, operation rules, and a mechanism for co-investment within the basins;
- (d) Activity 1.1.4. Strengthen and operationalize criteria for adaptation to climate change in institutional programme procedures;
- (e) Activity 1.2.1. Create a communication and awareness campaign to disseminate calls for proposals;
- (f) Activity 1.2.2. Formulate climate-smart multiannual investment plans based on micro-basin zoning plans and validated through free, prior and informed consent;
- (g) Activity 1.3.1. Implement capacity development programmes for extension workers, local facilitators and “brigades”;
- (h) Activity 1.3.2. Support and create forest protection brigades and local climate change resilience facilitators; and
- (i) Activity 1.3.3: Strengthen the *Promotorías forestales* (regional and local offices) of CONAFOR (the National Forestry Commission of Mexico).

Component 2: Climate change prioritization criteria applied to the allocation of rural development incentives for the climate-sensitive management of ecosystems, and climate-resilient systems and value chains (USD 73,496,484; GCF cost: USD 17,507,109 in loans and USD 9,462,362 in grants)

- 11. The outcome of component 2 is that forest landscapes are sustainably managed and better adapted to climate change.
- 12. The activities of component 2 are:
 - (a) Activity 2.1.1. Payment of environmental services;
 - (b) Activity 2.2.1. Climate-smart forest landscape restoration of micro-watersheds and strategic areas;
 - (c) Activity 2.2.2. Commercial forest and agroforestry plantations adapted to climate change;
 - (d) Activity 2.3.1. Sustainable forest management adapted to climate change, inclusion and access to markets;
 - (e) Activity 2.3.2. Support households to improve food security/nutrition; and
 - (f) Activity; 2.3.3. Brigades to strengthen adaptation capacity in agrarian communities (community forest monitoring).

Component 3: Development of a climate information system, and implementation of knowledge management (USD 3,648,721; GCF cost: USD 3,320,721 in grants)

- 13. The outcome of this component is that stakeholders in the Balsas Basin receive the required information for climate risk management and disaster prevention in a timely manner.
- 14. The activities of component 3 are:
 - (a) Activity 3.1.1. Development of a platform and applications to facilitate access to climate information and early warning systems;
 - (b) Activity 3.1.2. Strengthening the CONAFOR prediction and prevention systems for fire and forest phytosanitary risks;
 - (c) Activity 3.1.3. Strengthening the CONAFOR measurement, reporting and verification system;

- (d) Activity 3.1.4. Strengthening the CONAFOR environmental and social safeguard management system;
- (e) Activity 3.2.1. Production of training and systematization materials;
- (f) Activity 3.2.2. Exchange of experiences, and South–South and triangular cooperation; and
- (g) Activity 3.2.3. Monitoring and evaluation.

Project management (total cost: USD 3,716,070; GCF cost: USD 588,870 in grants)

- 15. This activity refers to the project management activities, such as conducting financial management, and procurement.

III. Assessment of performance against investment criteria

- 16. The programme scores well against GCF investment criteria. A detailed assessment of the programme's fit with GCF investment criteria is provided below.

3.1 Impact potential

Scale: Medium to high

- 17. The impact potential is expected to be medium to high, as the number of direct beneficiaries total 201,867 (members of agrarian communities, small producers, and Indigenous groups), including 40 per cent of women. Likewise, a large mitigation impact with 3,926,876 Mt CO₂ eq over 20 years will be achieved from the reduced degradation and therefore enhanced conservation of 324,578 hectares of forests.

- 18. Direct beneficiaries of the project are defined as small producers of agrarian communities from 121 of the most socially and climate vulnerable municipalities in the BALSAS project area, focusing on Indigenous communities (82 of the 121 municipalities). These direct beneficiaries are assessed with a focus on those impacted by land use cover change and value chain investments in component 2, as well as those benefiting from capacity-building and brigades across all components.

- 19. Indirect beneficiaries of the project, totalling 3,107,344 individuals, are defined as members of the households of the direct beneficiaries, as well as other community members from the 68 agrarian communities targeted. Indirect beneficiaries were primarily estimated as a proportion of the inhabitants residing in the prioritized watersheds, where investments in component 2 are scheduled to occur. Beneficiaries of early warning systems (component 3) correspond to the population of high-priority municipalities. However, as these municipalities largely overlap with the priority watersheds, they are not double counted in the total of indirect beneficiaries, reflecting a conservative approach.

- 20. While primarily adaptation focused (78 per cent), the cross-cutting project also includes a mitigation component (22 per cent). The project will contribute to Mexico's 2030 national target of reducing greenhouse gas (GHG) emissions by 22 per cent by implementing afforestation, reforestation and forest, farmland, and pasture management (land use, land-use change and forestry sector); and agricultural soil management activities and programmed soil burns (agriculture sector).

3.2 Paradigm shift potential

Scale: High

- 21. The programme has the potential to be scaled up by enhancing public finance flows aligned with climate objectives and establishing a sustainable environment for institutional coordination, knowledge and data collection, and skill development. It is estimated that the

average of USD 160 million per year in traditional forestry finance provided by CONAFOR – totalling more than USD 1.54 billion (net present value) over the project's 20-year lifespan – will be redirected towards climate change adaptation and mitigation finance to benefit Mexico's most vulnerable communities.

22. The project also has high potential for replication across the country through the operationalization of climate change criteria in the prioritization of rural development incentives and by strengthening of multisectoral coordination structures to re-align other sources of national development finance towards climate finance, as mentioned above.

3.3 Sustainable development potential

Scale: High

23. The programme contributes to key Sustainable Development Goals (SDGs): SDG 13 (Climate action), SDG 1 (No poverty); SDG 2 (Zero hunger); SDG 3 (Good health and well-being); SDG 5 (Gender equality); SDG 6 (Clean water and sanitation); and SDG 15 (Life on land).

24. The major environmental co-benefit is the retention, regulation and release of rainwater flows within the ecosystems and the landscapes as both prolonged drought and more severe and heavy rainfalls pose important hazards for local communities and especially for the most vulnerable rural livelihoods that depend on forest products and agriculture for which a regulated water flow and soil humidity is vital.

25. Another main environmental co-benefit is to promote the conservation of biodiversity and forest resources, resulting in increased availability of habitat for wildlife, the preservation of culturally and aesthetically important areas, and improved ecosystem service supplies, such as disease regulation.

3.4 Needs of the recipient

Scale: Medium to high

26. The population of 15.8 million people living in the BALSAS project area is extremely vulnerable to climate-change impacts due to considerably higher poverty rates and lower social development levels compared to the rest of Mexico. Inequities affecting women's access to services and assets have worsened under climate-change effects. Additionally, about 55 per cent of the population, mainly young people, lack economic opportunities and community decision-making involvement, reducing their resilience to climate change. Moreover, over 9.6 million Indigenous People living in poverty in the municipalities are particularly vulnerable.

27. Though the Government of Mexico continues to invest in the country's forestry sector, there are several obstacles preventing the flow of tailored funding for climate change adaptation and mitigation in the project area, such as not specifically focusing on the most vulnerable communities and ecosystems, limited access to credit for promoting the inclusion of small producers, and lack of inclusion of women and youth in ecosystem management and value chain activities.

28. The project provides financial and technical support to these vulnerable areas, and incorporates community needs through participatory planning.

3.5 Country ownership

Scale: High

29. This project is highlighted as the top priority in Mexico's pipeline by its national designated authority, the Mexican Ministry of Finance and Public Credit (Secretaría de Hacienda y Crédito Público, referred to as HACIENDA), and has been developed in collaboration with multiple institutions and stakeholders under the lead of CONAFOR and the guidance of the

national designated authority. It is listed as one of the 100 promises of the favourite candidate in the upcoming presidential elections.

30. A no-objection letter has been issued by the country.
31. The project will aim to use and strengthen existing rural extension and development channels.

3.6 Efficiency and effectiveness

Scale: High

32. The estimated total cost of the project is USD 91 million, of which USD 73.5 million will be used in investment activities in component 2.
33. The potential leverage ratio is 1:40 through mobilization of USD 1.5 billion over 20 years if scaled up across Mexico.
34. The economic and financial analysis shows that this project is economically viable as a whole. The main sources of quantifiable benefits result from the investments in component 2 and include the increase in productivity, resilience of the productive systems to climate change, and mitigation benefits from GHG emission reductions.
35. The economic results at the aggregate level of the project are estimated based on the number of agrarian communities and producer groups to be covered by the project. This step also includes the benefits related to climate-change mitigation through the impact on GHG emission reductions estimated with the EX-Ante Carbon Balance Tool (EX-ACT) for the different models.
36. With various scenarios of social carbon prices and discount rate, the economic internal rate of return of the project reaches a value between 16.6 per cent and 34.6 per cent and a positive economic net present value between USD 25 million and USD 216 million.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

37. **Project background.** The project aims to increase the climate resilience of the most vulnerable communities and ecosystems and reduce emissions from production systems in the BALSAS project area by enabling community-level investment in adaptation and mitigation initiatives, through the climate-focused transformation of rural development incentives for the conservation of forest resources and the strengthening of local institutions and capacities.
38. **Environmental and social (E&S) risk category and safeguards instrument.** The accredited entity (AE) has assigned an E&S risk category B rating to the project, given that the project will have potential limited adverse environmental and social impacts that are site specific and can easily be mitigated. This also takes account of the fact that the project's activities are focused on the protection, conservation and restoration of natural resources and promote models of sustainable use of resources, such as water, forests and their timber and non-timber products. The project will not be involved in actions and investments for the development of aquaculture or fishing, either on a medium or large scale, in ecologically sensitive areas and will not encourage the conversion of wetlands, changes in hydrology or the introduction of exotic marine species. The Secretariat concurs with the categorization and confirms that it is within the E&S risk category that the AE is accredited for. The environmental and social co-benefits include conservation of biodiversity and improved access by the communities to environmental services. An environmental and social management framework (ESMF) has been prepared, which indicates the processes and guidelines for the preparation,

implementation and supervision of E&S risk instruments designed for the project and its subprojects.

39. **Compliance with GCF's Environmental and Social Safeguards (ESS) Standards.** The following paragraphs describe how the project complies with the standards.

40. **ESS1: Assessment and management of environmental and social risks and impacts.** The environmental and social risks and impacts under the project are expected to be of moderate scale and can be readily managed and mitigated through the deployment of appropriate measures. Given the nature of the interventions, which are mostly related to capacity-building, environmental restoration, and targeted agricultural/agroforestry activities, there are no significant or cumulative impacts expected on biodiversity, natural habitats, land or water contamination, or on labour and working conditions. The ESMF follows the management procedures for identification, mitigation or compensation for any possible adverse environmental and social impacts derived from the execution of the BALSAS project. The ESMF will be used as guidance in examining the risks and impacts of the project and its subprojects when establishing the details of the exact locations and the potentially affected peoples. The ESMF establishes the principles, rules, guidelines and procedures for assessing the E&S risk and impacts of the interventions during implementation.

41. **ESS2: Labour and working conditions.** The activities of the project will mostly be implemented on the ground by the local inhabitants of the *ejidos* and rural and Indigenous communities that are present within the locality. It is expected that the labour of migrants or temporary workers will not be required. Some of the low-risk activities in the forest areas include occupational health and safety risks, such as heat stress and dehydration due to exposure to high temperatures, accidents involving forest machinery and exposure to biological hazards (e.g. allergic reactions to plants, pollen and insect bites and contracting vector-borne diseases). The project will provide occupational health and safety training and promote the use of safety equipment when conducting the activities. The project will ensure that international and national labour regulations, including the AE and GCF requirements, will be applied by those engaged by the project, including its beneficiaries.

42. **ESS3: Resource efficiency and pollution prevention.** The project intends to promote measures to reduce pollution and increase the sustainable use of resources, given that the project is focused on sustainable use of forest and water resources. Nevertheless, the activities could lead to generation of liquid and solid waste as a result of agricultural production and processing activities (e.g. crop residue, animal manure, agrochemical containers, harvesting and processing waste, packaging material) that could pollute land and water resources. Land and water contamination may also occur from the use of fuels and lubricants in machinery or the use of pesticides and/or agrochemicals. To mitigate these impacts, the project will support the implementation of agroecological best practices such as composting, mulching, recycling packaging materials and ensuring that the machinery is properly maintained and serviced. Farmers will also be trained on the safe use and handling of agrochemicals. The project will also promote measures to reduce water use, such as drop irrigation and rainwater harvesting which could enhance infiltration, reduce soil erosion, improve water quality and minimize silting of water bodies.

43. **ESS4: Community health, safety and security.** The AE has conducted consultations at the design stage, and the risk of conflicts among the targeted communities on the use of natural resources during implementation is expected to be minimized through the implementation of the free, prior and informed consent plan and by ensuring that meaningful and continuous consultation and engagement with project stakeholders is implemented. The AE has identified the risk of the presence of organized crime in the project region, which may affect project implementation and/or put project personnel at risk. The project plans to mitigate this by recruiting additional facilitators in the field who will be made aware of security protocols and

provided with proper communications equipment and advanced and intensive field security training.

44. **ESS5: Land acquisition and involuntary resettlement.** The project will exclude any activity requiring land acquisition. The exclusion list also includes “Activities that imply the economic displacement or physical resettlement of people”. The project will also ensure that “there is no risk of potential restriction of traditional access to land/livelihoods due to project activities”.

45. **ESS6: Biodiversity conservation and sustainable management of living natural resources.** The project seeks to improve the provision of ecosystem services, such as water capture, soil conservation and carbon sequestration. While it intends to conduct environmental restoration and promote biodiversity, some activities have the risk of introducing invasive species. As a mitigation measure, the ESMF states that “No invasive species are allowed on CONAFOR supported projects. Unwarranted introduction will lead to the suppression of invasive species individuals”.

46. **ESS7: Indigenous Peoples and GCF Indigenous Peoples Policy.** The project sets a target of 58 per cent of direct beneficiaries being Indigenous Peoples; thus the project has potential benefits including the promotion of Indigenous knowledge systems. Information on Indigenous Peoples and Afro-Mexican communities is mainstreamed throughout the project’s ESMF, focusing on strengthening their territorial governance and participation through the provision of technical support, demand-driven incentive payments and strengthening of interinstitutional coordination. The AE has undertaken meaningful consultations at the design phase, including with Indigenous women and youth. In line with their roles and functions, the Indigenous Peoples Advisory Group is available to provide advice to the AE and executing entities (EEs), and the AE is encouraged to share and disseminate good practices from project implementation.

47. **ESS8: Cultural heritage.** The project is not expected to adversely affect cultural heritage. The ESMF’s exclusion list also states that the project will not finance “Activities that involve directly or indirectly the conversion, losses or major or minor adverse impacts on the physical, cultural, religious, archaeological resources or intrinsic value designated by the local, rural and Indigenous population”.

48. **Sexual exploitation, sexual abuse and sexual harassment (SEAH) safeguarding.** The project is considered by the Secretariat to potentially face high SEAH risks and impacts, given the existing prevalence of gender-based violence (GBV) and social conflicts linked to non-compensatory water extraction practices, water access difficulties and a lack of employment opportunities in protected natural areas, which disproportionately affect rural women, youth and Indigenous Peoples in the Balsas Basin. As the project aims to strengthen local watershed governance and the participation of women, youth and Indigenous Peoples at the sub-basin level, the AE will trigger ESS standards related to labour and working conditions and community health and safety to prevent and mitigate GBV and SEAH-related risks and impacts related to the project. Moreover, the AE will further identify and assess specific GBV and SEAH risks as part of the subproject due diligence and through a social risk assessment to be conducted at the community level. Detailed SEAH risk assessments and relevant mitigation measures will be required as part of the subproject-specific environmental and social management plans. The AE will ensure the safety and security of women’s groups in the target communities, and will find age-, gender- and culturally-appropriate ways to facilitate the participation of women in the project. The AE will also enhance community awareness, capacities and strengths of project beneficiary communities, implementing partners and government staff in preventing and reducing the GBV and SEAH risks, and proactively map the existence of and gaps in services for survivors/affected people. In the case of an SEAH incident, the project will follow the reporting and response protocols already established by the AE that include specific SEAH-related procedures to provide essential services and redress to

survivors/affected people. SEAH cases can be reported through the project's grievance redress mechanisms, and SEAH complaints can also be received through the AE's complaints procedure to be forwarded to the AE's Ethics Office.

49. **Implementation arrangements.** CONAFOR as the EE will have the overall operational responsibility in implementing the project. A project coordination unit will be established, based at CONAFOR's headquarters, which will be responsible for day-to-day project management, coordination, implementation, and monitoring including on environmental and social performance. The unit will coordinate the screening of subprojects, assessment and management of potential risks and impacts, as well as conduct stakeholder engagement, capacity-building and disclosure of information. The AE will also provide training to the relevant staff on safeguards-related topics when the project is launched and will continue to provide support as necessary.

50. **Stakeholder engagement and information disclosure.** In preparation for the project, the AE has conducted consultations and workshops with representatives from national ministries, regional organizations, multisectoral bodies, Indigenous Peoples, as well as local government organizations and the private sector. A stakeholder engagement plan outlines the processes and mechanisms for continued consultations with different stakeholders, including disadvantaged and vulnerable groups, during project implementation. Engagement approaches will include consultations, meetings, workshops, distribution of accessible materials/leaflets on project activities and local radio, among others.

51. **Grievance redress mechanism.** The ESMF describes the existing institutional level grievance mechanism (the Citizen Attention Mechanism) that is in place at CONAFOR for complaints that may be received by the project. It also describes the AE's grievance mechanism including a mechanism for handling SEAH grievances. These will be communicated to the organizations and beneficiaries of the project by the EEs. In line with the GCF Indigenous Peoples Policy, the GCF Indigenous Peoples Focal Point will be available for assistance at any stage, including before a claim has been made.

4.2 Gender policy

52. The AE has provided a gender analysis and action plan, and therefore complies with the GCF Gender Policy. The project is designed to stimulate climate investments that contribute to reducing gender gaps and promote equality through climate-change mitigation and adaptation measures. Based on the findings of the gender analysis, recommendations were formulated into a gender action plan (GAP) to ensure women's participation in project activities and benefits by defining participation targets and capacity-building for gender equality for the different groups of project stakeholders. The GAP is elaborated on the basis of the project's expected results and outputs framework and defines activities, indicators and targets for women's participation along with an estimated budget.

53. The GAP aims to strengthen women's participation in decision-making, promote greater gender equity in institutions, meaningfully include women as beneficiaries of investment plans, build capacity through training on gender and inclusion, and use selection criteria for payment of environmental services projects that favour women's participation. Additionally, the project will build women's capacity to strengthen the systems of the farming communities, be trained in production practices and technologies adapted to climate change and variability, and benefit from the financial transfers made by the project. Climate-resilient sustainable forest management activities will open opportunities for women to participate in value chains, such as those linked to the sustainable use of and value addition to timber and non-timber forest products while also allowing the exchange of experiences and good practices implemented by women in the project.

54. The AE is strongly encouraged to both measure and promote women-led and managed proposals and projects related to integrated land and water resources management. This would further align with the aim of the project to strengthen women's leadership and further minimize the risk of women being marginalized.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

55. This funding proposal is a cross-cutting project (78 per cent adaptation and 22 per cent mitigation) to strengthen the climate resilience of vulnerable communities and reduce emissions in the Balsas Basin in Mexico. To achieve its objectives, the project aims to implement a climate-sensitive ecosystem management system, a climate-resilient productive system, and a climate information and risk management system in the country. GCF is considering a loan of USD 20 million and a grant of USD 19.5 million together with co-financing of USD 51.5 million in loan and grant from IFAD and the Government of Mexico. The co-financing ratio is 1:1.3 and accounts for 57 per cent of total financing.

4.3.2. Accredited entity/executing entity capability to execute the current programme (low risk)

56. IFAD is an international AE with over 40 years of development work in global agriculture and food security. To date, IFAD has provided over USD 23 billion in grants and low-interest rate loans to fund livelihoods in fragile and developing countries. The project's EEs are HACIENDA, NAFIN, and CONAFOR. As the focal point for international climate finance in the country, Ministry of Finance and Public Credit will sign the subsidiary loan agreement with IFAD, and NAFIN will act as the intermediary and fiduciary of the project loans. Responsibility for project execution will rest with CONAFOR, which also manages the country's National Forest Fund. Since its creation in 2001, CONAFOR has developed a track record of working on internationally funded forestry projects and programmes. CONAFOR has worked with IFAD for over 15 years, including on the implementation of loans and grants for forest areas and vulnerable populations.

4.3.3. Project-specific execution risks (medium risk)

57. In addition to the three EEs mentioned above (HACIENDA, NAFIN and CONAFOR), decisions are overseen by a steering committee that reports to political decision makers who approve the working plan and select implementation partners. CONAFOR is one of five steering committee members. Under the steering committee are two layers of technical coordination teams and units which provide technical advice. These are followed by regional committees and local co-implementing partners, and at the bottom are the grass-roots organizations, communities and forestry enterprises. Given the multilayered and cross-institutional arrangement, operational risk and potential delays due to poor coordination is a concern. This risk is partly mitigated by the project's initiative to strengthen intersectoral coordination and regular engagement with stakeholders. The project will also benefit from the support of the Office for Coordination of Advisers and HACIENDA. The funding proposal presents low credit risk given the investment grade BBB rating of Mexico as the debtor. Although a portion of the funding will be denominated in United States dollars to fund local currency generating activities, foreign currency risk is likewise limited because this will be backstopped by the government's investment grade rating. With regard to the risk of local currency devaluation after disbursement affecting the project budget, this will be managed by planning currency conversion as close as possible to expected disbursement.

4.3.4. Project viability and concessionality

58. The proposed project will develop plans to integrate climate change into government programmes to improve capacity and strengthen coordination. Other components will enhance sustainable forest landscapes while enabling timely climate information availability to manage climate risk. All these initiatives are well within the expertise of the AE and EEs given their track record in this space, and the project is expected to be viable. The requested participation from GCF will represent approximately 43 per cent of total programme size in a project funded partly by loans and partly by grants. The level of concessionality of one of the loans appears high in view of an economy as creditworthy and large as that of Mexico. However, it is intended to be blended to make a forest payment of environmental services scheme viable over a 40-year period. The lower concessionality loan is intended for revenue-generating activities that are more economically viable but that would not have been funded otherwise.

4.3.5. Compliance risk (medium risk)

59. Mexico faces a number of challenges in terms of internal conflict and social unrest. This is largely due to the presence of drug cartels and criminal gang violence, including in the targeted region. The proposal by the AE (IFAD) does address this issue and sets out measures to address and mitigate any consequences or impacts. The AE has also created its programme to assure appropriate financial integrity practices, such as preventing money-laundering, terrorist financing and prohibited practices. For any proposed disbursement of funds or commodities to beneficiaries directly, IFAD has established clear guidelines as to eligibility and measures to monitor and prevent abuse. Having considered the context in which the proposal will be operating, and the measures proposed by the AE, a medium rating is applied.

4.3.6. GCF portfolio concentration risk (low risk)

60. In case of approval, the impact of this proposal on the GCF portfolio concentration in terms of results area and single proposal is not material.

Summary risk assessment	
Overall project	Medium
Accredited entity/executing entity capability to implement this project	Low
Project-specific execution	Medium
GCF portfolio concentration	Low
Compliance risk	Medium

4.4 Fiduciary

61. CONAFOR's International Affairs and Financial Promotion Unit in conjunction with the project coordination unit will handle the project's financial management. This management structure adheres to the legal framework governing public administration in Mexico.

62. NAFIN will be the administrator for the project's financing. Responsibilities include following up on audits, managing disbursements and overseeing loan disbursement processes. The GCF funds will be channelled through the AE (IFAD) via NAFIN and the Treasury to CONAFOR.

63. There will be two methods for the disbursements of funds: reimbursement for GCF Loan A and B; and a replenishment mechanism based on advance payments for grants. Specific bank

accounts managed by NAFIN will facilitate these transactions, with funds transferring to appropriate bodies for loans and grants.

64. The flow of funds is to be rigorously monitored using CONAFOR's integrated support system, which ensures controlled disbursement to beneficiaries. In addition, IFAD's client portal will be used by NAFIN and CONAFOR for the submission of disbursement requests for loans and grants.

65. Internal audits are to be conducted by CONAFOR's office of internal control, while external audits are to be performed by private firms selected according to terms agreed with major international financial institutions and approved by IFAD. These audits aim to ensure compliance and evaluate financial management effectiveness.

66. IFAD will implement on-site supervision missions to assess financial management arrangements, identify areas for improvement and fiduciary risks, and propose mitigation measures.

4.5 Results monitoring and reporting

67. As a cross-cutting initiative, this project aims to generate results under GCF mitigation results area 4 and adaptation results areas 1, 2 and 4. It has an implementation period of eight years (six years under GCF funding) and a lifespan of 20 years. During implementation, it is expected to reach 201,867 direct beneficiaries (6.5 per cent of the population of the prioritized watersheds) and 3,107,344 indirect beneficiaries (9.25 per cent of the project area population). The estimated mitigation impact of the project during its lifespan is 3.9 Mt CO₂ eq.

68. The Secretariat worked with the AE over five rounds of review to improve results management under the project. This included refining the project theory of change to make the intended results more specific and measurable. The theory of change includes a description of the co-benefits expected under the project, as well as the underlying risks and barriers.

69. The logical framework has also been refined to align it with the GCF Integrated Results Management Framework and the guidance contained in the *GCF Results Handbook*. The AE has identified suitable paradigm shift and GCF outcome indicators. Moreover, project-level outcome and output indicators have been defined. The Secretariat provided the AE with relatively minor feedback on formatting GCF core and supplementary indicators to ensure smooth reporting and results aggregation at the Secretariat level.

70. The Secretariat also provided the AE with feedback on various reporting milestones listed in annex 5 (Implementation timetable) to bring it in alignment with GCF recommended standards.

4.6 Legal assessment

71. The Accreditation Master Agreement was signed with the Accredited Entity on 24 September 2018 (the "AMA"), and it became effective on 9 November 2018. The Accredited Entity's five-year accreditation term lapsed on 8 November 2023 but was extended by Decision B.37/18, paragraph (q) until the earlier of three years from the date of lapse of the five-year accreditation term or the date on which a revised accreditation framework is adopted by the Board. The Board approved the re-accreditation of the Accredited Entity pursuant to Decision B.37/18, and negotiation of the Amended and Restated Accreditation Master Agreement is ongoing.

72. The Accredited Entity has not provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

73. The proposed project will be implemented in the United Mexican States (“Mexico”), a country in which GCF is not provided with privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of Mexico most recently on 24 January 2023

74. The Heads of the Independent Redress Mechanism (IRM) and Independent Integrity Unit (IIU) have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where the GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by the GCF are made only after the GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities

75. With regard to the activities being financed by the GCF loans, the Accredited Entity has proposed that it will enter into a loan agreement with the Government of Mexico, acting through its Ministry of Finance (“**HACIENDA**”), with HACIENDA subsequently entering into a mandate and implementation agreement as required under Mexican law with Nacional Financiera, S.N.C., Banca de Desarrollo (“**NAFIN**”) and the National Forestry Commission (Comisión Nacional Forestal) (“**CONAFOR**”), assigning NAFIN to act as intermediary on behalf of the Government of Mexico and CONAFOR to act as the entity with overall responsibility for project execution and coordination. Although it has been agreed that HACIENDA, NAFIN and CONAFOR are Executing Entities for the project, under the Accredited Entity’s proposed contractual structure, the Accredited Entity will not have a direct contractual relationship with NAFIN and CONAFOR which are responsible for the channelling of GCF proceeds and project implementation respectively. For the Accredited Entity to pass down and enforce the relevant requirements of the AMA and funded activity agreement (“**FAA**”) on NAFIN and CONAFOR, the Accredited Entity has agreed in the term sheet that it will ensure that HACIENDA passes down, monitors, supervises and enforces such requirements against NAFIN and CONAFOR in a manner that allows the Accredited Entity to comply with the AMA and FAA requirements as if it had a direct contractual relationship with NAFIN and CONAFOR

76. Since the Accredited Entity’s direct counterparty is a sovereign state (as one of the Executing Entities), the Secretariat has agreed to disapply the provisions of the AMA allowing GCF to step in to the Accredited Entity’s contractual arrangements with the Executing Entity, i.e., the Government of Mexico as, due to the sovereign nature of the counterparty and the contracts, it is not feasible for the Accredited Entity to allow GCF or any other party to take over its contractual relationship with the Executing Entities under the subsidiary agreements. Accordingly, and similar to the arrangements for FP143, FP162, FP183, SAP012 and SAP017, as a condition for execution of the FAA, the Accredited Entity is required to inform the National Designated Authority (NDA) of the Host Country of this disapplication and its implications. The Accredited Entity has already done so for the proposed project and an express provision has been included in the term sheet to provide for such disapplication.

77. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Submission by the Accredited Entity to the Fund of a certificate or legal opinion, in form and substance satisfactory to the GCF Secretariat, within 120 days after Board approval confirming that the Accredited Entity has obtained all final internal approvals needed by it and has the capacity and authority to implement the proposed project;
- (b) Signature of the FAA in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval, or the date the Accredited Entity has provided a certificate or legal opinion confirming that it has obtained all final internal approvals, whichever is later; and

- (c) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP236

Proposal name:	Basin Approach for Livelihood Sustainability through Adaptation Strategies (BALSAS)
Accredited entity:	International Fund for Agricultural Development (IFAD)
Country:	Mexico
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1. This is the first submission of the funding proposal “Basin Approach for Livelihood Sustainability through Adaptation Strategies (BALSAS)” submitted to the independent Technical Advisory Panel (iTAP) under the standard proposal approval process. The accredited entity (AE) is the International Fund for Agricultural Development (IFAD). The executing entities (EEs) are the National Forestry Commission of Mexico (CONAFOR), the Nacional Financiera, S.N.C., Banca de Desarrollo (NAFIN) and the Ministry of Finance (HACIENDA).
2. This cross-cutting funding proposal is aimed at increasing the climate resilience of the most vulnerable community and ecosystems, while reducing greenhouse gas (GHG) emission from production systems (agriculture and forestry) in the Balsas Basin in Mexico. It prioritizes eight watersheds and two indigenous territories for a total of 121 municipalities, and responds to the needs of agrarian communities, small producers, women, youth, people with disabilities and indigenous peoples to adapt to climate change.
3. The expected adaptation outcome includes 201,867 direct and 3,107,344 indirect beneficiaries, and the expected mitigation outcome is 3,926,876 tonnes of carbon dioxide equivalent (t CO₂ eq) over the 20-year lifespan of the project.
4. The overall budget for this funding proposal is USD 90.95 million, with a funding request to GCF of USD 39.5 million, divided in USD 19.5 million as a grant and USD 20 million as a loan.
5. In its assessment the iTAP considered the documents submitted on 3 May 2024, the written answers to the iTAP and the further clarifications during the call with the AE on 22 May, as well as minor revisions regarding the mitigation impact potential received on 25 May 2024.

1.1 Impact potential

Scale: High

1.1.1. Adaptation potential

6. There are four ecoregions in the project area: (i) the inter-montane depressions; (ii) the southern Pacific coastal plains and hill landscape; (iii) the temperate mountain ranges of the Sierra Madre del Sur; and (iv) the transverse neovolcanic system.
7. The funding proposal includes an analysis of climate change and variability for the 4 ecoregions, using daily climatological data from 262 weather stations located in the project area. The analysis compares temperature and precipitation data for 1970–1999 as reference period, with data from the period 2000–2018. In order to contextualize the analysis, the observed changes were compared to the regional information provided by the interactive atlas of the Intergovernmental Panel on Climate Change. Future climatic trends were considered for

the period 2014–2060 and using Shared Socioeconomic Pathway (SSP) 4.5 and SSP5–8.5. The funding proposal includes an analysis of how increasing climate hazards will affect the main land uses in each of the four ecoregions, which are agriculture, forestry and grasslands.

8. With regard to the level of exposure and the adaptive capacity of the population, the funding proposal demonstrates the high overall vulnerability of the local stakeholders due to low education level, poverty, gender and intergenerational aspects as well as food insecurity. These factors contribute to a low (climate) adaptation capacity of the different stakeholder groups (indigenous people, women, Afro-Mexican communities, agricultural communities, etc). Existing projects or the expected levels of law enforcement will not change the situation.

9. Considering future climate scenarios, current and future vulnerability of the productive systems and expected evolution of the vulnerability of the sociocultural system, the funding proposal demonstrates that vulnerability to climate change will increase in future if adaptation measures are not implemented soon.

10. As a response to the increasing climate vulnerability, the funding proposal aims at achieving three main outcomes:

- (a) Outcome 1: Government programmes and community development plans integrate climate change criteria for allocation support, which will strengthen intersectoral coordination, improve local capacities and facilitate the preparation of climate-responsive investment plans. Of the total budget 11 per cent has been allocated to this outcome;
- (b) Outcome 2: Forest landscapes are sustainably managed and better adapted to climate change, where climate responsive systems (adaptation and mitigation) will be implemented. This outcome includes the implementation of land-use plans through multiannual payments for environmental services (PES), climate-smart forest landscape, restoration of micro-watersheds, and commercial forestry and agroforestry plantations. This has potential to avoid the deforestation of 4,652 hectares. This outcome will secure all environmental and social services for these forest ecosystem (adaptation impact), while reducing GHG emissions (mitigation impact). Further, the main systems for climate-smart restoration, which have already been designed considering future climate scenarios, will contribute to increase the overall landscape resilience. The agroforestry and forestry systems will be developed in a participative manner during the project implementation and then assessed against future climate scenarios. These agroforestry and forestry systems will also respond to needs of the stakeholders related to food production and production of marketable products. This outcome will allow the use of schemes for payments for ecosystem services that already exist in Mexico to facilitate implementation of the above-mentioned climate responsive systems by agrarian communities. In addition, activities included in this outcome will strengthen the participation of agrarian communities in forest and agroforestry value chains, as well as in monitoring activities. The implementation of these systems will also have a mitigation impact (see below for a detailed assessment). Of the total budget 81 per cent has been allocated to this outcome; and
- (c) Outcome 3: Producers, inhabitants and authorities in the project area have timely and relevant climate information for climate risk management and disaster prevention, where the project will contribute to improve access of local stakeholders to climate information and early warning systems; improve the capacity of the executive entity to prevent forest fires and pests; improve safeguard management; and promote knowledge management and exchange. Of the total budget 8 per cent is allocated to this outcome.

11. Several activities in this project will increase the capacity of local stakeholders in understanding the impacts of climate hazards over time and monitoring the progress achieved through the adaptation measures. This includes the creation of forest protection brigades and

the training of climate resilience facilitators, who will support the bottom up planning and monitoring of field activities. Potentially therefore, the project will have 201,867 direct and 3,107,344 indirect beneficiaries. Moreover, this funding proposal includes a dedicated activity for strengthening the adaptation capacity of the agrarian communities through their participation in community monitoring with an allocated budget of over USD 3 million. This activity and the link to the national reporting system demonstrate well how monitoring progress on climate change adaptation will be conducted in a participative manner and how the results will be integrated into the national reporting to the United Nations Framework Convention on Climate Change.

12. Considering the above, the iTAP assesses the adaptation potential of this funding proposal as high.

1.1.2. Mitigation potential

13. The funding proposal claims a mitigation impact potential of 1,644,348 t CO₂ eq during the eight-year implementation period and 3,926,876 t CO₂ eq over the 20-year lifespan of the project and for activities implemented under outcome 2 (see table 1 below).

Table 1: Mitigation impact potential in the BALSAS funding proposal

Outcome 2			CONAFOR		Total at 8 years	Total at 20 years
	Model	Unit	Total hectares	Ton CO2eq	Ton CO2eq	
Community Forest Management and Value Chains						
Activity 2.3.1 Sustainable Forest Management adapted to climate change, inclusion and access to markets	Sub-activity 2.3.1.1 Manage forests sustainably adapted to climate change	hectare	120,487	-419,137.13	-863,848.00	
Commercial Forest Plantations						
Activity 2.2.2 Forest Plantations Adapted to Climate Change	Sub-activity 2.2.2.1 Establish climate adapted planted forests	hectare	4,353	-404,890.61	-1,012,226.54	
	Sub-activity 2.2.2.2 Establish silvopastures	hectare	1,500	-159,005.93	-397,514.83	
Forest Restoration						
Activity 2.2.1 Restoration of Forest Landscapes adapted to climate change	Sub-activity 2.2.1.1 Establish climate resilient agroforestry systems	hectare	2,000	-93,552.00	-233,880.00	
	Sub-activity 2.2.1.2 Establish agave – forest water / soil retention terraces	hectare	1,000	-209,882.82	-524,707.05	
	Sub-activity 2.2.1.3 Establish forest– pasture water / soil retention terraces	hectare	1,400	-164,384.00	-410,960.00	
Payment for Environmental Services						
Activity 2.1.1 Payment of environmental services	Sub-activity 2.1.1.1 PES projects are approved and financed	hectare	193,838	-193,496.00	-483,740.00	
				-1,644,348.50	-3,926,876.41	

Source: Funding proposal.

14. The EX-Ante Carbon-balance Tool (EX-ACT) was used for estimating the mitigation impact potential, using either tier 1 or tier 2 data, when available, and the funding proposal includes justified explanations of all assumptions used in the quantification. Furthermore, when a sub-activity is to be implemented in two different ecological zones, the quantification of the mitigation impact potential was done for each zone.

15. In addition, the funding proposal will strengthen the Monitoring, Reporting and Verification system (MRV) implemented already by CONAFOR (activity 3.1.3) and will pilot community forest monitoring (activity 2.3.3).
16. Considering the above, the iTAP assesses the mitigation impact potential as high.

1.2 Paradigm shift potential

Scale: Medium to High

17. The BALSAS project can contribute to a paradigm shift towards resilient and sustainable land use in rural Mexico. First, the project will mainstream climate change and adaptation as decisive factors in the existing schemes for PES. Although Mexico has a long tradition in PES, climate vulnerability and adaptation were not systematically considered when preparing or implementing the territorial plans. The competences, selection criteria and mechanisms for institutional coordination that this project will use for mainstreaming adaptation to climate change into PES schemes in the project area can be used in future in other regions of Mexico, facilitating an upscaling effect. However, this potential paradigm shift is limited by two factors: first, the capacity of the Government of Mexico to mobilize funding for future PES in other regions; and second, the fact that the already ongoing PES activities in Mexico will not benefit from these innovations.
18. The replicability of mainstreaming adaptation to climate change in PES schemes in other countries depends on the specific scheme and on the regulatory framework of a given country. As GCF already co-finances other projects that mainstream adaptation to climate change into PES schemes, the iTAP highly recommends pursuing an exchange of knowledge and experiences between these projects, aimed at identifying success factors and common challenges that can be useful for future funding proposals.
19. Two additional features of the project can further contribute to a paradigm shift: the activities on community monitoring for adaptation, and the participation of agrarian communities in value chains related to timber and non-timber forest products from climate-resilient forestry and agroforestry. Community monitoring can reduce the cost of adaptation while increasing local competencies for a fast and appropriate response to climate hazards. Participation in new value chains can increase farmers' income, reduce their overall vulnerability and thus increase the willingness of other farmers to implement climate-resilient land-use options.
20. Considering the above, the iTAP assesses the paradigm shift potential of this funding proposal as medium to high due to the limiting factors explained in paragraph 18.

1.3 Sustainable development potential

Scale: High

21. In addition to its contribution to Sustainable Development Goal (SDG) 13 on climate action, implementation of the BALSAS project will directly contribute to achieve SDG 1: no poverty; SDG 2: zero hunger; SDG 5: gender equality; SDG 10: reduced inequalities; and SDG 15: life on land.
22. The project uses a basket approach aimed at diversifying farmers' income by combining PES with income from climate resilient agricultural systems, timber and non-timber forest products. While output 2.1 operationalizes multi-annual PES to agrarian communities, output 2.2 secures the production of timber and non-timber forest products and output 2.3 promotes the participation of these agrarian communities in the relevant markets and value chains. This diversity of options will have a positive impact on improving income for rural livelihoods.
23. On the environmental side, the project will preserve and increase environmental services in the BALSAS watershed through climate-smart agriculture and resilient forestry and

agroforestry systems, producing a positive impact on biodiversity, soil protection and water regulation services.

24. The project has a socially inclusive approach, with special consideration of the rights of and benefits for women, Indigenous Peoples and youth. Along with this approach, several activities highlight the importance of securing the participation of women, Indigenous Peoples and youth, including: strengthening “promotorías forestales” (a mechanism for promoting forestry activities using a participative approach); creating community brigades; preparing multi-annual investment plans; implementing climate-smart agriculture, resilient plantations or agroforestry systems. All these activities include corresponding monitoring indicators.

25. The project team conducted multiple consultations during the design of the funding proposal in explicit alignment with the GCF Indigenous Peoples Policy adopted by decision B.19/11 (see annexes 7a and 7b). The discussions and conclusions during these consultations were used as input for the project design and serve as the basis for the Stakeholder Engagement Plan (annex 7c). In addition, the funding proposal includes a detailed Gender Assessment and Action Plan (annex 8) aligned with the GCF Updated Gender Policy, adopted by decision B.24/12.

26. Considering the above, the iTAP assesses the sustainable development potential of this proposal as high.

1.4 Needs of the recipient

Scale: High

27. With a population of over 130 million, Mexico is among the 15 largest economies in the world, the second largest in Latin America, and belongs to the Organisation for Economic Co-operation and Development (OECD). According to the World Bank, the expected real gross domestic product growth for 2024 is 2.3 per cent. However, a Gini index of 43.5 (2022) indicates a significant level of inequality.

28. According to the funding proposal, the population in the project area suffers from poverty and exhibits low social development relative to the rest of the country. This is characterized by significant deficiencies in health, food, education, social security, housing, drinking water and basic sanitation, as well as limited income, access to decent employment, and a lack of productive options, causing overall vulnerability. In addition, the project area is suffering from increasing violence and insecurity, which affects both the investment priorities of the government and migration patterns, especially for young people.

29. Gender inequality is present in the project area, where women have less access to education, employment, credit and economic activities as well as lower representation in decision-making bodies, especially at community level. Consequently, women are even more vulnerable to increasing climate hazards. Despite the efforts made by the project team, participation of women in consultations for this funding proposal was far lower than that of men. This makes the gender lens used by the project even more relevant and highlights the importance of conducting monitoring of gender-related aspects during the implementation period, as proposed.

30. Indigenous and traditional peoples in Mexico have the lowest monetary income and access to services, and have high rates of food and nutritional insecurity. Seventy-five per cent of the indigenous people in Mexico are concentrated in the southern and south-eastern states, four of which are included in the project intervention area: Oaxaca, State of Mexico, Puebla and Guerrero. Furthermore, the majority of the Afro-Mexican or Afro-descendant population is concentrated in the states of Guerrero and Oaxaca. The funding proposal acknowledges this social diversity and includes multiple strategies for promoting the participation of Indigenous People and the Afro-Mexican population during the planning and implementation of the project.

31. As highlighted in the funding proposal, there are several aspects that prevent tailored funding for climate change adaptation in the project area: (i) lack of specificity of the current financial support on the most vulnerable communities and ecosystems; (ii) lack of consideration of future climate vulnerability as a criterion to support climate adaptive sustainable forest ecosystem management; (iii) limited access to credit for promoting the inclusion of small producers (especially women and youth who are usually not the owners of land in agrarian communities) in value chains related to timber, non-timber and agroforestry products; (iv) lack of inclusion of women and youth in ecosystem management and value chain activities; and (v) current low-level contribution to the maintenance of the relevant ecosystems, despite the efforts of the Government of Mexico.

32. Considering the above, the iTAP assesses the response to the needs of the recipient described above as high.

1.5 Country ownership

Scale: High

33. The funding proposal includes a no-objection letter signed 2 March 2023 by the Director General of the national designated authority of Mexico, Laura Elisa Aguirre Téllez. It also includes a detailed description of the alignment of the project with existing policies, programmes and projects related to climate change, including the nationally determined contributions (2020 update), the National Climate Change Strategy (2013), the Special Programme on Climate Change 2021–2024, the National REDD+ Strategy for Mexico, the National Forestry Programme 2020–2024, the General Law on Climate Change (2012), and the most recent National Development Plan 2019-2024.

34. IFAD's first accreditation term by GCF lasted from 9 November 2018 to 8 November 2023; the re-accreditation date was 25 October 2023, but the execution date of the accreditation master agreement is still pending at the time of this assessment.

35. The funding proposal mentions three EEs: HACIENDA, NAFIN and CONAFOR.

36. HACIENDA is the political and operational government focal point for international climate finance agencies. It also administers and defines the federal budget and expenditures for implementing public programmes, including those related to climate.

37. NAFIN is an accredited national direct access entity based in Mexico. It will act as the fiduciary of the loans of the project and will channel the funds from IFAD to CONAFOR, as CONAFOR cannot directly receive external loan financing. As such, NAFIN will be an EE, but it will not implement any funded activity.

38. The Secretariat of Environment and Natural Resources (SEMARNAT) has been designated to be formally responsible for implementing the project, and has delegated responsibility for project execution to CONAFOR, a decentralized public entity of SEMARNAT responsible for implementing the country's public policy on forestry and with extensive experience in operating projects with international financing.

39. Other organizations will play an important role during the implementation of the project. The Food and Agriculture Organization of the United Nations (FAO) in Mexico will act as advisory entity; COANGUA, SEMARNAT and the Ministry of Agriculture will act as implementation partners; and the regional committees for integrated management of the territory (RCIMT) will act as multisectoral platforms for regional coordination, transparency and accountability. The funding proposal includes an explanation of the specific roles of each of the above-mentioned organizations (see main document table 4 and figure 13).

40. As mentioned in the sections above, several stakeholder consultations were conducted as input for this funding proposal. Furthermore, the project will be implemented using integrative and participatory mechanisms.

41. Considering the above, the iTAP assesses the country ownership of this funding proposal as high.

1.6 Efficiency and effectiveness

Scale: Medium to High

42. GCF financing for the project is made up of loans worth USD 20 million, which consist of a USD 10 million loan with high concessionality (GCF Loan A) and a USD 10 million loan with low concessionality (GCF Loan B), and a USD 19.5 million grant. The total co-financing is USD 51.5 million, for a co-financing ratio of 1.3. This co-financing is made up of a USD 38.45 million loan from IFAD and two grants from the Government of Mexico, one for USD 7 million through CONAFOR and the second grant of USD 6 million through CONAGUA.

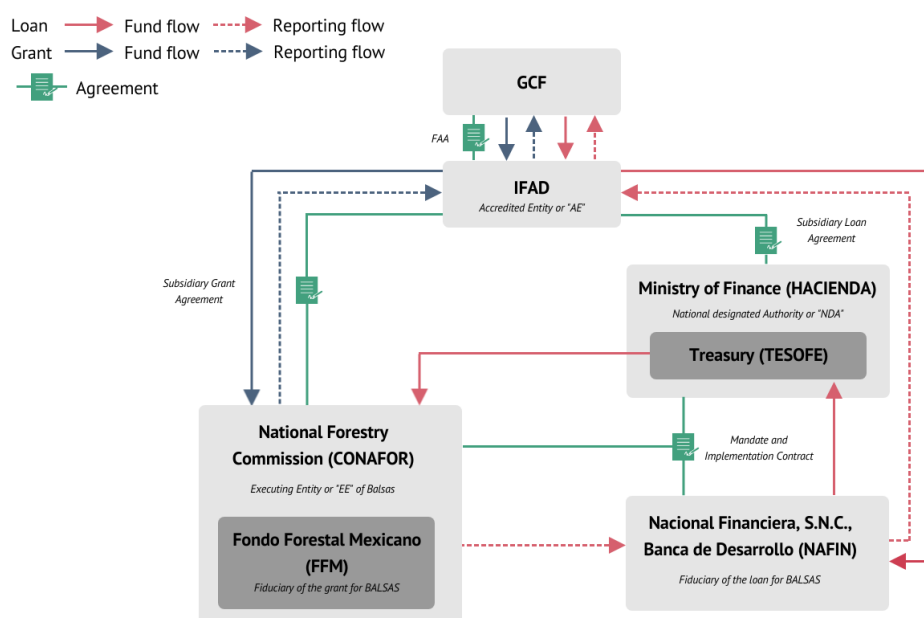
43. IFAD will enter into the following subsidiary agreements to channel the GCF proceeds to the EEs:

(a) HACIENDA for the subsidiary loans. HACIENDA will subsequently enter into a Mandate and Implementation Agreement to assign: (i) NAFIN to act as intermediary on behalf of the Federal Government, which will serve as the executing entity (EE2) and the fiduciary of the project's loans; and (ii) CONAFOR to act as the executing entity for the project (EE3) with overall responsibility for project execution and coordination. By designated authority from HACIENDA, NAFIN will channel funds to CONAFOR upon request. NAFIN will administer the GCF loans but will not execute any specific activities; and

(b) CONAFOR for the GCF grant.

44. Under both subsidiary agreements, CONAFOR will undertake all operational responsibility for implementation of the project in the country (see figure 1).

Figure 1: Schematic diagram of financial flows



Source: Funding proposal.

45. The senior loan (Loan A) from GCF has 40 years tenor with 10 years grace period and will be used for implementing the PES to climate-vulnerable agrarian communities through existing PES schemes (output 2.1). This is a highly concessional mechanism, effectively a zero-

interest loan (with long tenor and grace period for the capital repayment). According to the project proponents, removing barriers to bringing the existing PES schemes to the most vulnerable communities of the Balsas Basin incurs a high upfront cost, necessitating a highly concessional loan. Using a grant would not be considered justified, however, as this would fail to account for future avoided costs by the state as a result of PES investments that successfully prevent deforestation.

46. According to the AE, “using a GCF loan for a PES scheme is a key innovation that fosters improved incentives for scaling up forest establishment and maintenance activities as it brings co-responsibility for natural resources conservation.” The iTAP agrees that this is an innovative approach, and trusts that monitoring of project results will be able to determine the eventual degree of its success in removing one-time barriers to extension of state PES schemes to vulnerable communities and ecosystems, allowing these to be mainstreamed into the schemes in future.

47. The senior loan B from GCF has 20 years tenor with five years of grace period and 0.75 per cent pricing. The senior loan B will be used for activities related to integrating climate change adaptation as a decision criterion in the development plans (output 1.1), strengthening the capacities of local actors (output 1.3), and for strengthening the productive systems of the agrarian communities (output 2.2). The IFAD loan will be focused on improving all productive systems (outputs 2.1, 2.2 and 2.3). The GCF grant and the grants from the Government of Mexico will be distributed for implementing activities in all outputs.

48. Procurement activities will be carried out in accordance with the national procurement system of Mexico. The procurement methods will be those established in the regulations and the maximum thresholds per method will be those established for each fiscal year in the Federal Expenditure Budget.

49. The funding proposal includes a detailed economic and financial analysis of the productive activities included (annexes 3a and 3b). Table 2 summarizes the results.

Table 2: Summary of the results of the economic and financial analysis

Indicator by hectare/project	NPVf@6%	NPVf@10%	B/Cf@6%	B/Cf@10%	Year>@6%	Year>@10%	IRRf
Sub-activity 2.2.1.1 Establish climate resilient agroforestry systems	- 220	- 1,067	1.26	1.20	0	NA	5%
Sub-activity 2.2.1.2 Establish agave – forest water / soil retention terraces	1,487	- 2,000	1.57	1.33	18	NA	7%
Sub-activity 2.2.1.3 Establish forest – pasture water / soil retention terraces	641	75	1.76	1.65	12	18	11%
A. Activity 2.1.1 Payment of environmental services	- 179	- 167	0.22	0.24	0	NA	0%
Sub-activity 2.3.1.1 Manage forests sustainably adapted to climate change	48	20	1.09	1.19	19	22	15%
Sub-activity 2.2.2.1 Establish climate adapted planted forests	10,940	5,488	2.76	2.25	10	11	19%
Sub-activity 2.2.2.2 Establish silvopastures	1,782	379	1.30	1.15	12	18	12%
E. Activity 2.3.2 Supporting households to improve food security and nutrition	- 2,881	- 4,162	1.12	1.08	0	NA	0%



Sub-activity 2.3.1.2 Sustainable use and add value to timber forest products	200,420	101,767	1.21	1.18	7	8	16%
Sub-activity 2.3.1.4 Support women in benefiting from forest based value chains	12,212	3,895	2.09	1.99	5	5	12%

NPVf: Net present value at the activity level, IRRf: Internal rate of return at the activity level; B/Cf: Benefit-Cost ratio at the activity level¹

Source: Funding proposal.

50. Summarizing, the funding proposal provides a clear justification for the mix of financial instruments, and how the principle of minimum concessionality was applied in determining their application to various parts of the project. This includes the use of grants for non-revenue generating activities such as capacity development, and the use of the Loan B instrument for activities, such as plantation forestry, which are expected to see a good financial return on investment. However, after the implementation period of BALSAS, the PES scheme in Mexico will heavily depend on international grants or credits or on the re-alignment of own resources of the Government of Mexico. This creates a risk for the sustainability of the scheme, potentially reducing the long-term effectiveness of this project over its 20-year lifespan.

51. Considering the above, the iTAP assesses the efficiency and effectiveness of this funding proposal as medium to high.

II. Overall remarks from the independent Technical Advisory Panel

52. The iTAP acknowledges the high quality of this funding proposal.

53. The iTAP recommends that the Board approve this funding proposal..

¹ Annex 3a of the funding proposal includes the definition of these indicators as well as the explanations how these were calculated. The calculations are included in annex 3b.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP236)

Proposal name:	Basin Approach for Livelihood Sustainability through Adaptation Strategies (BALSAS)
Accredited entity:	International Fund for Agricultural Development (IFAD)
Country:	Mexico
Project/programme size:	Medium

Impact potential
IFAD takes note of the positive assessment.
Paradigm shift potential
IFAD takes note of positive the assessment and thanks ITAP for providing recommendations to i) further strengthen replicability through knowledge exchange across GCF projects that mainstream adaptation to climate change into PES schemes and ii) to strengthen the paradigm shift through the activities on community monitoring for adaptation, and the participation of agrarian communities in value chains related to timber and non-timber forest products from climate-resilient forestry and agroforestry
Sustainable development potential
IFAD takes note of positive the assessment.
Needs of the recipient
IFAD takes note of positive the assessment.
Country ownership
IFAD takes note of positive the assessment.
Efficiency and effectiveness
IFAD takes note of positive the assessment.

Overall remarks from the independent Technical Advisory Panel:

IFAD thanks the iTAP for its positive review of the proposal.



Mexico

Basin Approach for Livelihood Sustainability through Adaptation Strategies (BALSAS)

Annex 8. Gender Assessment and Action Plan

GCF Additional Financing

January 2023

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Executive summary

The Gender Assessment takes into account GCF key questions for the development of a gender analysis/assessment and, at this stage, is based on information from databases and national documents, mostly produced by the Mexican Government, and studies carried out by international research agencies and institutes.

The Government of Mexico has adopted international instruments on the human rights of women and together with its Member States has legislated extensively and developed institutional programs and initiatives in favor of rural women.

In the Mexican countryside, and specifically in the project area, women are more affected by poverty than men, (55.9% and 54.7% respectively)¹. The situation is even more adverse for indigenous women living in rural areas where 8 out of 10 live in poverty and 4 out of 10 in extreme poverty².

For centuries, stereotypes have been constructed in Mexican culture (as in other contexts) that characterize men as providers of the household, heads of the family and, to a certain extent, decision-makers. When productive work is considered a man's own responsibility, he is in a position to control and manage the economic and technological resources to which the exercise of power, both public and private, is strictly linked. In contrast, women's work within the household is for immediate consumption and is therefore invisible and not valued economically or socially³. Gender stereotypes and *machismo* are determining factors in women's living conditions and their possibilities for economic and social empowerment.

Gender violence, less access to education and information, as well as long working hours, lack of recognition of their role in agriculture, little representation in decision-making positions, limited access to land and services for production in the context of climate change, are gaps that women face. 34.7% of rural women over 15 years of age suffered gender violence in the last 12 months according to the National Survey on the Dynamics of Relationships in Households (ENDIREH 2021)⁴. General illiteracy in Mexico is 3.8% for men and 6.1% for women, but in the state of Guerrero women illiteracy is 13.6% and in Oaxaca 13.3% followed by Michoacán and Puebla is 8.3% in both cases⁵. In Mexico women are connected to the internet 2 hours per week less than men, but in the State of Oaxaca, the difference raises to 4 hours a week⁶.

On average, women work 6.2 hours more than men because of unpaid household work. In the project area the biggest differences are in the states of Guerrero and Oaxaca, where the gap is -9.6 and -9.1 hours respectively⁷. On the other hand, as for the labor force employed in agricultural work, 56% of men and only 29% of women are paid for their work⁸. According to

¹ INEGI. 2019. Women and men in México.

https://www.inegi.org.mx/contenido/productos/prod_serv/contenidos/espanol/bvinegi/productos/nueva_estruc/702825189990.pdf

² IFAD-FAO. 2018. Closing the Gaps: Public Policy Note for Gender and Intercultural Mainstreaming in Agriculture and Rural Development. <https://www.ifad.org/en/web/knowledge/publication/asset/41195962>

³ INAM. S.f. *The impact of stereotypes and gender roles in Mexico*.

⁴ ENDIREH 2021. *Principales Resultados*.

https://www.inegi.org.mx/contenidos/programas/endireh/2021/doc/endireh2021_presentacion_ejecutiva.pdf

⁵ Encuesta Intercensal, 2015. Consultado en CNDH. México. <https://desca.cndh.org.mx/indicadores/Educacion>

⁶ INEGI, INMUJERES. *Encuesta Nacional sobre Uso del Tiempo 2019*. Presentación de resultados. Segunda edición.

⁷ ENIGH. 2019. Encuesta Nacional de Ingresos y Gastos de los Hogares.

⁸ National Agrarian Survey 2017 Cited in Cubillas, A. sf. Women's rights in the Agrarian Law.

data from the National Agrarian Registry (RAN), women represent only 26% of landowners in the project area.

Women hold much fewer⁹ decision-making positions in *ejidos* and communities. At the national level only 22% of public decision makers are women, but in the project area, women account for only 6.3 percent of *ejido* commissariat presidencies. This figure goes down to 7.4% in Puebla and 2.8% in the State of Guerrero.

Only one third of women in agriculture are paid for their work and they have less access to credit and technical assistance. This occurs despite the reform of the Agrarian Law in 2016 incorporates an additional paragraph in article 37, which refers to "*Candidates for elected posts that make up the ejidal commissariat and the oversight council must be made up of no more than sixty percent of candidates of the same sex and may aspire to any of the posts indistinctly. For the commissions and auxiliary secretaries of the ejidal commissariat, the integration of women will be sought*"¹⁰.

In addition to the above, women are recognized to be more vulnerable to the effects of climate change than men. Household activities – for which women are largely responsible – are becoming more burdensome due to rising temperatures and scarcity (or excess) of water, increasing challenges related to washing, cleaning and cooking. The increasing occurrence of pathogenic diseases and the limited availability of food for the family also add to women's psychological and work burdens. *Traspatio* subsistence production and economic activities such as harvesting of non-timber forest products, which are also often carried out by women, are affected by the loss of vegetation. More women heads of households oversee production due to the labour migration of men, but the lack of recognition of their labour and land ownership, as well as the usual inequity in access to services and goods for production and resilience to climate change, affect women's performance and increase their vulnerability.

⁹ Mexican Institute for Competitiveness. IMCO.2020. Rural woman in the economy.

¹⁰ Chamber of Deputies of the Congress of the Union. 2016. *DECREE adding a second paragraph to Article 37 of the Agrarian Law*. Mexico.

https://www.diputados.gob.mx/sedia/biblio/prog_leg/Prog_leg_LXIII/108_DOE_19dic16.pdf

Introduction

1. Mexico has legislated extensively in favor of women and has been vigilant in its compliance through institutional programs and initiatives at the federal and member state levels. It has adopted international instruments on women's human rights, among which the United Nations Convention on the Elimination of All Forms of Discrimination against Women (CEDAW), ratified in 1981, and the Inter-American Convention to Prevent, Punish and Eradicate Violence against Women (better known as Belém do Pará), ratified in 1998 – both in force in accordance with the provisions of Constitutional Article 133. At the national level, there is a broad legislative framework that demonstrates an environment conducive to equality between men and women. From the Political Constitution of the Mexican States, The Official Mexican Standard (NOM-046-SSA"-2005) on Family, sexual and violence against women. the entry into force in 2008 of the General Law on Women's Access to a Life Free of Violence, the General Law to Prevent, Punish and Eradicate Crimes related to Human Trafficking and for the Protection and Assistance to Victims of these Crimes (2012) and the General Law to Prevent and Eliminate Discrimination (2013) ¹¹.

2. The National Development Plan 2019-2024 has the slogan “leave no one behind, leave no one out”, advocating substantive equality between men and women. The Plan refers to the government's work as a promoter of “equality as a guiding principle: effective equality of rights between women and men, between indigenous and mestizos, between young people and adults, and will commit to the eradication of discriminatory practices that have “perpetuated the oppression of entire population sectors” ¹². Within the framework of this Plan, the National Program for Equality between Women and Men (PROIGUALDAD) has been built for the period 2020-2024¹³. This is the special program of the Government that contains the main strategies and actions that 49 entities and 20 agencies of the Federal Public Administration must implement with the support of 3 Constitutional Bodies and 2 Powers of the Union to guarantee the rights of women and reduce violence against them, by addressing six strategic objectives: economic autonomy, health and well-being, care, non-violence, equal participation of women in all areas and the construction of safe and peaceful environments. Among the institutions responsible for the execution of this Plan is CONAFOR, which will contribute to the promotion of actions that facilitate access to credits, financing, subsidies or international cooperation for women producers in situations of marginalization or vulnerable conditions and to strengthen business coordination. of MSME production chains headed by women.

3. The Gender Assessment aims to consider the context and provide an overview of the gender gaps and problems faced by rural women in Mexico, as well as identify opportunities to close these gaps. This evaluation will be used to prepare the Project “BALSAS: Basin Approach for the Sustainability of Livelihoods through Adaptation Strategies”.

4. The Gender Assessment aims at addressing gender inequalities and identify opportunities provided by project activities to bridge the gender gaps. In addition to the Gender Diagnosis, a Gender Action Plan will provide detailed of measures which will ensure

¹¹ Women's Secretariat. Government of Mexico City. <https://www.semujeres.cdmx.gob.mx/conoce-tus-derechos>.

¹² Secretary of Government. Mexico. Official Journal of the Federation. National Development Plan 2019-2024.

¹³ National Women's Institute. INMUJERES. Government of Mexico. National Program for Equality between Men and Women 2020-2024. PROEQUALITY.

the participation of women in project activities, monitor progress, and evaluate the transformations achieved in favour of women.

5. Due to the context of the COVID-19 pandemic, it is pending to deepen and update the assessment and the GAP based on direct consultation with the beneficiaries.

Part I: Gender Assessment

6. The Gender Assessment takes into consideration GCF key questions for the development of a gender analysis/assessment and is at this stage based on a literature review of national and international sources.

7. The information collected and analysed is organised according to the following aspects: i) Poverty situation analysis, ii) vital statistics, iii) legal status of women, iv) political participation and decision-making, v) education, vi) division of labor, vii) employment status, viii) beliefs, perceptions and common stereotypes related to gender, ix) access to and control of resources and assets, x) access to information and communication technologies and xi) gender inequalities and climate change.

Poverty situation

8. In 2018, poverty and extreme poverty affected 41.9% (52.4 million) and 7.4% (9.3 million) of the total population, respectively. In rural areas, where a quarter of Mexicans live, poverty affected 55.3% (about 17 million people)¹⁴. Although poverty rates have not changed significantly in the last decade¹⁵, extreme rural poverty has decreased by almost 11 percentage points between 2008 and 2018 (27.1% and 16.4% respectively). Despite this reduction, it still represents one third of the total number of people living in poverty in rural areas¹⁶.

9. Economic growth has not been inclusive in its benefits, and reducing disparities remains a key challenge for the country's development policy. 88% of women living in rural areas of the country are highly marginalized¹⁷. In the Mexican countryside, women are more affected by poverty than men, (55.9% and 54.7% respectively)¹⁸. Indigenous peoples and the Afro-Mexican population have historically had the highest poverty rates and continue to be a highly vulnerable population due to their unequal situation. According to CONEVAL, in 2018, nearly 70% of the indigenous population, equal to more than 8 million people, experienced a situation of economic poverty and 28%, 3.4 million people suffered from a situation of extreme poverty¹⁹. The situation is even more adverse for indigenous women living in rural areas where 8 out of 10 live in poverty and 4 out of 10 in extreme poverty²⁰.

¹⁴ National Council for the Evaluation of Social Policy. CONEVAL. Rural Poverty in Mexico.

¹⁵ National Council for the Evaluation of Social Policy. CONEVAL. Findings of the Degree of Social Gap at urban AGEB level 2020

¹⁶ National Council for the Evaluation of Social Policy. CONEVAL. Rural Poverty in Mexico.

¹⁷ IFAD. MEXICO. Social Economy: Territory and Inclusion Project Detailed Design Report 2017.

¹⁸ INEGI. 2019. Women and men in México.

https://www.inegi.org.mx/contenido/productos/prod_serv/contenidos/espanol/bvinegi/productos/nueva_estruc/702825189990.pdf

¹⁹ Poverty measurement 2018. Población según pertenencia étnica, CONEVAL, Mexico, 2018.

https://www.coneval.org.mx/Evaluacion/PublishingImages/Pobreza_2018/POBLACION_PERTENENCIA_ETINICA.jpg

²⁰ IFAD-FAO. 2018. Closing the Gaps: Public Policy Note for Gender and Intercultural Mainstreaming in Agriculture and Rural Development. <https://www.ifad.org/en/web/knowledge/publication/asset/41195962>

Vital statistics

10. *Maternal mortality*: Since the start of the pandemic to date, more than 790 pregnant women have died from covid-19 in the country, the second worst figure in the Americas, only behind Brazil. Despite an intensive vaccination campaign, the coronavirus was the leading cause of maternal death in 2021. Other causes were directly related to uncontrolled conditions during pregnancy due to poor availability of health care. Edema, hypertensive disorders and obstetric hemorrhage were among the most frequent causes of death in this population group in 2021. Prior to the pandemic, in 2019, this rate stood at 31.1 per 100,000 births; during the coronavirus crisis, this rate increased to 53.1 per 100,000 births²¹.

11. *Infant mortality*: Third Government Report of the Federal Executive estimates infant mortality for the year 2021 at 14.4, a figure similar to those of 2013 and higher than those recorded between 2014 and 2020²².

12. *Life expectancy*: For 2019 the average life expectancy of the population was 75 years, being higher for women than for men. Women live an average of 78 years while men live an average of 72 years²³.

13. *Child marriage*: According to the United Nations Population Fund and the 2009 National Survey of Demographic Dynamics (ENADID 2009), Mexico's child marriage rate is the fifth highest in Latin America and the Caribbean; 22.9% of women aged 20-24 were married or entered into an informal union before the age of 18. This has not changed in almost 30 years. In Chiapas and Guerrero, the child marriage rate reaches the 'critical' level of 30%, as well as among the rural population in 14 states²⁴. The rate of child marriage among women who have only completed primary school ranges from 33.0% to 77.6%²⁵.

Legal status of women

14. In 1974, the Congress of the Union approved the reform of Article 4 of the Constitution, which establishes legal equality between men and women. There are also two main laws that protect the integrity of women: i) General Law for Equality between Women and Men, published in 2006, which aims to regulate and guarantee equality between women and men and to propose institutional guidelines and mechanisms to guide the nation towards the fulfilment of substantive equality in the public and private spheres, promoting the empowerment of women. Its provisions are of public order and social interest and of general observance throughout the national territory²⁶. ii) The General Law on Women's Access to a Life Free of Violence, issued in 2007, whose purpose is to establish coordination between the Federation, the federal entities, the territorial districts of Mexico City and the municipalities to prevent, punish and eradicate violence against women, adolescents and girls, as well as

²¹ EL PAÍS. Mexico. 2022. Maternal deaths in Mexico register the worst figures in two decades due to the coronavirus. <https://elpais.com/mexico/2022-02-10/la-muerte-materna-en-mexico-registra-las-peores-cifras-en-dos-decadas-debido-al-coronavirus.html>

²² Social Mexico. The Social Question in Mexico. Children and women: painful setbacks. <https://www.mexicosocial.org/ninas-ninos-y-mujeres-dolorosos-retrocesos/>

²³ INEGI. Tell me about Mexico. Population. <https://cuentame.inegi.org.mx/poblacion/esperanza.aspx?tema=P>. Accessed 11 July 2022

²⁴ Bonfil, P. UNESCO. 2020. Gender and Education in Mexico and Guatemala. Background paper prepared for the GEM 2020 Report. Citing Pérez Amador and Hernández 2015.

²⁵ Bonfil, P. UNESCO. 2020. Gender and Education in Mexico and Guatemala. Background paper prepared for the GEM 2020 Report. Citation to INSAID 2017

²⁶ Chamber of Deputies of the H. Congress of the Union. 2006. General Law for Equality between Women and Men. Mexico. <https://www.diputados.gob.mx/LeyesBiblio/pdf/LGIMH.pdf>

to establish the principles and mechanisms for full access to a life free of violence, and to guarantee the free exercise of their human rights and strengthen the democratic regime established in the Political Constitution of the United Mexican States²⁷.

15. In the agrarian sector, the General Law of Agrarian Reform was enacted in 1971 and recognized the possibility for women to hold agrarian rights on an equal footing with men. Article 200 of this law establishes that "Mexicans by birth, men or women over 16 years of age or of any age if they have dependents" have the right to land. Article 45 empowers women to participate in *ejido* assemblies, and Article 78 guarantees that women do not lose their agrarian rights by marrying another *ejidatario*. The Constitutional Reform of 1992 repealed the Federal Agrarian Reform Law and changed its name to the current Agrarian Law. Article 15 of this Law states that in order to become an *ejidatario*, it is mandatory to be a Mexican citizen of legal age or of any age if he or she has a dependent family or is the heir of an *ejidatario*. However, no explicit reference are made to gender and recognizes that when speaking of *ejidatarios*, there is no distinction between men and women²⁸. In 2017, a decree was approved reforming Article 71 and adding an eighth paragraph to Article 108 of the Agrarian Law, establishing the creation and granting of legal status to the Women's Industrial Agricultural Units (UAIM), for the purpose of carrying out and coordinating productive activities, mutual assistance, resource use, commercialisation or other activities not prohibited by law, which they develop in their agrarian nucleus²⁹.

Political participation and decision-making

16. According to data from January 2021, women occupy 50% of national parliamentary seats, but only 26% of the total number of state secretaries. In the case of state cabinets their percentage of participation is very variable, reaching more than 50% in the project area in the states of Morelos and Puebla (54% and 53% respectively) while in Guerrero and Jalisco the percentages are considerably low (5% and 16%)³⁰.

17. At the local level, the country reports 45% of women in the deliberative bodies, which ranks 11th out of 133 countries³¹ in the world.

18. In the case of the Agrarian Communities, in 2016 the Agrarian Law was reformed by incorporating an additional paragraph in article 37, in the section corresponding to the *ejido* bodies, which refers to "Candidates for elected posts that make up the *ejidal* commissariat and the oversight council must be made up of no more than sixty percent of candidates of the same sex, and may aspire to any of the posts indistinctly. For the commissions and auxiliary secretaries of the *ejidal* commissariat, the integration of women will be sought"³².

²⁷ Chamber of Deputies of the H. Congress of the Union. 2007. Ley General de Acceso de las Mujeres a una Vida Libre de Violencia (General Law on Women's Access to a Life Free of Violence). Mexico.

https://www.gob.mx/cms/uploads/attachment/file/209278/Ley_General_de_Acceso_de_las_Mujeres_a_una_Vida_Libre_de_Violencia.pdf

²⁸ Women in the territory. S.f. Government of Mexico. Mujeres por el acceso a la tierra. Aproximaciones a los retos que enfrentan en el ejercicio pleno de sus derechos agrarios.

²⁹ Chamber of Deputies of the Congress of the Union. 2017. DECREE reforming Article 71 and adding an eighth paragraph to Article 108 of the Agrarian Law. Mexico.

https://www.diputados.gob.mx/sedia/biblio/prog_leg/Prog_leg_LXIII/134_DOI_27mar17.pdf

³⁰ More Democracy. 2021. With information from state government websites.

³¹ Berevoescu, I. and Ballington, J. UN Women. 2021. "Women's representation in local government: a global analysis". Working paper. 2018 data for Mexico

³² Chamber of Deputies of the Congress of the Union. 2016. DECREE adding a second paragraph to Article 37 of the Agrarian Law. Mexico.

https://www.diputados.gob.mx/sedia/biblio/prog_leg/Prog_leg_LXIII/108_DOI_19dic16.pdf

This means that the minimum number of women representatives will be 40%. However, out of 14,531 *ejidos* and communities with representative bodies, only 22% have women representatives and only 7.4% were chaired by a woman, according to figures from the National Agrarian Registry 2019. In the project area, it is the state of Puebla that reports the highest percentage of women in *ejidal* positions (7.4%) while Guerrero is where they are least represented (2.8%)³³.

19. Despite the existence of legal spaces for greater participation of women in decision-making bodies, there are socio-cultural barriers that affect the recognition and support for women's ability to hold decision-making positions, which have traditionally been held by men, and women themselves sometimes question their ability to do so. The unequal distribution of work within the family, which overburdens women with long working hours, affects their availability of time to take on an additional job in addition to those they already hold. On the other hand, in the case of *ejidos*, there is the lack of land ownership.

20. Considering the existing legal protection for greater participation of women in decision-making bodies at all levels national, federal entities, localities and *ejidos*, as well as the existence of successful experiences of women's participation, it is possible to promote greater women's leadership. It will be necessary to address the limitations identified in relation to gender patterns and stereotypes that cast doubt on women's leadership capacity and frighten men about the possibility of ceding power, the strengthening of women's leadership capacities and access to opportunities for greater availability of information, especially through digital media.

Education

21. In rural areas of Mexico, school non-attendance is higher (17.1%) than in urban areas (13.5%)³⁴. Currently in Mexico, basic education coverage is close to 100%, mainly in primary education (from six to twelve years of age). Although the expansion of coverage has been an important educational achievement, there are several difficulties beyond the increase in school services. This has resulted in situations of vulnerability and social injustice, especially those belonging to rural and indigenous communities. In addition to the various shortages and difficulties faced by these communities (such as lack of basic services, few employment opportunities, high levels of marginalisation and poverty, lack of communication routes, among others), the complexity of the educational system established there must also be taken into consideration as it results in incomplete school organisation, decontextualised curricular proposals, insufficient and irrelevant teaching materials, etc. Given that these communities are often located in places that are difficult to access and where the population is small, it is considered unnecessary to assign more than one teacher per school, which is why there is a significant number of schools where there is lack of full teaching staff. This has led to the establishment of multi-grade schools, i.e. schools where one teacher attends students from different grades in the same classroom³⁵.

22. In addition, there are obstacles for children to stay in school: for example, domestic and family care work, early unions, and family and child labour are barriers to girls' access to and permanence in school, especially indigenous girls. External factors such as remoteness

³³ Mexican Institute for Competitiveness. IMCO.2020. *Rural women in the economy*.

³⁴ INEE. *Compulsory Education in Mexico. Report 2019*. Calculations based on the Intercensal Survey 2015 (INEGI, 2015a).

³⁵ Rojas, Enrique. 2018. *Conditions of rural education in Mexico, Findings from a multigrade school*. Revista Chakiñan de Ciencias Sociales y Humanidades, núm. 5, pp. 40-53. National University of Chimborazo. Ecuador.

from educational centres, especially at secondary level³⁶ should be mentioned. In the case of adults, *illiteracy* at the national level reaches 5.5% of the population, with women being the most affected (3.8% men and 6.1% women). In the states of the project area, the total rises to 13.6% in Guerrero and 13.3% in Oaxaca, followed by Michoacán and Puebla (8.3% in both cases). Three states are below the national average for female illiteracy: Jalisco, Michoacán and Morelos (5.5%, 5.6%, 6.1% respectively)³⁷. The illiteracy rate among indigenous women and men aged 15-29 reached 4.8% and 3.5% respectively, decreasing in the youth population³⁸. Rural women have 3.4 fewer years of schooling than urban women (7% vs. 10.5 respectively)³⁹.

Division of labour between women and men

23. The National Household Income and Expenditure Survey ENIGH 2019 shows that while women dedicate 67% of their total weekly working time to unpaid domestic work, men dedicate a similar percentage (69%) to market work. **On average, women work 6.2 hours more than men and, in the states where the project area is located, the greatest differences are found in the states of Guerrero and Oaxaca, where the gap is -9.6 and -9.1 respectively. while the state with the smallest gap is Morelos (4.3).** As a consequence, women choose to participate less in paid productive activities, participate less in management committees or take an active part in assemblies or other time-consuming decision-making bodies due to lack of time. This situation particularly affects women heads of household, who have the greatest responsibilities and must cover all the needs of their families, especially when the children are minors. 35% of all Mexican households are headed by women and these households receive three quarters of the income that households with a male head receive⁴⁰, aggravating their situation.

24. The functions of rural women include productive, communal and care work, representing a fundamental pillar of family survival. They fulfil their triple role generally in conditions of very high marginalization, which places them in one of the weakest links in the chain to strengthen and rescue the Mexican countryside. The gap between paid and unpaid work is one of the gender gaps that must be addressed if women are to be effectively incorporated in development in equal conditions, as active agents of development⁴¹.

25. On the other hand, it is important to highlight the relationship of young women with work and study. Of the young people living in rural areas of Mexico (4 out of 10), 25.2% of young women and 7.6% of young men neither study nor work, and the proportion engaged in agriculture has been decreasing. For women, the majority are engaged in household chores, which is associated with early marriage and childbearing. In addition to dropping out of school, young women relegated to the home have experienced reduction of opportunities for ~~per~~ exchange and access to public spaces in their communities.

³⁶ Bonfil, P. UNESCO. 2020. *GENDER AND EDUCATION IN MEXICO AND GUATEMALA. Indigenous girls and school inclusion in Mexico and Guatemala*. Background paper prepared for the GEM 2020 Latin America and the Caribbean Report.

³⁷ Intercensal Survey, 2015. Retrieved from CNDH. Mexico. <https://desca.cndh.org.mx/indicadores/Educacion>.

³⁸ Bonfil, P. UNESCO. 2020. *GENDER AND EDUCATION IN MEXICO AND GUATEMALA. Indigenous girls and school inclusion in Mexico and Guatemala*. Background paper prepared for the GEM 2020 Latin America and the Caribbean Report.

³⁹ INMUJERES. 2019. *Inequality in Figures. Rural women*. Year 5, Bulletin No. 11, November 2019.

⁴⁰ Instituto Mexicano de la Competitividad A.C. IMCO. How much we earn and what we spend on. July, 2023

⁴¹ FAO, IFAD. 2019. Closing the Gaps. Public Policy Note for Gender and Intercultural Mainstreaming in Agriculture and Rural Development.

Employment status

26. *Women's labour force participation* in Mexico was only 45% in 2019, compared to 77% for men, a gap of 32 percentage points. This is below the OECD and Latin American average⁴². Although there has been an increase in women's labour force participation in the economically active population in rural areas, it remains below the national average.

27. According to information from the 2019 National Occupation and Employment Survey, 35.6% of rural women aged 15-64 have a paid job, a lower percentage than urban women (50.2%). In addition, among employed women living in rural localities, 75.5% have no benefits, while this percentage is significantly lower in urban areas (49.6%).

28. Of employed rural women, 24.5% work in commerce, 23.4% in services, 20.1% in manufacturing industry and 13.3% in agriculture, livestock, forestry, hunting or fishing. Among women employed in agricultural and fishing activities, only 3.1% are exclusively engaged in paid work, while 96.9% are engaged in other unpaid activities⁴³.

29. The three main unpaid activities performed by women working in agricultural and fishing activities are household chores, caring for people and fetching firewood and water for their household⁴⁴.

30. Women's *unemployment* as a percentage of the female labor force in 2021 was 4.2% while for men it was 4.5%⁴⁵. As for the labor force employed in agricultural work, 56% of men and only 29% of women are paid for their work⁴⁶.

31. *Informal economy*: Mexico has the highest level of informal employment among the largest economies in Latin America with 60% according to the OECD, which translates into a population with low wages and low social security coverage, non-compliance with labor rights and no taxes. Women working in the informal economy are more vulnerable to living in poverty, yet their work contributes more than 9 million pesos daily to the country's economy, according to figures from the Laboratory for Analysis in Commerce, Economics and Business (LACEN). According to the World Bank, one of the effects of the pandemic has been that many unemployed women opted to enter the labor market as traders even though it is informal employment: "Informal employment, underemployment and unemployment are associated with lower levels of labor income and therefore a greater likelihood that the women and as well as their families will be poor or vulnerable"⁴⁷.

Common beliefs, perceptions and stereotypes related to gender

32. For centuries, stereotypes have been constructed in Mexican culture (as in other contexts) that characterize men as providers of the household, heads of the family and, to a certain extent, decision-makers. When productive work is considered a man's own responsibility, he is in a position to control and manage the economic and technological resources to which the exercise of power, both public and private, is strictly linked. In contrast, women's work in the home is for immediate consumption and is therefore invisible and not

⁴² World Bank Group. 2020. *Women's labour participation in Mexico*.

⁴³ INMUJERES. 2019. *Inequality in Figures. Rural women*. Year 5, Bulletin No. 11, November 2019.

⁴⁴ INMUJERES. 2019. *Inequality in Figures. Rural women*. Year 5, Bulletin No. 11, November 2019.

⁴⁵ ILO in World Bank database. <https://datos.bancomundial.org/indicador/SL.UEM.TOTL.MA.ZS?locations=MX>

⁴⁶ National Agrarian Survey 2017 Cited in Cubillas, A. sf. Women's rights in the Agrarian Law.

⁴⁷ *Forbes Women. 2021. Informal' women contribute 9 million pesos a day to GDP... and are more likely to be poor.*

valued economically or socially⁴⁸. These differences not only in what each person does, but also in the value socially assigned to these tasks, mark a difference that affects women's performance in the public sphere where men act and restrict them to the home and family private sphere.

33. *The* concept that men are in charge and make the decisions makes women "asking for permission" to act, which is often, denied, especially when it comes to authorizing women's participation in productive and recreation activities outside the family. This behavior also affects their possibilities to participate in assemblies or other decision-making activities.

34. These stereotypes, which are also present in institutional officials, limit women's access to goods and services that are not related to their culturally assigned roles. Thus, for example, services related to care and health are oriented towards women while those related to production and the market are directed towards men.

35. To overcome these limitations, CONAFOR has implemented Integral Productive Projects for women that incorporate affirmative actions by allocating up to 20% of the resources assigned to the project to pay tasks associated to caring.

Gender Violence

36. In 2022, 27 of Mexico's 32 states experienced increases in their rates of sexual violence and 23 experienced increases in their rates of family violence. According to the National Survey on the Dynamics of Relationships in Homes (ENDIREH) 2021, sexual violence constitutes around two-thirds of the violence that women experience in public spaces, and around two-thirds of these acts are committed by strangers. In 2022, the country reached a new record for the number of emergency calls reporting incidents of sexual violence, with 6,977 calls. This equates to a 13.1% increase from 2021 and is almost double the number in 2017. In 2022, 968 cases of femicides were reported, a 127% increase from 2015. Currently, approximately one in four murders of women in Mexico is classified as feminicide⁴⁹. Women in rural areas generally have fewer resources to report and get help, due to the lack of institutional presence and the permissiveness of sexist stereotypes rooted in rural communities. "Violence against women is an extreme manifestation of gender inequality which violates their human rights and undermines their physical and mental health and their ability to be productive members of the community⁵⁰.

Access to and control of resources and assets

37. Forestry and agricultural production activities play a fundamental role in the livelihood of the rural population in the project area. Indigenous *ejidos* and communities cover 65% (11.7 million ha) of the Basin area; however, *ejidatarios* and communal farmers generally own small plots (5 ha or less) with low productivity, mainly due to lack of access to means of production, technical assistance and financing. According to data from the National Agrarian Registry (RAN), of the 4.9 million people who own agrarian nuclei throughout the country, more than 3.6 million are men and only 1.3 million are women (26%). Of the federal entities in the project area, Guerrero has the highest percentage of women owners (33%) and Jalisco has the lowest percentage (24.7%)⁵¹. Their productive activities are concentrated in backyard

⁴⁸ INAM. S.f. *The impact of stereotypes and gender roles in Mexico*.

⁴⁹ IEP. Peace Index. Mexico 2023. *The increase in gender violence*.

Article. <https://www.indicedepazmexico.org/el-aumento-en-la-violencia-de-gnero>

⁵⁰ IFAD. *It is time to end violence against rural women*. November 2022

⁵¹ Mexican Institute for Competitiveness. IMCO.2020. *Rural women in the economy*.

production from which they obtain food for self-consumption, and collection and processing of non-timber forest products mainly from community areas.

38. Lack of land ownership is a constraint to accessing credit and other formal finance products that require land as collateral. This reality pushes women to use other types of informal financing such as loans from family, friends or acquaintances. According to data from the ENIGH 2019, 65% of women use this type of credit, which is generally for smaller amounts to solve emergencies.

39. Producers in general have limited access to technical assistance for production and less access to those practices related to climate change adaptation and ecosystem restoration. In this context, women face greater constraints than men due to the low recognition and value placed on their role as producers. In general, services are oriented towards men, even in those cases where women led production due to male migration or female-headed households.

40. Women's access to existing initiatives in the National Forestry Commission (CONAFOR) stands out, with women representing 27% of the beneficiaries of the Programme for Business Strengthening in Forest Landscapes (PROFOEM) and 35.3% of the beneficiaries of the Support Programme for Sustainable Forestry Development (PADFS) in 2021.

Access to information and communication technologies

41. In the country, women spend an average of 15.1 hours per week on mass media while men spend 16.1 hours. This difference is increased in communities with less than 100,000 inhabitants where the average is 12.8 hours for women and 14.0 hours for men. In most of the states of the project area, the average time spent on mass media is less than the national average, with the exception of Jalisco (15.7 hours) and Michoacán (16.6 hours). Oaxaca stands out as the state where the population spends the least amount of time with an average of 11.2 hours, probably related to the greater presence of the indigenous population⁵² and the difficulty of access to communication technologies either due to lack of Internet connectivity, lack of electricity or difficulties in purchasing equipment such as mobilephones and computers.

Gender inequalities and climate change

42. In general, the effects of climate change on the population are diverse, and serious issues that are already occurring may intensify due to climate change, as for example: water shortages; increase in diseases due to the transmission of pathogens such as dengue fever, diarrhea, asthma and others, internal migration and population growth concentrated in a small number of urban centers that are already facing environmental stress.

43. The project area is highly vulnerable to the effects of climate change due to its specific biogeographical conditions; the way natural resources are managed; social vulnerability existing inequalities, particularly gender inequalities. It is among the two main regions of the country with the highest climate vulnerability index. The impacts of climate change, including rising temperatures and heavy rains, aggravate the situation and increase the vulnerability of rural families.

⁵² INEGI, INMUJERES. *National Time Use Survey 2019*. Presentation of results. Second edition.

44. The reduction of productive resources and domestic supply, caused by climate change and the destruction of ecosystems, exacerbate the conditions of poverty and the unsatisfaction of families' basic needs. As a consequence, their members adopt various strategies where migration, redistribution of responsibilities and tasks on the farm and in the household, result in an increase in the number of female heads of household and an increase in the time and burden of their working hours.

45. There is sufficient evidence that the collection and use of water and fuel for washing, cleaning and cooking increases with the adverse effects of climate change. In rural areas, the main source of energy is fuelwood, and the effects of deforestation and desertification mean that it is sought further away from the household. In localities of less than 2,500 inhabitants, the percentage of households using firewood or charcoal is 49.2%, only 76.2% have piped water to the house⁵³. The effects of climate change mean that people have to travel longer distances to fetch firewood and water, requiring not only a greater investment of time, but also greater physical effort. In addition, the lack of drinking water has consequences for people's health, causing illnesses that in turn increase women's household tasks referring care work.

46. In terms of their productive role, women's reduced access to institutional resources to address the effects of climate change on their farms leads to a progressive deterioration or abandonment of the activity, with consequent effects on their well-being and on the well-being of their families.

47. In many rural and indigenous communities, when men migrate to the United States, women are the ones who work the land, but they are not recognized as having rights to it, even if the owners do not return. As they do not own the land, women cannot receive support from equipment or infrastructure programs, nor can they receive credit or economic support for payment of environmental services⁵⁴.

48. In the Balsas Basin, the usual inequity faced by women in accessing different services and goods has been aggravated by the effects of climate change. On the one hand, activities traditionally assigned to subsistence production in the backyard, housework and family care become more tiresome due to rising temperatures, water scarcity and the absence of public drinking water services. On the other hand, economic activities, such as the extraction and management of non-forest timber products (palm, aromatic and medicinal plants, fruits, etc.), which are also often carried out by women, are affected by the loss of vegetation due to droughts, fires or other extreme weather phenomena, such as hurricanes and torrential rains. Disasters, such as landslides and floods, affect houses and cause losses in fruit production, farm animals and household goods, situations that have to be tackled mainly by women in charge of their families. Gender gaps expressed in women's limited access to resources, restricted rights and lack of voice in decision-making, are the conditions that make them most vulnerable to climate change⁵⁵.

Main findings for gender equality in the project

49. This assessment shows gender **gaps** that limit women's participation in the project. It highlights that women are more affected than men by poverty in rural areas and that existing

⁵³ Government of the Republic. Mexico. *PROIGUALDAD*. National Programme for Equal Opportunities and Non-Discrimination against Women 2013-2018.

⁵⁴ Mexican Institute for Competitiveness. IMCO.2020. Rural women in the economy.

⁵⁵ UNAM. 2016. *Climate change, gender perspectives*.

gender stereotypes result in long working hours and make their participation in productive and environmental activities invisible. It also reveals the increase in gender violence against women, their higher levels of illiteracy, fewer links with the mass media, hold few decision-making positions in the *ejidos* and communities, only a third of women engaged in agricultural work are paid for their work, and have less access to land, credit and technical assistance than men. It is also evident that women are more vulnerable to climate change due to the roles they play, for example: there are more women headed households due to migration, more effort and time are required to carry out care work. It is also important to note that the usual inequity women face in accessing different services and goods for production has been aggravated by the effects of climate change,

50. On the other hand, it identifies **opportunities** from which the project can benefit. These include the existence of national laws and policies aimed at the full entitlement of women's human rights, gender equality and a life free of violence. In the agrarian sector, laws have been formulated recognizing women's right to land, to be elected to decision-making positions in *ejido* commissions and oversight councils, and to develop productive activities through their own organizations.

51. Although women hold barely a quarter of all land ownership certificates, there are mechanisms such as the Women's Industrial Agricultural Units (UAIM), CONAFOR's Common Interest Participatory Groups that provide opportunities for women to access land collectively through their own organisations, and the PES on collective lands that allows women to participate and benefit even if they are not direct *ejidatarias*.

52. As part of the programmatic framework of the forestry sector, the National Forestry Programme 2019-2024 details the commitment to the implementation of a strategy for the inclusion of indigenous peoples, youth and the gender perspective in the instruments and actions aimed at the forestry sector to favour their effective participation. This strategy is being implemented and shows significant progress in the adoption of measures that favour the participation of rural women in this sector, among them: progress has been made in the incorporation of the gender perspective in the ROPs of the PADFS, in the priority criteria for phytosanitary treatments, rural fire management brigades, general criteria for forest sanitation brigades, as well as the implementation of the concept of comprehensive productive projects for women's groups that include activities for the generation of added value to forest products as well as include resources to provide support care tasks.

53. There are also experiences of women's participation as beneficiaries of CONAFOR's Payment for Environmental Services Programme (PES), which show that women's land tenure is not necessarily a limitation for their participation as PES beneficiaries. Male chauvinist prejudices in the community, lack of information about the existence of community projects and the interest of some *ejido* leaders (*comisariado*) in benefiting only a sector made up mostly of family members and acquaintances, are factors that limit women's participation⁵⁶. This research conducted in the state of Guerrero shows that PES projects on collective lands in agrarian communities and *ejidos* are an opportunity for women to actively participate and benefit even though they are in the minority as *ejidatarias*. For example, the *Ejido* of San Vicente de Benitez in the state of Guerrero benefited from PES for 5 years starting in 2012, and despite having only 160 *ejidatarios*, 992 people (men and women) were

⁵⁶ Olea Reséndiz, María de Jesús. 2017. *Impact of payment for environmental services on women from three cultures in the state of Guerrero*. Thesis presented as a partial requirement for the Master's degree in Agricultural Sciences and Local Management. Autonomous University of Guerrero. Iguala, Gro. Mexico.
http://ri.uagro.mx/bitstream/handle/uagro/213/FP15152199_maestria.pdf?sequence=1&isAllowed=y

defined as beneficiaries, which was the total population of the five localities that make up the *ejido*. As a result, the participation of women was 89% and the number of daily wages contributed by citizens (*ejidatarios* or not) participating in the project, 2012-2015 was 1294 in total, of which 51% were from women and 49% from men. In two other *ejidos* that were part of the research, women participants were only 5.6% in Yolojóchitl and 16.7% in Copanatoyac, mainly due to the aforementioned factors related to machismo, limited access to information and the interests of the authorities to benefit specific groups in the communities and *ejidos*.

54. The openness towards women's participation in the *Ejido* of San Vicente de Benitez is not the most common behaviour in the *ejidos*, but the exception. However, it shows that even if women do not have their own land, they can be beneficiaries of a PES programme if the necessary social conditions are created.

Part II: Gender Action Plan (GAP)

55. The BALSAS project is fully committed to social inclusion, women's empowerment and the transformation towards a balanced relationship between men and women. Through its proposals for action, it is aligned to a mainstreamed transformative gender approach so that it is integrated throughout the project cycle and permeates its entire structure. It seeks to ensure that men and women have access to all the services offered and in particular to the resources and benefits aimed at increasing resilience to climate change and ~~reducing~~ the vulnerability of ecosystems in the Balsas River Basin. Mainstreaming therefore means that this GAP is not an add-on to the project but rather an integral part of its structure, as well as a responsibility of all actors in charge of its design, implementation, monitoring and evaluation.

56. Its preparation required i) an analysis of current national laws and policies, commitments made and signed at the international level related to gender by the Government of Mexico, as well as institutional policies and strategies, especially those of CONAFOR as project executing entity, ii) a review of the commitments of international financial agencies (International Fund for Agricultural Development (IFAD) and the Green Climate Fund (GCF)) expressed in their policies, strategies and actions to ensure gender equality in their projects, iii) a review of the progress in the design of the project "BALSAS: Basin Approach for Livelihood Sustainability through Adaptation Strategies", especially its Logical Framework and iv) the Gender Assessment contained in the first part of this document, which provides an overview of the gender gaps and issues women face in Mexico, based on a literature review of national and international sources. This assessment, however, faced shortcomings in the availability of quantitative and qualitative information on gender gaps and the situation of women, especially indigenous women, in the specific area of intervention of the project, an aspect that will have to be completed with the results of the respective consultations.

57. The Action Plan contains three sections: the first one refers to the identification of mechanisms and activities considered necessary to eliminate or reduce the gender gaps identified during the diagnosis; the second part contains the description of the specific activities addressed by the project for the achievement of the expected impact and results in terms of climate resilience, improvement of productive ecosystems and rural development, considering the quantitative and qualitative improvement of women's participation and gender equality; the third part presents a matrix according to the requirements of the Green Climate Fund that fits the logical framework of the project containing the activities by expected result and output, indicators and targets, the timeline of their implementation and responsibilities.

2.1. Mechanisms and activities identified to address gender gaps in the

BALSAS Project

58. Rural women are affected by an overload of work in the household and by stereotypical beliefs that see their lives as reduced to their supposed reproductive responsibilities, with little participation and decision-making power over the family's productive and economic resources. Additionally, their low levels of education and literacy, their lesser access to land, their limited access to paid work and information are obstacles to the recognition of their managerial, productive and economic roles and capacities. For this reason, in order to ensure their participation, mechanisms and activities have been defined in a way that allows the project as a whole and across its components to respond to the principle of gender equality mainstreaming. These mechanisms refer to the definition of percentage or participation goals, adapting to women's time availability, guaranteeing access to information, valuing and building on their knowledge and experience, capacity building including technical and soft skills, gender training for technical teams and communities, working with women role models, and existence of specialized gender human resources in the project.

- 2.1.1. Define quotas for women's participation as part of the target population of the project and the implementing teams.

59. Gender **quotas (minimum percentages for women participation)** have been an incentive to encourage the inclusion of women and to achieve equal opportunities for men and women. In Mexico, the allocation of quotas in various fields is already a practice. Their application in the project strengthens the participation of women in productive forestry and community activities. Due to the diversity of women's roles, they are agents widely interested in building resilience to the effects of climate change and the restoration of productive ecosystems, but unfortunately gender patterns and stereotypes tend to make their contribution and potential invisible. In the case of technical teams, the role of women in science, technology adoption and knowledge transfer in the climate field, ecosystem restoration and forestry activities is widely known.

Activities:

- ☐ It is defined that the participation of women as direct beneficiaries of the Balsas project will be 40%. This percentage is distributed within the different components and activities promoted by the project, considering the existing and potential coverage in CONAFOR and the possibilities offered by the BALSAS project. The specific targets by result and activity are detailed below.
- ☐ An increase in women's participation in all decision-making bodies of the project and its beneficiaries will be promoted. In this case, the project considers the current figures of women's participation in the representative bodies in *ejidos* and communities and will seek to increase these figures towards what is proposed by article 37 of the Agrarian Law regarding the existence of no more than sixty percent of representatives of the same gender in these bodies. Their role in the elaboration of multi-annual investment plans within basins, sub-basins and prioritized ethnic areas will be relevant.
- ☐ The limitations that women face in terms of land tenure will also be considered in those activities that require their availability for agroforestry and silvopastoral projects, seeking the participation of those who do have land titles or who can have access collectively through organizations.

- The project will use targeting criteria and methods that are inclusive of the different existing population groups (direct targeting and self-targeting) to ensure the identification and promotion of women's participation particularly women heads of household, and those in conditions of poverty and vulnerability. The proposal is based on using the priority criteria established in the Rules of Operation (ROP) of the Support Programme for Sustainable Forestry Development 2022 (PADFS) and adding direct targeting criteria to support the inclusion of groups in conditions of vulnerability.
- CONAFOR will seek to expand the participation of women in the technical teams, mainly in the *promotorias* supported by the project.

2.1.2. Facilitate women's availability of time to participate in project activities.

60. Both men and women are expected to actively participate in the project activities as both are affected by climate change in the performance of their productive initiatives for food and market purposes. However, given that women face longer work hours due to their domestic role, it will be sought that they can fully participate in the processes and activities of the project, exercising their rights, without this meaning a greater overload of tasks. With this objective in mind, the following is proposed:

- Assure socialization of the **domestic and care tasks within the family, as well as** the community organization by raising awareness of the inequality in the distribution of roles between men and women, and the importance of women having time for project activities and thus benefiting from project related services. To this end, the project's capacity-building spaces will be used, and dissemination materials will be developed.
- The **consideration of places and times suitable to the availability** of women for the development of project activities, so that distances, safety, mobility and the times are adapted to their needs. This way women can fully participate and their multiple tasks are not an obstacle to their benefiting from project services
- Identify and create partnerships with existing initiatives in institutions, communities and organizations that contribute to **alleviating the multiple domestic burdens** (childcare, fetching firewood and water and other tasks) to target or intensify their services for the households in the project area.
- Strengthen CONAFOR's initiative on integral productive projects for women that finance **affirmative actions** related to care work.

2.1.3. Ensuring access to information on participation opportunities and benefits offered by the project

61. Due to the existing constraints, the project will analyze the means used by people to get information. On one hand, it will strengthen the options of digital connectivity, and on the other hand, it will combine digital information with other mechanisms that are commonly used in the communities, such as radio.

62. A common problem is that the contents and means of dissemination of the products

and services offered do not reach women, limiting their demand. In view of this, the BALSAS project will take into consideration the media and forms of communication most used by women, as well as the design of contents that motivate and stimulate their active participation.

63. On the other hand, the digital divide, which mainly affects women, will be considered in the framework of climate information systems and options will be sought to bring women closer to digital technology and the use of complementary means of dissemination so that they can receive the required information. The activities will be as follows:

- A study on **women's** use of media (by age, ethnicity and place of residence) including information on the availability and use of internet and communication devices (radio, TV, smartphones, computers, etc.). This study could be included as part of the project baseline.
- Define a **communication strategy**, with messages that inform and promote women's interest in the services offered by the project, taking into account their access to the media, language and educational level.
- Create **digital education** opportunities **for women**, primarily leveraging smartphones to enable them to access climate information systems.

2.1.4. Valuing and drawing on women's and men's specific knowledge and experience to contribute to project outcomes and solutions.

64. The project focuses its activities and results on groups in conditions of greater vulnerability due to their situation of poverty and affectation due to climate change, but also recognises these groups for their constant struggle to face the difficulties it causes and for being holders of experiences and knowledge that have been effective and deserve to be taken up again as part of the solutions for mitigation and adaptation to climate change that the project proposes. Both men and women are in a position to make significant contributions to the search for solutions that combine practical knowledge with the science-based knowledge of technicians and professionals. In this respect, the project will carry out the following activities:

- **Systematisation of experiences** on climate-resilient and adapted agronomic techniques and technologies designed and implemented by women (mainly indigenous women), to be incorporated into project proposals and used in the framework of training for the different actors.
- Consider this **knowledge and experience as part of** adaptation, mitigation and restoration **strategies** and in the development of training materials for use by the teams and the beneficiary population.
- Create participatory spaces for the **exchange of knowledge and experiences** where women play a leading role.

2.1.5. Create a didactic offer aimed at strengthening women's capacities for empowerment and the development of successful climate-resilient productive initiatives

65. The use of the services and benefits provided by the project through its components

depends to a large extent on the capacities of women to face the challenges involved in leading and managing agro-climatic initiatives, managing the production and marketing of products in value chains and the appropriate use of information, among other issues. For this reason, the following activities are considered:

- Develop and execute training based on the didactic principles of the active methodology called **APA (*Aprendo, Practico y Aplico*)** which promotes the interaction of the participant with the learning process, so that, through guidance, he/she is able to become more autonomously involved in the training process. It favours the appropriation of knowledge through the construction and discussion of ideas individually or collectively, develops communicative abilities, develops skills for social coexistence and gives contextual value to their learning. The following topics would be addressed:
 - Gender focus
 - Leadership and management of organisations
 - Climate-resilient production management and marketing
 - Climate-smart value chains: concept, functioning, participation, benefits and challenges
 - Approach to digital technology (use of smartphones)

Some of these topics are already included in RO_22's catalogue of capacity building topics while others will need to be incorporated.

2.1.6. Gender training for technical teams, project contractors and communities

66. Gender stereotypes are generalised opinions or prejudices that are harmful when they - limit the capacity of men and women to develop their personal abilities, - impede the social recognition of their capacities, - affect their leadership possibilities, and - limit their access to goods and services that are not in accordance with their socially established image. The project will address these stereotypes through the following activities aimed at technical teams, contractors and communities:

- Through workshops, the gender capacities of the project's teams of professionals, technicians and contractors will be strengthened, according to their roles.

It will address

- i) gender stereotypes and their effect on the lives of men and women,
 - ii) Analysis on gender violence: prevention and care protocols
 - iii) the relevance of a gender equality approach to climate change mitigation and adaptation and ecosystem restoration,
 - iv) existing international commitments, regional and national policies and strategies, which are mandatory for the implementation of restoration actions with this approach,
 - v) experiences and lessons learned in climate change adaptation and mitigation and ecosystem restoration with women's participation,
 - (v) analysis of the commitments to the theme contained in the project design.
- How to address gender equality in the framework of the assigned activities.
 - Training will also be extended to communities, producers and producers through

mass-use educational materials such as posters, banners, and activities with audio-visual and easy-to-understand content that encourage everyone to participate as agents of change and take action against negative stereotypes and behaviours.

2.1.7. Prevention and channeling of complaints about violence against women

67. The increase in gender violence in the country is a concern of the federal government and of the states and municipalities in the project area, as well as of IFAD and GCF in all their actions. In response to this problem, the project will act:

- Coordination with the offices (secretaries, institutes, etc.) dedicated to the care of women in state and local (municipal) governments to identify and/or support the development of protocols with the mechanisms and route of actions for reporting. . and care for victims of gender violence.
- As part of the educational material produced for the training mentioned in 2.1.6, pamphlets will be published with the protocols for reporting and caring for victims of gender violence. These materials will also be distributed during the various events held by the project to achieve broad coverage.
- CONAFOR, through its Citizen Attention Mechanism (MAC), will act on internal complaints with the support of its Internal Control Body. This mechanism will act as the project's grievance mechanism.

2.1.8. Gender specialist staff for the implementation of the GAP and gender mainstreaming in the project.

68. Implementing strategies and actions that ensure gender mainstreaming in the project requires the participation of professionals specialised in gender. The support and advice of CONAFOR's specialised gender team is essential, as well as the strengthening of alliances with other public entities oriented towards gender equality such as the National Institute for Women (INMUJERES), the Ministry of Environment and Natural Resources (SERMANAT), institutions providing services related to health, education and care, as well as private sector companies and Non-Governmental Organisations (NGOs) that contribute to gender equality in the project area.

69. Considering also the dimensions of the project in terms of geographical coverage and beneficiary population, it will be necessary to have **human resources** assigned to provide continuous advice to the technical teams, implement concrete actions required for mainstreaming and provide continuous monitoring to ensure fulfilment of this GAP.

2.2. Women's participation in climate resilience, productive ecosystem enhancement and rural development activities

70. The BALSAS project expects to increase climate resilience, improvement of productive ecosystems and rural community development with the participation of women from the communities involved, as part of the direct and indirect actors as well as beneficiaries. Within the framework of its expected results and specific outputs, activities are identified, indicators are defined and compliance targets are established for monitoring and evaluation. Activities are incorporated to eliminate or reduce gender gaps that guarantee a quantitative and qualitative improvement of women's participation in the activities proposed by the project for the achievement of the expected results and outputs. Among these, access

to information and knowledge, capacity building, time use considerations, are considered fundamental for the active and transformative participation of women.

2.3. Activities by result

Outcome 1. Government programs and community development plans integrate climatechange criteria for attribution of supports

71. Output 1.1. Mechanisms for intersectoral coordination are strengthened to increase the technical and organizational capacities of small producers and facilitate synergies with the different levels of government and the private initiative. Within the framework of this output, women will be invited to participate in the basin, sub-basin and indigenous community committees. Their contribution will be relevant as bearers of knowledge and experiences that will feed back into the discussions and decisions regarding the achievement of agreements on the project's priority areas at the micro-basin level.

- Convene and strengthen the participation of women in decision-making in the RCIMT for the prioritization of intervention areas, as well as in the development of participatory multi-year investment plans. Based on women-oriented call and by opening spaces for their participation, it is expected that 40% of the members of these committees will be women.
- Considering the efforts carried out by the sector's institutions in relation to gender equality and the adjustments they have been making to adapt laws, policies, strategies and regulations for greater equity in participation between men and women, the project will support the adjustment of operational procedures in at least three programmes, including criteria for adaptation to climate change and positive actions for women. An analysis of the progress achieved to date and the identification of gaps in operational procedures that can be addressed under the project will be necessary.

72. Output 1.2: Participatory multiannual investment plans within prioritized basin, sub-basins and ethnic areas integrating mitigation instruments and adaptation to climate change.

- Women will be invited to participate in decision-making in relation to multi-annual investment plans. Consultation spaces will be created so that your needs and opportunities to present projects are considered, as well as to collect necessary information to guarantee your active participation in the project.
- The implementation of communication and awareness-raising campaigns to disseminate calls for projects will contain inclusive and gender-sensitive language and will take into account media and content (informative, motivational and empathetic messages) in accordance with the profiles of the target population.
- Considering the educational levels and the limited previous experience of women in the elaboration of project profiles, a simplified format will be designed to be easy to fill in and support will be provided by CONAFOR teams when necessary.
- With these measures, it is expected that at least 30% of people directly benefiting from the investment plans will be women.

73. Output 1.3: Strengthened capacities of local actors for climate change adaptation, resilience, and inclusiveness

- The project will strengthen CONAFOR's current capacity building programme by incorporating complementary themes to the existing ones. These include gender equality and the inclusion of women as well as life skills. In order to address these issues, experts will be involved in designing the training curriculum and the necessary support materials taking into account the contents proposed in section 2.1.6 of the chapter on Mechanisms and activities identified to address gender gaps in the BALSAS project.

This training will be aimed at extension workers, local facilitators and brigadistas, among whom women, especially young women, are expected to participate. The presence of women in these activities, besides opening a space of opportunity for them, is a strategic measure to facilitate project approaching to the female target population.

- The project will also strengthen CONAFOR's local promoters through training, equipment and materials for new promoters. Among the people recruited to carry out this work, at least one promoter (woman) will be hired per sub-basin. As in the case of facilitators and brigadistas, the attention of female beneficiaries will be favored by the strengthening of capacities on gender related issues for all promoters.

Outcome 2: Forest landscapes are sustainably managed and better adapted to climate change for the provision of environmental services and increase livelihood resilience.

74. Output 2.1 Agrarian communities conserve, restore and sustainably use forest landscape based multiannual investment plans through payment of environmental services (PES).

- Considering the experiences of women's participation as beneficiaries of PES, even without land titles, the project will promote changes in the limiting factors identified in 2017 (i.e. machismo in the community, women's lack of access to information, selection of beneficiaries for political and camaraderie purposes), to ensure their involvement in the projects within communities and *ejidos*. The project will inform communal and *ejido* authorities about the programme's focus on benefiting women, calls for participation in PES will use messages that appeal to both men and women, and project selection criteria that favor women's participation will be used.

75. Output 2.2. The productive systems of Agrarian Communities are strengthened and better adapted to climate change and variability

- In the framework of this product, women will be trained in production practices and technologies adapted to climate change and variability. This training will provide specific solutions to the specific needs of women according to their current and potential roles in the production systems in which they participate. The implementation of training activities will be defined according to women's schedules and, as far as possible, the selected locations should consider short journeys, taking into account the difficulties of commuting that they have.
- The agrarian communities will identify and incorporate women as beneficiaries of

financial transfers made by the project for the provision of technical assistance, of inputs and compensation of labor costs for their participation in landscape restoration activities, forest reforestation, establishment of silvopastoral systems and forest management adapted to climate change. It will be necessary to consider the actual and potential involvement of women in the promoted activities, rather than landowner status. Adequate dissemination of these opportunities will be key to motivate women and bring them closer to these initiatives.

76. Output 2.3: Participation of Agrarian Communities and Producer Organisations in value chains of forest and agroforestry products is strengthened and their livelihood resilience improved

- As part of the activity of Sustainable Forest Management adapted to climate change, inclusion and access to markets, value chains will be strengthened in a climate-sensitive manner, opening opportunities for greater participation of women's groups and organizations or with broad participation of women in the different segments of the chains, especially in economic activities linked to sustainable use and value addition to timber and non-timber forest products.
- In order to strengthen their capacities, gender training will be provided and women will learn about sustainable and climate-resilient production practices, organizational, managerial and marketing strengthening of enterprises within the framework of value chains. More details on the proposed methodology and contents are presented in section 2.1.5 of the previous section on mechanisms and activities identified to address gender gaps in the BALSAS project.
- The range of productive projects include affirmative actions with a gender perspective will be expanded, taking into account the successful experience developed by CONAFOR.
- Considering the backyard as a key space for food production and strategic for the subsistence of families, support will be provided for crop diversification and improved irrigation. Due to the proximity to the household, women will be the most involved and will receive technical assistance, necessary equipment and economic compensation for their work. The improvement in the quantity, quality and variety of food will allow for better consumption within the household, but can also generate surpluses for sale at the community and local level.

Outcome 3: The producers, inhabitants and authorities of the Balsas Basin have timely and relevant climate information for climate risk management, disaster prevention and knowledge management.

77. Output 3.1. Climate-hazards-related early warning and monitoring systems supported

- As part of the development of a platform and applications to facilitate access to climate information and early warning system, a participatory diagnosis will be needed to identify possible constraints in access to information that women may face so that measures can be taken to overcome these constraints.
- Considering the potential availability of smart phones, women will be trained in the use of these devices and in the use of the integrated climate information system. In addition, in response to the diagnosis developed, alternative ways of communicating information

through other media such as radio and written information will be developed if necessary.

- Regarding CONAFOR's plot monitoring and MRV system, capacities will be built to collect, analyse and report on quantitative and qualitative indicators related to gender and gender participation, social and environmental safeguards. To this end, mechanisms will be defined for the active participation of local communities in the collection and basic processing of information.

78. Output 3.2. Policy-relevant knowledge products elaborated from the systematic documentation of good practices and intervention approaches of the project

- Systematization of women's experiences and good practices on the use of climate resilient agronomic technologies and techniques that are relevant for knowledge sharing, adoption of new policies and institutional strategies and replication for scaling up will be undertaken.
- In order to disseminate the progress achieved and promote the exchange of knowledge and learning among beneficiaries, workshops will be held with women playing a leading role in sharing their experiences. These workshops will be accompanied by field technicians, who will be able to reinforce and reorient their work according to the findings. Field technicians, who will be able to reinforce and reorient their work according to the findings, will accompany these workshops and lessons learned that emerge from these spaces.
- .

2.4. Matrix of activities, indicators, goals, timeline and people responsible for them.

Impact: The climate resilience of the most vulnerable communities and ecosystems will be increased, and emissions from the Basin's production systems will be reduced with the participation of women as part of actors and direct beneficiaries of the project.				
Outcome 1. Government programs and community development plans integrate climate change and social inclusion criteria for attribution of supports.				
Output 1.1. The knowledge and experiences of women are considered in decision-making at the territorial level, due to their participation in the intersectoral coordination mechanisms strengthened by the project.				
Activities	Indicators	Targets	Timeline	Responsible
Activity 1.1.2. Women, as members of the RCIMT participate in the discussion making and contribute with their skills and knowledge to identify priority areas of attention at the micro-basin level.	% of the local members of the committees are women	At least 40	Y1-Y2	UCP*, CONAFOR, Promotorias
	% of women on committees who report influencing decision-making within the RCIMT framework	50		
Activity 1.1.4 Strengthen and operationalize criteria for adaptation to climate change and inclusion for women in institutional program procedures.	New operational proposals incorporate affirmative actions to improve the participation of women in the production of value chains	3	Y1-Y2-Y3	UCP, Promotorías, CONAFOR
Output 1.2: The processes for developing multi-year investment plans that contain instruments for mitigation and adaptation to climate change have 40% female participation and consider the needs and contributions of women.				
Activities	Indicators	Targets	Timeline	Responsible
1.2.1 Communication and awareness campaigns will be carried out to disseminate calls for project proposals with gender considerations.	Gender-sensitive campaigns and calls for proposals carried out	3	Y1-Y3-Y5	UCP, Promotorías, CONAFOR
	Simplified format for the submission of proposal profiles available to women	1	Y1	
1.2.2 Women participate in consultation and preparation of multiannual investment plans in basins, sub-basins and prioritized ethnic areas that integrate climate change mitigation and adaptation instruments with a gender perspective.	% of women population participating in updating the multiannual investment plans.	40	Y1-Y2	UCP, Promotorías, CONAFOR
	% of the direct beneficiaries of investment plans are women	40	Y1-Y6	
Output 1.3: Extension workers, promoters, local facilitators and “brigades” value the importance of women's participation for climate change adaptation, resilience and 40% of these groups are women.				
Activities	Indicators	Targets	Timeline	Responsible

Activity 1.3.1: Capacity development programs for extension workers, local facilitators and “brigades” with mechanism for the participation of women.	% of the participants are women	40	Y1 to Y6	UCP, Promotorias, CONAFOR
	<i>Module on gender equality is part of capacity development</i>	1 module	Y1 to Y6	
	<i>% of trained people (men and women) who consider that they have improved their attitude towards women's participation in the project.</i>	90	Y3, Y6	
Activity 1.3.2: Support and creation of forest protection brigades and local climate change resilience facilitators	<i>% of women beneficiaries incorporated in brigades and as a local climate change resilience facilitators</i>	40	Y1-Y6	UCP, Promotorias, CONAFOR
Activity 1.3.3: Strengthening the local CONAFOR Promotorias	<i>% de promotorías a cargo de mujeres</i>	50	Y1-Y6	UCP, Promotorias, CONAFOR

Outcome 2: Women access environmental services for sustainable management and better adapted to climate change of forest landscapes, improve their production and increase the resilience of their livelihoods.

Output 2.1. Men and women from agrarian communities conserve, restore and sustainably use forest landscape based multiannual investment plans through payment of environmental services (PES).

Activities	Indicators	Targets	Timeline	Responsible
Development of project selection criteria for women's participation in PES	<i>Manual incluye criterios de selección favorables a la participación de mujeres</i>	1	Y1	UCP, CONAFOR
2.1.1. Promotion of the participation of women landowners and members of groups and communities in the initiatives of PES	<i>% of beneficiaries from PES are women</i>	30	Y1 to Y5	UCP, Promotorias, CONAFOR

Output 2.2. The productive systems of Agrarian Communities are strengthened and better adapted to climate change and variability with the support of trained women who participate and benefit from RFM and PFC

Activities	Indicators	Targets	Timeline	Responsible
Activity 2.2.1: Incentive payments and support to Agrarian Communities to implement Climate Smart Forest Landscape Restoration of Micro-watersheds and Strategic Areas (RFM)	<i>% of women direct beneficiaries of RFM incentive payments and support</i>	30	Y1 to Y6	UCP, Promotorias, CONAFOR
Activity 2.2.2 Incentive payments and support to Commercial Forest and Agroforestry Plantations Adapted to Climate Change (PFC)	<i>% of direct beneficiaries of PFC incentive payments and support are women</i>	50	Y1 to Y6	UCP, Promotorias, CONAFOR
	<i>% of women who consider that their income has improved due to their participation in RFM and PFC</i>	80	Y1 to Y6	

Output 2.3: Women and men from Agrarian Communities and Producer Organizations participate in the value chains of forestry and agroforestry products and improve the resilience of their livelihoods.

Activities	Indicators	Targets	Timeline	Responsible
Activity 2.3.1 Sustainable Forest Management adapted to climate change, inclusion and access to markets (MFCCCV)	Total % of women beneficiaries	40	Y1-Y6	UCP, Promotorias, CONAFOR
	<i>% del total de propuestas/proyectos son dirigidos y/o gestionados por mujeres</i>	30	Y1-Y6	UCP, Promotorias, CONAFOR
Sub Activity 2.3.1.4 Support women in benefiting from forest-based value chains especially for production processes, fair trade and certified value chains	<i>Number of women are beneficiaries of the project in the value chains</i>	5000	Y1-Y6	UCP, Promotorias, CONAFOR
As part of Sub Activity 2.3.1.4 Comprehensive productive projects for women that include affirmative actions with a gender perspective will be implemented.	<i>Number of comprehensive projects for women executed</i>	50	Y2-Y4	UCP, Promotorias, CONAFOR
	<i>Number of women participate in comprehensive projects for women</i>	500		

Activity 2.3.2. Support households to improve food security and nutrition, including water harvesting projects	% of households are women who received technical assistance, supplies, and labor cost compensation	At least 60	Y1-Y5	UCP, Promotorias, CONAFOR
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Outcome 3: The producers, inhabitants and authorities (men and women) of the Balsas Basin have climate information and documentation of good practices and intervention approaches with a gender perspective of the project.

Output 3.1. Information from surveillance and early warning systems related to climate dangers is used by women.

Activities	Indicators	Targets	Timeline	Responsible
Activity 3.1.1 Carry out a participatory diagnosis to identify the necessary measures to guarantee access to and use of the climate information platform and early warning system. by women	<i>Participatory diagnosis carried out as part of the baseline of the project and identified barriers and opportunities</i>	1	Y1	UCP, MRV-CONAFOR
	<i>Measures adopted for greater access to information by women according to diagnosis</i>		Y1	
Strengthening the CONAFOR Monitoring Report and Verification System (MRV) including that local communities participate in the collection of information on gender and social and environmental safeguards.	Information on gender and social and environmental safeguards is collected		Y1, Y2	UCP, MRV-CONAFOR
Sub Activity 3.1.3.a. <i>Improved plot monitoring and MRV systems to allow the local facilitator to collect field information on gender and inclusion, environmental safeguards.</i>	<i>Plot monitoring systems and MRV have field information on gender</i>		Y2	UCP, MRV-CONAFOR
Sub Activity 3.1.3.b. Women will be trained in the use of the integrated climate information system and in the handling of smartphones to access information	% of women have access to climate information	40	Y2-Y5	UCP, MRV-CONAFOR

Output 3.2. Policy-relevant knowledge products elaborated from the systematic documentation of good practices and intervention approaches with a gender perspective of the project

Activities	Indicators	Targets	Timeline	Responsible
Activity 3.2.1. Systematize experiences of agronomic techniques and technologies designed and put into practice with a gender approach, so that they can be used in the framework of training for the different actors.	<i>systematizations carried out.</i>	2	Y3, Y5	UCP, CONAFOR
Participatory events for the exchange of knowledge and experiences where women have a leading role.	<i>national face to face events executed</i>	2	Y4, Y6	UCP, CONAFOR
Other activities	Indicators	Targets	Timeline	Responsible

Gender Baseline Study as part of the Project Baseline	<i>Baseline study with a gender perspective has been carried out at the beginning of the project</i>	1	Y1	M&E-UCP, CONAFOR	ETP-
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 *The UCP has a professional in gender and inclusion

2.5. Responses of the Gender Action Plan to the gender gaps identified in the Gender Assessment

79. Considering the results of the Gender Assessment, this GAP and the project in general, seek to provide concrete responses to the identified gaps that can be addressed within the framework of the "BALSAS: Basin Approach for Livelihood Sustainability through Adaptation Strategies" project, taking into account also the identified strengths, which are detailed in the last section of the Gender Assessment called Main findings for gender equality in the Project.

Gender gaps/barriers identified	Recommended actions
little participation in decision-making spaces	-Participation quotas as an incentive for the participation of women in decision-making bodies. -40% Participation in multisectoral decision-making bodies in the project (RCIMT). -The selection of project proposals led by women is privileged.
long work hours due to domestic overload	-Community water harvesting projects such as collector roofs. -Execution of comprehensive projects for women with attention to their domestic role and use of time. -Partnerships with actors in the territory for care and domestic solutions.
Increased poverty and low income	-Priority for women heads of household in projects on food and nutritional security. -Access to economic compensation for their participation in PES, RFM, CP, Value Chain, and food security and nutrition projects. -Improvements in income due to improvements in production.
Gender prejudices and stereotypes	- Training activities in gender equality for beneficiaries and project teams (men and women). -Preparation and dissemination of materials on the topic.
Gender violence	- Coordination with the offices (secretaries, institutes, etc.) dedicated to the care of women in state and local (municipal) governments. - Identify and/or support the development of protocols with the mechanisms and route of actions for reporting and care for victims of gender violence. - Production of educational material. Actuación del Internal Control Body del MAC de CONAFOR como mecanismo interno de quejas y reclamos
Invisibility of the participation in productive and environmental activities	-Training for project staff and community members on gender equality and the effects on discrimination against women. - Participation quotas for women in the different activities averaging 40% throughout the project
Less access and control of productive resources and assets	-access to technology and the necessary inputs in PES, RFM, CP, Value Chain, and food security and nutrition activities.

Vulnerability to climate change	-All Project activities allow for an increase in climate resilience and better living conditions for women and their families.
fewer links with the mass media	-Diagnosis to identify possible constraints in access to information from the climate information system. - Train women in the use of smart devices and in the use of the integrated climate information system.

2.6. Gender Action Plan Implementation, Monitoring and Evaluation

Implementation

80. The Balsas project has a Steering Committee at the federal level, which is the highest level in decision-making for the Project. Gender issues that will be on the Committee's agenda through the following mechanisms:

- The annual work plans and budget that are presented to the Committee for approval will include the activities and compliance goals, as well as the budgets allocated for gender activities in the project.
- With the support of the project's monitoring and evaluation system, information will be provided to the committee to monitor the results in relation to gender and the improvement measures that must be implemented.
- IFAD will make recommendations or draw attention to the consideration of gender and inclusion aspects, when required, in its capacity as a guest with voice but no vote.
- the Committee will be advised to invite representatives of the CONAFOR Liaison and Gender Unit, or other relevant actors to inform and support the Committee's decision-making in relation to gender and inclusion issues.

81. In the case of the Regional Committees for Integrated Territory Management (RCIMT), the Project will involve women in decision-making for prioritizing areas of intervention, as well as in the development of participatory multiparty alliances. Annual investment plans. Based on the call aimed at women and opening spaces for their participation, it is expected that 40% of the members of these committees will be women as mentioned in the GAP (Product 1.1).

82. For the implementation of this action plan, the project will have human resources specialized in gender to ensure the mainstreaming of the approach and the successful participation of women as executors and beneficiaries of the project. A professional in gender and inclusion will be hired to lead the implementation of this GAP, with the support of specialists in participatory planning in the basin and in social participation and safeguards. Additionally, it is expected that the 26 technicians that will be hired with project resources in the forestry promoters will have previous work experience under this approach. As part of the selection and hiring process, terms of reference will be drawn up that clearly define the commitments and responsibilities in gender and inclusion for the positions.

83. The strengthening of capacities in gender and inclusion will help the technical teams, forestry instructors, extension agents and the beneficiary population itself to direct efforts towards a greater participation of women, the young population and the indigenous population.

84. The support and advice of CONAFOR's specialized gender team and the strengthening of alliances with key actors in the territories are essential.

85. It is necessary to assign adequately prepared personnel to advise the teams, gender training for personnel, and the clear definition of monitoring and evaluation mechanisms for gender issues and women's participation.

Monitoring and Evaluation

86. As part of the project's monitoring and evaluation system, activity-based indicators are considered for each output and component.

87. The possibility of incorporating the results indicator on empowerment will be analyzed, measured by IFAD's Women's Empowerment Index that addresses 10 dimensions: economic autonomy, self-efficacy, attitudes towards domestic violence, participation in production decisions, ownership of land and assets, access to financial services decisions, control over the use of income, balance in the use of time, belonging to organized groups and belonging to influential groups. The result indicator is Beneficiary women showing an improvement in empowerment (Percentage). In the case of this project, the dimensions that make up the index can be analyzed in terms of the following dimensions.

88. The collection of the information required by the index is carried out through a questionnaire that will be applied as part of the baseline at the beginning of the project and at the end during its evaluation.

2.7. Budget

89. Of the project's general budget, it is estimated that around USD 30,874,935 will be allocated to the necessary arrangements to guarantee the provision of quality services with a gender and inclusion approach, as well as the direct allocation of project benefits to women. This amount is distributed according to expected results as follows:

	BUDGET USD
Outcome 1. Government programs and community development plans integrate climate change criteria for attribution of support.	3.488.930
Outcome 2: Forest landscapes are sustainably managed and better adapted to climate change for the provision of environmental services and increase livelihood resilience.	26.380.669
Outcome 3: The producers, inhabitants and authorities of the Balsas PA have timely and relevant climate information for climate risk management, disaster prevention and knowledge management.	1.005.336
TOTAL	30.874.935

90. Presupuesto detallado con enfoque de género según actividades se presenta a continuación:

Impact: The climate resilience of the most vulnerable communities and ecosystems will be increased, and emissions from the Basin's production systems will be reduced with the participation of women as part of actors and direct beneficiaries of the project.			
Outcome 1. Government programs and community development plans integrate climate change criteria for attribution of support.			
Output 1.1. Mechanisms for intersectoral coordination are strengthened to increase the technical and organizational capacities of small producers and facilitate synergies with the different levels of government and the private initiative.			
Activities	Indicators and Targets	Budget (USD)	Criteria
Activity 1.1.2. Women, as members of the Basin, Sub-basin and Ethnic committees participate in the discussion making and contribute with their skills and knowledge to identify priority areas of attention at the micro-basin level.	<i>At least 40% of the local members of the committees are women</i>	244.320	100% as they are committees that ensure women's participation & gender sensitive communication.
Activity 1.1.4 Strengthen and operationalize criteria for adaptation to climate change and inclusion for women in institutional program procedures.	<i>Three new operational proposals that incorporate affirmative actions to improve the participation of women in the production of value chains</i>	141.143	TORs of the Gender and Social Participation Specialist
Output 1.2: Participatory multiannual investment plans within prioritized basins, sub-basins and ethnic territories integrating mitigation instruments and adaptation to climate change.			
Activities	Indicators and Targets	Budget (USD)	Criteria
Women participate in the preparation of multiannual investment plans in basins, sub-basins and prioritized ethnic areas that integrate climate change mitigation and adaptation instruments with a gender perspective.	<i>40 % of women population participating in updating the multiannual investment plans.</i>	390.000	Gender perspective is integrated in 30% of planning instruments.
	<i>30% of the direct beneficiaries of investment plans are women</i>		
1.2.1 Communication and awareness campaigns will be carried out to disseminate calls for project proposals with gender considerations.	<i>Gender-sensitive campaigns carried out</i>	28.000	Gender-sensitive campaign
	<i>Simplified format for the submission of proposal profiles available to women</i>	N/A	No budgetary implication/Technical TORs/ CONAFOR Application forms
Output 1.3: Strengthened capacities of local actors for climate change adaptation, resilience, and inclusiveness.			
Activity 1.3.1: Capacity development programs for extensionists and local facilitators with measures for the participation of women.	40% of the participants are women	66.240	30% of the ongoing investment allocated to women

	<i>One module on gender equality is part of capacity development</i>	30.400	Gender-sensitive training
Activity 1.3.2: Support and creation of forest protection brigades and local climate change resilience facilitators	<i>300 women beneficiaries incorporated in brigades</i>	1.276.117	30% of the investment allocated to women (facilitators or beneficiaries of FP support)
Activity 1.3.3: Strengthening the local CONAFOR Promontories	<i>At least one promotora per sub-basin is run by a woman.</i>	1.267.711	Female promoters (40%)
	<i>The total number of promoters have strengthened their knowledge and skills in gender and social inclusion</i>	45.000	Gender training
Outcome 2: Forest landscapes are sustainably managed and better adapted to climate change for the provision of environmental services and increase livelihood resilience.			
Output 2.1. Agrarian communities conserve, restore and sustainably use forest landscapes based on land-use plans through payment of environmental services (PES).			
Activities	Indicators and Targets	Budget (USD)	Criteria
2.1.1. Promotion of the participation of women landowners and members of groups and communities in the initiatives of PES	30% of beneficiaries from PES are women	6.883.307	30% of PES investment is allocated to women
Output 2.2. The productive systems of Agrarian Communities are strengthened and better adapted to climate change and climate variability.			
Activities	Indicators and Targets		
Activity 2.2.1 Training in production practices and/or technologies	<i>40% of people trained on production practices and/or technologies are women.</i>	N/A	
Activity 2.2.1: identification and incorporation of women as beneficiaries of financial transfers to agricultural communities for their participation in landscape restoration activities, forest reforestation, establishment of silvopastoral systems and forest management adapted to climate change.	<i>Women are 30% of direct beneficiaries</i>	5.368.500	30% of investment in forest restoration and commercial forestry plantations goes to women
Output 2.3: Participation of Agrarian Communities and Producer Organizations in value chains of forest and agroforestry products is strengthened and their livelihood resilience improved.			
Activity 2.3.1 Sustainable Forest Management adapted to climate change, inclusion and access to markets (MFCCCV)		N/A	
Sub Activity 2.3.1.a. Strengthen value chains in a climate-sensitive manner, opening opportunities for the participation of women in the different levels, especially in economic activities linked to sustainable use and value addition to timber and non-timber forest products.	<i>1.458 Women are beneficiaries of the project in the value chains (30%)</i>	3.859.500	30% of the investment in value chain projects MFCCV 2 and 3

Sub Activity 2.3.1.b. Women will be prioritized but also men will be trained in gender equality, climate-resilient production practices, organizational, managerial and marketing strengthening of companies within the framework of value chains.	<i>60% of women beneficiaries in value chains strengthened their capacities with climate-resilient production practices, organisational, managerial, marketing and gender.</i>	541.200	30% of investment in capacity building goes to women
Sub Activity 2.3.1.c. Comprehensive productive projects for women that include affirmative actions with a gender perspective will be implemented.	<i>50 comprehensive projects for women executed</i>	1.500.000	Specific projects
	<i>500 women participate in comprehensive projects for women</i>	N/A	
Activity 2.3.2. Transformation of backyard plots through crop diversification and improved irrigation for better quality and variety of food for the family and for sale	<i>At least 80% of backyard ecosystems are led by women who received technical assistance, supplies, and labor cost compensation</i>	8.228.162	80% of investment in agroforestry/backyard systems
Outcome 3: The producers, inhabitants and authorities of the Balsas Basin have timely and relevant climate information for climate risk management, disaster prevention and knowledge management.			
Output 3.1. Climate hazards related early warning and monitoring systems supported.			
Activities	Indicators and Targets	Budget (USD)	Criteria
Activity 3.1.1 Carry out a participatory diagnosis to identify the necessary measures to guarantee access to and use of the climate information platform and early warning system. by women	<i>Participatory diagnosis carried out as part of the baseline of the project and identified barriers and opportunities</i>	N/A	TORs Gender Specialist
	<i>Measures adopted for greater access to information by women according to diagnosis</i>		
Strengthening the CONAFOR Monitoring Report and Verification System (MRV) including that local communities participate in the collection of information on gender and social and environmental safeguards.			
Sub Activity 3.1.3.a. <i>Improved plot monitoring and MRV systems to allow the local facilitator to collect field information on gender and social, environmental safeguards.</i>	<i>Plot monitoring systems and MRV have field information on gender</i>	489.000	The system is Gender sensitive
Sub Activity 3.1.3.b. Women will be trained in the use of the integrated climate information system and in the handling of smartphones to access information	<i>40% of women</i>		20% is dedicated to specific training for women in the use of smartphones.
Actividad 3.1.4 Sistema de Salvaguardas Sociales y Ambientales			

Consultas del sistema de salvaguardas		30.000	
Contratación de Especialista en Salvaguardas ambientales y sociales		126.336	
Output 3.2. Policy-relevant knowledge products elaborated from the systematic documentation of good practices and intervention approaches of the project			
Activity 3.2.1. Systematize experiences of agronomic techniques and technologies designed and put into practice with a gender approach, so that they can be used in the framework of training for the different actors.	<i>2 systematizations carried out.</i>	40.000	2 of 6 are gender specific
Activity 3.2.2 Participatory events for the exchange of knowledge and experiences where women have a leading role.	<i>2 national face to face events executed</i>	320.000	Exchanges of experiences and women's learning paths
TOTAL		30.874.935	

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